MINISTRY OF HEALTH OF THE REPUBLIC OF UZBEKISTAN
TASHKENT STATE MEDICAL UNIVERSITY
Department of Propaedeutics of Internal Medicine No. 2

PROPAEDEUTICS OF INTERNAL MEDICINE

complete study manual

A study manual for practical classes



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This manual is based on the 14 practical class topics of the calendar-thematic plan (semester V) of the course “Propaedeutics of Internal Medicine” approved for the 2026–2027 academic year. It covers history taking, general inspection, palpation, percussion and auscultation, the main syndromes in diseases of the respiratory and cardiovascular systems, and the symptoms of rheumatism, heart valve defects and infective endocarditis.

Each chapter consists of learning objectives, a theoretical part (with “Remember”, “Clinical significance”, “Memory aid” and “Caution” boxes), a clinical part, practical skill algorithms, review questions, 15 test questions, 3 clinical cases and a quick review table. The answers to the test questions are given at the end of the book.

The manual is based on the textbooks “Propaedeutics of Internal Diseases” by A. Gadayev, M.Sh. Karimov and X.S. Axmedov (2012) and “Internal Diseases” by N.M. Kamolov and M.U. Qo‘yliyev (1991), supplemented by the main Russian- and English-language propaedeutics textbooks (N.A. Mukhin, V.S. Moiseyev; A.L. Grebenev; Bates’ Guide; Macleod’s Clinical Examination) and current international clinical guidelines (GINA 2025, GOLD 2026, ESC 2023, ESC/EACTS 2025, AHA 2015). The figures are taken from the main textbooks, and the source is given under each figure. Labels inside some figures remain in the language of the original source.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks and sources listed. The compiler selected the material, defined its structure and checked it. Before clinical information is applied in practice, it is recommended to compare it with the main textbooks and the current clinical protocols.

The manual is intended for 3rd-year students of the faculties of general medicine of higher medical education institutions.

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Preface

Dear student! Propaedeutics of internal medicine is the gateway to clinical medicine. It is in this subject that you meet a patient face to face for the first time, listen to the complaints, percuss the chest and listen to the heart sounds. The skills mastered at this stage remain your main tools throughout your career as a physician: even the most modern equipment cannot replace careful history taking and attentive physical examination.

The manual covers the 14 practical class topics of the department's calendar-thematic plan approved for the 2026–2027 academic year:

- topics 1–3 — the general principles of examining a patient: history taking, the case history scheme, general inspection, fever, anthropometry and palpation;
- topics 4–9 — the respiratory system: complaints, inspection, palpation, percussion, auscultation and the main pulmonary syndromes (bronchial obstruction, emphysema, consolidation, fluid and air in the pleural cavity, a cavity in the lung);
- topics 10–14 — the cardiovascular system: inspection and palpation, percussion and the pulse, auscultation, rheumatism, mitral and aortic defects, infective endocarditis.

How should the manual be used? Every chapter has the same structure. First read the “Learning objectives” to find out what you should know and be able to do. The colored boxes in the theoretical part highlight the key points:

REMEMBER
Yellow box — information that must be memorized: normal values, classifications, key definitions.

CLINICAL SIGNIFICANCE
Blue box — the practical significance of the information, clinical examples.

MEMORY AID
Green box — techniques and abbreviations that make memorizing easier.

CAUTION
Orange box — common mistakes, discrepancies between textbooks and updated international standards.

The “Clinical part” shows how to apply the knowledge of the chapter at the bedside and how to record it in the case history. Practice the algorithms in the “Practical skills” section step by step in the practical class. The review questions, tests and clinical cases at the end of each chapter are intended to check your knowledge; the answers to the tests are given at the end of the book. The “Quick review” table is convenient for a short revision before an exam.

Some normal values differ slightly between the Uzbek textbook and the Russian and English literature. In such cases the manual cites all the sources and recommends the value that agrees with current international guidelines.

I wish you success!

The compiler

TOPIC 1

The tasks of the propaedeutics of internal medicine. The order of patient examination. The case history outline. History taking

Learning objectives

To introduce the student to the content of the propaedeutics of internal medicine, the order of the clinical examination of the patient and the outline for keeping the case history (medical record); to teach the student to collect the complaints, the history of present illness (anamnesis morbi) and the life history (anamnesis vitae) from the patient independently.

By the end of the session, the student should know:

- the concepts of propaedeutics, semiotics, symptom, syndrome and diagnosis;
- the main (physical) and additional methods of the clinical examination of the patient;
- the structure of the case history and what is written in each section;
- the principles of dividing complaints into main and additional groups;
- the plan of the anamnesis morbi and the anamnesis vitae;
- the principles of medical deontology and medical confidentiality.

The student should be able to:

- start the conversation with the patient correctly, introduce himself or herself and obtain consent;
- use open and closed questions appropriately;
- analyze each complaint in detail according to a plan;
- record the collected information in the case history competently.

1. The propaedeutics of internal medicine: the subject, its aim and tasks

Propaedeutics (Greek propaideuo — “to teach beforehand, to introduce into a science”) is the introductory stage of the clinical disciplines. The propaedeutics of internal medicine teaches the future physician the methods of examining the patient, how to detect and interpret the signs of disease and make a syndromic diagnosis, as well as medical ethics and deontology. This knowledge is needed not only in the senior years but throughout the physician’s whole career.

The discipline has three main parts:

1. **Diagnostic technique** — the methods of history taking, inspection, palpation, percussion and auscultation of the patient, as well as laboratory and instrumental investigations.
2. **Semiotics (semiology)** — the study of the signs of disease (symptoms), their mechanism of origin and diagnostic significance.
3. **Diagnostic methodology** — clinical reasoning, combining the signs into syndromes and the rules for formulating the diagnosis.

REMEMBER

The main tasks of propaedeutics:

- to teach a systematic and complete examination of the patient;
- to teach how to detect and interpret symptoms and syndromes;
- to teach how to make a syndromic (provisional) diagnosis;
- to teach how to keep the case history correctly;
- to instill medical ethics and deontology.

The concepts of health and disease

According to the Constitution of the World Health Organization (WHO), **health** is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.

Disease is a complex general response of the organism to the action of harmful factors, in which structural, functional and metabolic changes and a breakdown of adaptive processes are observed and the capacity for work is limited. The causes of disease are mechanical, physical, chemical, biological, psychological and hereditary (genetic) factors.

2. The concepts of symptom, syndrome and diagnosis

A **symptom (sign)** is a single manifestation of disease. Symptoms are classified according to several principles.

Table 1.1. *Classification of symptoms*

Basis of classification	Type	Example
Who detects it	Subjective (felt by the patient)	Pain, nausea, a sensation of shortness of breath
	Objective (detected by the physician)	Jaundice, liver enlargement, edema
Significance	General (nonspecific)	Fever, weakness, weight loss
	Local (organ-specific)	Cough — in respiratory diseases
	Pathognomonic (specific to one disease only)	A diastolic murmur and the “quail rhythm” — in mitral stenosis
Time of appearance	Early and late signs	Cyanosis — a late sign

A **syndrome** is a combination of symptoms that share a common pathogenesis. For example, the syndrome of lung tissue consolidation: a dull note on percussion, increased vocal fremitus and bronchial breathing. One syndrome may occur in several diseases (consolidation — pneumonia, pulmonary infarction, tumor).

A **diagnosis** is the physician’s brief and precise conclusion about the patient’s disease. The process of making a diagnosis proceeds in the following sequence.

Table 1.2. *Stages of making a diagnosis*

Stage	Content
1	Collecting data: history taking, physical examination, additional investigations
2	Analysis of the signs (assessing the significance of each sign)
3	Combining the signs into syndromes according to their pathogenetic connection (synthesis)

Stage	Content
4	Formulating the primary (provisional) diagnosis
5	Differential diagnosis
6	Final (clinical) diagnosis

Types of diagnosis

- **Primary (provisional)** — made after the first examination of the patient.
- **Clinical (main)** — made after clinical, laboratory and instrumental investigations, usually within three days.
- **Final** — made before the patient is discharged from hospital (or after death).

The structure of the clinical diagnosis: **the main disease**, **complications of the main disease**, **a competing disease**, **a background disease** (for example, diabetes mellitus) and **concomitant diseases**. A complete diagnosis includes anatomical (morphological), functional, pathogenetic and etiological components.

> **CLINICAL SIGNIFICANCE**
> An example of a complete clinical diagnosis: “Rheumatic heart disease: mitral stenosis (moderate). Complication: atrial fibrillation, permanent form. Chronic heart failure stage IIA, NYHA functional class III.” Here the morphology (the defect), function (the failure), complication (the arrhythmia) and etiology (rheumatic) are reflected in a single conclusion.

Methods of clinical reasoning

- **Induction** — moving from particular signs to a general conclusion: the signs are collected and combined into syndromes, and then a conclusion about the disease is drawn.
- **Deduction** — from general knowledge to the particular: the physician looks in the patient for the other signs characteristic of the disease he or she suspects and confirms or rejects the hypothesis.
- **Diagnosis ex juvantibus** — if treatment directed against the suspected disease is effective, the diagnosis is considered confirmed.

> **CAUTION**
> It is a mistake to base a diagnosis on a single sign. For example, ascites is not specific to liver cirrhosis: it also occurs in heart failure, nephrotic syndrome and tumors of the peritoneum. The combination of all the signs must always be analyzed.

3. Medical deontology and communication with the patient

Medical deontology (Greek deon — duty, logos — study) is the study of the physician’s professional duty and conduct. **Medical ethics** is a broader concept covering the relationships of the health worker with the patient, his or her relatives, colleagues and society.

Great contributions to the history of deontology were made by Hippocrates (“Do no harm” — primum non nocere), Avicenna (Ibn Sina) (“A physician should have the eyes of a falcon, the hands of a maiden, the wisdom of a serpent and the heart of a lion”) and, later, the Declaration of Geneva (1948), the International Code of Medical Ethics (1949) and the Declaration of Helsinki (1964).

REMEMBER

The basic rules of communicating with the patient:

- introduce yourself, address the patient by name and obtain consent for the examination;
- sit face to face with the patient, with your eyes at the same level, in a relaxed position;
- listen to the patient without interrupting and show your attention by nodding;
- hide your negative emotions and stay calm even if you detect a serious sign;
- keep medical confidentiality: information about the patient is used only for treatment.



Figure 1.1. *The position of the physician and the patient during the conversation (Gadayev A. et al., 2012, Fig. 1)*

Types of questions

Table 1.3. *Questions used in history taking*

Type of question	Purpose	Example
Open question	Lets the patient speak freely	“What is bothering you?”, “How did your illness begin?”
Closed question	Obtaining precise information (“yes” or “no”)	“Does a cough bother you at night?”
Clarifying question	Checking that the patient’s meaning has been understood correctly	“So the pain appears when you climb stairs, is that right?”
Leading question (not recommended)	Determines the answer in advance	“The pain spreads to your left arm, right?”

CAUTION

Leading (suggestive) questions push the patient toward a wrong answer and lead to diagnostic error. The conversation should begin with open questions and then move on to closed questions for clarification (the “funnel” technique).

4. Methods and order of the clinical examination of the patient

The examination of the patient is divided into **main (physical)** and **additional (laboratory and instrumental)** methods.

Table 1.4. *Methods of examining the patient*

Group	Method	Latin name	What is determined
Subjective	History taking	Interrogatio, anamnesis	Complaints, history of the illness and of the patient's life
Objective (physical)	Inspection	Inspectio	General condition, skin, body build, shape of the organs
	Palpation	Palpatio	Condition of the organs, pain, vibration, pulsation
	Percussion (tapping)	Percussio	Borders of the organs, density and air content of the tissues
	Auscultation (listening)	Auscultatio	Breathing, heart sounds, murmurs, intestinal peristalsis
Additional	Laboratory	—	Analysis of blood, urine, sputum and aspirates
	Instrumental	—	X-ray, ECG, US, endoscopy, spirometry, CT, MRI

REMEMBER
The order of the examination is always the same: **history taking → general inspection → inspection, palpation, percussion and auscultation of the organ systems → additional investigations**. The systems are usually examined in the following order: the respiratory, cardiovascular, digestive, urinary, endocrine, musculoskeletal and nervous systems.

In the opinion of experienced clinicians, history taking provides about half of the information needed for the diagnosis, the physical examination about 30 percent and laboratory and instrumental investigations about 20 percent. For example, the diagnosis of angina pectoris can in many cases be made from the characteristics of the pain.

MEMORY AID
The four methods of physical examination — **"IPPA"**: **I**nspection, **P**alpation, **P**ercussion, **A**uscultation. Never change the order: first we look, then we touch, then we tap, and last of all we listen.

5. Outline of the case history (medical record)

The case history is the main medical and legal document in which all the information about the patient is recorded. The student's educational case history is written according to the same outline.

Table 1.5. *Structure of the case history*

№	Section	Content
1	Passport section	Full name, age, sex, nationality, occupation, place of work, address, date of admission, referred by whom
2	Complaints (querelae)	Main and additional complaints, each in detail
3	History of present illness (anamnesis morbi)	When and how the disease began, its course, treatment, the latest deterioration

№	Section	Content
4	Life history (anamnesis vitae)	Growth and development, living conditions, occupation, harmful habits, past illnesses, allergy, heredity
5	Present status (status praesens)	General inspection and objective examination by systems
6	Local status (status localis)	Detailed information on the main affected organ
7	Primary diagnosis and its justification	The syndromes identified and the conclusion based on them
8	Plan of investigation	Laboratory and instrumental investigations
9	Results of additional investigations	The tests and their interpretation
10	Differential diagnosis	Distinguishing from similar diseases
11	Clinical diagnosis and its justification	Main disease, complications, concomitant diseases
12	Etiology and pathogenesis	With reference to this patient
13	Treatment plan	Regimen, diet, medicines (with prescriptions)
14	Daily progress notes (curation)	Daily dynamics of the condition
15	Discharge summary (epicrisis)	Brief conclusion, outcome, recommendations
16	References	Sources used

CAUTION
The case history is a legal document. Every entry made in it is confirmed with the date, time and signature; corrections are made only in the established manner. Giving information about the patient to third parties without his or her consent is a breach of medical confidentiality.

6. History taking (interrogatio)

Although history taking seems simple, it is the most complex and the most valuable method of examination. It consists of the following parts: passport data → complaints → history of present illness → life history.

6.1. Passport data

The patient’s age, sex, occupation and place of residence are important for the diagnosis. For example, dyspeptic symptoms that appear for the first time in old age raise suspicion of a tumor, whereas in a young person they are most often associated with a functional disorder. The occupation helps to suspect occupational diseases (pneumoconiosis, intoxication).

6.2. The patient’s complaints (querelae)

Complaints are divided into two groups:

- **Main (leading) complaints** — the signs that made the patient seek medical help and that are characteristic of the main affected organ (for example, cough and sputum, pain in the heart region).
- **Additional (general) complaints** — general weakness, easy fatigability, fever, sweating, loss of appetite, sleep disturbance, change in body weight.

Once the main complaints have been established, the **review of systems** is carried out: all the systems are briefly asked about (respiratory, circulatory, digestive, urinary, nervous, endocrine and musculoskeletal systems and the sense organs) to detect the signs that the patient has forgotten or not paid attention to.

Each complaint, especially pain, is analyzed in detail according to the following plan.

Table 1.6. *Plan for analyzing pain (a complaint)*

Feature	Question to ask	Example (angina pectoris)
Site	Where does it hurt?	Behind the sternum
Onset	When and after what did it begin?	On physical exertion
Character	What kind of pain is it?	Squeezing, pressing
Radiation (irradiation)	Where does it spread to?	To the left arm, shoulder blade, lower jaw
Duration	How long does it last?	2–10 minutes
Exacerbating and relieving factors	What relieves it?	Rest, nitroglycerin
Severity	How much on a 0–10 scale?	6–7 points
Associated symptoms	What else happens?	Shortness of breath, sweating, fear

MEMORY AID

In English-language textbooks (Macleod's), the analysis of pain is remembered as **SOCRATES**: **S**ite, **O**nset, **C**haracter, **R**adiation, **A**ssociations (accompanying symptoms), **T**ime course, **E**xacerbating/relieving factors, **S**everity.

6.3. History of present illness (anamnesis morbi)

The history of present illness reflects the course of the disease from its onset to the present day. It answers the following questions:

1. When, where and under what circumstances did the patient fall ill?
2. How did the disease begin — suddenly (acutely) or gradually?
3. In the patient's opinion, what caused the disease (chilling, physical or emotional strain, dietary errors, infection, occupational factors)?
4. What were the first signs?
5. When and where was medical help first provided, and with what result?
6. How has the disease progressed: the dynamics of the symptoms, periods of exacerbation and remission and their duration?
7. What investigations were carried out and with what results (outpatient card, discharge summaries, ECG, X-ray films)?
8. What treatment did the patient receive, and how effective was it?
9. What caused the latest deterioration, and what led to the present hospital admission?

6.4. Life history (anamnesis vitae)

The life history is the patient's medical biography. It helps to identify the factors predisposing to disease and the risk factors.

Table 1.7. *Plan of the life history*

Section	What is determined
General biography	Place of birth, which child in the family, growth and development in childhood, education
Living conditions	Housing, family status, regularity and quality of meals, physical activity
Occupational history	Since when the patient has worked, occupation, working conditions, occupational hazards (dust, chemicals, noise, vibration), military service
Harmful habits	Smoking (from what age, how many a day), alcohol, drugs
Past illnesses	In chronological order from childhood: infections, tuberculosis, hepatitis, sexually transmitted diseases, injuries, operations, blood transfusions
Allergy history	Reactions to drugs, foods and household allergens: urticaria, angioedema (Quincke's edema), anaphylactic shock
Family history	Arterial hypertension, IHD, diabetes mellitus, tumors, tuberculosis and mental illness in close relatives
Gynecological and obstetric history (in women)	Age at menarche, regularity of menstruation, pregnancies, deliveries, abortions, menopause
Epidemiological history	Contact with an infectious patient, travel, contact with animals, injections
Insurance and disability	Disability group, duration of temporary incapacity for work

REMEMBER

The smoking index (**"pack-years"**) = number of cigarettes smoked per day × years of smoking / 20. When the index exceeds 10 pack-years, the risk of chronic obstructive pulmonary disease (COPD) rises considerably.

CLINICAL SIGNIFICANCE

The key to the diagnosis often lies in the life history: frequent tonsillitis and joint pain in childhood raise suspicion of rheumatic heart disease; mining — of silicosis; prolonged fever after a tooth extraction — of infective endocarditis; frequent injections — of hepatitis and endocarditis.

7. A brief history of the development of internal medicine and propaedeutics

The methods of examining the patient have taken shape over thousands of years. Knowing their history helps to understand the meaning of each method more deeply.

Table 1.8. *Important milestones in the history of internal medicine*

Period	Scientist	Contribution
5th–4th centuries BC	Hippocrates	Made observation and questioning of the patient the basis of medicine; described direct auscultation (with the ear placed on the chest), the "Hippocratic face", "Hippocratic fingers" and the "succussion splash" of fluid and air in the pleural cavity
2nd century	Galen	Study of the pulse, anatomical and physiological observations
10th–11th centuries	Avicenna (Ibn Sina)	"The Canon of Medicine": more than 10 types of pulse, inspection of urine and stool, systematization of the signs of disease; precepts on medical ethics
1761	L. Auenbrugger	Discovered the method of percussion ("Inventum novum")
1808	J. Corvisart	Introduced percussion widely into practice and developed the semiotics of heart disease

Period	Scientist	Contribution
1816–1819	R. Laennec	Invented the stethoscope and created indirect auscultation and the nomenclature of breath sounds
1828	P. Piorry	Proposed the pleximeter
Mid-19th century	J. Škoda	Developed the physical basis of percussion and auscultation
19th century	S.P. Botkin, G.A. Zakharyin	Russian clinical schools: Botkin — viewing disease as a disease of the whole organism, and clinical reasoning; Zakharyin — raised history taking to the level of an art
1896–1905	S. Riva-Rocci, N.S. Korotkov	Method of measuring blood pressure (Korotkov sounds, 1905)
1887–1903	A. Waller, W. Einthoven	Electrocardiography
1909	V.P. Obraztsov, N.D. Strazhesko	Deep sliding methodical palpation of the abdominal organs; among the first to describe the clinical picture of myocardial infarction (1909)
1895	W. Röntgen	Discovery of X-rays — the beginning of the era of imaging the internal organs

The formation of the clinical school of internal medicine in Uzbekistan is linked to the founding of the medical faculty of Turkestan (later Central Asian) State University in the 1920s. The Tashkent school of therapy was founded by Professor A.N. Kryukov; later, many Uzbek clinicians emerged and developed the schools of propaedeutics, cardiology, pulmonology and gastroenterology in the republic.

CLINICAL SIGNIFICANCE

History shows that every physical method was “high technology” in its time. Even though CT and MRI exist today, the stethoscope and percussion remain the only examination methods that are used at every patient’s bedside, are free and safe and can be applied anywhere.

8. Review of systems

The review of systems (anamnesis per systemae, review of systems) is carried out to detect the signs that the patient has not mentioned. It is performed after the main complaints and the history of present illness, with short, precise questions. Concomitant diseases are often discovered at this stage.

Table 1.9. *Plan of the review of systems*

System	Main signs to ask about
General condition	Weakness, easy fatigability, fever, chills, night sweats, change in body weight (over what period, how many kg), appetite, thirst
Skin and lymph nodes	Rash, itching, change in color, changes in the hair and nails, enlargement of the lymph nodes
Respiratory system	Cough, sputum, hemoptysis, dyspnea, choking attacks, chest pain, snoring and pauses in breathing during sleep
Cardiovascular system	Pain in the heart region, palpitations and skipped beats, dyspnea, orthopnea, nocturnal choking attacks, edema, syncope, leg pain on walking
Digestive system	Appetite, difficulty swallowing, heartburn, belching, nausea, vomiting, abdominal pain, bowel movements (frequency, character, admixture of blood), jaundice

System	Main signs to ask about
Urinary system	Frequency of urination and pain on urination, nighttime urination (nocturia), color and amount of urine, low back pain, edema
Endocrine system	Thirst, polyuria, intolerance of heat or cold, sweating, change in weight, enlargement of the neck
Musculoskeletal system	Joint pain, swelling, morning stiffness and its duration, muscle pain, low back pain
Nervous system and mental state	Headache, dizziness, syncope, seizures, sensory and motor disturbances, sleep, mood, anxiety
Blood system	Bleeding, bruising, bleeding gums, frequent infections
Vision and hearing	Decreased vision, double vision, tinnitus, decreased hearing

CAUTION

The review of systems is not a “formality”. For example, in a young woman who complains only of joint pain, questioning may reveal a facial rash and hair loss, pointing to systemic lupus erythematosus; in a patient admitted with a respiratory disease, the discovery of nocturia and edema may lead to heart or kidney disease.

9. Special situations in analyzing complaints

9.1. Features related to age, sex and occupation

The same complaint is interpreted differently in different patients:

- **Age:** chest pain in a 20-year-old man is most often due to a musculoskeletal or pleural cause, whereas in a 60-year-old male smoker it is due to angina pectoris; constipation or weight loss appearing for the first time in the elderly raises suspicion of a tumor.
- **Sex:** in women, angina pectoris often runs an atypical course (weakness, dyspnea, nausea); iron deficiency anemia, thyroid diseases and systemic lupus erythematosus are more common in women.
- **Occupation:** silicosis in miners and stonecutters; byssinosis and the effects of pesticides in cotton growing; diseases of the vocal apparatus in teachers and singers; low back pain, hemorrhoids and ischemic heart disease in drivers; viral hepatitis and tuberculosis in health workers.

9.2. Pain rating scales

Because the intensity of pain is subjective, expressing it in numbers makes it easier to follow its dynamics.

Table 1.10. *Pain rating scales*

Scale	Content
Numeric rating scale (NRS)	0 — no pain, 10 — the worst pain imaginable; 1–3 — mild, 4–6 — moderate, 7–10 — severe
Visual analog scale (VAS)	The patient marks the intensity of pain on a 10 cm line
Faces scale (Wong–Baker)	Pictures of faces for children and patients who cannot speak

9.3. The patient's "words" and medical terms

Patients express their sensations in everyday expressions. The student must interpret them correctly.

Table 1.11. *Patients' expressions and their possible meaning*

Patient's expression	What needs clarifying	Possible medical meaning
"My heart is being squeezed"	Is it pain or anxiety? Is it related to exertion?	Angina pectoris, anxiety disorder
"My heart is jumping"	Is it fast beating or skipped beats?	Tachycardia, extrasystole
"I can't get enough air"	When, on what exertion? Does it get worse when lying down?	Dyspnea, orthopnea
"My insides burn"	Behind the sternum or in the epigastrium? Is it related to food?	Heartburn, gastritis
"My bowels are blocked"	How often are the bowels opened?	Constipation
"My head is spinning"	Do the surroundings spin, or does the vision go dark?	Vestibular vertigo, or hypotension, arrhythmia
"I have too much blood"	Has the blood pressure been measured?	Arterial hypertension

10. Modern principles of medical ethics

Modern medical ethics (biomedical ethics) rests on four basic principles (T. Beauchamp and J. Childress):

Table 1.12. *The basic principles of biomedical ethics*

Principle	Content	Example in practice
Nonmaleficence	No intervention should cause more harm than benefit	Refraining from unjustified invasive investigations
Beneficence	Putting the patient's interests first	Choosing the most effective, evidence-based treatment
Patient autonomy	The patient has the right to make decisions about his or her own health	Informed consent, respecting the right to refuse treatment
Justice	Equal and impartial treatment of all patients	Equal care regardless of nationality, religion or social status

Informed consent is voluntary consent obtained after the purpose, benefits, risks and alternatives of an investigation or treatment have been explained to the patient in understandable language. Before invasive procedures (puncture, endoscopy, surgery) it is documented in writing.

REMEMBER

When working with a patient, the student:

- introduces himself or herself as a student and obtains the patient's consent;
- does not give the patient independent information about the diagnosis or prognosis — this is the task of the attending physician;
- does not share information about the patient (photos, case history) on social media;
- respects the patient's decision if he or she refuses.

A modern model of the consultation

According to the **Calgary–Cambridge model**, widely used in international practice, the consultation consists of the following stages:

1. **Initiating the session** — introductions, identifying the reason for the patient's visit, agreeing on the plan of the conversation.
2. **Gathering information** — moving from open to closed questions, identifying the patient's ideas, concerns and expectations.
3. **Physical examination.**
4. **Explanation and planning** — explaining the findings in understandable language and making decisions together with the patient.
5. **Closing the session** — summarizing, explaining the next steps and what to do if "red flag" symptoms appear.

11. Typical errors in clinical reasoning

Most diagnostic errors arise not from a lack of knowledge but from errors in reasoning.

Table 1.13. *Cognitive errors in clinical reasoning*

Error	Content	Example
Anchoring	Clinging to the first impression and ignoring new information	A patient who came with a cough is labeled "bronchitis", and the hemoptysis and weight loss found later are ignored
Premature closure	Stopping the search once a diagnosis has been made	Anemia is found, but its cause (intestinal bleeding) is not sought
Confirmation bias	Looking only for signs that confirm one's own hypothesis	Thinking of angina pectoris and failing to hear the signs of pericarditis (a friction rub)
Availability error	Recalling only the most recently seen or vivid cases	Regarding any fever as influenza during an influenza epidemic
Diagnostic overshadowing	Not taking seriously the complaints of a patient with a mental illness or alcoholism	Missing a head injury in an intoxicated patient

MEMORY AID
Protection against errors is the question **"What else could it be?"** After every diagnosis, list at least two alternative diseases and look for the signs that rule them out or confirm them.

12. General semiotics: common general signs

Some signs are not specific to a single organ but occur in many diseases. Although they are "nonspecific", they are often the first and only manifestation of the disease.

Table 1.14. *General signs and their main causes*

Feature	Main causes	What to clarify during history taking
Weakness, easy fatigability	Anemia, hypothyroidism, diabetes mellitus, heart failure, chronic infections, tumors, depression	Since when, in the morning or in the evening, does it pass after rest

Feature	Main causes	What to clarify during history taking
Weight loss (more than 5 % in 6–12 months)	Tumors, thyrotoxicosis, diabetes mellitus, tuberculosis, HIV, malabsorption, depression	Is the appetite preserved (in thyrotoxicosis and diabetes the appetite is good)
Night sweats	Tuberculosis, lymphomas, chronic infections, infective endocarditis, menopause	Does it force the patient to change the nightshirt
Fever	Infections, tumors, connective tissue diseases	Type, duration, with or without chills
Itching	Cholestasis, chronic kidney disease, Hodgkin lymphoma (lymphogranulomatosis), erythremia (after bathing), allergy	With or without a rash
Thirst and polyuria	Diabetes mellitus, diabetes insipidus, hypercalcemia, diuretics	How much fluid the patient drinks a day, whether he or she gets up at night

CLINICAL SIGNIFICANCE

“B symptoms” — fever (above 38 °C), night sweats and loss of more than 10 % of body weight in 6 months — are important prognostic signs in lymphomas. The presence of all three signs together always requires a thorough investigation.

12.1. Pathophysiological types of pain

Table 1.15. *Types of pain*

Type	Source	Features	Example
Somatic (superficial and deep)	Skin, muscles, bones, parietal pleura and peritoneum	Well localized, sharp, worse on movement	Pleurisy, rib fracture
Visceral	Internal organs (distension of hollow organs, ischemia, inflammation)	Vague, diffuse, squeezing or colicky, with nausea and sweating	Angina pectoris, colic
Referred (radiating)	The internal organ and the skin area are innervated from the same spinal cord segment	Felt in a distant skin area	The left arm in angina pectoris, the right shoulder blade in gallbladder disease
Neuropathic	Damage to nerve fibers	Burning, “like an electric shock”	Postherpetic neuralgia, diabetic neuropathy

Zakharyin–Head zones — pain and increased sensitivity (hyperesthesia) in certain areas of the skin in diseases of the internal organs. For example, for the heart — the left shoulder girdle, the inner surface of the left arm and the region of the left shoulder blade (segments CIII–CIV and ThI–ThV); for the liver and gallbladder — the right hypochondrium and the region of the right shoulder blade.

13. An overview of laboratory and instrumental methods of investigation

Besides the physical methods, the discipline of propaedeutics also studies the purpose and capabilities of the main additional investigations.

Table 1.16. *The main additional methods of investigation*

Method	Principle	Diagnostic capabilities
Complete blood count	Formed elements, hemoglobin, ESR	Anemia, inflammation, leukemias, allergy (eosinophilia)
Blood chemistry	Glucose, bilirubin, enzymes, creatinine, lipids, electrolytes	Disorders of the liver, kidneys, heart and metabolism
Urinalysis	Physical properties, protein, glucose, sediment	Diseases of the kidneys and urinary tract, diabetes
X-ray (radiography)	Different absorption of X-rays by the tissues	Lungs, cardiac shadow, bones, intestinal obstruction
Ultrasound (US)	Reflection of ultrasound waves	Liver, gallbladder, kidneys, thyroid gland, heart (echocardiography), pleural fluid
Electrocardiography (ECG)	Recording the electrical activity of the heart	Arrhythmias, ischemia, infarction, hypertrophy
Endoscopy	Viewing hollow organs with an optical instrument	Gastrointestinal tract, bronchi; taking a biopsy
CT and MRI	Layer-by-layer images	Tumors, hemorrhages, thromboembolism, the brain
Functional tests	Measuring the function of an organ	Spirometry, bicycle ergometry, Holter monitoring
Radioisotope methods	Accumulation of radioactive isotopes	Thyroid gland, myocardial perfusion, bones

CAUTION

Every additional investigation is ordered **to answer a clinical question**. Ordering many tests “just in case” exhausts the patient, increases costs and, because of incidental findings (“incidentalomas”), may lead to unnecessary invasive procedures. A correctly collected history and a meticulous physical examination reduce the number of investigations needed.

14. Medical records and legal responsibility

The case history (the medical record of the inpatient) and the outpatient card are **legal documents** that confirm the patient’s rights, the physician’s work and the quality of medical care. The following requirements apply to them:

- entries are made **on time**, accurately and legibly, with the date and time indicated and with a signature;
- the results of investigations, the treatment prescribed and the patient’s consent (or refusal) are recorded;
- corrections are made in the established manner (without erasing the old entry: it is crossed out, dated and signed);
- information about the patient is **medical confidentiality** and is not given to third parties except in cases established by law.

The Law of the Republic of Uzbekistan “On the Protection of the Health of Citizens” establishes the patient’s rights — respectful treatment, receiving information about his or her health, giving consent to medical intervention or refusing it, and the preservation of medical confidentiality.

15. Special situations in history taking

Table 1.17. *History taking in difficult situations*

Situation	Recommendations
Severely ill or unconscious patient	Information is obtained from relatives, ambulance staff and documents; the vital functions are assessed first
Elderly patient	Speak slowly, clearly and loudly; take impaired hearing and vision into account; find out whether the patient takes many medicines (polypharmacy); if memory is impaired, talk to the relatives
Language barrier	Through an interpreter (a professional rather than a relative wherever possible); short, simple questions
Anxious or depressed patient	Allow more time, acknowledge the patient's feelings, use open questions
Aggressive patient	Stay calm, keep a safe distance, stand on the side of the door, do not raise your voice
Children and adolescents	With the parents; adolescents are also questioned alone for part of the interview
Sensitive topics (harmful habits, sexual life)	Begin with an explanation ("We ask all patients these questions"), do not judge

MEMORY AID

The **"3A" rule** of a good history: **A**ttend (do not interrupt the patient), **A**nalyze (each complaint according to the plan), **A**ffirm (repeat back what you heard and get the patient's confirmation).

16. Clinical part

Algorithm of independent history taking

From the moment the student enters the patient's ward, he or she follows this order:

1. Cleans the hands and checks the gown and mask (if necessary).
2. Greets the patient and introduces himself or herself: "Hello, I am a 3rd-year student ..., may I ask you about your illness?"
3. Establishes the patient's passport data.
4. Asks an open question: "What is bothering you?" — and listens to the patient without interrupting for 1–2 minutes.
5. Identifies the main complaints and asks about each in detail according to the plan.
6. Identifies the additional (general) complaints, then carries out a brief review of systems.
7. Collects the history of present illness in chronological order.
8. Collects the life history according to the plan.
9. Briefly repeats what he or she has heard and gets the patient's confirmation: "Have I understood you correctly?"
10. Thanks the patient and explains the next stage — the objective examination.

Sample case history entry

Complaints: cough, production of up to 50 ml of mucopurulent sputum a day, a rise in body temperature to 38.5 °C, stabbing pain in the right side of the chest on deep inspiration. Additional

complaints: general weakness, sweating, loss of appetite.

History of present illness: considers himself ill for 5 days. The illness began suddenly after chilling, with chills and a rise in temperature to 39 °C. The next day a cough and right-sided pain were added. He took paracetamol at home without effect. Because his condition worsened, he was admitted to hospital by ambulance.

Life history: 45 years old, the third child in the family; his growth and development kept pace with his peers. A driver; working conditions involve cold and dampness. Has smoked 20 cigarettes a day for 25 years (smoking index 25 pack-years). Denies tuberculosis, hepatitis and venereal diseases. No drug allergy. His father had arterial hypertension.

> **CAUTION**
> In the case history, write not the patient's own expressions ("my insides burn", "my heart is being squeezed") but their medical interpretation; however, do not arbitrarily exaggerate or play down the patient's words. For example, "my heart is being squeezed" may not be written down as "angina pectoris" without clarifying what it means.

17. Practical skills

Skill: detailed analysis of the patient's complaint

1. Seat the patient comfortably and sit face to face with him or her.
2. Ask an open question: "What is bothering you?"
3. For each complaint, ask according to the plan: site, onset, character, radiation, duration, exacerbating and relieving factors, severity, associated symptoms.
4. Divide the complaints into main and additional groups.
5. Carry out a brief review of systems.
6. Briefly repeat the information obtained to the patient and get confirmation.
7. Record the result in the case history in medical terms.

Skill: collecting the life history

1. Begin with the biography (place of birth, growth, education).
2. Establish the living and dietary conditions.
3. Ask about the occupational history and occupational hazards.
4. Identify harmful habits and calculate the smoking index.
5. Ask about past illnesses, operations and blood transfusions in chronological order.
6. Establish the allergy history, especially drug allergy.
7. Collect the family and epidemiological history and, in women, the gynecological history.

Review questions

1. What does the term propaedeutics mean, and what are the main tasks of the discipline?
2. Explain the difference between a symptom and a syndrome with examples.
3. What do you understand by a pathognomonic symptom?
4. When are the primary, clinical and final diagnoses made?
5. What components does the clinical diagnosis consist of?
6. What is the difference between medical deontology and medical ethics?

7. What are the advantages and disadvantages of open and closed questions?
8. List the physical methods of examining the patient and their sequence.
9. What sections does the case history consist of?
10. How do main and additional complaints differ?
11. According to what plan should pain be analyzed?
12. What questions does the history of present illness answer?
13. What sections does the life history consist of?
14. How is the smoking index calculated, and what is its significance?
15. Why must the allergy history always be collected?

Test questions

1. What does the term “propaedeutics” mean in translation from Greek?
   A) The study of the physician’s duty
   B) Treatment of disease
   C) Teaching beforehand, introduction into a science
   D) Prevention of disease
   E) The study of the signs of disease

2. What is a combination of symptoms with a common pathogenesis called?
   A) Anamnesis
   B) Nosology
   C) Semiotics
   D) Diagnosis
   E) Syndrome

3. Which of the following is an objective symptom?
   A) Nausea
   B) Liver enlargement
   C) Weakness
   D) Headache
   E) Pain in the heart region

4. Within what period after hospital admission is the clinical (main) diagnosis usually made?
   A) Within 1 hour
   B) Within 1 day
   C) Within 3 days
   D) Within 10 days
   E) At discharge

5. Indicate the correct sequence of the physical methods of examination.
   A) Inspection, auscultation, palpation, percussion
   B) Auscultation, percussion, palpation, inspection
   C) Inspection, palpation, percussion, auscultation
   D) Palpation, inspection, auscultation, percussion
   E) Percussion, auscultation, inspection, palpation

6. Which of the following questions is an open question?
   A) “Do you smoke?”
   B) “Do you take any medicines?”
   C) “Tell me how your illness began.”
   D) “Have you had a temperature?”
   E) “Does the pain spread to your left arm?”

7. What does “diagnosis ex juvantibus” mean?
   A) The diagnosis is confirmed by the effect of treatment
   B) The diagnosis is made on the basis of a laboratory test
   C) The diagnosis is made with the help of X-rays
   D) The diagnosis is made on the basis of the history
   E) The diagnosis is made after death

8. What is written in the “status praesens” section of the case history?
   A) Treatment plan
   B) The history of the development of the disease
   C) The patient’s complaints
   D) Discharge summary (epicrisis)
   E) The results of the objective examination

9. How is the smoking index calculated?
   A) Years of smoking × 20
   B) Age / years of smoking
   C) Cigarettes per day / years
   D) Cigarettes per day × years / 20
   E) Number of packs × 365

10. Which complaint belongs to the additional (general) complaints?
    A) Retrosternal pain
    B) Attacks of dyspnea
    C) Hemoptysis
    D) General weakness and sweating
    E) Cough with sputum

11. To whom does the principle “Primum non nocere” (“First, do no harm”) belong?
    A) S. Botkin
    B) Hippocrates
    C) R. Virchow
    D) G. Zakharyin
    E) Avicenna (Ibn Sina)

12. Which of the following can serve as an example of a pathognomonic sign?
    A) Weakness — in anemia
    B) Fever — in pneumonia
    C) Cough — in bronchitis
    D) Edema — in heart failure
    E) A diastolic murmur and the “quail rhythm” — in mitral stenosis

13. To which section of the life history does the fact that the patient worked in mining for 20 years belong?
    A) Allergy history
    B) Living conditions
    C) Epidemiological history
    D) Occupational history
    E) Family history

14. What does the history of present illness (anamnesis morbi) include?
    A) The patient’s development in childhood
    B) The onset and course of the present disease
    C) Harmful habits
    D) Drug allergy
    E) Diseases in relatives

15. What is the main disadvantage of leading (suggestive) questions?
    A) They push the patient toward a wrong answer
    B) They force the patient to speak freely
    C) They have no disadvantage
    D) They are given only in writing
    E) They take a lot of time

Clinical cases

Case 1

A 58-year-old man presented saying “my heart is being squeezed”. The student immediately asked, “The pain spreads to your left arm, right?”, the patient answered “yes”, and the student wrote “angina pectoris” in the case history.

Questions: What mistakes did the student make? How should the complaint have been analyzed?

Answer: The student asked a leading question and wrote a diagnosis without clarifying the patient’s words. The correct way is to start with an open question (“Describe the squeezing”) and to analyze the pain according to the plan: site, onset (on exertion?), character, radiation, duration, the effect of nitroglycerin and rest, associated symptoms. The complaint is written in the case history in medical terms after the analysis, and the diagnosis is made only after all the information has been collected.

Case 2

A 35-year-old woman complains of palpitations and dyspnea. From the life history: in childhood she often had tonsillitis, and at the age of 14 she had “migrating” pain in the knee and ankle joints and was treated in hospital.

Questions: Which information in the life history is important? Which disease can be considered?

Answer: Recurrent tonsillitis in childhood and migrating polyarthritis are characteristic of acute rheumatic fever. The present palpitations and dyspnea give grounds to think of rheumatic heart disease (most often mitral stenosis) and its complications (atrial fibrillation, heart failure). This example shows how important it is to collect the life history in full.

Case 3

After a patient was admitted to hospital, penicillin injections were prescribed. After the first injection the patient developed dizziness, flushing of the skin and a sharp fall in blood pressure. It later turned out that the patient had wanted to say that he had previously “come out in a rash” from penicillin, but nobody had asked.

Questions: Which stage of history taking was missed? What are the consequences of such an error?

Answer: The allergy history was not collected. As a result, the patient developed drug anaphylaxis (anaphylactic shock), a life-threatening condition. Every patient must be asked about reactions to drugs, foods and other allergens, and the result is recorded separately on the first page of the case history.

Quick review

Table 1.18. *Key facts on Topic 1*

Question	Answer
Propaedeutics	Introduction to the clinical disciplines: examination methods, semiotics, diagnostic methodology
Symptom	A single sign of disease (subjective or objective)
Syndrome	A combination of symptoms with a common pathogenesis
Stages of diagnosis	Data collection → analysis → synthesis → primary → differential → final diagnosis
Physical methods	Inspection → palpation → percussion → auscultation
Parts of history taking	Passport data → complaints → anamnesis morbi → anamnesis vitae
Analysis of a complaint	Site, onset, character, radiation, duration, factors, severity, associated symptoms
Smoking index	Cigarettes per day × years / 20 (pack-years)
Deontology	The study of the physician’s duty and conduct

TOPIC 2

General inspection of the patient: general condition, consciousness, position, body build. Inspection of the head, face and neck

Learning objectives

To teach the student the rules of performing the general inspection, how to assess the patient's general condition, consciousness, position and constitutional type, and the diagnostic signs detected on inspection of the head, face, eyes, oral cavity and neck.

By the end of the session, the student should know:

- the conditions and order of performing the general inspection;
- the grades of the general condition and their criteria;
- the quantitative and qualitative types of impaired consciousness and the Glasgow Coma Scale;
- the active, passive and forced positions of the patient and their causes;
- the signs of the constitutional types (normosthenic, hypersthenic, asthenic);
- the facial appearances characteristic of diseases (facies);
- the eye signs of thyrotoxicosis and the changes in the tongue and neck.

The student should be able to:

- perform the general inspection according to a set plan;
- assess consciousness and calculate the score on the Glasgow scale;
- determine the epigastric angle and establish the constitutional type;
- inspect the face, eyes, tongue and neck and record the result in the case history.

1. Rules for performing the general inspection

However advanced laboratory and instrumental methods become, the general inspection (inspectio) remains one of the main methods of examination. An experienced physician often draws an important conclusion at the very first sight of the patient: a patient with heart failure sitting in the orthopnea position, a thyrotoxic patient with protruding eyes, and so on.

REMEMBER

Conditions of the inspection:

- the room should be warm (so that the patient does not get chilled) and quiet;
- the light should be **natural, diffuse daylight**: jaundice and pallor are poorly seen in artificial (electric) light;
- the light should fall from the side: protrusions and depressions on the body surface and pulsation are then well seen;
- the patient's body is uncovered step by step, and the parts not being examined are kept covered;
- the symmetrical parts of the body are always compared.

The inspection is performed according to a plan: first the **general inspection** (general condition, consciousness, position, habitus — body build, posture and gait), then the **inspection of the body regions** (head, face, neck, trunk, limbs), and then the skin, subcutaneous fat, lymph nodes, muscles and joints are assessed.

2. Assessment of the general condition

The general condition is assessed according to the patient's consciousness and position, the degree of impairment of organ function and the presence of a threat to life.

Table 2.1. *Grades of the general condition*

Grade	Criteria	Examples
Satisfactory	Consciousness clear, position active, the function of the vital organs almost unimpaired	Mild diseases, the period of remission
Moderately severe	Consciousness clear or slightly clouded, position active or forced, organ function markedly impaired, but no immediate threat to life	Heart failure IIA, moderately severe pneumonia
Severe	Consciousness often impaired (obtundation, sopor), position often passive or forced, the function of the vital organs sharply impaired	Lobar pneumonia at its height, status asthmaticus
Extremely severe	Coma, profound impairment of the vital functions (respiration, circulation), urgent resuscitation measures needed	Cardiogenic shock, uremic coma
Agonal (terminal)	The state preceding death	The state preceding clinical death

3. Assessment of consciousness

Consciousness may be **clear** (the patient answers questions quickly and correctly and is oriented in time, place and person) or **impaired**. Impairments are divided into **quantitative** (depression of consciousness) and **qualitative** (change in the content of consciousness) types.

Table 2.2. *Degrees of depression of consciousness*

Grade	Definition	Signs
Obtundation (obnubilation, "oglushenie")	Mild depression of consciousness	The patient answers questions with delay and briefly, is drowsy and indifferent to the surroundings
Stupor	Deeper depression	The patient responds only briefly to a loud voice or a painful stimulus and then "falls asleep" again
Sopor	Pathologically deep sleep	Makes defensive movements only in response to a strong painful stimulus, does not answer questions, reflexes preserved
Coma	Complete loss of consciousness	No response to any stimulus, reflexes depressed or absent, vital functions impaired

Qualitative disturbances of consciousness: delirium — disorientation in time and place, hallucinations, agitation (alcohol withdrawal, severe infections, high fever); hallucinations and illusions; amentia.

> **CLINICAL SIGNIFICANCE**
>
> English-language textbooks (Bates, Macleod) use the **AVPU** scale for rapid assessment of consciousness: **A**lert; **V**oice — responds to voice; **P**ain — responds only to pain; **U**nresponsive — no response. The **Glasgow Coma Scale** (GCS) is used for detailed assessment.

Table 2.3. *The Glasgow Coma Scale*

Parameter	Response	Score
Eye opening	Spontaneous	4
	To voice (to speech)	3
	To pain	2
	None	1
Verbal response	Oriented	5
	Confused speech	4
	Inappropriate words	3
	Incomprehensible sounds	2
	None	1
Motor response	Obeys commands	6
	Localizes pain	5
	Withdraws from pain	4
	Abnormal flexion (decorticate)	3
	Abnormal extension (decerebrate)	2
	None	1

REMEMBER
The total GCS score ranges from 3 to 15: **15** — clear consciousness; **13–14** — obtundation; **9–12** — sopor (stupor); **8 or less** — coma (the airway needs protecting).

Types of coma and their distinguishing signs

Table 2.4. *Common comas*

Coma	Main cause	Distinguishing signs
Diabetic (ketoacidotic)	Decompensation of diabetes mellitus	Dry skin, odor of acetone on the breath, Kussmaul breathing, soft eyeballs
Hypoglycemic	Insulin overdose, missed meals	Moist skin, profuse sweating, seizures, rapid onset
Uremic	End-stage kidney failure	Skin pale gray, dry, scratch marks, odor of ammonia on the breath, Kussmaul breathing
Hepatic coma	Liver cirrhosis, acute liver dystrophy	Jaundice, “liver” (sweetish-musty) odor from the mouth, “spider angiomas”
Alcoholic coma	Alcohol poisoning	Cyanotic face, dilated pupils, odor of alcohol on the breath
Apoplectic (stroke)	Intracerebral hemorrhage	Flushed face, deep and noisy breathing, hemiparesis
Epileptic	Epilepsy	Tonic-clonic seizures, bitten tongue, involuntary passage of urine and stool

MEMORY AID

Coma by the odor of the breath: **“Acetone — Ammonia — Alcohol — Liver”**. Acetone — diabetic coma, ammonia (urine) — uremic coma, alcohol — alcoholic coma, sweetish-musty “liver odor” (fetor hepaticus) — hepatic coma.

4. The patient’s position

The patient’s position may be of three kinds: **active**, **passive** and **forced**.

- **Active position** — the patient can assume any position he or she wishes and moves freely (mild diseases).
- **Passive position** — the patient cannot change the position independently; the head and limbs hang down (coma, extreme weakness, spinal cord injury).
- **Forced position** — a position the patient assumes himself or herself to relieve his or her condition (to lessen pain, dyspnea or cough).



Figure 2.1. *Forced position of the patient: sitting leaning forward (Gadayev A. et al., 2012, Fig. 1)*

Table 2.5. *Forced positions and their diagnostic significance*

Forced position	In which disease	Mechanism
Sitting with the legs lowered (orthopnea)	Left ventricular failure, cardiac asthma	Part of the blood pools in the leg veins, and congestion in the pulmonary circulation decreases
Sitting leaning forward with the hands propped on a table or the edge of the bed	Attack of bronchial asthma, exacerbation of COPD	The shoulder girdle is fixed and the accessory respiratory muscles come into play
Lying on the affected side	Dry and exudative pleurisy, lung abscess, bronchiectasis	Movement of the pleural layers decreases and the pain lessens; the healthy lung takes part freely in breathing; pus does not flow into the bronchus, so the cough does not increase
Lying on the healthy side	Rib fracture	Lying on the affected side increases the pain
Sitting leaning forward, the “prayer position” (knee-elbow)	Exudative pericarditis	The fluid shifts downward and the pressure on the base of the heart and the great vessels decreases

Forced position	In which disease	Mechanism
Lying motionless on the back with the legs bent	Acute abdomen (peritonitis, perforated ulcer)	Movement of the peritoneum and its pain decrease
Lying on the side with the head thrown back and the legs drawn up to the abdomen (the “pointing dog” position)	Meningitis	Tension of the meninges decreases
Stopping suddenly and standing still	Attack of angina pectoris, acute ischemia, intermittent claudication	When movement stops, the oxygen demand of the myocardium (muscle) decreases
Restless tossing from side to side	Renal and biliary colic	The pain is very severe; no comfortable position can be found

CAUTION

In lung abscess and bronchiectasis, the patient lies on the affected side, because on turning onto the healthy side, pus flows through the draining bronchus into the trachea and increases the cough and sputum production. However, this very feature is used in treatment: for **postural drainage**, the patient is placed in the position in which the sputum drains out.

5. Body build (constitution)

Constitution is the combination of the morphological and functional features of the organism shaped by heredity and the environment. According to the classification of M.V. Chernorutsky, three types are distinguished. The main distinguishing parameter is the **epigastric angle** (the angle between the costal arches, determined by placing the palms along the arches).



Figure 2.2. *Normosthenic type (Gadayev A. et al., 2012, Fig. 2)*



Figure 2.3. *Hypersthenic type (same source)*



Figure 2.4. *Asthenic type (same source)*

Table 2.6. *Constitutional types*

Feature	Normosthenic	Hypersthenic	Asthenic
Epigastric angle	About 90°	More than 90° (obtuse)	Less than 90° (acute)
Ratio of height to trunk	Proportional	Height average or short, trunk broad	Height tall, trunk slender
Chest	Conical	Broad and short, large anteroposterior diameter	Long, narrow, flat
Neck and limbs	Proportional	Short neck, short limbs	Long neck, long and thin limbs
Muscles and fat layer	Well developed	Strongly developed, prone to obesity	Poorly developed
Position of the diaphragm and heart	Average	Diaphragm high, heart horizontal (“lying”)	Diaphragm low, heart vertical (“hanging”)
Predisposed diseases	—	Arterial hypertension, atherosclerosis, IHD, obesity, diabetes mellitus, gallstone disease	Tuberculosis, peptic ulcer disease, hypotension, prolapse of the internal organs (splanchnoptosis)

MEMORY AID

The epigastric angle — **“Hyper — Wide, Asthenic — Narrow”**: in the HYPERsthenic type the angle is wide (>90°), in the ASTHENIC type it is narrow (<90°), and in the normosthenic type it is a right angle (90°).

6. Inspection of the head

The shape and size of the head are assessed. **Macrocephaly** (enlargement of the head) is seen in hydrocephalus, and **microcephaly** in developmental delay. In acromegaly, the skull bones, lower jaw, nose and ears enlarge. Rhythmic **nodding** of the head in time with the heartbeat — **de Musset’s sign** — is characteristic of aortic regurgitation. Head tremor is seen in Parkinson’s disease.

7. Inspection of the face

The facial expression (facies) often provides important information for the diagnosis.



Figure 2.5. *Appearance of the face in diffuse toxic goiter (Gadayev A. et al., 2012, Fig. 3)*



Figure 2.6. *Facies mitralis (same source, Fig. 4)*

Table 2.7. *Facial appearances characteristic of diseases*

Name	Signs	Disease
Febrile face (facies febrilis)	Flushed face, shining eyes, restless expression	Infections with high fever; in lobar pneumonia, flushing of the cheek on the affected side and herpes on the lips
Mitral face (facies mitralis)	Bluish-red patches on the cheeks (the “mitral butterfly”); the lips and tip of the nose cyanotic	Mitral stenosis
Corvisart’s face (facies Corvisari)	Puffy, yellowish-pale, cyanotic, mouth half open, eyes dull	Severe chronic heart failure
Hippocratic face (facies Hippocratica)	Sunken eyes, pinched nose, pale gray skin, drops of cold sweat	Peritonitis, severe dehydration, agony
Nephritic face (facies nephritica)	Pale, puffy, the eyelids especially swollen	Glomerulonephritis, nephrotic syndrome
Basedow’s face (facies Basedowica)	Protruding (exophthalmos), shining eyes, an expression of fear	Diffuse toxic goiter
Myxedematous face	Puffy, coarse, apathetic, the outer part of the eyebrows lost	Hypothyroidism
“Moon” face	Round, flushed, shiny; beard and mustache growth in women	Cushing’s syndrome, long-term glucocorticoid therapy
Acromegalic face	Enlarged lower jaw, nose, lips and brow ridges	Acromegaly (pituitary adenoma)
“Butterfly” rash	Red erythema over the bridge of the nose and the cheeks	Systemic lupus erythematosus (SLE)
Masklike face	Immobile, without facial expression	Parkinson’s disease, systemic sclerosis



Figure 2.7. *Face of a patient with systemic lupus erythematosus (Gadayev A. et al., 2012, Fig. 6)*

8. Inspection of the eyes, nose, mouth and tongue

Eyes and eyelids

- **Eyelid edema** — kidney diseases, hypothyroidism, allergic edema (angioedema).
- **Xanthelasmas** — yellowish flat patches on the eyelids (lipid metabolism disorders, atherosclerosis, cholestasis).
- **Yellowing of the sclerae** — the earliest sign of jaundice.
- **Pallor of the conjunctiva** — anemia.
- **Pupils**: constriction (miosis) — morphine poisoning, uremia, intracerebral hemorrhage; dilation (mydriasis) — atropine poisoning, coma, agony; anisocoria — neurological damage.
- **Horner's triad** (ptosis, miosis, enophthalmos) — damage to the cervical sympathetic fibers, including a tumor of the lung apex (Pancoast tumor).

Table 2.8. *Eye signs in thyrotoxicosis*

Feature	Content
Exophthalmos	Forward protrusion of the eyeball
Graefe's sign	On looking down, the upper eyelid lags behind the eyeball, and a strip of sclera is visible above the iris
Stellwag's sign	Infrequent blinking
Möbius' sign	Inability to converge the eyes on a near object (weakness of convergence)
Dalrymple's sign	Widening of the palpebral fissure, sclera visible above the iris
Kocher's sign	On looking up, the upper eyelid rises ahead of the eyeball

Nose

Movement of the nasal alae during inspiration ("flaring" of the nostrils) is a sign of severe respiratory failure and lobar pneumonia. A saddle nose — congenital syphilis, granulomatosis with polyangiitis.

Cyanosis of the tip of the nose — heart failure.

Oral cavity and tongue

The color of the lips (pallor — anemia, cyanosis — heart and respiratory failure), cracks at the corners of the mouth (**angular stomatitis** — iron and vitamin B_2 deficiency), herpetic eruptions (lobar pneumonia, influenza), bleeding of the gums and the state of the teeth are assessed.



Figure 2.8. *Inspection of the tongue: from left to right — normal, in peptic ulcer disease, in scarlet fever, in severe intoxication (Gadayev A. et al., 2012, Fig. 7)*

Table 2.9. *Diagnostic significance of changes in the tongue*

Appearance of the tongue	Disease or condition
Moist, pink, papillae distinct, no coating	Normal
Dry, covered with a brown coating	Severe intoxication, dehydration, uremia, peritonitis
White coating	Gastritis, dyspepsia, candidiasis
Smooth, shiny, red (“lacquered tongue”, Hunter’s glossitis)	Vitamin B_{12} deficiency anemia
Raspberry-colored, papillae prominent	Scarlet fever
Enlarged (macroglossia), with tooth marks at the edges	Hypothyroidism, acromegaly, amyloidosis
Atrophied papillae, fissures	Iron deficiency, pellagra

9. Inspection of the neck

On inspection of the neck, attention is paid to the following:

- **Shape and symmetry** — enlargement of the thyroid gland (goiter), enlargement of the lymph nodes, tumors.
- **Neck veins** — distension and pulsation (right ventricular failure, tricuspid regurgitation, pericarditis).
- **Strong pulsation of the carotid arteries** — the “carotid dance” (aortic regurgitation).
- **Use of the accessory muscles** — tensing of the sternocleidomastoid muscles during breathing (bronchial asthma, COPD).
- **“Stokes’ collar”** — edema of the neck, face and shoulder girdle with dilated veins (compression of the superior vena cava, tumors of the clavicular region).

CLINICAL SIGNIFICANCE

To assess the thyroid gland on inspection, the patient is asked to swallow a sip of water: during swallowing the gland rises together with the larynx. If a swelling in the neck moves with swallowing, it belongs to the thyroid gland; a lymph node, by contrast, hardly moves on swallowing.

CAUTION

When assessing distension of the neck veins, the patient lies at an angle of 45°. If the upper level of the internal jugular venous pulsation is **more than 3 cm above** the sternal angle, the central venous pressure is considered raised.

10. Additional methods of assessing the constitution

In addition to the epigastric angle, anthropometric indices are used to determine the constitutional type.

Table 2.10. *Indices for assessing the constitution*

Index	Calculation	Interpretation
Pignet index	Height (cm) − [weight (kg) + chest circumference (cm)]	Less than 10 — hypersthenic; 10–30 — normosthenic; more than 30 — asthenic
Ratio of chest circumference to height	Chest circumference / height × 100	50–55 — normosthenic; more than 55 — hypersthenic; less than 50 — asthenic
Ratio of limb length to trunk length	Arm span / height	Normally about 1.0; more than 1.05 — Marfan syndrome suspected

CLINICAL SIGNIFICANCE

Tall stature, long thin limbs and fingers (**arachnodactyly**), chest deformity, dislocation of the lens and joint hypermobility are signs of **Marfan syndrome**. Such a patient has a high risk of aortic root dilation, aortic dissection and aortic regurgitation, so echocardiography is necessary.

11. Posture, gait and movements

The patient’s posture and gait are observed from the moment he or she enters the room.

Table 2.11. *Types of gait*

Type of gait	Signs	Cause
Hemiparetic (“circumduction”)	The paralyzed leg is held straight and brought forward in a semicircle from the side; the arm is flexed	Hemiparesis after a stroke
Parkinsonian	Trunk bent forward, small shuffling steps, no arm swing, difficulty stopping	Parkinson’s disease
Ataxic (“drunken gait”)	Feet placed wide apart, swaying	Cerebellar damage, alcohol intoxication
“Waddling gait”	The trunk sways from side to side	Bilateral hip joint disease, myopathies
“Steppage gait” (“high-stepping gait”)	The leg is lifted high, the foot drops	Common peroneal nerve palsy, polyneuropathy
Antalgic (limping)	The time of weight bearing on the painful leg is shortened	Joint and bone diseases, intermittent claudication

Involuntary movements: hand tremor — fine and rapid in thyrotoxicosis; “pill-rolling” at rest in Parkinson’s disease; in alcoholism; in liver failure, a **“flapping tremor”** (asterixis — a wing-beating movement of the hands when the arms are stretched forward); seizures; chorea (ARF, Huntington’s disease).

12. Additional signs on inspection of the head and face



Figure 2.9. *Face characteristic of acromegaly: enlarged lower jaw, nose and lips (Gadayev A. et al., 2012, Fig. 7)*

Table 2.12. *Additional signs on the face and scalp*

Feature	Content	Diagnostic significance
Facial edema	On the eyelids in the morning	Kidney diseases, hypothyroidism, angioedema
Cyanotic edema of the face and neck	Dilated veins (“Stokes’ collar”)	Compression of the superior vena cava (tumor, mediastinal lymphoma)
“Skull-like” appearance of the face	Sunken temples and cheeks, thin skin	Cachexia (tumors, severe chronic diseases)
Yellowing under the tongue and of the gums	Together with the sclerae	Jaundice (raised bilirubin)
Cracks at the corners of the mouth (cheilosis)	Angular stomatitis	Deficiency of iron and vitamins B_2 and B_6
Gingival hypertrophy and bleeding	—	Scurvy, leukemia, taking phenytoin and nifedipine
Dental caries and periodontal disease	Focus of infection	Source of infective endocarditis, lung abscess (aspiration)
Enlarged tonsils	With purulent plugs on them	Chronic tonsillitis — linked with ARF and glomerulonephritis

Detailed inspection of the eyes



Figure 2.10. *Horner's triad: on the right, drooping of the eyelid (ptosis), constriction of the pupil (miosis) and retraction of the eyeball (Gadayev A. et al., 2012, Fig. 5)*

- **Pupillary reaction to light:** normally the pupils are equal and constrict briskly to light (direct and consensual reaction). Absence of the reaction — deep coma, severe brain damage, atropine poisoning.
- **Argyll Robertson sign** — no pupillary reaction to light, but the reaction to convergence is preserved (neurosyphilis, diabetes mellitus).
- **Conjunctival petechiae** (Lukin–Libman sign) — infective endocarditis.
- **Bluish sclerae** — iron deficiency, osteogenesis imperfecta.
- **A gray ring at the edge of the cornea** (arcus senilis) — if it appears before the age of 50, a sign of hypercholesterolemia; the **Kayser–Fleischer ring** (brownish-green) — Wilson's disease (Wilson–Konovalov disease).
- **Exophthalmos** — if it is unilateral, an orbital tumor must also be excluded.

13. Inspection of the neck: assessment of the central venous pressure

The filling of the neck veins reflects the pressure in the right atrium (the central venous pressure — CVP). Its assessment is important in detecting heart failure and pericarditis and in judging fluid balance.

Method: the patient lies at an angle of 45° with the head turned slightly to the left. The highest point of pulsation of the right internal jugular vein is identified. The vertical distance from it to the **sternal angle** (angle of Louis) is measured. Because the sternal angle lies about 5 cm above the right atrium, CVP ≈ the measured distance + 5 cm.

> **REMEMBER**
> Normally, the venous pulsation rises no higher than **3 cm above** the sternal angle (CVP up to 8 cm H_2O). At 4 cm or more the CVP is raised: right ventricular failure, tricuspid regurgitation, cardiac tamponade, constrictive pericarditis, hypervolemia. If the neck veins are full with the patient sitting upright, the CVP is markedly raised.

Table 2.13. *Differences between the pulsation of the neck veins and of the carotid artery*

Feature	Venous pulsation	Arterial pulsation
Appearance	Soft, undulating, two waves in each cardiac cycle	Abrupt, a single strong beat
Palpation	Not palpable	Easily palpable
Pressure above the clavicle	The pulsation disappears	Unchanged

Feature	Venous pulsation	Arterial pulsation
Change of position	Falls on standing, rises on lying down	Unchanged
Respiration	Falls on inspiration	Unchanged
Pressure on the abdomen	Rises (hepatojugular reflux)	Unchanged

14. Qualitative disturbances of consciousness and assessment of the mental state

Qualitative disturbances of consciousness often occur in the severe course of somatic diseases and should alert the physician.

Table 2.14. *Qualitative disturbances of consciousness*

Condition	Signs	Causes
Delirium	Acute onset, worse at night; disorientation in time and place, hallucinations (most often visual), agitation	Alcohol withdrawal (“delirium tremens”), severe infections, sepsis, hypoxia, medicines in the elderly
Oneiroid state	Dreamlike fantastic experiences	Encephalitis, intoxications
Amentia	Disorganized thinking, incoherent speech, disorganized movements	Terminal stage of severe somatic diseases
Twilight state (“sumerechnoye”)	Begins suddenly and ends suddenly; followed by amnesia	Epilepsy

CAUTION

Sudden confusion and restlessness in an elderly patient are often not “old age” but a sign of **pneumonia, urinary tract infection, hypoglycemia, dehydration or an adverse drug effect**. In such a situation the cause must be actively sought.

15. Additional examples of assessing the patient’s position



Figure 2.11. *Forced sitting position — orthopnea (Gadayev A. et al., 2012)*

Table 2.15. *Additional types of forced position*

Condition	Disease
Squatting	“Blue” congenital heart defects (tetralogy of Fallot) — peripheral resistance increases and the right-to-left shunt decreases
Head thrown back, neck muscles rigid	Meningitis

Condition	Disease
Lying on the side with the legs drawn up to the abdomen	Acute pancreatitis (sometimes sitting leaning forward)
Lying with the leg flexed at the hip joint	Inflammation of the iliopsoas muscle (psoitis), paranephritis
Sitting with the legs hanging over the edge of the bed	Severe stage of obliterative disease of the leg arteries (the nocturnal pain lessens)

16. Recognizing endocrine diseases by external appearance

Many endocrine diseases change the patient's appearance so much that the diagnosis can be suspected at first sight.

Table 2.16. *External appearance in endocrine diseases*

Disease	Face and head	Trunk and skin	Other signs
Diffuse toxic goiter (thyrotoxicosis)	Exophthalmos, shining eyes, Graefe's and Stellwag's signs	Thin; skin moist, warm, soft	Hand tremor, tachycardia, goiter in the neck
Hypothyroidism	Puffy, pale yellowish, apathetic face; the outer third of the eyebrows lost; enlarged tongue	Skin dry, thick, cold; the edema is firm and does not pit	Hoarse voice, bradycardia, slowness, drowsiness
Cushing's syndrome	"Moon", flushed face; hirsutism in women	Central obesity, thin limbs, "buffalo hump", broad purple striae, thin skin, bruising	Arterial hypertension, muscle weakness, osteoporosis
Addison's disease	Hyperpigmentation of the skin and mucous membranes (gums, lips)	Bronze-colored skin, especially in the skin folds and scars	Hypotension, weight loss, weakness, craving for salt
Acromegaly	Enlarged jaw, nose, lips, ears and brow ridges; widened spaces between the teeth	Enlarged hands and feet (shoe size increases)	Sweating, headache, narrowing of the visual field
Diabetes mellitus	Often unchanged; "diabetic rubeosis" (redness of the cheeks)	Obesity in type 2 diabetes mellitus; skin infections, xanthomas, acanthosis nigricans	Thirst, polyuria

17. Odor from the mouth and its diagnostic significance

Table 2.17. *Odors from the mouth and breath*

Odor	Cause
Acetone (rotten apples)	Diabetic ketoacidosis, prolonged starvation
Ammonia, urine	Uremia (chronic kidney disease)
Sweetish-musty ("liver odor", foetor hepaticus)	Liver failure, hepatic coma
Putrid, foul	Lung abscess and gangrene, bronchiectasis, diseases of the oral cavity, esophageal diverticulum
Alcohol	Alcohol poisoning (other comas must also be excluded)

Odor	Cause
Garlic	Poisoning with organophosphates, arsenic
Bitter almonds	Cyanide poisoning

18. Differential diagnosis of swellings in the neck

Table 2.18. *Swellings in the neck*

Swelling	Location	Features
Enlargement of the thyroid gland (goiter)	Anterior surface, below the larynx	Moves upward on swallowing
Thyroid nodule	In one lobe	Moves on swallowing; a hard, fixed nodule — suspicion of a tumor
Lymph nodes	The anterior and posterior triangles of the neck, submandibular region	Do not move on swallowing; often several
Median cyst of the neck (thyroglossal duct cyst)	In the midline, at the level of the hyoid bone	Rises when the tongue is protruded
Lateral cyst of the neck	At the anterior border of the sternocleidomastoid muscle	Soft, fluctuant
Carotid artery aneurysm or carotid body tumor	Lateral surface of the neck	Pulsates; a bruit may be heard over it
Enlargement of the salivary glands	Submandibular, in front of the ear	Mumps (parotitis) — swelling in front of the ear; pain increasing when eating sour food (a stone)

19. Algorithm for performing the general inspection “from head to toe”

1. **First impression (as the patient enters the room or in bed):** general condition, consciousness, position, facial expression, breathing, color, body build, gait.
2. **Vital signs:** pulse, BP, respiratory rate, body temperature, SpO_2, pain score.
3. **Anthropometry:** height, weight, BMI, waist circumference.
4. **Head:** shape, hair, scalp.
5. **Face:** expression, symmetry, color, edema, rashes.
6. **Eyes:** eyelids, conjunctiva, sclerae, pupils, position of the eyeballs.
7. **Nose and ears.**
8. **Oral cavity:** lips, gums, teeth, tongue, palate, tonsils, odor.
9. **Neck:** shape, thyroid gland, lymph nodes, veins, arterial pulsation, position of the trachea.
10. **Hands:** palms, fingers, nails, joints, tremor, skin.
11. **Trunk:** skin, chest, breasts, abdomen (followed by the systematic examination).
12. **Legs:** edema, veins, skin, nails, joints, peripheral pulses.

MEMORY AID

Inspection of the hands is **“a mirror that tells a great deal”**: the nails (clubbing, koilonychia, Terry’s nails), the palms (erythema, pallor, Dupuytren’s contracture), the fingers (Osler’s nodes, Heberden’s and Bouchard’s nodes, rheumatoid deformities), tremor (thyrotoxicosis, asterixis), temperature and moisture. That is why it is convenient to begin the objective examination by taking the patient’s hand and counting the pulse.

20. Clinical part

Sample case history entry

General condition moderately severe. Consciousness clear (GCS 15), oriented in time and place. Position forced — sitting with the legs lowered (orthopnea). Body build hypersthenic, epigastric angle 100°. The face is puffy, the lips and tip of the nose are cyanotic (acrocyanosis). The sclerae are not subicteric, the conjunctivae pink. The tongue is moist, covered with a white coating. The neck is symmetrical, the thyroid gland is not visible, and the neck veins are distended and pulsate even in the upright position.

Conclusion: orthopnea, acrocyanosis and distended neck veins raise suspicion of heart failure with congestion in the systemic and pulmonary circulations.

21. Practical skills

Skill: determining the epigastric angle and establishing the constitutional type

1. Undress the patient to the waist with him or her standing.
2. Place the palms of both hands along the costal arches so that the tips of the thumbs touch the xiphoid process.
3. Assess the angle between the thumbs: 90° — normosthenic, more than 90° — hypersthenic, less than 90° — asthenic.
4. Assess the additional signs: the shape of the chest, the length of the neck and limbs, muscular development.
5. Write the conclusion in the case history.

Skill: assessing consciousness with the Glasgow Coma Scale

1. Address the patient by name and observe eye opening (4–1 points).
2. Ask simple questions: “What is your name? Where are you now? What month is it?” (5–1 points).
3. Give a command: “Raise your hand.” If it is not obeyed, apply a painful stimulus (pressure on the sternum, nail-bed pressure) and assess the response (6–1 points).
4. Add up the points and record the result, for example: “GCS 12 (E3 V4 M5)”.

Review questions

1. Under what conditions should the general inspection be performed? Why is artificial light unsuitable?
2. What grades of the general condition are there, and by what criteria are they determined?
3. List and define the degrees of depression of consciousness.

4. What are the components of the Glasgow Coma Scale and its range of scores?
5. By what signs can diabetic, hypoglycemic and uremic coma be distinguished?
6. What do the active, passive and forced positions of the patient mean?
7. In which disease and by what mechanism is the orthopnea position observed?
8. Why does a patient with pleurisy lie on the affected side?
9. Describe the constitutional types and the epigastric angle.
10. What is the difference between facies mitralis and facies Corvisari?
11. What eye signs are observed in thyrotoxicosis?
12. What does Horner’s triad consist of, and what does it indicate?
13. Which changes of the tongue are characteristic of which diseases?
14. What signs are looked for on inspection of the neck?
15. Which disease are de Musset’s sign and the “carotid dance” characteristic of?

Test questions

1. In what light is it correct to perform the general inspection?
   A) In strong electric light
   B) The light does not matter
   C) In diffuse daylight
   D) In blue lamp light
   E) In a dark room with a flashlight

2. If a patient makes defensive movements only in response to a strong painful stimulus, does not answer questions and has preserved reflexes, how is consciousness assessed?
   A) Delirium
   B) Sopor
   C) Coma
   D) Obtundation
   E) Clear

3. Which score on the Glasgow Coma Scale corresponds to coma?
   A) 15 points
   B) 13–14 points
   C) 10–12 points
   D) 8 points or less
   E) 20 points

4. For which coma are the odor of acetone on the breath and Kussmaul breathing characteristic?
   A) Epileptic
   B) Hypoglycemic
   C) Ketoacidotic (diabetic)
   D) Apoplectic
   E) Alcohol

5. In which condition is the orthopnea position most often observed?
   A) Left ventricular failure
   B) Meningitis
   C) Renal colic
   D) Dry pleurisy
   E) Peritonitis

6. What position does a patient with exudative pericarditis assume?
   A) Lies on the healthy side
   B) Sits leaning forward or assumes the knee-elbow position
   C) Lies tossing restlessly
   D) Lies on the back with the legs stretched out
   E) Lies with the head thrown back

7. What is the epigastric angle in the hypersthenic constitution?
   A) Less than 90°
   B) Equal to 90°
   C) More than 90°
   D) Equal to 45°
   E) Cannot be determined

8. Which facial appearance is characterized by bluish-red patches on the cheeks and cyanosis of the lips?
   A) Facies Basedowica
   B) Facies febrilis
   C) Facies mitralis
   D) Facies Hippocratica
   E) Facies nephritica

9. What is the lagging of the upper eyelid behind the eyeball on looking down called?
   A) Dalrymple’s sign
   B) Stellwag’s sign
   C) Kocher’s sign
   D) Möbius’ sign
   E) Graefe’s sign

10. What does Horner’s triad include?
    A) Ptosis, miosis, enophthalmos
    B) Mydriasis, enophthalmos, strabismus
    C) Miosis, exophthalmos, nystagmus
    D) Exophthalmos, mydriasis, ptosis
    E) Ptosis, mydriasis, exophthalmos

11. Which condition is a smooth, shiny, red “lacquered tongue” characteristic of?
    A) Scarlet fever
    B) Uremia
    C) Gastritis
    D) Vitamin B_{12} deficiency anemia
    E) Hypothyroidism

12. Which valvular defect is nodding of the head in time with the heartbeat (de Musset’s sign) characteristic of?
    A) Aortic regurgitation
    B) Pulmonary artery stenosis
    C) Mitral regurgitation
    D) Tricuspid stenosis
    E) Mitral stenosis

13. To which diseases are people with an asthenic constitution more predisposed?
    A) Arterial hypertension and atherosclerosis
    B) Gout
    C) Obesity and diabetes mellitus
    D) Tuberculosis and peptic ulcer disease
    E) Gallstone disease

14. Which coma is characterized by profuse sweating, moist skin and seizures of rapid onset?
    A) Alcohol
    B) Hepatic coma
    C) Uremic
    D) Ketoacidotic
    E) Hypoglycemic

15. If a swelling in the neck rises during swallowing, what does it most often belong to?
    A) The carotid artery
    B) A lymph node
    C) The subcutaneous fat
    D) A salivary gland
    E) The thyroid gland

Clinical cases

Case 1

A 24-year-old man was brought to the emergency department unconscious. His skin is dry, the eyeballs are soft, his breathing is deep and noisy, and an odor of acetone is noticeable in the room. His relatives said that he had been on insulin for 5 years and that in the last few days he had had influenza, was not eating and had therefore stopped his insulin injections.

Questions: Which level of consciousness and which type of coma can be considered? What is this type of breathing called?

Answer: Coma. The clinical signs (dry skin, soft eyeballs, odor of acetone, development after stopping insulin and against the background of an infection) are characteristic of ketoacidotic diabetic coma. Deep, noisy, slow breathing is Kussmaul breathing; it serves to compensate for metabolic acidosis.

Case 2

A 62-year-old woman is sitting on the bed in the ward with her legs lowered, propping herself on the edge of the bed with her hands. When an attempt is made to lay her down, her dyspnea increases. There are bluish-red patches on her cheeks, and her lips and the tip of her nose are cyanotic.

Questions: Assess the patient’s position and facial appearance. Which disease can be considered?

Answer: Forced position — orthopnea; facial appearance — facies mitralis. These signs are characteristic of mitral stenosis and of heart failure with congestion in the pulmonary circulation. In the orthopnea position, part of the blood pools in the leg veins, congestion in the lungs decreases and the dyspnea is relieved.

Case 3

A 30-year-old woman complains of palpitations, sweating, hand tremor and weight loss. On inspection: her eyes are protruding and shining, the upper eyelid lags behind on looking down, and she blinks infrequently. On the anterior surface of the neck there is a visible swelling that rises on swallowing.

Questions: What signs have been found? Which disease are they characteristic of?

Answer: Exophthalmos, Graefe's sign (lagging of the eyelid on looking down), Stellwag's sign (infrequent blinking) and enlargement of the thyroid gland (a swelling that moves on swallowing) have been found. Together with the complaints, these are characteristic of diffuse toxic goiter (Graves' disease). The facial appearance is facies Basedowica.

Quick review

Table 2.19. *Key facts on Topic 2*

Question	Answer
Conditions of the inspection	Diffuse daylight, a warm room, light from the side
General condition	Satisfactory, moderately severe, severe, extremely severe, agonal
Depression of consciousness	Obtundation → stupor → sopor → coma
GCS	15 — clear; 13–14 — obtundation; 9–12 — sopor; 8 or less — coma
Condition	Active, passive, forced
Orthopnea	Left ventricular failure, cardiac asthma
Lying on the affected side	Pleurisy, lung abscess, bronchiectasis
Constitution	Normosthenic (90°), hypersthenic (>90°), asthenic (<90°)
Facies mitralis	Mitral stenosis
Graefe, Stellwag, Möbius	Thyrotoxicosis
Horner's triad	Ptosis, miosis, enophthalmos
De Musset's sign	Aortic regurgitation

TOPIC 3

General inspection: skin, subcutaneous fat, muscles, bones, joints. Fever. Anthropometry. Palpation. Palpation of the lymph nodes and thyroid gland

Learning objectives

To teach the student to inspect the skin, subcutaneous fat, muscles, bones and joints, to measure body temperature and assess the types of fever, to determine the anthropometric parameters and to examine the lymph nodes and thyroid gland by palpation.

By the end of the session, the student should know:

- the changes in skin color (pallor, cyanosis, jaundice, hyperpigmentation) and their causes;
- the primary and secondary elements of skin rashes;
- the types of edema and the difference between cardiac and renal edema;
- the normal values of body temperature and the grades and types of fever;
- the normal values of the body mass index and waist circumference;
- the types of palpation, the groups of lymph nodes and the criteria for assessing them;
- the grades of thyroid enlargement (WHO).

The student should be able to:

- determine skin turgor, moisture and edema;
- measure body temperature in the axilla and fill in the temperature chart;
- measure height, weight and waist circumference and calculate the body mass index;
- palpate the peripheral lymph nodes and the thyroid gland.

1. Inspection of the skin

The skin is regarded as a “mirror” reflecting the state of the whole organism. On inspection of the skin, its **color**, **moisture**, **turgor (elasticity)**, **rashes**, **vascular changes**, **scars**, **edema** and the state of the hair and nails are assessed.

1.1. Skin color

Table 3.1. *Changes in skin color*

Change	Mechanism	Causes
Pallor	Decreased hemoglobin or constriction of the skin vessels	Anemia, blood loss, shock, aortic regurgitation, glomerulonephritis
Redness (hyperemia)	Dilation of the skin vessels, increased red cells	Fever, arterial hypertension, erythremia, alcoholism
Central (diffuse, “warm”) cyanosis	Insufficient oxygen saturation of the arterial blood	Lung diseases (respiratory failure), “blue” congenital heart defects
Peripheral (“cold”) cyanosis — acrocyanosis	Slowing of blood flow, increased extraction of oxygen by the tissues	Heart failure (lips, tip of the nose, earlobes, fingertips)
Jaundice	Raised bilirubin in the blood (visible above 34 µmol/l)	Liver diseases, obstruction of the bile ducts, hemolysis

Change	Mechanism	Causes
Bronze hyperpigmentation	Increased melanin	Adrenocortical insufficiency (Addison's disease), hemochromatosis
Depigmentation (vitiligo)	Loss of melanin	Autoimmune diseases

REMEMBER

In central cyanosis the skin is **warm**, and the tongue and lips are also blue; in peripheral cyanosis the skin is **cold**, the tongue is usually pink, and the blueness is found in the parts of the body farthest from the heart (acral). Cyanosis becomes visible when the amount of reduced hemoglobin in the blood exceeds **50 g/l**; for this reason it appears late in severe anemia.

CAUTION

Distinguishing jaundice: in **true jaundice** the sclerae, the palate and the mucosa under the tongue also turn yellow; in **false jaundice** (from eating carrots, carotene, taking quinacrine) only the palms and soles turn yellow, while the sclerae remain white. Assess jaundice only in daylight.

1.2. Skin moisture and turgor

Skin moisture increases in the fall of a high fever (crisis), thyrotoxicosis, hypoglycemia and tuberculosis (night sweats). Dry skin — dehydration, hypothyroidism, diabetes mellitus, chronic kidney disease.

Skin turgor is determined by picking up a fold of skin on the back of the hand or on the chest between the thumb and index finger and releasing it: normally the fold smooths out at once; in dehydration, old age and cachexia it smooths out slowly.

1.3. Skin rashes

Table 3.2. *Elements of skin rashes*

Element	Definition	In which condition
Roseola	A red spot up to 1–5 mm in diameter that disappears on pressure	Typhoid fever, syphilis
Erythema	A larger (over 1 cm) area of redness	Allergy, erysipelas, the "butterfly" in SLE
Erythema marginatum	Pink rings with a pale center	Acute rheumatic fever
Petechia	Pinpoint hemorrhage that does not disappear on pressure	Thrombocytopenia, vasculitis, infective endocarditis
Purpura	Hemorrhages of 2–5 mm	Hemorrhagic vasculitis, thrombocytopenia
Ecchymosis	A large hemorrhage (bruise)	Hemophilia, overdose of anticoagulants
Papule	A nodular element raised above the skin	Allergic dermatitis
Vesicle	A cavity element containing clear fluid	Herpes, chickenpox
Pustule	A pus-filled vesicle	Pyogenic skin infections
Urticaria (hives)	Itching, migrating wheals	Allergic reactions

1.4. Vascular changes in the skin and other signs

- **"Spider angiomas"** (telangiectasias) — on the face, neck, chest and shoulders; **liver palms** (palmar erythema) — redness of the thenar and hypothenar eminences. Both are signs of liver cirrhosis.
- **White and pink lines (striae)** — obesity, pregnancy; **broad purple-red striae** — Cushing's syndrome.
- **Subcutaneous nodules** — rheumatic (acute rheumatic fever), rheumatoid and gouty nodules (tophi).
- **Osler's nodes** — painful pea-sized nodules on the fingertips; **Janeway lesions** — painless hemorrhagic spots on the palms and soles (infective endocarditis).

1.5. Nails and hair



Figure 3.1. *Changes in the nails: A — "watch-glass" nails; B — Terry's nails (Gadayev A. et al., 2012, Fig. 11)*

- **Clubbing** (Hippocratic fingers) — bulbous enlargement of the terminal phalanges of the fingers with nails convex like a "watch glass". It occurs in chronic hypoxia and suppurative lung diseases (bronchiectasis, lung abscess, lung cancer), in "blue" congenital heart defects, infective endocarditis and liver cirrhosis.
- **Koilonychia** (spoon-shaped nails) — iron deficiency anemia.
- **Terry's nails** (whitening of the nail plate with a pink band at the tip) — liver cirrhosis, chronic kidney disease.
- **Hair loss (alopecia)** — SLE, hypothyroidism, iron deficiency; male-pattern hair growth in women (hirsutism) — hyperandrogenism.

> **MEMORY AID**
> The causes of clubbing — **"LHL"**: **L**ungs (suppurative diseases, cancer), **H**eart ("blue" congenital defects, infective endocarditis), **L**iver (cirrhosis) and inflammatory bowel diseases.

2. Edema

Edema is an excess accumulation of fluid in the interstitial spaces. If a pit remains after pressing with a finger for 5–10 seconds, the presence of edema is confirmed (it is usually tested on the anterior surface of the lower leg, around the ankle and over the sacrum). Edema of the whole body is **anasarca**; accumulation of fluid in the body cavities is ascites, hydrothorax, hydropericardium.

Table 3.3. *Differences between cardiac and renal edema*

Feature	Cardiac edema	Renal edema
Location	First in the legs (in ambulatory patients); in bedridden patients over the sacrum and lower back	First on the face, especially the eyelids
Time of appearance	Worse in the evening	Worse in the morning
Spread	From below upward, gradually	Spreads rapidly, to the whole body
Skin	Bluish (cyanotic), cold, firm	Pale, warm, soft
Associated symptoms	Dyspnea, acrocyanosis, liver enlargement, distended neck veins	Urinary changes (proteinuria), arterial hypertension

3. Subcutaneous fat, muscles, bones and joints

The development of the **subcutaneous fat** is assessed by the thickness of a skinfold (next to the umbilicus, below the shoulder blade, on the back of the upper arm). Normally the skinfold is 1–2 cm thick. Decreased fat tissue is weight loss, and its sharp decrease is **cachexia** (tumors, tuberculosis, thyrotoxicosis, severe chronic diseases). Excess accumulation is obesity: uniform (alimentary), in the abdominal region (“apple” type, metabolic syndrome), on the thighs and buttocks (“pear” type).

The **muscular system** is assessed on inspection by the degree of development, symmetry, atrophy and hypertrophy, tone and strength. Muscle atrophy is seen in damage to the nervous system, prolonged immobility and cachexia.



Figure 3.2. *Muscle atrophy (Gadayev A. et al., 2012, Fig. 14)*



Figure 3.3. *Varicose veins of the leg (same source, Fig. 15)*

The **bones** are assessed on inspection for deformity, thickening and pain: the “rosary” and bowed legs after rickets, enlargement of the bones in acromegaly, pain on tapping the bones in myeloma and leukemias.

On inspection of the **joints**, their shape (swelling, deformity, defiguration), the color and temperature of the overlying skin, the range of movement and pain are assessed. Acute rheumatic fever is characterized by “migrating” inflammation of the large joints (knee, ankle, elbow), and rheumatoid arthritis by symmetrical involvement and deformity of the small joints of the hands and feet.

On inspection of the **limbs**, varicose veins, trophic ulcers, edema, tremor and contractures are detected.

4. Body temperature and fever

4.1. Measuring body temperature

Body temperature is measured with mercury thermometers (used less often today for safety reasons), electronic (digital) and infrared (ear, forehead) thermometers. The main method is **measurement in the axilla**: the axilla is wiped dry, the tip of the thermometer is placed in the center of the axillary fossa and the arm is pressed against the chest; with a mercury thermometer the measurement takes **10 minutes**. The temperature is measured at least twice a day (in the morning at 7–8 a.m. and in the evening at 17–19 h) and recorded on the **temperature chart**.

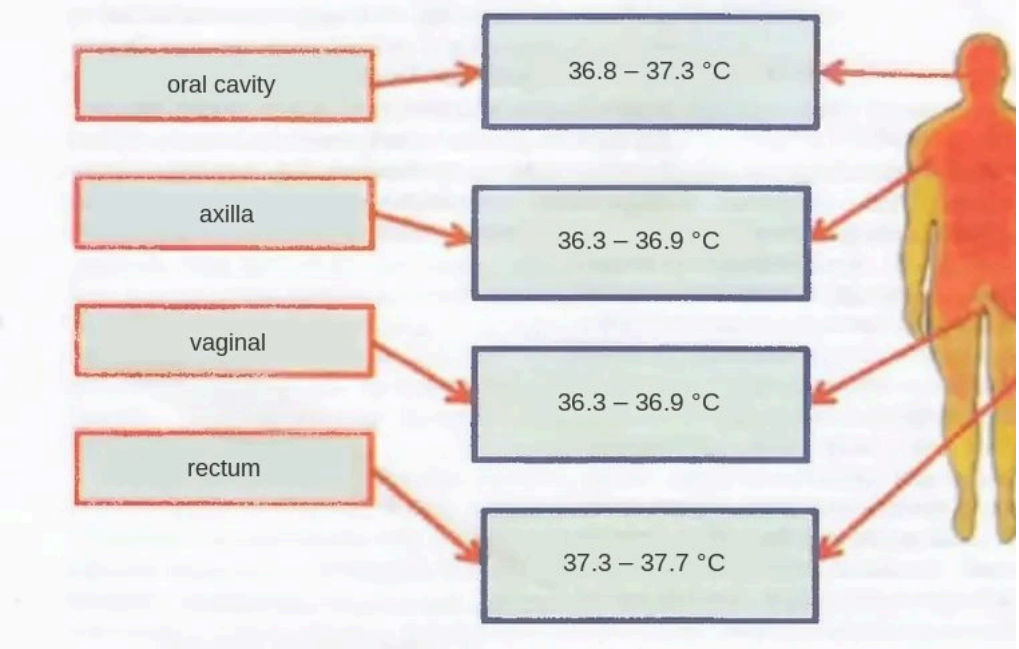


Figure 3.4. *Normal values of body temperature at various sites (Gadayev A. et al., 2012, Fig. 23)*

REMEMBER

Normal temperature: in the axilla **36.4–36.8 °C** (36.3–36.9 °C in the textbook); in the mouth **0.3–0.5 °C** higher than in the axilla; in the rectum **0.5–1.0 °C** higher. In a healthy person the daily fluctuation does not exceed **1 °C**: the lowest temperature is between 3 and 6 in the morning, and the highest between 17 and 21 h in the evening.

4.2. Fever (febris) and its grades

Fever is a rise in body temperature resulting from an upward shift of the “set point” of the thermoregulatory center in the hypothalamus under the action of pyrogens. It must be distinguished from **hyperthermia**: in hyperthermia (heatstroke) the thermoregulatory center has failed and heat loss is insufficient.

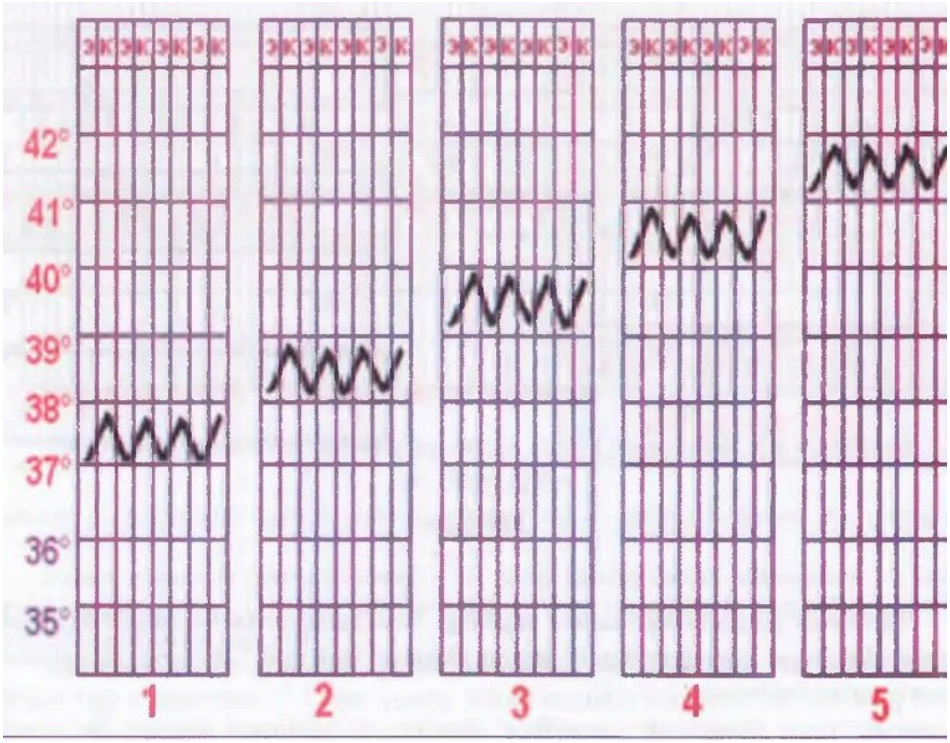


Figure 3.5. *Grades of fever: 1 — low-grade; 2 — moderate; 3 — high; 4 — hyperpyretic; 5 — extremely high (Gadayev A. et al., 2012, Fig. 24)*

Table 3.4. *Grades of fever (axillary)*

Grade	Temperature
Low-grade (subfebrile)	37.1–38.0 °C
Moderate	38.1–39.0 °C
High	39.1–41.0 °C
Hyperpyretic	Above 41 °C

Stages of fever:

1. **Stage of rising temperature** (stadium incrementi) — heat production exceeds heat loss: chills, shivering, pale and cold skin ("goose flesh").
2. **Stage of high temperature** (stadium fastigii) — heat production and heat loss are balanced at a high level: a feeling of heat, flushed and hot skin, headache, tachycardia.
3. **Stage of falling temperature** (stadium decrementi) — heat loss predominates, with profuse sweating. If the temperature falls sharply within a few hours, it is a **crisis** (risk of vascular insufficiency — collapse); if it falls gradually over several days, it is **lysis**.

CLINICAL SIGNIFICANCE

For every 1 °C rise in temperature, the heart rate increases by about **8–10 beats** per minute. If the temperature is high but the pulse is relatively slow (**relative bradycardia**), typhoid fever, brucellosis and legionellosis should be considered. In lobar pneumonia, collapse may develop during the critical fall: the patient sweats, turns pale and the blood pressure falls.

4.3. Types of fever (by the temperature curve)

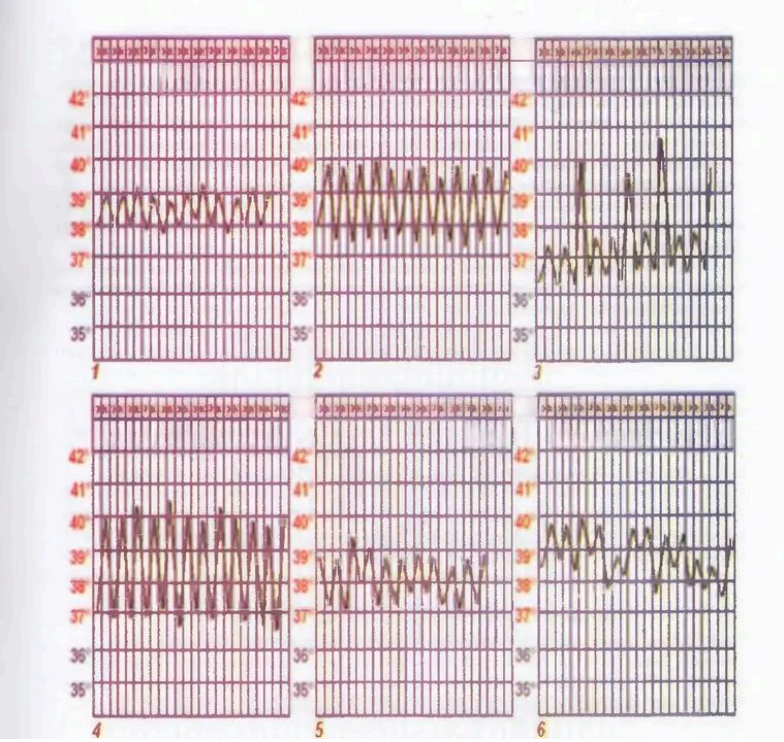


Figure 3.6. *Types of fever according to the change in temperature over the day: 1 — continuous; 2 — remittent; 3 — intermittent; 4 — hectic; 5 — inverse; 6 — irregular (Gadayev A. et al., 2012, Fig. 26)*

Table 3.5. *Types of fever*

Type	Latin name	Daily fluctuation	In which diseases
Continuous	Febris continua	Does not exceed 1 °C; the temperature is constantly high	Lobar pneumonia, typhoid fever
Remittent	Febris remittens	More than 1 °C, but does not fall to normal	Focal pneumonia, pleurisy, viral infections
Intermittent	Febris intermittens	Periods of high temperature alternate with periods of normal temperature	Malaria, pyelonephritis
Hectic (exhausting)	Febris hectica	3–5 °C, with profuse sweating	Sepsis, lung abscess, tuberculous cavity
Inverse	Febris inversa	The morning temperature is higher than the evening temperature	Tuberculosis, sepsis
Irregular	Febris irregularis	Variable, without a pattern	Rheumatism, endocarditis, influenza
Relapsing	Febris recurrens	Periods of fever lasting several days alternate with fever-free periods	Relapsing fever
Undulant	Febris undulans	The temperature rises gradually and falls gradually; the waves recur	Brucellosis, Hodgkin lymphoma (lymphogranulomatosis)

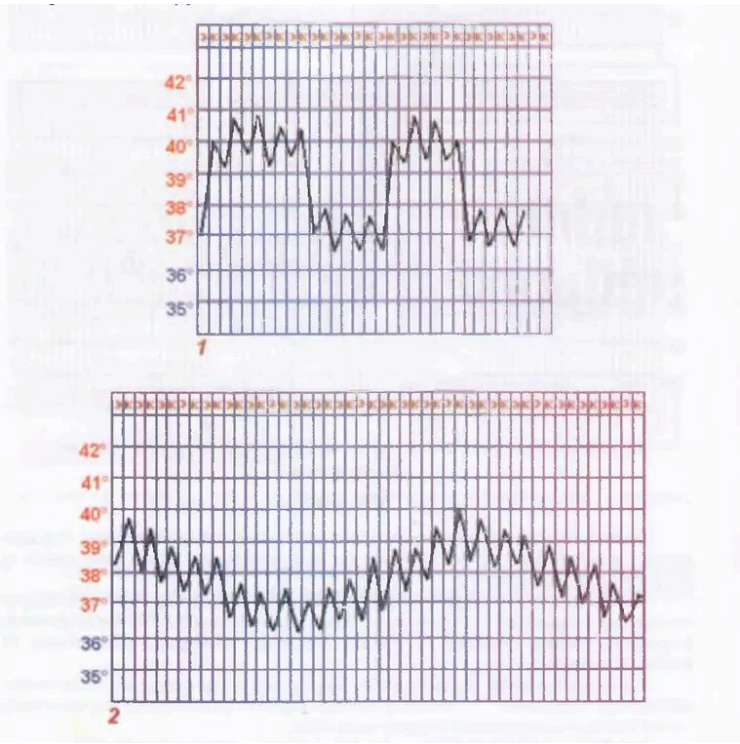


Figure 3.7. *Temperature curves in relapsing (1) and undulant (2) fevers (Gadayev A. et al., 2012, Fig. 27)*

MEMORY AID

"Continuous — Lobar": in lobar pneumonia the fever is continuous (continua) and ends with a crisis.
"Hectic — Pus": a hectic fever always makes one think of a suppurative process (abscess, sepsis).

5. Anthropometry

Anthropometry is the measurement of the body's dimensions (height, weight, chest and waist circumference). Height is measured in the morning, without shoes, with a stadiometer; weight is measured on an empty stomach, in light clothing.

Body mass index (BMI, Quetelet index) = weight (kg) / height^2 (m^2).

Table 3.6. *Classification by body mass index (WHO)*

BMI, kg/m^2	Interpretation
Less than 18.5	Underweight
18,5–24,9	Normal weight
25,0–29,9	Overweight (pre-obesity)
30,0–34,9	Obesity class I
35,0–39,9	Obesity class II
40 or more	Obesity class III (morbid)

The **waist circumference** is measured at the level midway between the lower edge of the costal arch and the iliac crest. The criterion of abdominal (visceral) obesity: **more than 94 cm** in men and **more than 80 cm** in women (for Europid populations, IDF); the level of a further increased cardiometabolic risk is more than 102 cm in men and 88 cm in women. A waist-to-hip ratio greater than 0.9 in men and 0.85 in women is also regarded as abdominal obesity.

CAUTION

BMI misjudges fat tissue in athletes with strongly developed muscles and in patients with edema. That is why the waist circumference must always be measured together with the BMI: abdominal obesity may be present even with a normal BMI.

6. Palpation as a method of objective examination

Palpation is a method of examination based on the tactile sensitivity of the palm and fingers. It is used to determine the location, size, shape, consistency, surface, mobility and tenderness of organs, as well as pulsation and vibration.

Table 3.7. *Types of palpation*

Type	Content	Application
Superficial (orienting)	Palpation with light pressure of the whole palm, comparing symmetrical areas	Skin temperature and moisture, muscle tension, tender areas, the abdominal wall
Deep	Palpation with deeper pressure of the fingers	Abdominal organs (the Obraztsov–Strazhesko method), kidneys
Bimanual	With two hands, holding the organ from two sides	Kidneys, gynecological examination
Ballottement	With short pushes	Detecting the liver and spleen in ascites
Static	Keeping the palm placed over the organ	Chest vibration (vocal fremitus), the cardiac impulse

REMEMBER

General rules of palpation: the physician’s hands should be **warm**, the nails short and the movements **gentle**; the examination begins in a painless area, and the painful area is palpated last; the patient should be in a comfortable position with the muscles relaxed; symmetrical areas are always compared.

7. Palpation of the peripheral lymph nodes

Normally the peripheral lymph nodes are not palpable, or in some groups (inguinal, axillary) they are found up to **1 cm** in diameter, soft-elastic, painless, mobile, and not adherent to each other or to the skin.



Figure 3.8. *Palpation of the cervical lymph nodes (Gadayev A. et al., 2012, Fig. 13)*

Order of palpation (from above downward, symmetrically on both sides at the same time): occipital → postauricular → preauricular → submandibular → submental → anterior and posterior triangles of the neck → supraclavicular and infraclavicular → axillary → cubital → inguinal → popliteal.

Table 3.8. *Criteria for assessing the lymph nodes*

Criterion	Normal	Pathology and possible cause
Size	Up to 1 cm	Enlarged (lymphadenopathy)
Consistency	Soft-elastic	Stony hard — tumor metastasis; firm-elastic — lymphoma
Tenderness	Painless	Tender — acute inflammation (lymphadenitis)
Mobility	Mobile	Fixed to the surrounding tissues — metastasis, tuberculosis
Relation to each other	Separate	Conglomerate (packet) — tuberculosis, Hodgkin lymphoma, metastases
Overlying skin	Unchanged	Red, hot — lymphadenitis; fistulas — tuberculosis

CLINICAL SIGNIFICANCE

Virchow's node — an enlarged hard node in the left supraclavicular fossa — is the classic sign of metastasis of gastric (and other abdominal) cancer. The right supraclavicular nodes most often enlarge in tumors of the lung and esophagus.

8. Palpation of the thyroid gland

The thyroid gland is located on the anterior surface of the neck, below the thyroid cartilage of the larynx, and consists of two lobes and the isthmus connecting them. Normally a lobe is no larger than the terminal phalanx of the patient's thumb.

Method of palpation (from in front of the patient): the patient sits or stands with the head bent slightly forward. Standing opposite the patient, the physician places the thumbs of both hands below the thyroid cartilage at the inner edge of the sternocleidomastoid muscles. The patient is asked to swallow a sip of water: during swallowing the gland slides upward under the fingers, and its lobes and isthmus are palpated.

Method of palpation (from behind the patient): the physician stands behind the patient, places fingers 2–4 of both hands over the gland and palpates the lobes during swallowing.

The following are assessed: size, consistency (soft, firm, hard), surface (smooth, uneven), the presence of nodules, tenderness, mobility. A systolic bruit heard on auscultation over the gland indicates the increased blood flow of diffuse toxic goiter.

Table 3.9. *Grades of goiter (WHO, 2001)*

Grade	Definition
0	No goiter: the lobes are smaller than the terminal phalanx of the thumb of the person examined
I	The goiter is palpable but not visible with the neck in the normal position
II	The goiter is visible with the neck in the normal position

CAUTION

Older textbooks (the classification of O.V. Nikolaev) distinguished **grades 0–V** of goiter. Present-day practice uses the simplified **grade 0–II** classification recommended by the WHO, and the gland volume is determined precisely by US (normal up to 18 ml in women and 25 ml in men).

9. Additional aspects of the skin examination

9.1. Technique of assessing skin turgor and moisture



Figure 3.9. *Picking up a skinfold (Gadayev A. et al., 2012, Fig. 9)*



Figure 3.10. *Observing the smoothing out of the fold (same source, Fig. 9)*

A skinfold is picked up with the thumb and index finger on the back of the hand or on the skin of the chest or abdomen and released. If the fold persists for more than 2 seconds, turgor is reduced (dehydration, old age, cachexia). Skin moisture is assessed with the back of the hand: dry, normally moist, moist, wet (profuse sweating).

9.2. Additional types of skin rashes



Figure 3.11. *Skin lesions in a patient with psoriasis (Gadayev A. et al., 2012, Fig. 10)*

Table 3.10. *Important skin signs in internal diseases*

Feature	Appearance	Internal disease
Psoriatic plaques	Red plaques with silvery scales (elbows, knees, scalp)	Psoriatic arthritis
Erythema nodosum	Tender red-bluish nodules on the anterior surface of the lower leg	Sarcoidosis, tuberculosis, streptococcal infection, inflammatory bowel diseases
Pyoderma gangrenosum	Deep ulcers with bluish edges	Ulcerative colitis, Crohn’s disease
Acanthosis nigricans	Dark velvety skin in the axillary and neck folds	Insulin resistance, diabetes mellitus, sometimes tumors
Xanthomas	Yellow nodules (tendons, elbows)	Familial hypercholesterolemia
Thickening and tightness of the skin	Masklike face, thin, immobile fingers	Systemic sclerosis
Heliotrope rash	Violaceous erythema of the eyelids	Dermatomyositis
“Liver palms” and “spider angiomas”	Redness of the palms, spider telangiectasias	Liver cirrhosis
Rash of hemorrhagic vasculitis	Symmetrical, raised purpura on the legs that does not disappear on pressure	Henoch–Schönlein purpura (IgA vasculitis)

9.3. Hair and nails



Figure 3.12. *Alopecia areata (Gadayev A. et al., 2012, Fig. 12, A)*



Figure 3.13. *Alopecia totalis (same source, Fig. 12, B)*

Other changes of the nails: **Beau's lines** (transverse grooves — a temporary arrest of nail growth after a severe illness), **leukonychia** (white spots), **onycholysis** (separation of the nail plate — thyrotoxicosis, psoriasis), **splinter hemorrhages** under the nail (infective endocarditis), **nail pitting** (the "thimble" sign — psoriasis).

10. Pathogenetic types of edema

Edema is the accumulation of fluid in the interstitial spaces and develops as a result of a disturbance in the balance of the Starling forces: the hydrostatic pressure in the capillaries drives fluid out, and the plasma oncotic pressure (mainly albumin) retains it inside the vessel.

Table 3.11. *Pathogenetic types of edema*

Type	Mechanism	Examples
Hydrostatic (congestive)	Increased venous pressure	Heart failure, venous thrombosis, varicose veins
Hypo-oncotic (hypoproteinemic)	Decreased albumin (below 30 g/l)	Nephrotic syndrome, liver cirrhosis, starvation ("hunger edema"), enteropathies
Membranogenic	Increased capillary permeability	Inflammation, allergy (angioedema), burns, sepsis
Lymphatic (lymphedema)	Impaired lymph flow	After removal of lymph nodes, filariasis, tumors; firm, does not pit
Hormonal	Retention of sodium and water	Hypothyroidism (myxedema — firm, does not pit), hyperaldosteronism, glucocorticoids

CLINICAL SIGNIFICANCE

Unilateral leg edema with pain and redness may be a sign of deep vein thrombosis; because of the risk of **pulmonary embolism**, this condition requires urgent Doppler ultrasound. Bilateral symmetrical edema is most often linked with a systemic cause (heart, kidneys, liver, hypoalbuminemia).

11. Examination of the muscles and joints

Muscle strength is assessed on a 0–5 point scale.

Table 3.12. *Scale for assessing muscle strength (MRC)*

Score	Definition
0	No muscle contraction
1	A flicker of contraction is felt, no movement
2	Movement with gravity eliminated (in the horizontal plane)
3	Movement against gravity, but not against resistance
4	Movement against resistance, strength slightly reduced
5	Normal strength

The sequence of **examining the joints**: inspection (shape, symmetry, swelling, skin color, deformity) → palpation (temperature, tenderness, consistency of the swelling, fluid — "ballottement" of the patella, crepitus) → determining the range of active and passive movements (with a goniometer). Joint pain may be **inflammatory** (morning stiffness lasting more than 30 minutes, lessening with movement — rheumatoid arthritis) or **mechanical** (worse on loading, relieved by rest, stiffness for less than 30 minutes — osteoarthritis).

Table 3.13. *Characteristic joint changes in internal diseases*

Disease	Nature of involvement
Acute rheumatic fever	Large joints, migrating, without deformity
Rheumatoid arthritis	Metacarpophalangeal and proximal interphalangeal joints, symmetrical; "walrus flipper" (ulnar deviation) and "swan neck" deformities
Gout	Often an acute, extremely painful arthritis of the first joint of the big toe (at night), tophi
Osteoarthritis	Knee and hip joints, distal (Heberden's) and proximal (Bouchard's) interphalangeal nodes
Ankylosing spondylitis	The spine and sacroiliac joints, the "beggar's posture"

12. Thermometry: additional information



Figure 3.14. *Glass mercury thermometer (Gadayev A. et al., 2012, Fig. 21)*



Figure 3.15. *Electronic thermometer (same source, Fig. 22)*

Table 3.14. *Methods of measuring body temperature*

Method	Normal	Advantages and disadvantages
In the axilla	36.4–36.8 °C	Most widely used, safe; sensitive to external factors, less accurate

Method	Normal	Advantages and disadvantages
In the mouth (under the tongue)	36.8–37.3 °C	More accurate; wait 15 minutes after a hot or cold drink
In the rectum	37.0–37.7 °C	Closest to the core temperature; in newborns and severely ill patients
In the ear (tympanic, infrared)	36.8–37.6 °C	Fast (1–2 seconds); inaccurate if there is earwax in the ear canal
On the forehead (infrared)	Depends on the device	Contactless, convenient for screening; the least accurate

12.1. Pathogenesis of fever

Exogenous pyrogens (bacterial endotoxin, viruses) and tissue breakdown products activate leukocytes and macrophages → **endogenous pyrogens** (interleukin-1, interleukin-6, tumor necrosis factor alpha) are released → **prostaglandin E_2** is formed in the hypothalamus → the "set point" of the thermoregulatory center rises. Antipyretic drugs (paracetamol, NSAIDs) act precisely by blocking prostaglandin synthesis.

The protective and harmful sides of fever: fever enhances the immune response and slows the multiplication of many microbes; however, high fever increases oxygen consumption and places a burden on the heart (tachycardia), the nervous system (seizures in children) and metabolism.

12.2. Hypothermia and fever of unknown origin

Hypothermia — a core temperature below 35 °C: exposure to cold, hypothyroidism (myxedema coma), the severe stage of sepsis, alcohol intoxication, adrenal insufficiency.

Fever of unknown origin — a temperature above 38.3 °C lasting more than 3 weeks whose cause has not been found even after one week of investigation in hospital. The main groups of causes: **infections** (tuberculosis, infective endocarditis, occult abscesses), **tumors** (lymphomas, kidney cancer), **connective tissue diseases** (SLE, vasculitides, Still's disease), **drug fever**.

> **MEMORY AID**
> The three large groups of fever of unknown origin — **"INC"**: **I**nfection, **N**eoplasm (tumor), **C**onnective tissue (rheumatic) diseases.

13. Causes of lymph node enlargement

Table 3.15. *The main causes of lymphadenopathy*

Group	Causes	Features of the nodes
Local infection	Tonsillitis, dental caries (submandibular), leg injuries (inguinal)	Tender, soft, mobile, regional
Systemic infections	Infectious mononucleosis, HIV, toxoplasmosis, brucellosis	Generalized, moderately firm
Tuberculosis	Cervical nodes more often	Matted packets, fistulas
Lymphomas	Hodgkin lymphoma, non-Hodgkin lymphomas	Firm-elastic ("rubbery"), painless, large packets
Leukemias	Chronic lymphocytic leukemia, acute leukemias	Generalized, soft-elastic

Group	Causes	Features of the nodes
Tumor metastases	Virchow’s (left supraclavicular), axillary (breast cancer)	Stony hard, fixed, painless
Autoimmune diseases	SLE, rheumatoid arthritis, sarcoidosis	Moderately enlarged

CAUTION

If a lymph node is **larger than 2 cm**, **hard**, **fixed**, located **in the supraclavicular region**, or remains enlarged for **more than 4 weeks** and the patient is older than 40, there is a strong indication for biopsy. It is a mistake to keep observing such a node as “inflammation”.

14. General methods of palpation and palpation of the abdomen

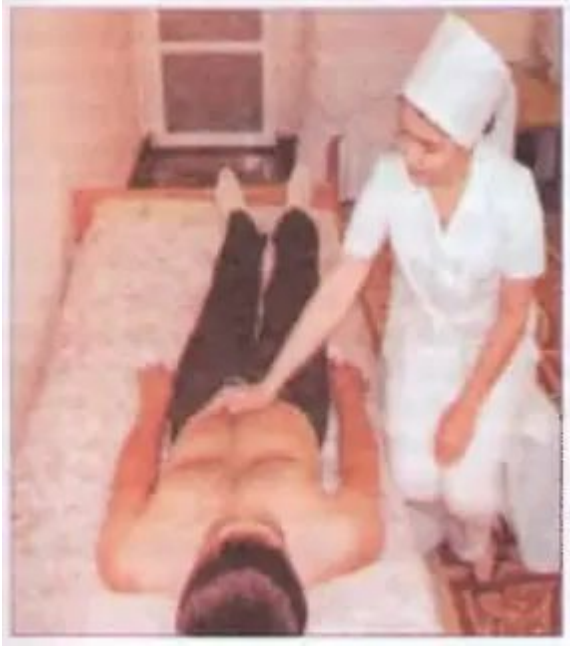


Figure 3.16. *Superficial palpation of the abdomen (Gadayev A. et al., 2012, Fig. 16)*

Palpation is one of the oldest methods of propaedeutics, in which the physician’s hand serves as a “measuring instrument”. Different parts of the hand are sensitive to different sensations: **the fingertips** — small structures and consistency; **the base of the palm and the region of the metacarpophalangeal joints** — vibration (vocal fremitus, the “cat’s purr”); **the back of the hand** — temperature.

In the abdomen, **superficial orienting palpation** detects tension of the abdominal wall, tender areas, hernial openings and superficial masses, whereas **deep sliding methodical palpation** (Obraztsov–Strazhesko) makes it possible to examine the intestines, stomach, liver, spleen and kidneys in succession. This method is studied in detail in the next semester in the section on the digestive organs.

15. Clinical assessment of thyroid function

Along with palpation of the thyroid gland, its **function** must be assessed by clinical signs.

Table 3.16. *Clinical signs of thyrotoxicosis and hypothyroidism*

System	Thyrotoxicosis	Hypothyroidism
General	Weight loss (even with a good appetite), heat intolerance, sweating	Weight gain, cold intolerance, weakness
Skin	Warm, moist, soft	Dry, thick, cold, yellowish, firm edema
Cardiovascular system	Tachycardia, atrial fibrillation, raised systolic BP, wide pulse pressure	Bradycardia, muffled heart sounds, pericardial fluid
Nervous system	Nervousness, hand tremor, insomnia, brisk reflexes	Slowness, drowsiness, memory impairment, slowed reflexes
Digestive system	Diarrhea	Constipation
Eyes	Exophthalmos and eye signs (in Graves' disease)	Eyelid edema
Laboratory	TSH decreased, T_4 and T_3 raised	TSH raised (in primary hypothyroidism), T_4 decreased

16. Location of the lymph nodes and their drainage areas

The location of an enlarged lymph node helps to suggest where an infection or tumor originates.

Table 3.17. *Groups of lymph nodes and their drainage areas*

Group	Drainage area	Causes of enlargement
Occipital	Scalp	Scalp infections, rubella
Postauricular and preauricular	Ear, eye, face	Conjunctivitis, otitis, rubella
Submandibular and submental	Tongue, mouth, teeth, lips	Dental caries, stomatitis, tonsillitis, oral tumors
Anterior triangle of the neck	Pharynx, tonsils, thyroid gland	Tonsillitis, pharyngitis, thyroid tumor, tuberculosis
Posterior triangle of the neck	Scalp, nasopharynx	Infectious mononucleosis, lymphomas, tuberculosis
Supraclavicular (right)	Mediastinum, lungs, esophagus	Lung and esophageal cancer, lymphomas
Supraclavicular (left) — Virchow's	Abdominal cavity (via the thoracic duct)	Cancer of the stomach, pancreas, intestine, ovary
Axillary	Arm, chest wall, breast	Arm infections, breast cancer, lymphomas, cat-scratch disease
Cubital	The hand and the inner part of the forearm	Hand infections, lymphomas, syphilis
Inguinal	Leg, external genitalia, perineum, anus	Leg infections, sexually transmitted diseases, tumors

17. Additional methods of assessing body weight

Broca index — ideal weight = height (cm) − 100 (for people 155–165 cm tall); at 165–175 cm, − 105; above 175 cm, − 110. Exceeding the ideal weight by more than 10 % is overweight.

Skinfold thickness is measured with a special instrument (caliper) at several points (the back of the upper arm — triceps, below the shoulder blade, next to the umbilicus, over the iliac crest), and the

proportion of fat in the body is calculated. Modern methods are **bioimpedance analysis** and **DEXA**.

CAUTION

A BMI below 18.5 kg/m^2 and marked weight loss over a short period are signs of **protein-energy malnutrition**. Malnutrition occurs in 20–40 percent of hospital patients, and it increases the risk of infectious complications, slow wound healing and death.

18. Additional features of conditions accompanied by fever

Table 3.18. *Combinations of fever with other signs*

Combination	Suspected condition
Fever + a new heart murmur	Infective endocarditis
Fever + migrating polyarthritis (after tonsillitis)	Acute rheumatic fever
Fever + stabbing chest pain + cough	Pneumonia, pleurisy
Fever + chills + low back pain + dysuria	Acute pyelonephritis
Fever + jaundice + pain in the right hypochondrium	Cholangitis (Charcot’s triad)
Fever + lymphadenopathy + splenomegaly	Infectious mononucleosis, lymphomas, leukemias
Fever + facial “butterfly” rash + arthritis	Systemic lupus erythematosus
Fever + headache + neck stiffness	Meningitis

Pel–Ebstein fever — periods of fever lasting several days alternate with fever-free periods — is characteristic of Hodgkin lymphoma (a variety of undulant fever).

19. Secondary elements of skin rashes

Secondary elements form as a result of changes in the primary elements (macule, papule, vesicle, pustule and others).

Table 3.19. *Secondary skin elements*

Element	Definition
Scale (squama)	Shedding of the horny layer (psoriasis, ichthyosis)
Crust (crusta)	Drying of the fluid of a vesicle or pustule
Erosion	A superficial defect within the epidermis that heals without a scar
Ulcer (ulcus)	A defect reaching the dermis and deeper layers that heals with a scar (trophic ulcers — venous insufficiency, diabetes, arterial insufficiency)
Excoriation	Scratch marks (cholestasis, uremia, allergy)
Scar	Replacement by connective tissue (surgery, injury, tuberculous fistulas)
Pigmentation	A change in color left at the site of an element

CLINICAL SIGNIFICANCE

The location of a trophic ulcer points to its cause: around the **medial malleolus**, with pigmentation and edema — **venous**; on the toes, the heel and pressure points, painful, with pale cold skin — **arterial**; under the heel and the metatarsophalangeal joints, painless, with impaired sensation — **neuropathic (diabetic)**.

20. Clinical part

Sample case history entry

The skin is pale yellowish, dry, with reduced turgor. Isolated “spider angiomas” on the chest and shoulders, erythema of the palms. No rashes or scars. The nails are unchanged. **The subcutaneous fat** is poorly developed; the skinfold next to the umbilicus is 0.8 cm thick. Soft edema of the lower legs. **Lymph nodes:** the peripheral groups are not palpable. **The thyroid gland** is not palpable (grade 0). **The muscles** are atrophied, with reduced tone. The joints are of normal shape, with a full range of movement. **Body temperature** 37.4 °C. Height 172 cm, weight 61 kg, BMI 20.6 kg/m^2.

Conclusion: jaundice, “spider angiomas” and liver palms suggest a chronic liver disease (cirrhosis).

21. Practical skills

Skill: measuring body temperature in the axilla

1. Wash your hands and explain the procedure to the patient.
2. Check the thermometer (shake the reading of a mercury thermometer down below 35 °C; switch on an electronic thermometer).
3. Wipe the axillary fossa with a dry napkin.
4. Place the tip of the thermometer in the center of the axillary fossa and press the patient’s arm against the chest.
5. Wait 10 minutes with a mercury thermometer, or until the signal sounds with an electronic thermometer.
6. Read the result, record it on the temperature chart (morning and evening) and draw the curve.
7. Disinfect the thermometer.

Skill: calculating the body mass index

1. Measure height in the morning, without shoes, with the heels, buttocks and shoulder blades touching the wall.
2. Measure weight on an empty stomach, in light clothing.
3. Calculate according to the formula BMI = weight (kg) / height^2 (m^2). Example: 85 kg / (1.70 m)^2 = 85 / 2.89 = 29.4 kg/m^2 — overweight.
4. Measure the waist circumference and compare it with the criteria of abdominal obesity.

Skill: palpation of the peripheral lymph nodes

1. Position the patient sitting, with the neck and shoulder girdle uncovered.
2. Palpate symmetrical areas at the same time with fingers 2–4 of both hands, using circular movements.
3. Order: occipital, postauricular, preauricular, submandibular, submental, anterior and posterior triangles of the neck, supraclavicular and infraclavicular.
4. When palpating the axillary nodes, the patient raises the arm slightly to the side; the physician inserts the fingers deep into the axillary fossa and, after the patient lowers the arm, slides them downward along the chest wall.
5. Palpate the cubital, inguinal and popliteal nodes.

6. Describe the nodes found according to the criteria (size, consistency, tenderness, mobility, matting, skin).

Review questions

1. What is the difference between central and peripheral cyanosis?
2. How can true and false jaundice be distinguished?
3. What elements of skin rashes do you know? What is the difference between a petechia and roseola?
4. What skin changes are observed in liver cirrhosis?
5. In which diseases does clubbing occur?
6. By what signs do you distinguish cardiac and renal edema?
7. What are the normal values of body temperature at various sites?
8. What is the difference between fever and hyperthermia?
9. List the grades and stages of fever.
10. What are crisis and lysis? What is the danger of a crisis?
11. What types of fever are there, and which diseases are they characteristic of?
12. How is the body mass index calculated and interpreted?
13. What types of palpation are there?
14. By what criteria are the lymph nodes assessed? What is Virchow’s node?
15. How is the thyroid gland palpated, and what are the grades of goiter?

Test questions

1. Which sign is characteristic of peripheral cyanosis?
   A) Blueness only on the face
   B) Warm skin, blue tongue
   C) Only the sclerae are blue
   D) Cold skin, blueness in the acral areas
   E) Flushed skin

2. Which sign is observed in false jaundice (from eating carotene)?
   A) The mucosa under the tongue turns yellow
   B) The sclerae turn yellow
   C) The stool becomes pale
   D) The urine becomes dark
   E) Only the palms and soles turn yellow; the sclerae are white

3. Which disease are painful pea-sized nodules on the fingertips (Osler’s nodes) characteristic of?
   A) Infective endocarditis
   B) Gout
   C) Diabetes mellitus
   D) Liver cirrhosis
   E) Rheumatoid arthritis

4. Which sign is characteristic of cardiac edema?
   A) It begins on the eyelids
   B) It spreads rapidly to the whole body
   C) It increases in the legs in the evening; the skin is cyanotic
   D) The skin is pale and warm
   E) It appears on the face in the morning

5. In what range is the normal body temperature measured in the axilla?
   A) 35.5–36.0 °C
   B) 36.4–36.8 °C
   C) 37.0–37.5 °C
   D) 37.5–38.0 °C
   E) 38.0–38.5 °C

6. What is a constantly high fever whose daily fluctuation does not exceed 1 °C called?
   A) Febris continua
   B) Febris inversa
   C) Febris remittens
   D) Febris intermittens
   E) Febris hectica

7. In which condition is a hectic fever most often observed?
   A) Hypothyroidism
   B) Bronchial asthma
   C) Lobar pneumonia
   D) Lung abscess and sepsis
   E) Influenza

8. By approximately how much does the pulse rate increase per minute when the temperature rises by 1 °C?
   A) 2–3 beats
   B) 8–10 beats
   C) 20–25 beats
   D) 30 beats
   E) Unchanged

9. How is a BMI of 32 kg/m^2 interpreted?
   A) Normal weight
   B) Obesity class I
   C) Obesity class II
   D) Overweight
   E) Obesity class III

10. What is the waist circumference criterion of abdominal obesity in women (IDF)?
    A) More than 70 cm
    B) More than 80 cm
    C) More than 94 cm
    D) More than 102 cm
    E) More than 110 cm

11. What is a hard enlarged lymph node in the left supraclavicular fossa called, and what does it indicate?
    A) Strazhesko’s node — tuberculosis
    B) Heberden’s node — osteoarthritis
    C) Bouchard’s node — rheumatoid arthritis
    D) Osler’s node — endocarditis
    E) Virchow’s node — metastasis of gastric cancer
12. What does a stony hard, painless lymph node fixed to the surrounding tissues most often indicate?
    A) An allergic reaction
    B) A viral infection
    C) A tumor metastasis
    D) A normal node
    E) Acute lymphadenitis
13. According to the WHO classification, which grade is a goiter that is palpable but not visible with the neck in the normal position?
    A) 0
    B) I
    C) II
    D) III
    E) IV
14. Which condition are spoon-shaped nails (koilonychia) characteristic of?
    A) Psoriasis
    B) Liver cirrhosis
    C) Iron deficiency anemia
    D) Vitamin B_{12} deficiency anemia
    E) Bronchiectasis
15. What is a sharp fall of the temperature from a high level to normal within a few hours called, and what is its main danger?
    A) Remission — safe
    B) Crisis — collapse
    C) Lysis — hyperthermia
    D) Lysis — seizures
    E) Crisis — hypertensive crisis

Clinical cases

Case 1

A 28-year-old patient has had a body temperature of 36.8 °C in the morning rising to 40.1 °C in the evening for 7 days. When the temperature falls, he sweats profusely and changes his shirt several times. A tooth was extracted two weeks ago.

Questions: What is the type of fever? Which processes is such a fever characteristic of? What signs should be looked for on the skin and nails?

Answer: A daily fluctuation of more than 3 °C with profuse sweating is a hectic (exhausting) fever. It is characteristic of suppurative processes and sepsis. Its development after a tooth extraction raises

suspicion of infective endocarditis, so one should look for petechiae, Osler's nodes, Janeway lesions, clubbing and pallor of the skin ("café au lait" color) and examine the heart.

Case 2

A 55-year-old man complains that he has lost 9 kg in the last 3 months and has no appetite. On examination, a 2 cm hard, painless, fixed lymph node is palpated in the left supraclavicular fossa.

Questions: Describe the node found. What might it indicate? What investigations are needed?

Answer: A stony hard, painless enlarged node fixed to the surrounding tissues is Virchow's node. Together with the weight loss and loss of appetite, it gives grounds to suspect metastasis of gastric (or other abdominal) cancer. Esophagogastroduodenoscopy, abdominal US/CT and biopsy of the node are needed.

Case 3

A 45-year-old woman is 160 cm tall, weighs 82 kg and has a waist circumference of 98 cm. Her blood pressure is 150/95 mm Hg.

Questions: Calculate and interpret the BMI. What type of obesity is this? What risk does this condition pose?

Answer: BMI = 82 / $(1.60)^2$ = 82 / 2.56 = 32.0 kg/m^2 — obesity class I. A waist circumference of 98 cm (more than 88 cm in women) indicates abdominal (visceral) obesity. Abdominal obesity together with arterial hypertension is a sign of the metabolic syndrome and increases the risk of type 2 diabetes mellitus, atherosclerosis and ischemic heart disease.

Quick review

Table 3.20. *Key facts on Topic 3*

Question	Answer
Central cyanosis	Warm skin, blue tongue — respiratory failure
Peripheral cyanosis	Cold skin, acrocyanosis — heart failure
Jaundice	When bilirubin exceeds 34 µmol/l; first in the sclerae
Liver signs	"Spider angiomas", liver palms, Terry's nails
Cardiac edema	Evening, legs, cyanotic, cold
Renal edema	Morning, face, pale, warm
Axillary temperature	36.4–36.8 °C; daily fluctuation up to 1 °C
Grades of fever	37.1–38 / 38.1–39 / 39.1–41 / above 41
Febris continua	Fluctuation <1 °C — lobar pneumonia
Febris hectica	Fluctuation 3–5 °C — abscess, sepsis
BMI	18.5–24.9 normal; 25–29.9 overweight; 30 or more — obesity
Waist circumference	Men >94 cm, women >80 cm — abdominal obesity
Normal lymph node	Up to 1 cm, soft-elastic, painless, mobile
Goiter (WHO)	0 — none; I — palpable; II — visible

TOPIC 4

Methods of examining patients with respiratory diseases. History taking. Inspection and palpation of the chest

Learning objectives

To teach the student to identify and analyze the main complaints in respiratory diseases, to perform static and dynamic inspection of the chest, and to assess tender areas, chest wall resistance and vocal fremitus by palpation.

By the end of the session, the student should know:

- the main complaints in respiratory diseases: cough, sputum, dyspnea, choking attacks, hemoptysis, chest pain;
- the types of sputum and their diagnostic significance;
- the types of dyspnea (inspiratory, expiratory, mixed);
- the normal and pathological shapes of the chest;
- the types, rate, depth and rhythm of breathing and the pathological types of breathing;
- the causes of increased and decreased vocal fremitus.

The student should be able to:

- take the history of a patient with a respiratory disease according to a plan;
- perform static and dynamic inspection of the chest;
- count the respiratory rate without the patient noticing;
- determine chest wall resistance and vocal fremitus and interpret the result.

1. Brief anatomy and physiology of the respiratory organs

The respiratory system consists of the upper airways (nose, pharynx, larynx), the trachea, the bronchi and the lungs. At the level of the IV–V thoracic vertebrae, the trachea divides into the right and left main bronchi (the bifurcation). The right lung has three lobes and the left lung two; each lung is divided into 10 segments. The main structure of gas exchange is the **acinus** (respiratory bronchiole, alveolar ducts and alveoli).

The three stages of external respiration are **ventilation** (the movement of air into and out of the alveoli), **diffusion** (the exchange of O_2 and CO_2 across the alveolar-capillary membrane) and **perfusion** (blood flow in the pulmonary capillaries). Impairment of any of them leads to respiratory failure.

2. History taking: the main complaints

2.1. Cough (tussis)

Cough is a protective reflex that clears the airways of sputum, foreign bodies and other irritants. Cough receptors are most numerous in the larynx, the tracheal bifurcation, the large bronchi and the pleura; there are no cough receptors in the alveoli.

Table 4.1. *Characteristics of cough*

Feature	Type	Diagnostic significance
Presence of sputum	Dry (unproductive)	Onset of bronchitis, dry pleurisy, tracheitis, bronchial asthma, lung tumor, foreign body, taking ACE inhibitors
	Wet (productive)	Chronic bronchitis, pneumonia, bronchiectasis, rupture of an abscess
Time of appearance	Morning	Chronic bronchitis, bronchiectasis (the sputum accumulated overnight is expelled)
	Evening and nocturnal	Bronchial asthma, left ventricular failure (cardiac asthma), gastroesophageal reflux
Character of the sound	Barking (“like a dog’s bark”)	Laryngeal involvement (false croup)
	Bitonal	Compression of a large bronchus by a tumor or enlarged lymph nodes
	Paroxysmal, with a deep “whoop”	Pertussis (whooping cough)
	Clearing the throat (hacking)	Chronic pharyngitis, early tuberculosis, smokers
Duration	Acute (up to 3 weeks)	Acute respiratory infections, pneumonia
	Chronic (more than 8 weeks)	Chronic bronchitis, COPD, asthma, tuberculosis, lung cancer

2.2. Sputum

When sputum is described, its **amount** (single and daily), **character**, **color**, **odor**, **consistency**, **separation into layers** and **admixtures** (blood) are determined.

Table 4.2. *Types of sputum*

Type	Appearance	Diseases
Mucoid	Colorless, clear, viscous (“glassy”)	Onset of acute bronchitis, bronchial asthma (with Curschmann spirals)
Mucopurulent	Yellowish, turbid	Chronic bronchitis, focal pneumonia
Purulent	Yellow-green, often foul-smelling	Bronchiectasis, rupture of a lung abscess
Serous	Liquid, frothy, pink	Pulmonary edema
“Rusty”	Viscous, rust-colored (brownish-red)	Lobar pneumonia (stage of red hepatization)
“Raspberry jelly” type	Mucus mixed with blood	Lung cancer
Blood-stained	Streaks of blood or pure blood	Tuberculosis, lung cancer, bronchiectasis, pulmonary infarction, mitral stenosis

REMEMBER

When purulent sputum is left to settle in a container, it separates into layers. According to the textbook, **two layers** (pus and serous fluid) are characteristic of lung abscess, and **three layers** (pus at the bottom, serous fluid in the middle, frothy mucus on top) of bronchiectasis and cavitary tuberculosis.

CAUTION

In Russian-language textbooks (Grebenev, Mukhin), three-layered sputum is given as a common sign of **lung abscess, lung gangrene and bronchiectasis**. Hence the number of layers does not strictly distinguish an abscess from bronchiectasis; the decisive investigation is radiography and CT.

2.3. Dyspnea

Dyspnea is a disturbance of the rate, depth and rhythm of breathing in which the patient feels short of air. It may be **subjective** (felt by the patient), **objective** (detected by the physician) or **mixed**.

Table 4.3. *Types of dyspnea*

Type	Content	Causes
Inspiratory	Inspiration is difficult	Narrowing of the upper airways (larynx, trachea): edema, foreign body, tumor, diphtheria
Expiratory	Expiration is difficult and prolonged	Obstruction of the small bronchi: bronchial asthma, COPD, pulmonary emphysema
Mixed	Both phases are difficult	Reduction of the respiratory surface: pneumonia, pleurisy, pneumothorax; heart failure

The dyspnea scale of the Medical Research Council (mMRC):

Table 4.4. *The mMRC dyspnea scale*

Grade	Definition
0	Breathless only with strenuous exercise
1	Breathless when hurrying or walking up a slight hill
2	Walks slower than people of the same age because of breathlessness, or has to stop for breath when walking at own pace on level ground
3	Stops for breath after walking about 100 m or after a few minutes on level ground
4	Too breathless to leave the house, or breathless when dressing

2.4. Choking attacks (asthma)

A **choking attack** is a suddenly developing attack of severe dyspnea. Two types are distinguished:

- **An attack of bronchial asthma** — expiratory dyspnea, wheezes audible at a distance, expectoration of glassy viscous sputum at the end of the attack;
- **Cardiac asthma** — the result of acute left ventricular failure, usually at night; inspiratory or mixed dyspnea, the orthopnea position, moist rales in the lower parts of the lungs, and in severe cases pink frothy sputum (pulmonary edema).

2.5. Hemoptysis (haemoptysis)

Hemoptysis is the expectoration of blood with sputum on coughing. If more than 50 ml of blood is expectorated per day, it is called **pulmonary hemorrhage**. Causes: tuberculosis, lung cancer, bronchiectasis, lung abscess, pulmonary infarction, mitral stenosis, lobar pneumonia.

Table 4.5. *Differences between pulmonary and gastric bleeding*

Feature	Pulmonary bleeding	Gastric bleeding
How it is expelled	With coughing	With vomiting

Feature	Pulmonary bleeding	Gastric bleeding
Color	Bright red, frothy	Dark red or “coffee grounds”
Reaction (pH)	Alkaline	Acidic
Admixtures	Sputum	Food remnants
Subsequent signs	Blood-stained sputum for several days	Black stool (melena)
Anamnesis	Lung and heart diseases	Peptic ulcer disease, liver cirrhosis

2.6. Chest pain

In respiratory diseases, pain arises mainly from irritation of the **parietal pleura** (the lung tissue has no pain receptors).

> **CLINICAL SIGNIFICANCE**
> **Pleuritic pain** is stabbing, increases with deep inspiration and coughing, and lessens when the patient lies on the affected side or presses the chest with the hand (dry pleurisy, lobar pneumonia, pulmonary infarction, pneumothorax). In **intercostal neuralgia**, the pain spreads along the intercostal space and increases when the trunk bends to the healthy side. **The pain of angina pectoris**, by contrast, does not depend on breathing and appears on physical exertion.

2.7. Features of the history

In the history of present illness, the link of the disease with chilling or influenza and the periods of exacerbation and remission are established. In the life history, **smoking** (pack-years), **occupational hazards** (dust, silica, asbestos, chemicals), **allergic diseases** (pollinosis, eczema), past tuberculosis, contact with a tuberculosis patient and bronchial asthma in the family must always be asked about.

3. Inspection of the chest

The chest is inspected **statically** (shape, symmetry, supraclavicular and infraclavicular fossae, position of the shoulder blades, direction of the ribs, epigastric angle) and **dynamically** (type, rate, depth and rhythm of breathing, equal participation of both sides in breathing, use of the accessory muscles). The patient stands or sits, undressed to the waist.

3.1. Shape of the chest

The normal shapes correspond to the constitutional type: **normosthenic** (conical), **hypersthenic** (cylindrical, broad and short) and **asthenic** (long, narrow, flat).

Table 4.6. *Pathological shapes of the chest*

Shape	Signs	Causes
Emphysematous (barrel-shaped)	The anteroposterior diameter equals the transverse diameter, the ribs are horizontal, the intercostal spaces are wide, the epigastric angle is obtuse, the supraclavicular fossae bulge, and the chest is fixed “in the position of inspiration”	Pulmonary emphysema, COPD, long-standing bronchial asthma
Paralytic	An extreme form of the asthenic shape: flat, narrow, with winged shoulder blades and atrophied muscles	Tuberculosis, pulmonary fibrosis, cachexia
Rachitic (keeled, “pigeon chest”)	The sternum protrudes forward like the keel of a boat; a “rosary” at the costochondral junctions	Rickets in childhood

Shape	Signs	Causes
Funnel-shaped (“cobbler’s chest”)	A depression in the lower part of the sternum	Congenital defect, prolonged pressure in cobblers
Scaphoid (boat-shaped)	A boat-shaped depression in the upper and middle parts of the sternum	Syringomyelia (a disease of the spinal cord)
Kyphoscoliosis	Curvature of the spine backward (kyphosis) and to the side (scoliosis), deformity of the chest	Rickets, tuberculous spondylitis, ankylosing spondylitis (Bekhterev’s disease); pulmonary ventilation is impaired



Figure 4.1. *Emphysematous chest (Gadayev A. et al., 2012, Fig. 5)*



Figure 4.2. *Paralytic chest (same source, Fig. 6)*



Figure 4.3. *Scaphoid chest (Gadayev A. et al., 2012, Fig. 7)*

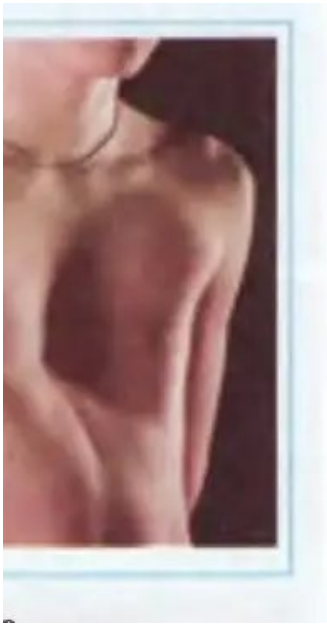


Figure 4.4. *Funnel chest (same source, Fig. 9)*

3.2. Asymmetry of the chest

- **Enlargement (bulging) of one side** — accumulation of a large amount of fluid (exudative pleurisy, hydrothorax) or air (pneumothorax) in the pleural cavity. On this side the intercostal spaces are smoothed out or bulge, and the side lags behind in breathing.
- **Reduction (retraction) of one side** — obstructive atelectasis, removal of a lung (pneumonectomy), pleural adhesions (schwartes), pneumosclerosis. The shoulder is lowered, the shoulder blade protrudes backward, and the intercostal spaces are narrowed.

3.3. Type, rate, depth and rhythm of breathing

Types of breathing: the thoracic (costal) type — mainly in women; the abdominal (diaphragmatic) type — mainly in men; the mixed type — in the elderly. A change from the thoracic to the abdominal type is seen in pleurisy, rib fractures and intercostal neuralgia; a change from the abdominal to the thoracic type in diaphragmatic paralysis, peritonitis, ascites and meteorism.

The respiratory rate is counted for one minute without the patient noticing (for example, as if counting the pulse), with the hand placed on the chest or abdominal wall.

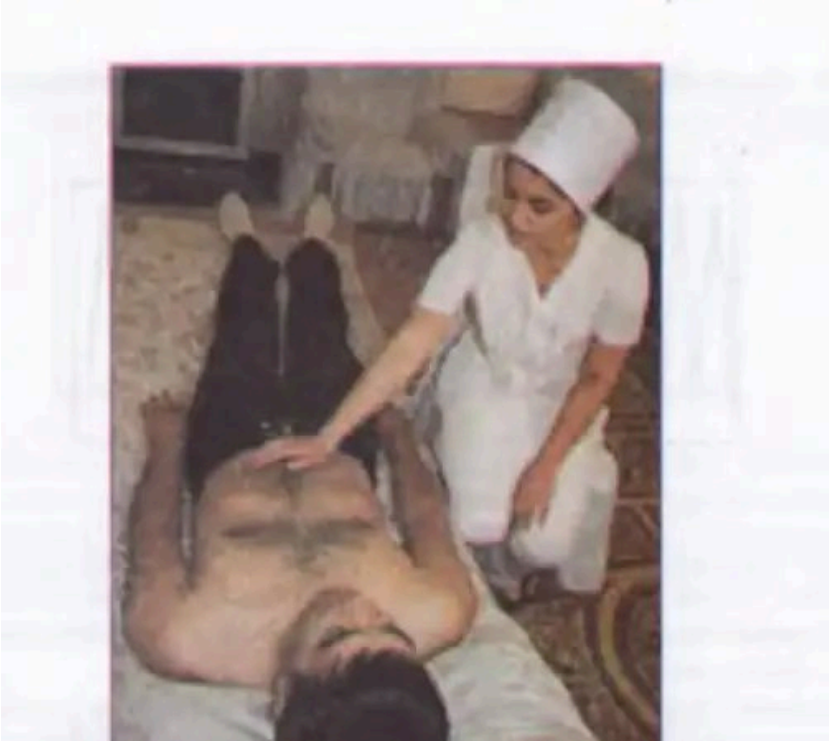


Figure 4.5. *Counting the respiratory rate (Gadayev A. et al., 2012, Fig. 10)*

> **REMEMBER**
> In a healthy adult at rest, the respiratory rate is **16–20** per minute (12–14 during sleep), and in a newborn 40–45. A respiratory rate of more than 20 per minute is **tachypnea**, less than 12 is **bradypnea**, and cessation of breathing is **apnea**. The ratio of the respiratory rate to the pulse rate is normally **1:4**.

3.4. Pathological types of breathing

Table 4.7. *Pathological types of breathing*

Type of breathing	Definition	Causes
Kussmaul breathing	Slow, deep, noisy, rhythmic (“great breathing”)	Metabolic acidosis: diabetic ketoacidosis, uremia, hepatic coma

Type of breathing	Definition	Causes
Cheyne–Stokes breathing	The depth of breathing gradually increases and then decreases, followed by apnea of 5–30 seconds; the cycle repeats	Decreased excitability of the respiratory center: severe heart failure, cerebrovascular disorders, intoxication
Biot breathing	Several breaths of equal depth alternate with long pauses	Meningitis, encephalitis, brain tumors and injuries, the agonal state
Grocco breathing ("undulating")	Similar to Cheyne–Stokes, but with shallow breathing instead of the pause	The initial stage of Cheyne–Stokes breathing, severe intoxication

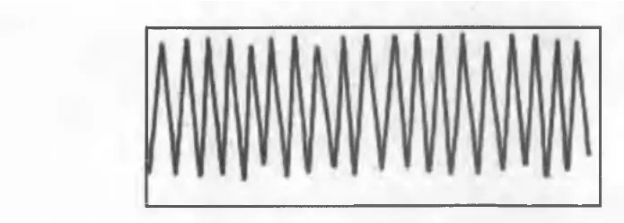


Figure 4.6. *Kussmaul breathing (Gadayev A. et al., 2012, Fig. 11)*

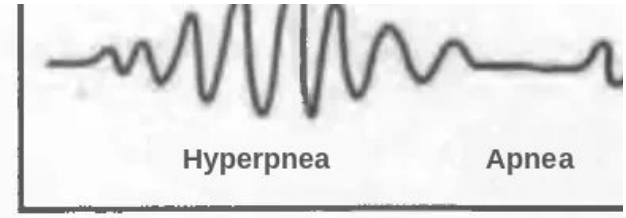


Figure 4.7. *Cheyne–Stokes breathing (same source, Fig. 12)*

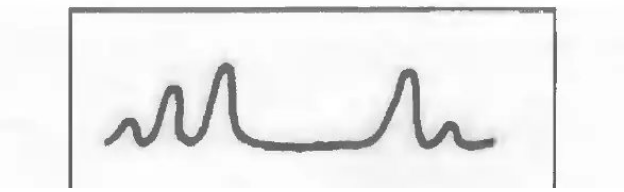


Figure 4.8. *Biot breathing (Gadayev A. et al., 2012, Fig. 13)*



Figure 4.9. *Grocco breathing (same source, Fig. 14)*

MEMORY AID

"Kussmaul — Ketoacidosis": deep, noisy breathing in acidosis. **"Cheyne–Stokes — Crescendo-decrescendo wave + Cessation"**. **"Biot — Bursts of equal Breaths, Blank pauses"**.

4. Palpation of the chest

Palpation determines the following: tender areas, the resistance (elasticity) of the chest, the equal participation of both sides in the respiratory movements and **vocal fremitus** (fremitus vocalis).

4.1. Detecting tender areas

The chest is palpated with the palms of both hands over symmetrical areas from above downward. Localized tenderness is found in rib fractures (a "grating" on pressure — crepitus), intercostal neuralgia (Valleix points: next to the spine, on the axillary line, next to the sternum) and muscle inflammation.



Figure 4.10. *Palpation of the chest from the front (Gadayev A. et al., 2012, Fig. 15)*

4.2. Determining the resistance (elasticity) of the chest

Standing beside the patient, the physician places one palm on the sternum and the other on the interscapular region and compresses the chest from front to back; then the two hands are placed on the sides and the chest is compressed from the sides. Normally the chest is elastic — it yields when compressed.



Figure 4.11. *Determining the elasticity of the chest in the anteroposterior direction (Gadayev A. et al., 2012, Fig. 17)*

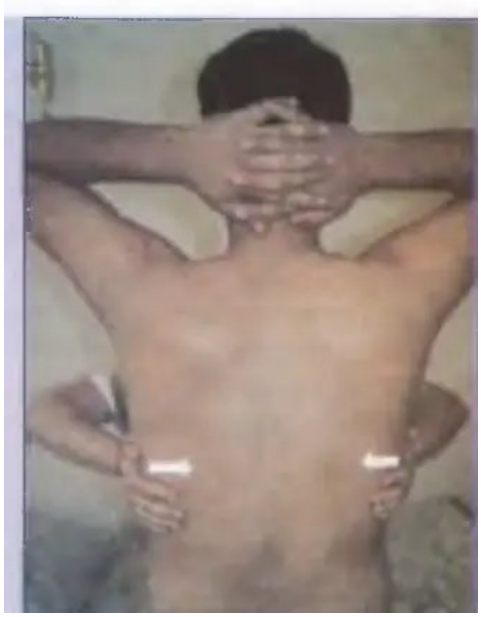


Figure 4.12. *Assessing the participation of the chest in breathing from behind (same source, Fig. 18)*

Causes of increased resistance (a “rigid” chest): pulmonary emphysema, a large amount of fluid in the pleural cavity, a pleural tumor, calcification of the costal cartilages in old age.

4.3. Determining vocal fremitus

Vocal fremitus is the transmission of the sound vibrations generated in the larynx during speech through the bronchi and lung tissue to the chest wall. The patient is asked to say, in a low (bass)

voice, words containing the letter “r” (“thirty-three”, “ninety-nine”, “tractor”). The physician places the palms on symmetrical areas, compares the fremitus on the two sides, then swaps the hands and examines again.

Areas examined: in front — the supraclavicular and infraclavicular regions and the lateral surfaces of the chest; behind — the suprascapular, interscapular and infrascapular regions.



Figure 4.15. *In the infrascapular region (same source, Fig. 23)*



Figure 4.13. *Determining vocal fremitus from the front (Gadayev A. et al., 2012, Fig. 20)*



Figure 4.14. *In the suprascapular region (same source, Fig. 21)*

Normally vocal fremitus is equal on both sides; it is stronger in men with a low voice and in thin people and weaker in women, children and obese people. Over the right apex it is somewhat stronger than on the left (because the right main bronchus is short and wide).

Table 4.8. *Changes in vocal fremitus*

Change	Mechanism	Causes
Increased	Consolidated lung tissue conducts sound well (bronchial patency preserved)	Lobar pneumonia, pulmonary infarction, compression atelectasis
	A superficially located air-filled cavity communicating with a bronchus acts as a resonator	Lung abscess (after emptying), tuberculous cavity

Change	Mechanism	Causes
Decreased or absent	Fluid or air in the pleural cavity damps the vibrations	Exudative pleurisy, hydrothorax, pneumothorax
	The bronchus is blocked, and sound does not pass into the lung	Obstructive atelectasis (tumor, foreign body)
	Increased air content of the lung tissue	Pulmonary emphysema
	Thickening of the chest wall	Obesity, subcutaneous edema

MEMORY AID

Vocal fremitus is **increased over solid tissue and dampened by fluid and air**: “Solid — Strong, Fluid and Air — Faint”.

CAUTION

For vocal fremitus to be increased, bronchial patency must be preserved. If the bronchus of the consolidated area is blocked by a tumor or a foreign body (obstructive atelectasis), the fremitus is **decreased** even though the tissue is dense.

5. Anatomical and physiological basis of the respiratory system

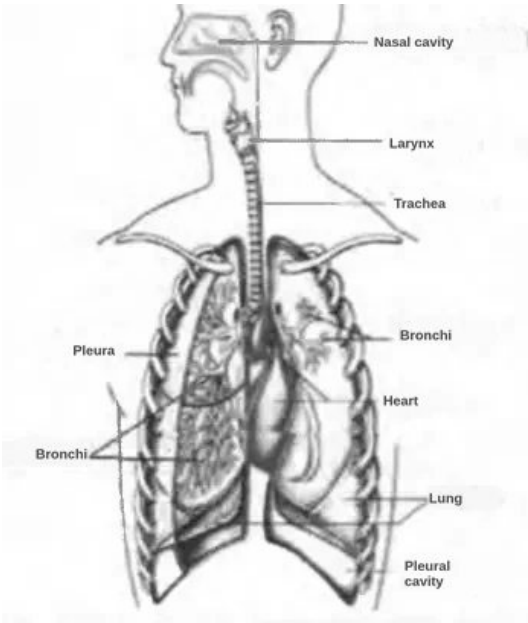


Figure 4.16. *General structure of the respiratory system (Gadayev A. et al., 2012, Fig. 1)*

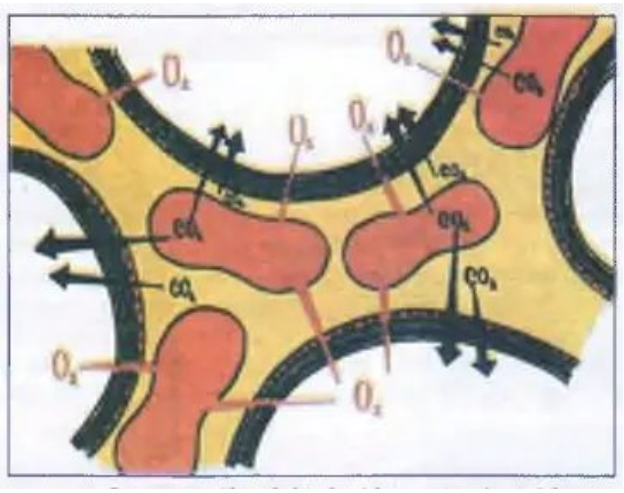


Figure 4.17. *Gas exchange between the air in the alveoli and the red cells (same source, Fig. 2)*

The airways are divided into the **conducting** part (from the nasal cavity to the terminal bronchioles — it does not take part in gas exchange; the “anatomical dead space”, about 150 ml) and the **respiratory** part (respiratory bronchioles, alveolar ducts and alveoli). The two lungs contain about **300 million alveoli** with a total surface area of **70–100 m^2**. The alveolar-capillary membrane is only **0.5 µm** thick, so gas diffusion is very rapid.

The pleura is a serous membrane consisting of two layers: the **visceral** layer (covers the lung and has no pain receptors) and the **parietal** layer (lines the inner surface of the chest wall, the diaphragm and the mediastinum and is rich in pain receptors). Between the layers there is 5–15 ml of fluid, and the pressure between them is negative — this keeps the lung in the expanded state.

Table 4.9. *Lung volumes and capacities (in an average young man)*

Parameter	Definition	Normal
Tidal volume (TV)	Air inhaled during quiet breathing	500 ml
Inspiratory reserve volume	Air that can be additionally inhaled after a quiet inspiration	2500–3000 ml
Expiratory reserve volume	Air that can be additionally exhaled after a quiet expiration	1000–1500 ml
Residual volume (RV)	Air remaining in the lungs after a maximal expiration	1000–1500 ml
Vital capacity (VC)	TV + the inspiratory and expiratory reserve volumes	3500–5000 ml
Total lung capacity	VC + RV	5000–6500 ml
Minute ventilation	TV × respiratory rate	6–8 l/min

Regulation of breathing: the respiratory center in the medulla oblongata responds to the levels of CO_2 (central chemoreceptors) and O_2 (peripheral chemoreceptors in the carotid bodies) in the blood. A rise in blood CO_2 (hypercapnia) is the strongest stimulus to breathing.

6. Pathogenesis of the main complaints

6.1. Mechanisms of dyspnea

Dyspnea is a multifactorial sensation that arises through the following mechanisms:

1. **Impaired gas exchange** — hypoxemia and hypercapnia stimulate the chemoreceptors and excite the respiratory center.
2. **Impaired respiratory mechanics** — increased bronchial resistance (obstruction) or decreased lung compliance (fibrosis, congestion) requires greater work of the respiratory muscles; the “mismatch between the effort required and the result obtained” is felt as dyspnea.
3. **Stimulation of receptors in the lungs** — the interstitial J receptors (pulmonary edema, fibrosis) and stretch receptors.
4. **Metabolic acidosis** — Kussmaul breathing.
5. **Psychological factors** — anxiety, panic attacks (hyperventilation syndrome: “unable to take a full breath”, sighing).

Table 4.10. *Causes of dyspnea by system*

System	Causes
Respiratory system	COPD, bronchial asthma, pneumonia, pleurisy, pneumothorax, pulmonary fibrosis, lung cancer
Cardiovascular system	Heart failure, valvular heart disease, arrhythmias, pericarditis, pulmonary embolism
Blood system	Anemia (oxygen transport decreased)
Metabolism	Diabetic ketoacidosis, uremia (acidosis), thyrotoxicosis, obesity
Neuromuscular system	Diaphragmatic paralysis, myasthenia gravis, Guillain–Barré syndrome
Functional	Hyperventilation syndrome, neuroses

CLINICAL SIGNIFICANCE

Dyspnea of sudden onset — pneumothorax, pulmonary embolism, pulmonary edema, an asthma attack, aspiration of a foreign body; increasing over several days — pneumonia, pleurisy, decompensated heart failure; increasing gradually over months and years — COPD, pulmonary fibrosis, valvular heart disease, anemia.

6.2. Causes and severity of hemoptysis

Table 4.11. *Causes of hemoptysis*

Group	Causes
Infections	Tuberculosis (one of the most common causes), bronchiectasis, lung abscess, pneumonia (especially lobar and Klebsiella), aspergilloma
Tumors	Bronchogenic cancer, bronchial carcinoid, metastases
Cardiovascular system	Mitral stenosis, left ventricular failure (pulmonary edema), pulmonary embolism and pulmonary infarction
Others	Chest injury, foreign body, hemorrhagic diatheses, anticoagulants, Goodpasture's syndrome, granulomatosis with polyangiitis

CAUTION

Loss of **more than 100–200 ml** of blood per day (500 ml in some sources), or bleeding accompanied by airway obstruction and hemodynamic instability, is **massive pulmonary hemorrhage**; it is life-threatening and requires urgent bronchoscopy and bronchial artery embolization. The patient is laid on the bleeding side (so that blood does not flow into the healthy lung).

6.3. Differential diagnosis of chest pain

Table 4.12. *Causes of chest pain related to breathing*

Cause	Character of the pain	Additional signs
Dry pleurisy	Stabbing, worse with breathing and coughing, lessens when lying on the affected side	Pleural friction rub, low-grade fever
Lobar pneumonia	Stabbing, sharp, from the first days	High fever, rusty sputum
Spontaneous pneumothorax	Sudden, acute, radiating to the shoulder	Severe dyspnea, tympanic note
Pulmonary embolism	Sudden, pleuritic or retrosternal	Dyspnea, tachycardia, hemoptysis, venous thrombosis in the leg
Intercostal neuralgia	Along a rib, worse on bending	Tenderness at the Valleix points, respiratory findings normal
Shingles (herpes zoster)	Burning, unilateral, along a segment	A vesicular rash after a few days
Mediastinal tumor	Constant, dull	Weight loss, compression of the superior vena cava

7. Additional aspects of the inspection of the chest

7.1. Topographic lines of the chest

Conventional vertical lines are used to localize changes on the chest precisely: the **anterior midline**, the **sternal line**, the **parasternal line** (midway between the sternal and midclavicular lines), the

midclavicular line (through the middle of the clavicle), the **anterior, middle and posterior axillary lines**, the **scapular line** (through the angle of the scapula), the **paravertebral line** and the **posterior midline**.

Counting the ribs: in front — the II rib joins the sternum at the level of the sternal angle (the angle of Louis, where the manubrium joins the body of the sternum), and counting proceeds downward from it; behind — with the arms lowered, the angles of the shoulder blades correspond to the VII rib; the spinous process of the VII cervical vertebra is the most prominent when the head is bent forward.

7.2. Additional examples of chest deformities



Figure 4.18. *Paralytic chest with cachexia (Gadayev A. et al., 2012, Fig. 6)*



Figure 4.19. *Kyphosis (same source, Fig. 8)*

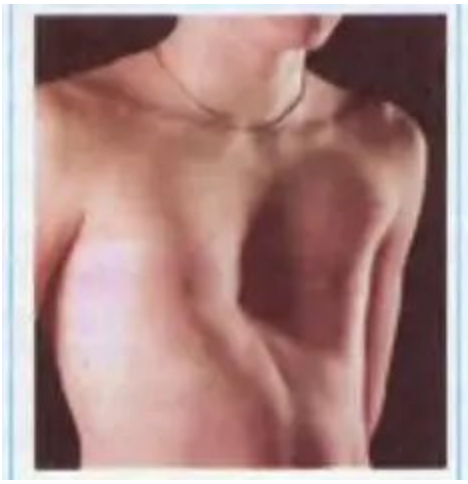


Figure 4.20. *Scoliosis and asymmetry of the chest (same source, Fig. 9)*

CLINICAL SIGNIFICANCE

In severe kyphoscoliosis the volume of the chest is sharply reduced and the respiratory muscles work under unfavorable conditions — **restrictive respiratory failure** and, later, **cor pulmonale** (the "kyphoscoliotic heart") develop. In such patients it is also harder to determine the borders of the heart and lungs.

7.3. Signs of respiratory failure detected on inspection

- Tachypnea (more than 25–30 per minute), use of the accessory muscles (sternocleidomastoid, trapezius, abdominal muscles) during inspiration;

- indrawing of the intercostal spaces during inspiration;
- **paradoxical abdominal movement** — indrawing of the abdominal wall during inspiration (diaphragmatic fatigue, severe respiratory failure);
- movement of the nasal alae with breathing;
- diffuse (central) cyanosis, sweating, tachycardia;
- in hypercapnia — warm and moist skin, headache, drowsiness, a “flapping” tremor of the hands, impaired consciousness (hypercapnic encephalopathy).

Table 4.13. *Grades of respiratory failure (by blood gases and pulse oximetry)*

Grade	PaO_2, mm Hg	SpO_2, %
Normal	80–100	95–98
I (mild)	60–79	90–94
II (moderate)	40–59	75–89
III (severe)	Less than 40	Less than 75

Clinically, the textbook divides respiratory failure into three grades: **I** — dyspnea on heavy or moderate physical exertion; **II** — on slight exertion, with the compensatory mechanisms working even at rest; **III** — dyspnea and cyanosis at rest.

REMEMBER

Pulse oximetry is the painless measurement of the oxygen saturation of arterial blood (SpO_2) and the pulse with a sensor placed on a finger. Normal is **95–98 %**. An SpO_2 below 90 % is hypoxemia (an indication for oxygen therapy). In patients with COPD, the target SpO_2 is 88–92 %.

8. Detailed description of the palpation technique

8.1. Palpation of the back and lateral regions



Figure 4.21. *Palpation of the chest from behind (Gadayev A. et al., 2012, Fig. 16)*



Figure 4.22. *Determining the elasticity of the chest from the side (same source, Fig. 17)*

To assess the equal participation (excursion) of both sides of the chest in breathing, the physician stands behind the patient and places the palms of both hands below the angles of the shoulder blades, with the thumbs close together beside the spine. When the patient takes a deep breath, the thumbs

should move the same distance away from the spine; a lag of one side indicates pathology on that side (pleurisy, pneumonia, atelectasis, pneumothorax).



Figure 4.23. *Assessing chest excursion with the thumbs (Gadayev A. et al., 2012, Fig. 18)*

8.2. Examining vocal fremitus from behind



Figure 4.24. *Determining vocal fremitus in the interscapular region (Gadayev A. et al., 2012, Fig. 22)*

MEMORY AID

What is examined by palpation — **"TREV"**: **T**enderness, **R**esistance, **E**xpansion (participation in breathing), **V**ocal fremitus.

CAUTION

Vocal fremitus is easy to examine in men with a low voice, whereas in women and children with a high-pitched voice it is felt only weakly. That is why **comparing the two sides** is important: it is not the absolute strength of the fremitus but its **asymmetry** that has diagnostic significance.

9. Lung diseases related to occupational and environmental factors

In respiratory diseases, a careful occupational history is often the key to the diagnosis.

Table 4.14. *Occupational lung diseases*

Disease	Harmful factor	Occupations
Silicosis	Dust of free silicon dioxide	Miners, stonecutters, foundry workers, glass and ceramics makers
Asbestosis and pleural mesothelioma	Asbestos fibers	Construction, insulation, shipbuilding, manufacture of brake linings
Anthracosis (coal workers' pneumoconiosis)	Coal dust	Coal miners
Byssinosis	Cotton and flax dust	Textile industry, cotton-ginning plants
Extrinsic allergic alveolitis ("farmer's lung", "bird fancier's lung")	Organic dust, molds, bird proteins	Agriculture, poultry farming
Occupational bronchial asthma	Isocyanates, flour, wood dust, latex, chemicals	Painters, bakers, carpenters, health workers
COPD	Dust, gases and fumes, smoke from household fuel	Many branches of industry; people who cook on wood and coal at home

CLINICAL SIGNIFICANCE

When occupational asthma is suspected, the most important question is: **"Do the symptoms lessen at weekends and on holiday?"** Symptoms that get worse at work and lessen at home point to an occupational factor. The diagnosis is confirmed by serial PEF measurements at work and during time off.

10. Detailed analysis of cough and sputum

10.1. Mechanism of the cough reflex

When the cough receptors are stimulated (mechanically, chemically, thermally, by inflammation), impulses travel along the vagus nerve to the cough center in the medulla oblongata. A cough consists of three phases: **a deep inspiration → closure of the glottis and contraction of the expiratory muscles** (the intrathoracic pressure rises to 100–300 mm Hg) → **sudden opening of the glottis** and expulsion of air at high speed (hundreds of km per hour) together with the secretions.

Table 4.15. *Causes of chronic cough outside the respiratory system*

Cause	Features
Gastroesophageal reflux	At night, after meals, on lying down; with heartburn
Postnasal drip	A feeling of something running down the throat, frequent throat clearing; rhinitis, sinusitis
ACE inhibitors (enalapril, lisinopril and others)	Dry, tickling; resolves 1–4 weeks after the drug is stopped
Left ventricular failure	Worse on lying down, with dyspnea
Psychogenic cough	Disappears during sleep, worse with excitement

10.2. The amount of sputum and its significance

- **A small amount** (a few spittings a day) — acute bronchitis, bronchial asthma, the onset of pneumonia;
- **a moderate amount** (50–100 ml) — chronic bronchitis, pneumonia;
- **a large amount** (more than 100–150 ml, sometimes up to 1 l) — lung abscess (after rupture), bronchiectasis, lung gangrene, pulmonary edema, cavitary tuberculosis.

> **REMEMBER**
> The production of a large amount of sputum at once ("a mouthful") is a sign of the **rupture of a cavity in the lung into a bronchus** (abscess) or of **bronchiectasis**; an increase in sputum in the morning and its dependence on position are characteristic of bronchiectasis.

11. Measuring the dimensions of the chest

The chest circumference is measured with a centimeter tape: behind, the tape passes below the angles of the shoulder blades, and in front, at the level of the nipples in men and at the junction of the IV rib with the sternum in women. Normally the circumference equals half the height or is slightly greater. The **anteroposterior** and **transverse** diameters of the chest are determined with a special caliper (pelvimeter); normally the ratio of the anteroposterior to the transverse diameter (the chest index) is **0.65–0.75**; in emphysema it approaches 1.0.

12. Questionnaires for assessing patients with COPD and asthma

Table 4.16. *Standard tools for assessing symptoms*

Tool	Content	Interpretation
mMRC scale	Grades 0–4, the effect of dyspnea on daily activity	2 or more — many symptoms
CAT (COPD Assessment Test)	8 questions (cough, sputum, chest tightness, breathlessness, activity, confidence leaving home, sleep, energy), each scored 0–5	10 or more — many symptoms (GOLD group B or E)
ACT (Asthma Control Test)	5 questions, each scored 1–5	20–25 — controlled; 16–19 — partly controlled; 15 or less — uncontrolled

13. Features of the respiratory system in the elderly

With age, the chest becomes rigid (the costal cartilages calcify), the strength of the respiratory muscles and the elasticity of the lungs decrease, mucociliary clearance weakens and the cough reflex diminishes. As a result:

- chest excursion decreases and resistance increases;
- the lower borders of the lungs descend slightly ("senile emphysema"), and vesicular breathing is diminished;
- pneumonia often runs an **asymptomatic or atypical** course: without fever, with only weakness, confusion, loss of appetite, falls;
- isolated crackles in the lower parts of the lungs that disappear after coughing may be normal (basal atelectasis).

CAUTION

In an elderly patient, a respiratory rate of **more than 25** per minute is often the earliest and most sensitive sign of pneumonia. That is why the respiratory rate must be counted accurately in every elderly patient.

14. Clinical part

Diagnostic interpretation of the main complaints

Table 4.17. *Combinations of complaints and the likely disease*

Combination of complaints	Suspected disease
Sudden chills, fever of 39–40 °C, stabbing pain, “rusty” sputum	Lobar pneumonia
Many years of morning productive cough, gradually increasing dyspnea, a smoker	COPD, chronic bronchitis
Attacks of expiratory choking, glassy sputum at the end of an attack, a history of allergy	Bronchial asthma
A large amount of foul-smelling purulent sputum in the morning, “a mouthful”, clubbing	Bronchiectasis
Hectic fever, then the sudden expectoration of a large amount of purulent sputum with improvement of the condition	Lung abscess (stage of rupture)
Sudden severe stabbing pain and dyspnea in a young, tall, thin man	Spontaneous pneumothorax
Low-grade fever, night sweats, weight loss, hemoptysis	Pulmonary tuberculosis
Prolonged cough, hemoptysis, weight loss in an elderly man who has smoked 40 pack-years	Lung cancer

Sample case history entry

The chest is emphysematous in shape and symmetrical. The epigastric angle is obtuse, the ribs run horizontally, the intercostal spaces are widened, and the supraclavicular fossae bulge. Breathing is of the mixed type, respiratory rate 24 per minute, rhythmic, shallow. The accessory muscles take part in inspiration. Both sides take part equally in breathing. **On palpation** the chest is painless, with increased resistance. Vocal fremitus is symmetrically decreased on both sides.

Conclusion: signs characteristic of the syndrome of increased air content of the lung tissue (emphysema).

15. Practical skills

Skill: counting the respiratory rate

1. Lay or seat the patient in a comfortable position.
2. Hold the patient’s hand as if counting the pulse (so that the patient does not consciously change his or her breathing).
3. Place your other hand on the lower part of the chest or on the abdominal wall.
4. Count the respiratory movements for one minute (one inspiration and one expiration — one movement).
5. At the same time, assess the type, depth and rhythm of breathing.
6. Record the result on the temperature chart.

Skill: determining vocal fremitus

1. Undress the patient to the waist, standing or sitting.
2. Place the palms of both hands lightly on symmetrical areas of the chest, with the fingers directed along the ribs.
3. Ask the patient to say “thirty-three” or “ninety-nine” in a low voice.
4. Compare the fremitus on the two sides; then swap the hands and examine again.
5. Examine the areas in turn: supraclavicular, infraclavicular, lateral surfaces, suprascapular, interscapular, infrascapular.
6. Interpret the result: equal, increased or decreased (in which area).

Skill: measuring chest excursion

1. Pass the centimeter tape behind below the angles of the shoulder blades, and in front at the level of the nipples in men and below the breasts in women.
2. Measure the chest circumference during quiet breathing, at maximal inspiration and at maximal expiration.
3. The difference between the circumferences at maximal inspiration and expiration is the excursion, normally 6–8 cm. It is reduced in emphysema, pleural adhesions and ankylosing spondylitis.

Review questions

1. What are the main complaints in respiratory diseases?
2. In which diseases are dry and wet cough observed? What does a bitonal cough indicate?
3. What types of sputum are there? Which disease is “rusty” sputum characteristic of?
4. In which diseases does purulent sputum separate into layers?
5. What are the causes of inspiratory, expiratory and mixed dyspnea?
6. What is the difference between bronchial and cardiac asthma?
7. How do you distinguish pulmonary from gastric bleeding?
8. What are the features of pleuritic pain?
9. What pathological shapes of the chest are there? Describe the emphysematous chest.
10. What are the causes of asymmetry of the chest?
11. How is the respiratory rate counted, and what is its normal value?
12. What are the differences between Kussmaul, Cheyne–Stokes and Biot breathing, and their causes?
13. How is the resistance of the chest determined, and when does it increase?
14. How is vocal fremitus determined?
15. In which conditions is vocal fremitus increased and decreased?

Test questions

1. Which disease is “rusty” sputum characteristic of?
   A) Bronchiectasis
   B) Lobar pneumonia
   C) Pulmonary edema
   D) Bronchial asthma
   E) Acute bronchitis

2. In which condition is pink, frothy serous sputum produced?
   A) Tuberculosis
   B) Lung abscess
   C) Chronic bronchitis
   D) Pulmonary edema
   E) Lung cancer

3. Which condition is expiratory dyspnea characteristic of?
   A) A foreign body in the trachea
   B) Laryngeal edema
   C) Pneumothorax
   D) Bronchial asthma
   E) Exudative pleurisy

4. Which sign is characteristic of pulmonary bleeding?
   A) Mixed with food remnants
   B) Bright red, frothy blood, with coughing
   C) Acidic reaction
   D) Melena appears afterward
   E) Blood the color of “coffee grounds”

5. What is the normal respiratory rate of an adult at rest?
   A) 8–10
   B) 12–14
   C) 16–20
   D) 24–28
   E) 30–35

6. What is slow, deep, noisy breathing called, and in which condition is it observed?
   A) Grocco — intoxication
   B) Kussmaul — diabetic ketoacidosis
   C) Cheyne–Stokes — heart failure
   D) Tachypnea — fever
   E) Biot — meningitis

7. What is breathing in which the depth gradually increases, then decreases, followed by a pause, called?
   A) Bradypnea
   B) Stridor
   C) Kussmaul breathing
   D) Biot breathing
   E) Cheyne–Stokes breathing

8. In which disease is a barrel chest observed?
   A) Rickets
   B) Syringomyelia
   C) Pulmonary emphysema
   D) Pneumosclerosis
   E) Tuberculosis

9. Which condition is characterized by enlargement of one side of the chest and its lagging behind in breathing?
   A) The state after pneumonectomy
   B) Pneumosclerosis
   C) Exudative pleurisy
   D) Pleural adhesions
   E) Obstructive atelectasis

10. In which condition is vocal fremitus increased?
    A) Pneumothorax
    B) Exudative pleurisy
    C) Pulmonary emphysema
    D) Lobar pneumonia
    E) Obstructive atelectasis

11. In which condition is vocal fremitus decreased?
    A) Lobar pneumonia
    B) Pulmonary infarction
    C) Compression atelectasis
    D) An emptied lung abscess
    E) Hydrothorax

12. What does a bitonal cough most often indicate?
    A) Inflammation of the larynx
    B) Pertussis (whooping cough)
    C) Compression of a large bronchus by a tumor or lymph nodes
    D) Bronchial asthma
    E) Chronic pharyngitis

13. Which sign is characteristic of pleuritic pain?
    A) It is relieved by nitroglycerin
    B) It radiates to the left arm
    C) It appears on physical exertion and passes with rest
    D) It is related to eating
    E) It increases with deep inspiration and coughing

14. Which condition is characterized by sudden choking at night, orthopnea and moist rales in the lower parts of the lungs?
    A) Cardiac asthma
    B) Dry pleurisy
    C) Bronchial asthma
    D) Acute bronchitis
    E) Pneumothorax

15. What is the normal ratio of the respiratory rate to the pulse rate?
    A) 1:1
    B) 1:2
    C) 1:4
    D) 1:8
    E) 1:10

Clinical cases

Case 1

A 60-year-old man has complained of a morning productive cough for 15 years and, in recent years, of dyspnea on climbing stairs. He has smoked a pack of cigarettes a day for 40 years. On inspection: the chest is barrel-shaped, the supraclavicular fossae bulge, the accessory muscles take part in inspiration, and he exhales through pursed lips. On palpation the chest is rigid, and vocal fremitus is decreased on both sides.

Questions: Calculate the smoking index. What type of dyspnea is it? Which syndrome are the inspection and palpation findings characteristic of?

Answer: Smoking index = 20 × 40 / 20 = 40 pack-years. The dyspnea is expiratory (bronchial obstruction). A barrel chest, increased resistance and symmetrically decreased vocal fremitus are characteristic of the syndrome of increased air content of the lung tissue (emphysema). The clinical picture suggests COPD (chronic obstructive bronchitis and emphysema).

Case 2

A 35-year-old patient developed sudden chills, a fever of 39.8 °C and stabbing pain on the right side that increased on deep inspiration 3 days ago. Today he began to expectorate viscous rust-colored sputum. Vocal fremitus is increased in the right infrascapular region.

Questions: Explain the character of the sputum and of the pain. Why is vocal fremitus increased?

Answer: The rust-colored sputum is due to the breakdown of red cells with the formation of hemosiderin and is characteristic of lobar pneumonia. Stabbing pain that increases on deep inspiration is pleuritic pain (inflammation of the parietal pleura). Increased vocal fremitus indicates consolidation of the lung tissue: the alveoli are filled with exudate, bronchial patency is preserved, and the dense tissue conducts sound well.

Case 3

On inspection of a 42-year-old woman, the left side of the chest is enlarged, the intercostal spaces are smoothed out, and the left side lags behind in breathing. On palpation, vocal fremitus is not detected in the lower part of the left side.

Questions: In which conditions is such a picture observed? What investigations are needed?

Answer: Enlargement of one side of the chest, its lagging behind in breathing and the absence of vocal fremitus are characteristic of the accumulation of fluid (exudative pleurisy, hydrothorax) or air (pneumothorax) in the pleural cavity. Changes in the lower parts suggest fluid rather than air. Percussion (a dull note and the Damoiseau line), auscultation, radiography, pleural US and pleural puncture are needed.

Quick review

Table 4.18. *Key facts on Topic 4*

Question	Answer
“Rusty” sputum	Lobar pneumonia
Pink frothy sputum	Pulmonary edema
“Raspberry jelly”	Lung cancer
Three-layered sputum	Bronchiectasis, abscess, gangrene, cavitary tuberculosis
Inspiratory dyspnea	Narrowing of the upper airways
Expiratory dyspnea	Bronchial obstruction (asthma, COPD)
Respiratory rate	16–20/min; respiration:pulse = 1:4
Kussmaul	Slow, deep — acidosis
Cheyne–Stokes	Waxing and waning + apnea — heart failure, cerebral ischemia
Biot	Equal breaths + long pause — meningitis, brain injury
Barrel chest	Emphysema
Increased vocal fremitus	Consolidation (pneumonia), superficial cavity
Decreased vocal fremitus	Fluid, air, obstruction, emphysema
Excursion	6–8 cm

TOPIC 5

Percussion as a method of objective examination. Comparative and topographic percussion of the lungs

Learning objectives

To teach the student the physical basis, methods and rules of percussion, the types of percussion notes, and how to perform comparative and topographic percussion of the lungs in a healthy person and in respiratory diseases and interpret the results.

By the end of the session, the student should know:

- the physical basis of percussion and the history of its discovery;
- the methods of direct and indirect (finger-on-finger) percussion;
- the general rules of performing percussion;
- the features of the main percussion notes (resonant, dull, tympanic) and their intermediate types;
- the points of comparative percussion and the causes of changes;
- the normal values for the height of the lung apices, Krönig's isthmus, the lower borders of the lungs and their mobility.

The student should be able to:

- perform finger-on-finger percussion with the correct technique;
- perform comparative percussion of the lungs and assess the result;
- perform topographic percussion of the lungs, mark the borders and compare them with the normal values.

1. General information on percussion

Percussion (Latin percussio — tapping) is a method of determining the physical state, borders and size of the underlying organs by tapping the body surface and assessing the characteristics of the resulting sound.

The method was discovered in 1761 by the Austrian physician **Leopold Auenbrugger** ("Inventum novum"). He was inspired by his father, a wine producer, who determined the amount of wine in barrels by tapping them. In 1808 the French physician **J. Corvisart** introduced the method widely into practice. After percussion with a pleximeter (1828, P. Piorry) and a hammer, **finger-on-finger percussion** eventually prevailed; Russian clinicians (G. Sokolov, F. Yanovsky, V. Obraztsov) played a large role in its development.

1.1. Physical basis of percussion

When tapped, the body tissues vibrate and produce a sound. The character of the sound depends on the **density**, **air content** and **elasticity** of the tissue.

Table 5.1. *Physical properties of the percussion note*

Property	Physical basis	Interpretation
Intensity (loudness)	Amplitude of the vibrations	Loud — soft (quiet)
Duration	Duration of the vibrations	Long — short
Pitch (frequency)	Frequency of the vibrations	Low — high

Property	Physical basis	Interpretation
Tympanic tinge	Uniformity of the vibrations (harmonicity)	Tympanic — nontympanic

REMEMBER
Percussion sets the tissues into vibration to a **depth of 5–7 cm** and **2–4 cm to the side** of the point struck. That is why deep-seated foci or foci smaller than 2–3 cm in diameter are not detected by percussion.

2. Methods of percussion

Table 5.2. *Methods of percussion*

Method	Content	Present-day significance
Direct (Auenbrugger)	Striking the body surface directly with the fingertips	Used partly over the clavicles and the lung apices
Yanovsky's method	A light direct blow with the terminal phalanx of one finger	For gentle percussion
Obraztsov's method ("flicking" blow)	Striking by sliding the middle finger off the index finger	Rarely used
With a pleximeter and hammer	The pleximeter is placed on the body surface and struck with a hammer	Hardly used today
Finger-on-finger (indirect)	The middle finger of the left hand (pleximeter) is placed on the body and struck with the middle finger of the right hand (plexor)	The main method



Figure 5.1. *Technique of finger-on-finger percussion (Gadayev A. et al., 2012, Fig. 17)*

2.1. Technique of finger-on-finger percussion

1. The **pleximeter finger** (the middle finger of the left hand) is applied closely to the body surface, but without excessive pressure; the other fingers are slightly separated and do not touch the body.

2. The **plexor finger** (the middle finger of the right hand) is bent and strikes the **middle phalanx** of the pleximeter perpendicularly, using movement at the **wrist joint** only.
3. The blow is short and sharp, and the finger is withdrawn at once after striking (it "bounces back").
4. **Two** blows of equal strength are delivered at each point.
5. The strength of the blows is kept the same throughout the examination.



Figure 5.2. *Performing percussion (Gadayev A. et al., 2012, Fig. 18)*

CAUTION

Common mistakes: striking with movement at the elbow or shoulder joint (the blow becomes heavy and imprecise), keeping the plexor finger on the pleximeter (the sound is damped), other fingers touching the body under the pleximeter, and long nails (the nail of the plexor finger must be short).

2.2. Strength of percussion

- **Loud (strong, deep) percussion** — for examining deep-seated organs (up to 7 cm); used in comparative percussion.
- **Moderate percussion** — the strength most often used in comparative percussion.
- **Quiet (weak) percussion** — for determining the borders of organs (topographic percussion).
- **The quietest (threshold) percussion** — Goldscheider's method; for determining the absolute cardiac dullness, with sound at the threshold of hearing.

3. General rules of performing percussion

1. The room should be warm and quiet.
2. The patient is in a comfortable position (standing, sitting or lying), with the muscles relaxed.
3. The physician's hands are warm and the nails short.
4. In **comparative percussion**, symmetrical points are compared alternately.

5. In **topographic percussion**, percussion proceeds from the resonant note (clear lung note) toward the dull note; the plexiimeter is placed **parallel** to the border being determined, and each step does not exceed **1–2 cm**.
6. The border is marked **on the side of the resonant note** (along the edge of the plexiimeter facing the resonant side).

MEMORY AID

The rule of topographic percussion: **"From resonant to dull, parallel, gently, the border on the resonant side"**.

4. Percussion notes

Table 5.3. *The main percussion notes*

Note	Character	Where it is heard (normally)	In pathology
Clear lung (resonant) note	Loud, long, low-pitched, nontympanic	Over healthy lung	—
Dull (thigh note)	Soft, short, high-pitched	Over airless organs: thigh muscles, liver, heart	Fluid in the pleura, consolidation of the lung (stage II of lobar pneumonia), lung tumor, atelectasis
Tympanic	Loud, long, harmonic (like the sound of a drum)	Air-filled hollow organs: stomach, intestines, Traube's space	Pneumothorax, a large smooth-walled cavity (cavern, emptied abscess)

Table 5.4. *Intermediate percussion notes*

Note	Character	Causes
Hyperresonant (box-like)	Louder than the clear lung note, with a tympanic tinge ("like tapping an empty box")	Pulmonary emphysema — increased air content and decreased elasticity of the lung tissue
Impaired (shortened) lung note	Softer and shorter than the clear lung note	Partial consolidation of the lung tissue: focal pneumonia, pneumosclerosis, pleural thickening
Dull-tympanic	A combination of the dull and tympanic notes	Stages I and III of lobar pneumonia (air and fluid together in the alveoli), compression atelectasis, the onset of pulmonary edema
Metallic note	Loud, tympanic, with a metallic ring	A superficially located smooth-walled cavity more than 6–8 cm in diameter; open pneumothorax
"Cracked-pot" sound	Rattling, clinking	A superficially located cavity communicating with a bronchus through a narrow opening

CLINICAL SIGNIFICANCE

In lobar pneumonia the percussion note changes with the stage: **stage I** (congestion) — dull-tympanic; **stage II** (hepatization) — dull; **stage III** (resolution) — dull-tympanic again, and then it returns to the clear lung note. These changes are explained by the ratio of air to exudate in the alveoli.

5. Comparative percussion of the lungs

Purpose — to detect a pathological focus by comparing the percussion note at symmetrical points of the chest. Percussion is performed with moderate or loud strength.

5.1. Order of comparative percussion

In front (the patient stands or sits with the arms lowered):

1. The supraclavicular fossae (pleximeter parallel to the clavicle).
2. The clavicles themselves (direct percussion — the finger strikes the middle part of the clavicle).
3. The I and II intercostal spaces along the midclavicular line.
4. Because the heart lies lower down on the left, comparative percussion is performed symmetrically only down to the III intercostal space.

From the side (the patient places the hands on the head): starting from the axillary fossa, the IV and V intercostal spaces along the mid-axillary line.

From behind (the patient crosses the arms over the chest and bends the head slightly — the shoulder blades move apart):

1. The suprascapular regions (pleximeter parallel to the spine of the scapula).
2. The interscapular region — the pleximeter is placed **vertically** (parallel to the spine), at three points from above downward.
3. The infrascapular regions — in the VII, VIII and IX intercostal spaces below the angle of the scapula, with the pleximeter parallel to the ribs.



Figure 5.3. *Points of comparative percussion in front (Gadayev A. et al., 2012, Fig. 26)*



Figure 5.4. *Comparative percussion from the side (same source, Fig. 27)*



Figure 5.5. *Comparative percussion from behind: the suprascapular region (Gadayev A. et al., 2012, Fig. 28)*

REMEMBER

Even in a healthy person, the note may differ slightly at symmetrical points:

- over the right apex the note is somewhat duller than on the left (the right apex lies lower, and the muscles of the right shoulder girdle are better developed);
- on the left in the II–III intercostal spaces it is somewhat duller because of the proximity of the heart;
- on the right in the lower parts it is dulled because of the proximity of the liver;
- on the left below it has a tympanic tinge because of Traube's space (the fundus of the stomach).

5.2. Changes on comparative percussion

Table 5.5. *Pathological notes on comparative percussion and their causes*

Note	Mechanism	Diseases and syndromes
Dull	No air in the alveoli, the tissue is consolidated	Lobar pneumonia (hepatization), lung tumor, obstructive atelectasis, pneumosclerosis (extensive)
	Fluid in the pleural cavity	Exudative pleurisy, hydrothorax, hemothorax
Impaired	Partial consolidation or decreased air content	Focal pneumonia, pulmonary edema, pleural adhesions
Tympanic	Air in the pleural cavity	Pneumothorax
	A large air-filled cavity in the lung	Cavern, emptied abscess
Hyperresonant (box-like)	Air content of the lung increased, elasticity decreased	Pulmonary emphysema, an attack of bronchial asthma

6. Topographic percussion of the lungs

Purpose: to determine the height and width of the lung apices (Krönig's isthmus), the lower borders of the lungs and the respiratory mobility of their lower edges. Percussion is performed with **quiet** strength.

6.1. Determining the height of the lung apices

In front: the pleximeter is placed in the supraclavicular fossa parallel to the clavicle, and percussion proceeds from the middle of the clavicle upward and medially (toward the sternocleidomastoid

muscle) until a dull note appears. Normally the apex is **3–4 cm** above the clavicle.

Behind: the plexímeter is moved horizontally from the middle of the spine of the scapula toward the spinous process of the VII cervical vertebra. Normally the apex is at the level of the **spinous process of the VII cervical vertebra**.



Figure 5.7. *Determining the height of the lung apex from behind (same source, Fig. 32)*



Figure 5.6. *Determining the height of the lung apex from the front (Gadayev A. et al., 2012, Fig. 31)*

6.2. Determining Krönig's isthmus

Krönig's isthmus is the width of the area of clear lung note over the lung apex along the anterior edge of the trapezius muscle. The plexímeter is placed perpendicular to the middle of the edge of the trapezius, and percussion proceeds first medially (toward the neck) and then laterally (toward the shoulder) until a dull note appears. The distance between the two points is measured.



Figure 5.8. *Determining Krönig's isthmus (Gadayev A. et al., 2012, Fig. 33)*

REMEMBER

The normal width of Krönig's isthmus is **5–6 cm** (individual variation 3.5–8 cm). The textbook gives the normal value as **3–8 cm**.

Changes:

- Increased height of the apices and widening of Krönig's isthmus — **pulmonary emphysema**.
- Decreased height of the apices and narrowing of Krönig's isthmus — **tuberculous infiltrate or pneumosclerosis** of the apex, shrinkage of the apex. A unilateral change is especially significant.

6.3. Determining the lower borders of the lungs

Percussion proceeds along the topographic lines from above downward, starting from the II–III intercostal space (behind, from the angle of the scapula), until the clear lung note changes to a dull note (on the right, liver dullness) or, on the left, to a tympanic note (Traube's space). The plexiметer is placed in the intercostal space, parallel to the ribs. The rib or intercostal space to which the border corresponds is counted.

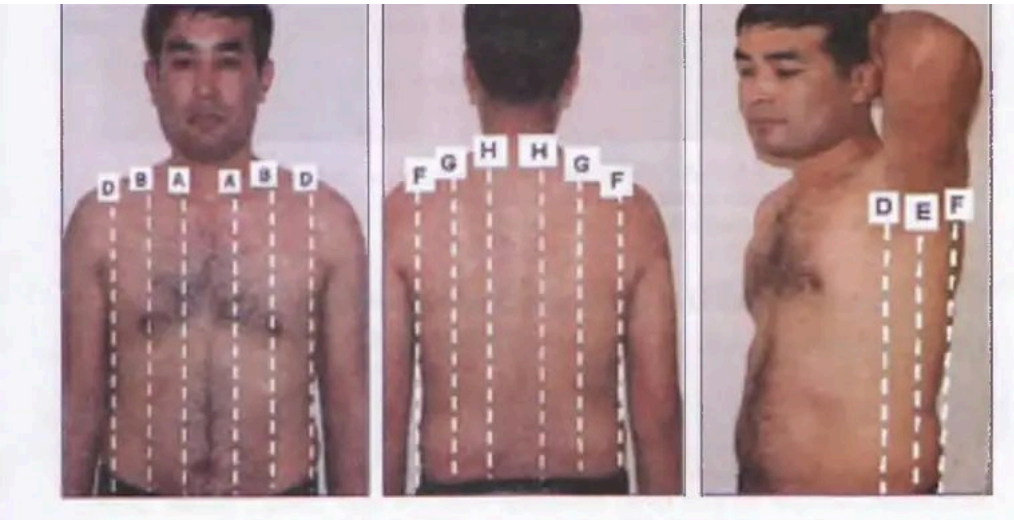


Figure 5.9. *Topographic lines used to determine the lower borders of the lungs (Gadayev A. et al., 2012, Fig. 36)*

Table 5.6. *Lower borders of the lungs in a normosthenic person*

Topographic line	Right lung	Left lung
Parasternal	V intercostal space	— (heart)
Midclavicular	VI rib	— (heart)
Anterior axillary	VII rib	VII rib
Mid-axillary	VIII rib	IX rib
Posterior axillary	IX rib	IX rib
Scapular	X rib	X rib
Paravertebral	Spinous process of the XI thoracic vertebra	Spinous process of the XI thoracic vertebra

MEMORY AID

Remember the lower border of the right lung from front to back: **"5-6-7-8-9-10-11"** — the parasternal line in the V intercostal space, the midclavicular VI, the anterior axillary VII, the mid-axillary VIII, the posterior axillary IX, the scapular X, and the paravertebral line at the XI vertebra. On the left along the mid-axillary line, it is one rib lower — **IX**.

In hypersthenic persons the border is one rib higher, and in asthenic persons one rib lower.

Table 5.7. *Causes of displacement of the lower border of the lung*

Displacement	Unilateral	Bilateral
Descent	Vicarious (compensatory) emphysema — when the other lung has stopped functioning	Pulmonary emphysema, an attack of bronchial asthma, prolapse of the internal organs (splanchnoptosis)
Elevation	Fluid or air in the pleural cavity, obstructive atelectasis, pneumosclerosis, diaphragmatic paralysis, marked enlargement of the liver or spleen	Ascites, meteorism, the last months of pregnancy, obesity, air in the abdominal cavity (perforated ulcer)

6.4. Determining the mobility of the lower edge of the lung

Along the midclavicular (on the right only), mid-axillary and scapular lines, the lower border is first determined during quiet breathing. Then the patient takes a deep breath and holds it — the border is determined again and marked. Next the patient exhales maximally and holds the breath — the border is determined once more. The distance between the borders at maximal inspiration and expiration is measured.



Figure 5.10. *Assessing the mobility of the lower border of the lung (Gadayev A. et al., 2012, Fig. 37)*

REMEMBER
The total mobility of the lower edge of the lung: **4–6 cm along the midclavicular and scapular lines**, **6–8 cm along the mid-axillary line** (3–4 cm each on inspiration and on expiration).

Causes of decreased mobility: decreased elasticity of the lung tissue (emphysema), inflammatory infiltrate, pulmonary congestion, pneumosclerosis, pleural adhesions, diaphragmatic paralysis. **Absent mobility:** a large amount of fluid or air in the pleural cavity, complete fusion of the pleural layers (obliteration).

7. Physical basis of percussion: a deeper analysis

According to their acoustic properties, the body tissues fall into three groups:

1. **Airless and dense tissues** (muscles, liver, spleen, heart, fluid) — damp vibrations quickly and give a low-amplitude, short sound (dull, the thigh note).
2. **Air-containing tissue with elastic walls** (the lung) — the alveoli with their tense walls hold the air; the vibrations are irregular (of different frequencies), so the sound is loud and long but not tympanic (the clear lung note).

3. **Large air-filled cavities with smooth walls** (stomach, intestines, a large cavern, pneumothorax) — the vibrations are of a single frequency (harmonic), hence the tympanic note.

The character of the percussion note depends on the ratio of the amount of air to the tension and elasticity of the tissue. For example, **when the tension of the lung tissue decreases** (the onset of pulmonary edema, stage I of lobar pneumonia, compression atelectasis), the alveolar walls slacken even though air is preserved, and a **dull-tympanic** note appears. **When the tension increases** and the air increases (emphysema), a **hyperresonant (box-like)** note results.

CLINICAL SIGNIFICANCE

Škoda's tympany is the dull-tympanic note heard in exudative pleurisy above the fluid level, over the compressed lung (in Garland's triangle and above it). When the lung tissue is compressed, its tension decreases, while air is partly preserved in the alveoli.

8. Technique of comparative percussion: in detail

8.1. Anterior surface



Figure 5.12. *Comparative percussion below the clavicle (same source, Fig. 25)*



Figure 5.11. *Direct percussion of the clavicle itself (Gadayev A. et al., 2012, Fig. 24)*

Percussion begins in the supraclavicular fossae (pleximeter parallel to the clavicle); then the clavicles themselves are tapped directly (with a fingertip), and then the I and II intercostal spaces along the midclavicular line, with the pleximeter placed along the intercostal space. In women, the breasts are moved upward and outward.



Figure 5.13. *Direct percussion (Gadayev A. et al., 2012, Fig. 17, A)*

8.2. Posterior surface



Figure 5.15. *The plexímeter in the vertical position in the interscapular region (same source, Fig. 29)*



Figure 5.14. *Percussion in the suprascapular region (Gadayev A. et al., 2012, Fig. 28)*



Figure 5.16. *Comparative percussion in the infrascapular regions (Gadayev A. et al., 2012, Fig. 30)*

REMEMBER

Percussion is not performed over the scapula: the thick bone and muscles damp the sound, and the result is unreliable. In the interscapular region the plexímeter is placed **vertically**, and in the infrascapular regions it is again placed parallel to the ribs (horizontally).

8.3. Typical mistakes in comparative percussion

1. Choosing symmetrical points incorrectly (one over a rib, the other over an intercostal space).
2. Percussing one side strongly and the other quietly.
3. Incorrect position of the patient: trunk turned, shoulders at different heights, muscles tense.
4. Pressing the plexímeter finger with varying pressure.
5. Examining only the anterior surface and missing the posterior and lateral regions — whereas pneumonia and pleurisy are most often located in the **lower posterior** regions.

9. Additional aspects of topographic percussion

9.1. The lung apices and Krönig's isthmus



Figure 5.17. *Determining the lung apex from the front (Gadayev A. et al., 2012, Fig. 31)*



Figure 5.18. *Determining the lung apex from behind (same source, Fig. 32)*



Figure 5.19. *The medial and lateral borders of Krönig's isthmus (Gadayev A. et al., 2012, Fig. 34)*

CAUTION

Textbooks give different normal values for Krönig's isthmus: Gadayev — **3–8 cm**, Grebenev and Mukhin — **5–6 cm** (individual variation 3.5–8 cm). In practice, **comparing the two sides** matters more than the absolute figure: unilateral narrowing (for example, 3 cm on the right and 6 cm on the left) raises suspicion of apical tuberculosis or pneumosclerosis.

9.2. Technique of determining the mobility of the lower edge of the lung



Figure 5.20. *Along the midclavicular line (Gadayev A. et al., 2012, Fig. 37)*



Figure 5.21. *Along the mid-axillary line (same source)*

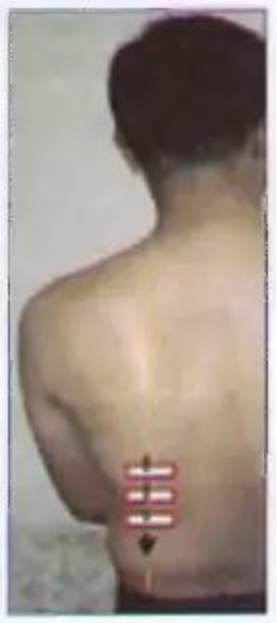


Figure 5.22. *Along the scapular line (same source)*

9.3. Traube's space

Traube's space (the semilunar tympanic area) is the area of tympanic note on the left in the lower part of the chest produced by the gas bubble of the fundus of the stomach. Its borders: above — the lower edge of the left lung; on the right — the left lobe of the liver; on the left — the spleen; below — the costal arch.

Table 5.8. *Changes in Traube's space*

Change	Causes
Reduced or absent (dullness instead of tympany)	Left-sided exudative pleurisy (the most important cause), marked enlargement of the spleen, enlargement of the left lobe of the liver, a large amount of pericardial fluid, a full stomach

Change	Causes
Enlarged	Distension of the stomach with gas, aerophagia

CLINICAL SIGNIFICANCE
Traube's space helps to distinguish left-sided exudative pleurisy from **left lower lobe pneumonia**: when there is fluid in the pleura, the space is "filled" with a dull note, whereas in pneumonia it is usually preserved.

10. Changes in the lung borders under various conditions

Table 5.9. *Lung borders in physiological states*

Condition	Change
With the patient lying down	The lower borders rise slightly (the abdominal organs push the diaphragm upward), especially on the dependent side
Lying on the side	The lower border of the dependent lung rises by 2–3 cm, and the border of the upper lung descends
On deep inspiration	The lower borders descend by 2–4 cm
In old age	The lower borders descend slightly, and mobility decreases (senile emphysema)
The last months of pregnancy	The lower borders rise
Asthenic constitution	One rib lower
Hypersthenic constitution	One rib higher

Relation between the lung and liver borders

The lower border of the right lung along the midclavicular line is at the same time the **upper border of the relative liver dullness**. Therefore:

- when the lung border **descends** (emphysema), the liver is also pushed downward and may be palpable below the costal arch — this is **false hepatomegaly**;
- **disappearance** of liver dullness (tympany over the liver on percussion) is a sign of free air in the abdominal cavity (perforation of a gastric or intestinal ulcer), a surgical emergency.

MEMORY AID
The lower border: **"up — pushed from below, down — the lung is inflated"**: its rise is most often due to pressure from below (ascites, meteorism, an enlarged liver) or to shrinkage of the lung; its descent is a sign of increased lung volume (emphysema).

11. Comparing percussion findings with modern methods

The accuracy of percussion is limited: it detects superficial (5–7 cm) changes larger than 2–3 cm. Fluid in the pleural cavity is usually detected by percussion when it exceeds **300–500 ml**; on the radiograph, blunting of the costophrenic sinus appears at 200–300 ml, whereas **on US even 50 ml** of fluid is visible.

Table 5.10. *Sensitivity of the physical methods and modern investigations (approximate)*

Condition	Physical examination	X-ray (radiography)	US / CT
Fluid in the pleura	From 300–500 ml	From 200–300 ml (frontal view), 50–100 ml (lateral view)	50 ml or less
Focus of pneumonia	A lobe or a large segment	From 1–2 cm	CT: a few mm
Emphysema	Moderate and severe	Moderate and severe	CT: the initial stage

CAUTION

The absence of changes on percussion does not mean the absence of disease. If there is clinical suspicion, radiography, US or CT is performed. However, percussion and auscultation are the first step: they direct the investigation correctly and make it possible to follow the dynamics at the bedside.

12. Features of percussion in children and the elderly

In children, because the chest wall is thin and elastic, percussion is performed **more quietly** (often directly or with a fingertip); a strong blow also sets neighboring areas into vibration and masks small foci. In children the lung borders are one rib higher than in adults, and the percussion note may have a slight tympanic tinge.

In the elderly, the chest is rigid and the air content of the lungs is increased (senile emphysema) — the note has a box-like tinge, the lower borders have descended, and mobility is decreased. A functional investigation is needed to distinguish these changes from pathological emphysema.

13. Additional methods of percussion

13.1. Auscultatory percussion

The stethoscope is placed over the organ, and a sound is produced around it by light flicking (or by rubbing with a finger). The sound changes sharply on crossing the border of the organ. The method is used to determine the borders of the liver, stomach and heart, as well as in bedridden patients and in situations where percussion is difficult.

13.2. Additional methods of detecting pleural fluid

- **Positional percussion:** free fluid (transudate) shifts when the patient’s position changes — dullness appears at the bottom when sitting and on the dependent side when lying on the side.
- **Independence of the fluid border from breathing:** the fluid level does not change with breathing, whereas the lung border shifts.

14. Drawing a conclusion from percussion

Percussion findings are never assessed in isolation. For example, a dull note is found in three different conditions — consolidation of the lung, pleural fluid and atelectasis — and to distinguish them, **inspection** (size of the hemithorax), **palpation** (vocal fremitus), **auscultation** (type of breath sounds) and **the position of the mediastinum** are analyzed together.

Table 5.11. *Combinations of the percussion note with other physical signs*

Percussion note	Vocal fremitus	Respiration	Conclusion
Dull	Increased	Bronchial	Consolidation of the lung tissue (pneumonia)
Dull	Absent	Absent	Fluid in the pleura
Dull	Absent	Absent, mediastinum shifted to the affected side	Obstructive atelectasis
Tympanic	Absent	Absent	Pneumothorax
Tympanic	Increased	Amphoric	A large cavity (emptied abscess, cavern)
Hyperresonant (box-like)	Decreased (bilaterally)	Diminished vesicular	Emphysema
Dull-tympanic	Slightly increased	Diminished vesicular, crepitation	Stage I or III of lobar pneumonia

MEMORY AID

The “triad” of physical signs: **Percussion — Fremitus — Breath sounds**. It is enough to remember each syndrome by these three parameters: consolidation — “Dull, Increased, Bronchial”; fluid — “Dull, Absent, Absent”; air — “Tympanic, Absent, Absent”; emphysema — “Hyperresonant, Decreased, Decreased”.

15. The influence of the diaphragm and abdominal organs on the lung borders

- **Unilateral elevation of the diaphragm** — phrenic nerve paralysis (tumor, surgery), subphrenic abscess, marked enlargement of the liver or spleen, lobar atelectasis. On percussion the lower border of the lung is raised on the affected side, and mobility is absent or paradoxical (the diaphragm rises on inspiration).
- **Subphrenic abscess** — the lower border of the right lung rises, and reactive pleural fluid sometimes appears; fever, pain in the right hypochondrium, pain radiating to the shoulder.
- **Meteorism and ascites** — bilateral elevation, shallow breathing.

CLINICAL SIGNIFICANCE

After abdominal operations, patients do not breathe deeply because of pain; diaphragmatic movement is restricted, and **atelectasis** and **hypostatic pneumonia** develop in the lower parts. Percussion reveals dulling and decreased mobility in the lower posterior regions. Prevention — early mobilization, breathing exercises, adequate pain relief.

16. Clinical part

Percussion findings in the main syndromes

Table 5.12. *Percussion findings in respiratory syndromes*

Syndrome	Comparative percussion	Topographic percussion
Consolidation of the lung tissue	Dull or impaired note	Decreased mobility of the lower edge on the affected side

Syndrome	Comparative percussion	Topographic percussion
Fluid in the pleural cavity	Dull note over the fluid, with an oblique upper border (the Damoiseau line)	The lower border cannot be determined, mobility absent
Air in the pleural cavity (pneumothorax)	Tympanic note	The border cannot be determined, mobility absent
Increased air content of the lung tissue (emphysema)	Hyperresonant (box-like) note (bilateral)	Height of the apices increased, Krönig's isthmus widened, lower borders lowered, mobility markedly decreased
An air-filled cavity in the lung	Tympanic note, sometimes with a metallic tinge or a "cracked-pot" sound	Often unchanged
Obstructive atelectasis	Dull note	Border raised on the affected side, mobility decreased

Sample case history entry

On comparative percussion a hyperresonant (box-like) note is found over all symmetrical areas of the lungs. **On topographic percussion**: the lung apices are 5 cm above the clavicle in front and 1 cm above the VII cervical vertebra behind. Krönig's isthmus is 9 cm on both sides. The lower borders of the lungs are lowered by one rib (on the right, the VII rib along the midclavicular line and the XI rib along the scapular line). The mobility of the lower edge of the lung along the mid-axillary line is 2 cm.

Conclusion: the syndrome of increased air content of the lung tissue (emphysema).

17. Practical skills

Skill: comparative percussion of the lungs

1. Undress the patient to the waist, standing or sitting; stand in front of the patient.
2. Begin in front: supraclavicular fossae → clavicles (directly) → I and II intercostal spaces along the midclavicular line.
3. At each level, percuss first the right and then the left side (or vice versa) with the same strength and compare the notes.
4. Move to the lateral regions: the patient places the hands on the head; the IV–V intercostal spaces, starting from the axilla.
5. Move to the back: the patient crosses the arms in front of the chest. The suprascapular regions, the interscapular region (pleximeter vertical, three points), the infrascapular regions (VII–IX intercostal spaces).
6. Record the result: what note over which area, symmetrical or asymmetrical.

Skill: determining the lower borders of the lungs and their mobility

1. On the right, percuss quietly downward along the parasternal line from the II intercostal space, and mark the place where the resonant note changes to a dull note along the upper edge of the pleximeter.
2. In the same way, determine the border along the midclavicular line, the anterior, mid- and posterior axillary lines (with the patient's hands on the head), and the scapular and paravertebral lines.

3. On the left, do the same, starting from the anterior axillary line.
4. Compare the borders found with the normal values by counting the ribs and intercostal spaces (start counting the ribs in front from the II rib — next to the sternal angle, and behind from the angle of the scapula — the VII rib).
5. To determine mobility, find the border during quiet breathing along the mid-axillary line, then have the patient hold the breath at deep inspiration and at full expiration and determine the border again.
6. Measure the distance between the points of maximal inspiration and expiration with a centimeter tape and compare it with the normal value.

Review questions

1. Who discovered percussion, and when?
2. What is the physical basis of percussion? To what depth does percussion provide information?
3. Explain the technique of finger-on-finger percussion.
4. What types of percussion are there by strength, and when are they used?
5. Compare the features of the clear lung, dull and tympanic notes.
6. In which condition is a hyperresonant (box-like) note heard, and why?
7. In which conditions does a dull-tympanic note appear?
8. In what order is comparative percussion performed?
9. Why may the note differ slightly at symmetrical points in a healthy person?
10. List the rules of topographic percussion.
11. How is the height of the lung apices determined in front and behind, and what are the normal values?
12. What is Krönig’s isthmus, and what is its normal width?
13. State the lower borders of the right and left lungs along the topographic lines.
14. In which conditions does the lower border of the lung descend and rise?
15. How is the mobility of the lower edge of the lung determined, and what is its normal value?

Test questions

1. Who discovered the method of percussion?
   A) R. Laennec
   B) S. Botkin
   C) J. Corvisart
   D) L. Auenbrugger
   E) Hippocrates
2. In finger-on-finger percussion, movement at which joint delivers the blow?
   A) The shoulder joint
   B) The wrist joint
   C) The elbow joint
   D) The whole arm
   E) The interphalangeal joints

3. To what depth from the point struck does percussion set the tissues into vibration?
   A) 1–2 cm
   B) 5–7 cm
   C) 10–12 cm
   D) 15 cm
   E) 20 cm

4. What note is heard over healthy lung?
   A) Dull-tympanic
   B) Dull
   C) Clear lung (resonant) note
   D) Hyperresonant (box-like)
   E) Tympanic

5. Which condition is a hyperresonant (box-like) note characteristic of?
   A) Pulmonary emphysema
   B) Atelectasis
   C) Pulmonary edema
   D) Exudative pleurisy
   E) Lobar pneumonia

6. What note is found on percussion in the hepatization stage of lobar pneumonia?
   A) Hyperresonant (box-like)
   B) Dull
   C) Metallic
   D) Tympanic
   E) Clear lung (resonant) note

7. What note is found on percussion in pneumothorax?
   A) Clear lung (resonant) note
   B) Dull
   C) Impaired
   D) Thigh note
   E) Tympanic

8. On which side is the border marked in topographic percussion?
   A) Between the two notes
   B) On the side of the dull note
   C) Always over a rib
   D) It does not matter
   E) On the side of the resonant (clear) note

9. At what level is the lung apex normally located from behind?
   A) The I thoracic vertebra
   B) The spinous process of the VII cervical vertebra
   C) The V cervical vertebra
   D) The spine of the scapula
   E) The III thoracic vertebra

10. What is the normal width of Krönig’s isthmus?
    A) 1–2 cm
    B) 5–6 cm
    C) 10–12 cm
    D) 15 cm
    E) 20 cm

11. Where is the lower border of the right lung along the midclavicular line normally located?
    A) The IV rib
    B) V intercostal space
    C) VI rib
    D) VII rib
    E) VIII rib

12. Where is the lower border of the left lung along the mid-axillary line normally located?
    A) VII rib
    B) VIII rib
    C) IX rib
    D) X rib
    E) The XI rib

13. Which condition is characterized by bilateral descent of the lower border of the lungs?
    A) Ascites
    B) Meteorism
    C) Pulmonary emphysema
    D) Pneumosclerosis
    E) Exudative pleurisy

14. What is the normal mobility of the lower edge of the lung along the mid-axillary line?
    A) 1–2 cm
    B) 2–3 cm
    C) 6–8 cm
    D) 10–12 cm
    E) 15 cm

15. How is the pleximeter finger placed in the interscapular region during comparative percussion?
    A) Horizontally
    B) It does not matter
    C) Parallel to the ribs
    D) Vertically (parallel to the spine)
    E) Obliquely

Clinical cases

Case 1

In a 65-year-old patient, comparative percussion revealed a hyperresonant (box-like) note over all areas. On topographic percussion, Krönig’s isthmus was 10 cm, the lower border of the right lung along the midclavicular line was at the VII rib, and the mobility along the mid-axillary line was 2 cm.

Questions: Interpret the percussion findings. Which syndrome is involved?

Answer: A hyperresonant note, widening of Krönig's isthmus (normal 5–6 cm), descent of the lower border by one rib (normal VI rib) and a sharp decrease in mobility (normal 6–8 cm) are characteristic of the syndrome of increased air content of the lung tissue, that is, pulmonary emphysema.

Case 2

In a 40-year-old patient, a dull note is found on the right below the infrascapular region. The upper border of the dullness runs along an oblique line: it rises upward and laterally from the spine, reaches its highest point at the posterior axillary line and then descends forward. In this area the mobility of the lower edge of the lung cannot be determined.

Questions: What do these findings indicate? What is the oblique line of the border called?

Answer: Accumulation of fluid in the pleural cavity (exudative pleurisy or hydrothorax). The oblique line of the upper border of the fluid is the Damoiseau line (Ellis–Damoiseau–Sokolov line); it is characteristic of an exudate. The absent mobility shows that the edge of the lung is compressed by the fluid. The diagnosis is confirmed by radiography, US and pleural puncture.

Case 3

In a tall, thin 25-year-old man, severe pain on the left side and dyspnea appeared suddenly after lifting a heavy load. On percussion there is a tympanic note on the left, and the lower border of the left lung cannot be determined.

Questions: Which condition can be considered? Interpret the percussion findings.

Answer: Spontaneous pneumothorax. As a result of air entering the pleural cavity, the lung collapses and a layer of air forms under the chest wall — that is why a tympanic note is heard and the lung border and its mobility cannot be determined. Urgent radiography is performed for confirmation.

Quick review

Table 5.13. *Key facts on Topic 5*

Question	Answer
Discovered by	L. Auenbrugger, 1761
Depth of percussion	5–7 cm
Clear lung (resonant) note	Loud, long, low-pitched, nontympanic
Dull	Fluid, consolidation, atelectasis
Tympanic	Pneumothorax, a large cavity
Hyperresonant (box-like)	Emphysema
Dull-tympanic	Stages I and III of lobar pneumonia
Height of the apex	In front, 3–4 cm above the clavicle; behind, CVII
Krönig's isthmus	5–6 cm (3.5–8 cm)
Right lower border	V intercostal space, VI, VII, VIII, IX, X rib, ThXI
Left lower border	Mid-axillary — IX rib
Mobility	Midclavicular and scapular — 4–6 cm; mid-axillary — 6–8 cm

TOPIC 6

Auscultation as a method of objective examination. Auscultation of the lungs: the main and added breath sounds

Learning objectives

To teach the student the history, technique and rules of auscultation, the main breath sounds heard on auscultation of the lungs (vesicular, bronchial) and their changes, the added breath sounds (dry and moist rales, crepitation, pleural friction rub), and how to examine and interpret bronchophony.

By the end of the session, the student should know:

- direct and indirect auscultation and the structure of the stethoscope and phonendoscope;
- the rules of auscultation and the points of auscultation of the lungs;
- the mechanism of production, features and sites of audibility of vesicular and bronchial breathing;
- the physiological and pathological changes of vesicular breathing;
- the types of pathological bronchial breathing and their causes;
- the mechanism of production and distinguishing features of rales, crepitation and pleural friction rub;
- bronchophony and its diagnostic significance.

The student should be able to:

- perform auscultation of the lungs in accordance with the rules;
- identify and assess the main breath sound;
- detect added sounds and apply the methods of distinguishing them from one another;
- examine bronchophony.

1. General information on auscultation

Auscultation (Latin auscultatio — listening) is a method of assessing the state of organs by listening to the sounds produced during their activity.

Direct auscultation (with the ear placed directly on the body) has been known since the time of Hippocrates. **Indirect auscultation** was discovered in 1816 by the French physician **René Laennec**: while examining a young woman, he rolled a sheet of paper into a tube, placed it on her chest and found that the heart sounds were clearly audible. He then made a wooden stethoscope and in 1819 published the book “On Mediate Auscultation”, in which many terms still in use today (rales, bronchial breathing, egophony) were described.



Figure 6.1. *Stethoscope (Gadayev A. et al., 2012, Fig. 19)*

Table 6.1. *Listening instruments*

Instrument	Structure	Advantage
Stethoscope (bell chest piece)	A funnel without a membrane	Conducts low-frequency sounds well (S3 and S4, the diastolic murmur of mitral stenosis)
Phonendoscope (diaphragm chest piece)	The funnel is closed by a membrane	Amplifies high-frequency sounds (vesicular breathing, the murmur of aortic regurgitation, fine rales)
Stethophonendoscope	Double-sided chest piece: bell and diaphragm	Universal, the most widely used
Electronic stethoscope	Electronic amplification and recording of sound	Phonocardiography, teaching, telemedicine

2. General rules of auscultation

1. The room should be warm and quiet.
2. The patient stands or sits, undressed to the waist; severely ill patients are examined lying down, being turned onto each side in turn.
3. The stethoscope head is applied to the skin closely but without excessive pressure; it must not be placed over clothing (friction sounds are produced).
4. The stethoscope head should be warm (a cold head causes muscle tremor).
5. Thick chest hair produces a crackling that resembles added sounds; it can be moistened or greased.
6. The patient breathes through the mouth, somewhat deeply and evenly.
7. At least 2–3 respiratory cycles are listened to at each point.
8. Symmetrical points are compared, and the examination follows the points of comparative percussion.
9. The main breath sound is assessed first, and then the added sounds.

Points of auscultation of the lungs

Auscultation is performed at the points where comparative percussion was carried out: **in front** — the supraclavicular fossae and the I–II (on the right, down to the III–IV) intercostal spaces along the midclavicular line; **from the side** — the axillary fossae and the IV–VI intercostal spaces along the mid-axillary line; **behind** — the suprascapular, interscapular (three points) and infrascapular regions.



Figure 6.2. *Points of auscultation in front and from the side (Gadayev A. et al., 2012, Fig. 38)*



Figure 6.3. *Points of auscultation from behind (same source, Fig. 39)*

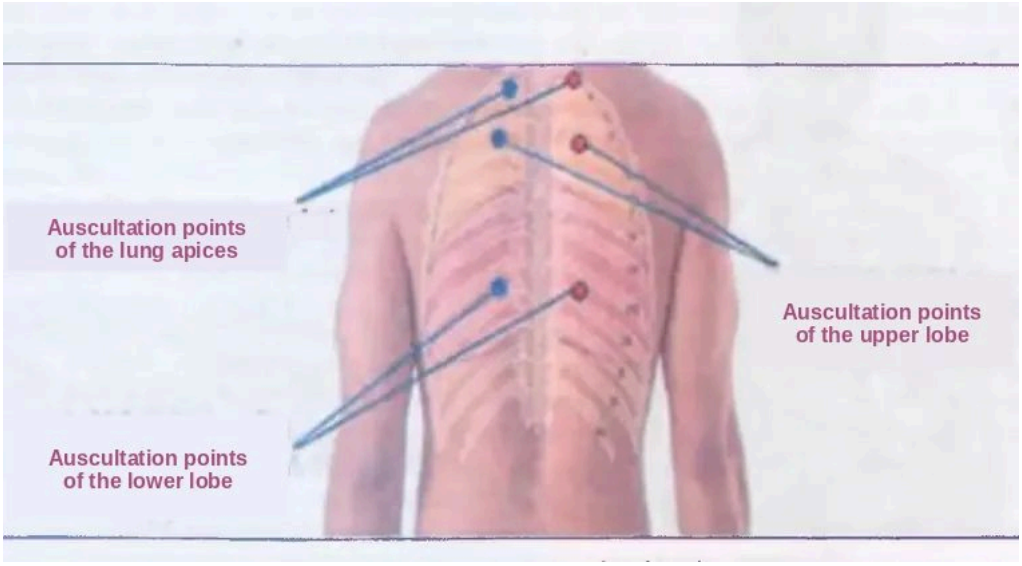


Figure 6.4. *Sites for listening to the parts of the lungs (Gadayev A. et al., 2012, Fig. 40)*

3. The main breath sounds

3.1. Vesicular (alveolar) breathing

Mechanism of production: during inspiration, air enters the alveoli and rapidly tenses their walls — the vibrations of the walls of millions of alveoli add up to produce vesicular breathing. At the beginning of expiration it continues briefly owing to the relaxation of the alveolar walls.

Features: a soft, blowing sound, resembling the sound produced when saying the letter “f” or drawing air inward. It is heard **throughout** inspiration and only during the **first third** of expiration.

REMEMBER

In vesicular breathing the ratio of the inspiratory to the expiratory phase is **3:1** (inspiration long and loud, expiration short and soft). It is best heard over the anterior lower parts of the chest, the axillary regions and the infrascapular regions (where the layer of lung tissue is thickest).

Table 6.2. *Changes of vesicular breathing*

Change	Type	Causes
Increased	Physiological	In children (“puerile” breathing), in thin people, during physical exertion
	Pathological (compensatory)	In the unaffected lung when the other lung does not take part in breathing (pleurisy, pneumothorax)
Harsh breathing	Pathological	Inflammation and unevenness of the bronchial mucosa (bronchitis, focal pneumonia): expiration is also loud and prolonged, and the sound is coarse
Diminished	Physiological	Obesity, strong development of the chest wall muscles
	Pathological — decreased number and elasticity of the alveoli	Pulmonary emphysema
	Pathological — decreased entry of air into the alveoli	The onset and resolution stage of lobar pneumonia (swollen alveolar walls), bronchial obstruction, the onset of atelectasis
	Pathological — impaired sound conduction	Fluid or air in the pleural cavity, pleural thickening
	Pathological — restriction of respiratory movements	Rib fracture, dry pleurisy (pain), diaphragmatic paralysis, weakness
Absent	Pathological	A large amount of fluid or air (the lung completely compressed), complete obstructive atelectasis
Interrupted (saccadic)	Pathological	Uneven narrowing of the small bronchi, uneven contraction of the respiratory muscles (chilling, nervousness)
Prolonged expiration	Pathological	Bronchial obstruction: bronchial asthma, COPD

3.2. Bronchial (laryngotracheal) breathing

Mechanism of production: when air passes through the narrow glottis in the larynx, the airflow becomes turbulent and a sound is produced. Because the glottis is narrower during expiration, the expiratory phase of bronchial breathing is louder, harsher and longer.

Features: harsh and loud, resembling the sound made when saying the letter “h”; it is heard **throughout** inspiration and expiration, and the expiratory phase is stronger and longer.

REMEMBER

In a healthy person, bronchial breathing is heard only **over the larynx and trachea and in the region of the manubrium**, and behind in the interscapular region at the level of the **VII cervical to the III–IV thoracic vertebrae**. If bronchial breathing is heard over other parts of the chest, it is considered **pathological**.

Pathological bronchial breathing is heard when the lung tissue has come to conduct sound well.

Table 6.3. *Types of pathological bronchial breathing*

Type	Mechanism	Diseases
Bronchial breathing (in tissue consolidation)	The alveoli are airless, and the consolidated tissue conducts the laryngotracheal sound in the bronchi well to the chest wall	The hepatization stage of lobar pneumonia, pulmonary infarction, compression atelectasis
Amphoric breathing	Resonance in a smooth-walled cavity more than 5–6 cm in diameter communicating with a bronchus; resembles the sound of blowing across the mouth of an empty bottle	A large cavern, an emptied abscess
Metallic breathing	Very loud, with a metallic ring	Open pneumothorax
Stenotic bronchial breathing	Narrowing of the trachea or a large bronchus, a loud sound	Tumor, foreign body

CAUTION

In obstructive atelectasis (a bronchus blocked by a tumor), bronchial breathing is **not heard** even though the lung tissue is dense, because there is no airflow — breath sounds are markedly diminished or absent. In compression atelectasis (when fluid compresses the lung), bronchial patency is preserved, so soft bronchial breathing is heard over the upper border of the fluid.

3.3. Mixed (bronchovesicular) breathing

It combines the features of vesicular and bronchial breathing: the inspiratory phase resembles vesicular and the expiratory phase bronchial breathing. It is heard when the lung tissue is consolidated in small foci, deep down or unevenly (focal pneumonia, tuberculous infiltrate, pneumosclerosis). In a healthy person it may also be heard over the right apex.

4. Added (adventitious) breath sounds

The added breath sounds include **rales** (dry and moist), **crepitation** and the **pleural friction rub**. They are heard against the background of the main breath sound and are always considered pathological.

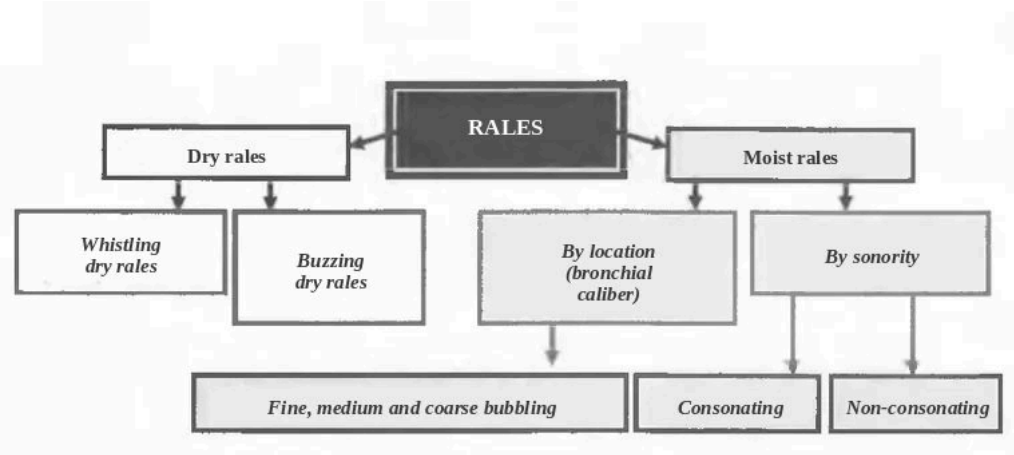


Figure 6.5. *Classification of rales (Gadayev A. et al., 2012, Fig. 41)*

4.1. Dry rales (rhonchi sicci)

Mechanism: narrowing of the bronchial lumen (mucosal edema, bronchospasm, viscous sputum) leads to acceleration and turbulence of the airflow; viscous sputum stretched like threads across the

bronchial lumen vibrates in the airflow like a “string”.

Table 6.4. *Types of dry rales*

Type	Where they arise	Character	Diseases
Whistling (high-pitched, sibilant)	Small bronchi and bronchioles	High, thin, like a whistle; most often during expiration	Bronchial asthma, COPD, bronchiolitis
Humming (low-pitched, sonorous)	Medium and large bronchi	Low, buzzing, like snoring	Acute and chronic bronchitis, tracheobronchitis

Characteristic features of dry rales: they are changeable — after coughing they may increase, decrease or change location; they are heard in both phases of respiration. In an attack of bronchial asthma, whistling rales are sometimes heard even at a distance (“distant wheezes”).

4.2. Moist rales (rhonchi humidi)

Mechanism: liquid secretions (sputum, blood, transudate) accumulate in the bronchial lumen; as air passes through them, bubbles form and burst — a sound “like blowing into water through a straw”.

Table 6.5. *Types of moist rales*

Classification	Type	Content
By the caliber of the bronchus where they arise	Fine-bubbling	In the small bronchi and bronchioles; short, numerous
	Medium-bubbling	In bronchi of medium caliber
	Coarse-bubbling	In the large bronchi, the trachea and cavities (cavern, abscess); longer, low-pitched
By resonance	Consonating (resonant)	The surrounding lung tissue is consolidated or there is cavity resonance — the sound is conducted well and heard “right under the ear”: pneumonia, cavern, abscess
	Nonconsonating	The lung tissue is air-filled and the sound is damped, heard “from afar”: bronchitis, pulmonary congestion (heart failure), pulmonary edema

CLINICAL SIGNIFICANCE

Consonating fine-bubbling moist rales over a limited area are an important sign of focal pneumonia. Bilateral, symmetrical **nonconsonating fine-bubbling** rales in the lower parts of the lungs indicate congestion in the pulmonary circulation (left ventricular failure). In pulmonary edema, coarse-bubbling rales are heard at a distance, like a “boiling samovar”.

4.3. Crepitation (crepitatio)

Mechanism: when there is a sticky secretion (exudate) on the inner surface of the alveoli, the alveolar walls stick together during expiration; at the **very end** of a deep inspiration, air enters and separates them simultaneously — a faint “crackling” is heard. This sound is likened to rubbing a lock of hair between the fingers next to the ear or to throwing salt onto a fire.

Causes: stages I (crepitatio indux) and III (crepitatio redux) of lobar pneumonia, pulmonary infarction, pulmonary congestion and the onset of pulmonary edema, fibrosing alveolitis.

4.4. Pleural friction rub (frictio pleurae)

Mechanism: normally the smooth, moist layers of the pleura slide silently during breathing. In inflammation of the pleura (fibrin deposits), its drying (dehydration, uremia), tumors and tuberculous tubercles, the surfaces of the layers become rough — and a sound is produced when they rub together. This sound resembles the crunching of snow underfoot or the creaking of new leather.

Causes: dry (fibrinous) pleurisy, lobar pneumonia (the pleura is also involved), pulmonary infarction, pleural tumors and tuberculosis, uremia, severe dehydration. In exudative pleurisy, it reappears during the period of fluid resorption.

The pleural friction rub is most often heard **in the lower part of the axillary region** (where the lung moves most during breathing). When the pleura next to the heart is inflamed, a **pleuropericardial rub** related to both the heartbeat and breathing is heard.

5. Distinguishing the added sounds from one another

Table 6.6. *Differences between moist rales, crepitation and the pleural friction rub*

Feature	Moist rales	Crepitation	Pleural friction rub
Site of origin	Bronchi	Alveoli	Pleural layers
Phase of respiration	In both phases	Only at the end of a deep inspiration	In both phases
After coughing	Changes (decreases, disappears or increases)	Unchanged	Unchanged
On pressing with the stethoscope	Unchanged	Unchanged	Increases
On “sham breathing” with the abdomen, with the mouth and nose closed	Not heard (no airflow)	Not heard	Heard (the pleural layers move)
Pain	Usually absent	Absent	Often painful, may be felt on palpation
Character of the sound	Bursting of bubbles	A uniform burst of fine “crackling”	Walking on snow, crunching

MEMORY AID

Three tests distinguish the three sounds: **“Cough — Pressure — Sham breathing”**. If the sound changes after coughing, it is **rales**; if it increases when pressing with the stethoscope and is heard even during sham breathing, it is a **pleural friction rub**; if it occurs only at the end of inspiration and does not change after coughing, it is **crepitation**.

CAUTION

In modern English-language textbooks (ATS/ERS nomenclature, Bates, Macleod) the terms are different: **crackles** are divided into fine and coarse types and cover moist rales and crepitation together; **wheezes** are high-pitched dry rales; **rhonchi** are low-pitched dry rales; **stridor** is a loud inspiratory sound in narrowing of the upper airways. The “crepitation” of the Russian and Uzbek school roughly corresponds to the English fine end-inspiratory crackles. Keep this difference in mind when reading the literature.

6. Bronchophony

Bronchophony is the audibility with a stethoscope, over the chest wall, of words whispered by the patient. It has the same mechanism as vocal fremitus but is more sensitive in detecting smaller foci.

Method: the stethoscope is placed on symmetrical points, and the patient is asked to whisper words containing hissing sounds (“cup of tea”, “sixty-six”, “hush”). Normally the words are not heard clearly; only an indistinct murmur is perceived.

Increased bronchophony (the words are heard clearly) — in consolidation of the lung tissue and in a cavity communicating with a bronchus (the same conditions as increased vocal fremitus). **Decreased** — fluid or air in the pleural cavity, obstructive atelectasis, emphysema.

> **CLINICAL SIGNIFICANCE**
>
> English-language textbooks distinguish two types of bronchophony: **whispered pectoriloquy** — clear audibility of whispered words, and **egophony** — the sound “ee” heard like “ay”, resembling the bleating of a goat; egophony is observed at the upper border of fluid (over the compressed lung) and in consolidation.

7. Structure and correct use of the phonendoscope



Figure 6.6. *Stethophonendoscope (Gadayev A. et al., 2012, Fig. 19)*



Figure 6.7. *Performing auscultation (same source, Fig. 20)*

A modern stethophonendoscope consists of the following parts: **earpieces** (soft; they must fit snugly into the ear canal), **ear tubes** (they should be angled forward, toward the nose, because the external ear canal is directed forward), **flexible tubing** (50–70 cm long; long tubing attenuates the sound) and the **chest piece** (diaphragm and bell).

Table 6.7. *Differences between the diaphragm and the bell*

Property	Diaphragm	Bell
Frequency	High-frequency sounds (above 100–1000 Hz)	Low-frequency sounds (30–150 Hz)
Pressure on the skin	Pressed firmly	Applied very lightly (if pressed hard, the skin acts as a diaphragm and the low frequencies are lost)
What is heard	Vesicular and bronchial breathing, rales, the I and II heart sounds, the murmur of aortic regurgitation, pericardial friction rub	The III and IV heart sounds, the diastolic murmur of mitral stenosis, bruits over vessels

CAUTION

"Artifact" sounds on auscultation: rubbing of the stethoscope head against clothing or hair (resembles crepitation), muscle tremor (in a cold room, with fear — a low rumble), the patient swallowing, the tubes touching each other, intestinal peristalsis (in the left lower region). If a doubtful sound is detected, press the stethoscope more firmly, ask the patient not to move and listen again.

8. Acoustic description of the breath sounds

Table 6.8. *Acoustic properties of the main breath sounds (after Bates)*

Sound	Duration of the phases	Severity	Pitch	Where heard normally
Vesicular	Inspiration longer than expiration	Soft	Low	Over most of the lungs
Bronchovesicular (mixed)	About equal	Average	Average	The I–II intercostal spaces in front, the upper interscapular region
Bronchial	Expiration longer	Loud	High	Over the manubrium
Tracheal	About equal	Very loud	High	Over the trachea in the neck

REMEMBER

The easiest way to distinguish bronchial from vesicular breathing is **the pause between the phases of respiration**: in bronchial breathing there is a short silence between inspiration and expiration, whereas in vesicular breathing there is no pause (inspiration passes without a break into expiration).

8.1. Conditions for the occurrence of pathological bronchial breathing

For bronchial breathing to be heard over the chest wall, three conditions must be met:

1. the lung tissue must be **consolidated** (airless);
2. the consolidated area must be connected with a **patent bronchus**;
3. the area of consolidation must be **close** to the chest wall and sufficiently **large**.

That is why bronchial breathing is rarely heard in focal pneumonia (small, deep foci), whereas in lobar pneumonia it is always heard.

9. Modern classification and mechanisms of the added sounds

9.1. Timing of crackles

Table 6.9. *Crackles and their timing*

Type	Phase of respiration	Mechanism	Diseases
Early inspiratory coarse	At the beginning of inspiration	Secretions in the large and medium bronchi, opening of the small bronchi	COPD, bronchiectasis, bronchitis
Late inspiratory fine, the "Velcro" sound	At the end of inspiration	Sudden opening of collapsed small airways and alveoli	Pulmonary fibrosis (idiopathic pulmonary fibrosis), interstitial diseases, congestion, the onset of pneumonia

Type	Phase of respiration	Mechanism	Diseases
Pan-inspiratory and expiratory	Throughout respiration	A large amount of fluid	Pulmonary edema, bronchiectasis

CLINICAL SIGNIFICANCE

The **late inspiratory fine crackles** heard over the basal areas in pulmonary fibrosis resemble the "Velcro" sound (the sound of a strip of Velcro being pulled apart). They do not change after coughing and often occur together with **clubbing**. In Russian-language textbooks this sound is described as "crepitation" or "consonating fine-bubbling rales".

9.2. Wheezes and stridor

- **Polyphonic** wheezes (of different pitches at the same time) — diffuse narrowing of many bronchi (bronchial asthma, COPD).
- **Monophonic** (a single pitch, always in the same place) — local narrowing of a single bronchus (tumor, foreign body). If such a wheeze does not change after coughing, bronchoscopy is needed.
- **Stridor** is a loud, harsh sound heard at a distance, mainly **during inspiration**, in narrowing of the upper airways (larynx, trachea). It is a life-threatening sign: angioedema, foreign body, laryngeal tumor, epiglottitis, false croup.

9.3. Other added sounds

- **Pleuropericardial rub** — a sound related to the heartbeat and to breathing when the pleura next to the heart is inflamed; it diminishes when the breath is held.
- **"Falling drop" sound** — a resonant dripping sound heard when the patient changes position in a large cavity containing air and fluid (hydropneumothorax, a large cavern).
- **Hippocratic succussion splash** — heard in hydropneumothorax when the patient is shaken by the shoulders.
- **"Mediastinal crunch"** (Hamman's sign) — a crunching sound heard in time with the heartbeat in mediastinal emphysema.

10. Voice-related auscultatory phenomena

Table 6.10. *Bronchophony and related phenomena*

Phenomenon	Method	Normal	Pathological result
Bronchophony	The patient says "thirty-three" in a normal voice	The words are an indistinct murmur	The words are heard clearly and loudly — consolidation
Whispered pectoriloquy	The patient whispers "one, two, three" or "cup of tea"	Very quiet, indistinct	The whisper is heard clearly — consolidation (the most sensitive sign)
Egophony	The patient says a prolonged "ee-ee-ee"	The sound "ee" is heard	Like "ay" or "ah", with a nasal quality ("the bleating of a goat") — consolidation, compressed lung above fluid

MEMORY AID

Vocal fremitus, bronchophony and egophony obey the same rule: **they are increased in consolidation and decreased in fluid, air, obstruction and emphysema**. All three confirm one another.

11. Points of auscultation on the anterior surface and special methods

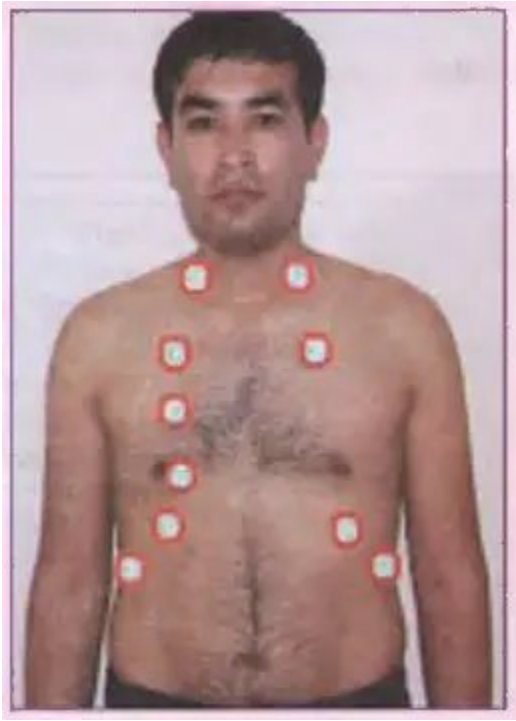


Figure 6.8. *Points of auscultation of the lungs from the front (Gadayev A. et al., 2012, Fig. 38)*

Special methods:

- **Listening after a forced cough** — to detect changes in rales (bronchial secretions) and to "awaken" latent crepitation.
- **Deep breathing and forced expiration** — to detect latent bronchial obstruction (wheezes).
- **Listening with the patient turned onto the side** — to examine the lower posterior regions in bedridden patients (hypostatic pneumonia, congestion).

> **CLINICAL SIGNIFICANCE**
> In modern practice, **lung ultrasound** is becoming widespread as a rapid bedside method: B-lines indicate pulmonary edema and the interstitial syndrome, the absence of the "lung sliding" sign indicates pneumothorax, and consolidation indicates pneumonia. However, it does not replace auscultation but complements it.

12. Standard form for recording auscultatory findings

The result of auscultation is recorded in the following order:

1. The type and intensity of the main breath sound (vesicular, diminished, increased, harsh, bronchial, mixed, amphoric).
2. The ratio of the respiratory phases (whether expiration is prolonged or not).
3. Added sounds: type (dry — whistling or humming; moist — fine-, medium- or coarse-bubbling, consonating or nonconsonating; crepitation; pleural friction rub), amount (isolated, numerous), location, phase of respiration, change after coughing.
4. Bronchophony. **Example:** "Vesicular breathing over the upper parts of the lungs. On the right in the infrascapular region, below the VII rib, breath sounds are diminished; in this area numerous consonating fine-bubbling moist rales are heard at the end of inspiration, unchanged after coughing. Bronchophony is increased in the right infrascapular region."

13. Dynamics of the auscultatory picture in various diseases

Auscultatory signs change during the course of a disease; observing them daily helps to assess the effectiveness of treatment.

Table 6.11. *Auscultation in the dynamics of diseases*

Disease	Onset	Height of the disease	Recovery period
Lobar pneumonia	Diminished vesicular breathing, crepitation (indux), pleural friction rub	Bronchial breathing, increased bronchophony	Crepitation (redux), consonating moist rales, vesicular breathing returns
Acute bronchitis	Harsh breathing, dry rales	Dry and nonconsonating moist rales	The rales decrease and disappear
Exudative pleurisy	Pleural friction rub	No breath sounds (over the fluid)	The friction rub reappears
Lung abscess	Diminished or bronchial breathing	After rupture, amphoric breathing and coarse-bubbling rales	The rales decrease and breath sounds are restored
Asthma attack	Wheezes, prolonged expiration	Numerous rales or a “silent chest” (severe)	The rales gradually decrease
Pulmonary edema	Crepitation and fine rales in the lower parts	Medium- and coarse-bubbling rales over the whole lung, “bubbling”	The rales decrease from below upward

CLINICAL SIGNIFICANCE

In pulmonary edema, **how high** the rales extend indicates the severity of the edema: only in the lower parts — initial; up to the angle of the scapula — moderate; over the whole lung — severe (alveolar edema). This is a very convenient indicator for following the effect of treatment.

14. Changes in breath sounds related to complications

- **When pleurisy complicates pneumonia** — a pleural friction rub appears; later, if fluid accumulates, breath sounds disappear over this area.
- **Transformation of pneumonia into an abscess** — amphoric breathing in place of bronchial breathing, numerous coarse-bubbling rales; the sputum increases.
- **Development of pneumothorax** (for example, in a patient with COPD) — breath sounds suddenly disappear on one side.
- **Atelectasis** (blockage by sputum) — breath sounds suddenly disappear; they may return after coughing and expectoration.

15. Typical errors in describing the results of auscultation in the case history

1. Writing “breathing vesicular, no rales” without listening to the lower posterior regions.
2. Not distinguishing crepitation from moist rales (not examining after coughing).
3. Not describing the rales: type, caliber, resonance, amount, location.
4. Confusing a pleural friction rub with a pericardial rub — not examining with the breath held.
5. Confusing bronchial breathing with “harsh breathing” (in bronchial breathing the expiratory phase is long and loud, and there is a pause).

MEMORY AID

On auscultation, always keep to the order **“Type — Intensity — Phases — Added sounds — Bronchophony”**: first the type of breathing, then its intensity, the ratio of the respiratory phases, the added sounds and, last of all, bronchophony.

16. Clinical part

Auscultatory findings in the main syndromes

Table 6.12. *Auscultatory findings in respiratory syndromes*

Syndrome	Main breath sound	Added sounds	Bronchophony
Bronchial obstruction	Harsh breathing, prolonged expiration	Dry whistling rales (wheezes)	Unchanged or decreased
Emphysema	Symmetrically diminished vesicular breathing	Often absent	Decreased
Lobar pneumonia (hepatization)	Bronchial breathing	Consonating moist rales, pleural friction rub	Increased
Lobar pneumonia (stages I and III)	Diminished vesicular breathing	Crepitation	Moderately increased
Focal pneumonia	Mixed or harsh breathing	Consonating fine-bubbling rales	Increased
Fluid in the pleural cavity	Breath sounds diminished or absent; soft bronchial breathing at the upper border	None (during resorption — a friction rub)	Decreased or absent
Pneumothorax	Breath sounds markedly diminished or absent; in open pneumothorax, amphoric, metallic	Absent	Decreased
Cavity (cavern, abscess)	Bronchial or amphoric breathing	Coarse-bubbling consonating moist rales	Increased
Left ventricular failure	Vesicular breathing	Bilateral nonconsonating fine-bubbling rales in the lower parts, crepitation	Unchanged

Sample case history entry

On auscultation, bronchial breathing is heard over the right infrascapular region of the lung, and in this area crepitation at the end of inspiration and a pleural friction rub are detected. Bronchophony is increased in the right infrascapular region. Over the other areas, vesicular breathing without added sounds.

Conclusion: the syndrome of consolidation of the lung tissue in the lower lobe of the right lung with inflammatory involvement of the pleura — characteristic of right-sided lower lobe lobar pneumonia.

17. Practical skills

Skill: auscultation of the lungs

1. Check the stethophonendoscope, warm the chest piece and put on the earpieces facing forward (matching the ear canals).

2. Undress the patient to the waist, standing or sitting; ask him or her to breathe evenly and somewhat deeply through the mouth.
3. Begin in front: the supraclavicular fossae, the I–II intercostal spaces; listen to 2–3 respiratory cycles at each point and compare with the symmetrical point.
4. Listen over the lateral regions: the patient places the hands on the head; from the axillary fossae downward.
5. Listen over the posterior regions: the patient crosses the arms in front of the chest; the suprascapular, interscapular and infrascapular regions.
6. At each point, assess first the main breath sound (type, intensity, ratio of the phases) and then the added sounds.
7. If an added sound is detected, perform the distinguishing tests: have the patient cough, press with the stethoscope, perform the sham-breathing test.
8. Record the result in the case history.

Skill: examining bronchophony

1. Place the stethoscope on symmetrical points in turn.
2. Ask the patient to whisper “cup of tea” or “sixty-six”.
3. Compare the clarity of the words on the two sides.
4. Interpret the result: normal, increased or decreased.

Review questions

1. Who discovered auscultation, and when?
2. What is the difference between the stethoscope and the phonendoscope, and how are they used?
3. List the main rules of auscultation.
4. How is vesicular breathing produced, and what are its features?
5. What are the causes of physiological and pathological intensification of vesicular breathing?
6. In which conditions is vesicular breathing diminished?
7. What is the mechanism of production of bronchial breathing, and where is it normally heard?
8. In which conditions is pathological bronchial breathing heard? What is amphoric breathing?
9. How are dry rales produced, and what types are there?
10. When are the consonating and nonconsonating types of moist rales heard?
11. How is crepitation produced, and in which diseases is it heard?
12. What are the causes of the pleural friction rub?
13. How do you distinguish moist rales, crepitation and the pleural friction rub?
14. What are the added sounds called in the modern English nomenclature?
15. How is bronchophony examined, and when is it increased?

Test questions

1. Who discovered indirect auscultation?
   A) V. Obraztsov
   B) N. Korotkov
   C) L. Auenbrugger
   D) J. Corvisart
   E) R. Laennec

2. What is the ratio of the inspiratory to the expiratory phase in vesicular breathing?
   A) 1:1
   B) 1:3
   C) 3:1
   D) 2:1
   E) 1:2

3. Where is bronchial breathing heard in a healthy person?
   A) It is not heard anywhere
   B) Over the larynx and trachea and in the upper part of the interscapular region
   C) In the infrascapular regions
   D) In the lower parts of the lungs
   E) In the axillary regions

4. How does vesicular breathing change in pulmonary emphysema?
   A) Increases
   B) It becomes amphoric
   C) It changes to bronchial breathing
   D) Unchanged
   E) It is symmetrically diminished

5. What breathing is heard in the hepatization stage of lobar pneumonia?
   A) Bronchial
   B) Vesicular
   C) Diminished vesicular
   D) Amphoric
   E) Saccadic

6. In which condition is amphoric breathing heard?
   A) A large smooth-walled cavity (cavern)
   B) Bronchial asthma
   C) Bronchitis
   D) Exudative pleurisy
   E) Pulmonary emphysema

7. Where are whistling dry rales produced?
   A) In the alveoli
   B) In the large bronchi
   C) In the pleura
   D) In the small bronchi and bronchioles
   E) In the trachea

8. Where is crepitation produced?
   A) In the trachea
   B) In the small bronchi
   C) In the large bronchi
   D) In the alveoli
   E) In the pleura

9. In which phase of respiration is crepitation heard?
   A) Only on coughing
   B) At the beginning of expiration
   C) At the very end of inspiration
   D) In both phases
   E) During the respiratory pause

10. Which sound increases on pressing with the stethoscope and is heard even during sham breathing?
    A) Pleural friction rub
    B) Crepitation
    C) Bronchial breathing
    D) Dry rales
    E) Moist rales

11. In which condition are consonating fine-bubbling moist rales most often heard?
    A) Pulmonary congestion
    B) Focal pneumonia
    C) Bronchial asthma
    D) Emphysema
    E) Acute bronchitis

12. What do bilateral nonconsonating fine-bubbling rales in the lower parts of the lungs indicate?
    A) Dry pleurisy
    B) Bronchial asthma
    C) Left ventricular failure (congestion)
    D) Pneumothorax
    E) Lobar pneumonia

13. Which added sound changes after coughing?
    A) Pleural friction rub
    B) Rales
    C) Bronchial breathing
    D) Crepitation
    E) Pleuropericardial rub

14. In which condition is bronchophony increased?
    A) Pulmonary emphysema
    B) Exudative pleurisy
    C) Obstructive atelectasis
    D) Pneumothorax
    E) Consolidation of the lung tissue

15. To which sound does the term “wheezes” in the English nomenclature correspond?
    A) Crepitation
    B) Coarse-bubbling moist rales
    C) High-pitched dry (whistling) rales
    D) Pleural friction rub
    E) Bronchial breathing

Clinical cases

Case 1

A 30-year-old patient suddenly developed a choking attack. He is sitting leaning forward with his hands propped on the edge of a table. On auscultation, harsh breathing is heard over the whole lung with markedly prolonged expiration and numerous whistling dry rales in both phases; they are also heard at a distance.

Questions: Interpret the auscultatory findings. Where and by what mechanism are the rales produced?

Answer: Prolonged expiration and whistling (high-pitched) dry rales are characteristic of obstruction of the small bronchi (bronchospasm, mucosal edema, viscous sputum). The airflow through the narrowed bronchi accelerates and becomes turbulent, producing a high-frequency sound. The clinical picture corresponds to an attack of bronchial asthma.

Case 2

On auscultation of a 70-year-old patient, a moist sound was heard over the right infrascapular region. The sound did not change after the patient coughed; it is heard only at the end of a deep inspiration as a uniform fine crackling and does not increase on pressing with the stethoscope.

Questions: Which added sound is this? How is it produced, and in which conditions is it heard?

Answer: Crepitation. It is produced by the simultaneous separation, at the end of inspiration, of alveoli that stuck together during expiration because of exudate adhering to the alveolar walls. Its not changing after coughing distinguishes it from moist rales, and its not increasing under pressure of the stethoscope and its being heard only at the end of inspiration distinguish it from the pleural friction rub. The onset or resolution stage of lobar pneumonia, pulmonary congestion and alveolitis should be considered.

Case 3

A 45-year-old patient has stabbing pain on the left that increases on deep inspiration. In the lower part of the left axillary region, a sound resembling the crunching of snow underfoot is heard in both phases of respiration; it increases on pressing with the stethoscope and does not change after coughing.

Questions: Which sound has been detected? What is its cause, and what is the confirmatory test?

Answer: A pleural friction rub — characteristic of dry (fibrinous) pleurisy. To confirm it, the “sham breathing” test is performed: the patient closes the mouth and nose and makes breathing movements with the abdominal wall — even without airflow, the pleural layers move and the rub is heard (whereas rales are not). The pain also confirms pleural involvement.

Quick review

Table 6.13. *Key facts on Topic 6*

Question	Answer
Discovered by	R. Laennec, 1816
Vesicular breathing	In the alveoli; the whole of inspiration, 1/3 of expiration; ratio 3:1
Bronchial breathing	In the larynx; both phases, expiration longer; normally over the larynx and trachea, CVII–ThIII
Pathological bronchial breathing	Consolidation, compression atelectasis, cavity
Amphoric breathing	A large smooth-walled cavity
Dry rales	Bronchial narrowing; whistling (small bronchi) and humming (large bronchi)
Moist rales	Liquid secretions in the bronchi; fine, medium, coarse; consonating and nonconsonating
Crepitation	Alveoli; at the end of inspiration; does not change after coughing
Pleural friction rub	In both phases; increases on pressure; heard during sham breathing
Increased bronchophony	Consolidation, cavity

TOPIC 7

The syndrome of impaired bronchial patency. Acute and chronic bronchitis. The syndrome of increased air content of the lung tissue. Bronchial asthma. Pulmonary emphysema

Learning objectives

To teach the student the pathogenesis and clinical signs of the syndromes of bronchial obstruction and increased air content of the lung tissue, the symptomatology of acute and chronic bronchitis, bronchial asthma and pulmonary emphysema, as well as how to examine patients with obstructive diseases and interpret the results of spirometry and peak flowmetry.

By the end of the session, the student should know:

- the main mechanisms of bronchial obstruction (reversible and irreversible components);
- the physical signs of the syndromes of bronchial obstruction and emphysema;
- the definition, causes and clinical signs of acute and chronic bronchitis;
- the definition, the periods of an attack, and the physical and laboratory signs of bronchial asthma, and status asthmaticus;
- the main signs of COPD and pulmonary emphysema;
- the main parameters and normal values of spirometry and peak flowmetry (GINA 2025, GOLD 2026).

The student should be able to:

- take the history of and physically examine a patient with an obstructive disease;
- identify the syndromes of bronchial obstruction and emphysema on the basis of physical signs;
- perform peak flowmetry and interpret the result;
- interpret the conclusion of a spirogram.

1. The syndrome of impaired bronchial patency (bronchial obstruction)

Bronchial obstruction is an increase in resistance to airflow and impaired ventilation of the lungs as a result of narrowing of the bronchial lumen. Clinically it is manifested mainly by expiratory dyspnea and dry rales.

Table 7.1. *Mechanisms of bronchial obstruction*

Component	Mechanism	Example
Reversible (functional)	Spasm of the bronchial smooth muscle	Bronchial asthma
	Inflammatory edema and infiltration of the mucosa	Acute bronchitis, asthma
	Increased viscous secretion (hyper- and dyscrinia)	Chronic bronchitis, asthma
Irreversible (organic)	Fibrosis and deformity of the bronchial wall	Chronic obstructive bronchitis
	Expiratory collapse of the small bronchi (when lung elasticity is lost)	Pulmonary emphysema

Component	Mechanism	Example
	Blockage of the bronchial lumen by a tumor or foreign body (local obstruction)	Lung cancer

REMEMBER

The small bronchi **less than 2 mm** in diameter, which have no cartilaginous skeleton, are held open by the elastic recoil of the lung tissue. When elasticity is lost in emphysema, they collapse and close during expiration (**expiratory collapse**) — air is “trapped” in the alveoli.

Signs of the syndrome of bronchial obstruction

Table 7.2. *Signs of the syndrome of bronchial obstruction (bronchospasm)*

Method of examination	Signs
Complaints	Expiratory dyspnea or choking attacks, cough with viscous, hard-to-expectorate sputum, a feeling of “tightness” in the chest
Inspection	Forced position (sitting propped on the hands), use of the accessory muscles, prolonged expiration, exhalation through pursed lips, diffuse cyanosis, the neck veins distend during expiration; the chest is “in the position of inspiration”
Palpation	Vocal fremitus symmetrically decreased, resistance increased
Percussion	Hyperresonant (box-like) note, lower borders lowered, mobility decreased
Auscultation	Harsh or diminished vesicular breathing, prolonged expiration, numerous dry whistling rales (especially during expiration)
Functional investigation	FEV_1, the Tiffeneau index (FEV_1/FVC) and PEF decreased

CLINICAL SIGNIFICANCE

The forced expiratory time test is a simple bedside test: the patient takes a deep breath and breathes out as fast and as completely as possible with the mouth open, while the physician places the stethoscope over the trachea and measures the time. Normally expiration takes **less than 4 seconds**; more than 6 seconds indicates bronchial obstruction.

2. Acute bronchitis

Acute bronchitis is an acute diffuse inflammation of the bronchial mucosa. It most often develops as a manifestation of acute respiratory viral infections and accounts for 20–25 percent of acute nonspecific respiratory diseases.

Causes: viruses (respiratory syncytial virus, rhinoviruses, adenoviruses, influenza), bacteria (pneumococcus, Haemophilus influenzae), physical and chemical factors (cold air, dust, acid and alkali vapors). Predisposing factors: chilling, smoking, alcohol, foci of infection in the nasopharynx.

Clinical picture: the disease often begins with a runny nose, low-grade fever and weakness. A scratchy feeling behind the sternum and a dry or scantily productive cough appear; after 2–3 days the cough loosens and mucoid or mucopurulent sputum is produced. In the uncomplicated course there is no dyspnea; its appearance indicates involvement of the small bronchi and bronchioles (bronchiolitis).

Physical signs: a clear lung note on percussion; on auscultation, harsh breathing, dry (humming and whistling) rales and, when the sputum increases, nonconsonating moist rales.

Laboratory and X-ray: no changes in the blood, or mild leukocytosis and a raised ESR (15–20 mm/h); no changes on X-ray, or an accentuated lung pattern.

Course: usually 10–14 days, ending in complete recovery. The most common complication is focal pneumonia. Fever lasting more than 3–4 days and a sharp rise in leukocytosis indicate that pneumonia has developed.

> **CAUTION**
> In acute bronchitis the auscultatory findings are diffuse and symmetrical. If **consonating** fine-bubbling moist rales or crepitation appear over a limited area, this is a sign not of bronchitis but of pneumonia — radiography is mandatory.

3. Chronic bronchitis

Chronic bronchitis is a diffuse, long-standing inflammation of the bronchi accompanied by hypersecretion of mucus and impaired drainage function.

> **REMEMBER**
> By the WHO criterion, chronic bronchitis is a productive cough **for at least 3 months a year in at least 2 consecutive years**, with other causes of cough (tuberculosis, bronchiectasis, tumor and others) excluded.

Causes: tobacco smoking (the main cause in about 80 percent of patients), occupational factors (cotton, coal and cement dust, chemical vapors), air pollution, recurrent respiratory infections, difficulty breathing through the nose.

Classification: by the nature of the exudate — catarrhal and purulent; by external respiratory function — obstructive and nonobstructive; by the course — phases of exacerbation and remission.

Clinical picture: the main sign is **cough**. At first it occurs in the morning after waking, with a small amount of sputum (the “smoker’s cough”); it gets worse in the cold, damp season. Later the sputum becomes mucopurulent and purulent. When obstruction and emphysema develop, **exertional dyspnea** appears and gradually increases.

Physical signs: in the uncomplicated course there are no changes on percussion; on auscultation, harsh breathing, dry rales and nonconsonating moist rales. In the obstructive form, prolonged expiration and wheezes; when emphysema develops, its signs appear.

Complications: pulmonary emphysema, respiratory failure, pulmonary hypertension and chronic cor pulmonale, pneumonia, bronchiectasis.

4. Chronic obstructive pulmonary disease (COPD)

In current international practice, chronic obstructive bronchitis and emphysema are considered within a single disease — **COPD**.

> **REMEMBER**
> According to GOLD 2026, spirometry is mandatory for the diagnosis of COPD: the **post-bronchodilator FEV_1/FVC ratio must be less than 0.70**. The severity of obstruction is assessed by FEV_1 as a percentage of the predicted value.

Table 7.3. *Severity of airflow limitation in COPD (GOLD, when FEV_1/FVC is less than 0.70)*

Grade	Definition	FEV_1, % of predicted
GOLD 1	Mild	80 % or more

Grade	Definition	FEV_1, % of predicted
GOLD 2	Average	50–79 %
GOLD 3	Severe	30–49 %
GOLD 4	Extremely severe	Less than 30 %

Patients are divided into groups **A, B and E** according to their symptoms (mMRC, CAT questionnaire) and the risk of exacerbations. In GOLD 2026, even a single moderate or severe exacerbation is assessed as a factor increasing the risk of further exacerbations.

CAUTION
Older textbooks (including the textbook used here) grouped chronic bronchitis, bronchial asthma, bronchiectasis and emphysema together as "chronic obstructive lung diseases". In modern classifications, **bronchial asthma and bronchiectasis are separate diseases**, and COPD is only a condition with persistent (not fully reversible) airflow limitation.

5. Bronchial asthma

Bronchial asthma is a heterogeneous disease with chronic inflammation of the airways. It is characterized by respiratory symptoms that vary over time and in intensity (wheezing, dyspnea, chest tightness, cough) and by **variable expiratory airflow** (GINA 2025).

Etiology: exogenous allergens (plant pollen, house dust and mites, particles from pets' fur, molds, foods), infections, drugs (aspirin and other NSAIDs, beta-blockers), occupational factors, physical exertion, cold air, psychological factors, hereditary predisposition.

Pathogenesis: it is based on sensitization of the organism and an IgE-dependent immediate (type I) allergic reaction. When the allergen binds to IgE fixed on the mast cells of the bronchial mucosa, histamine, leukotrienes and other mediators are released, causing **bronchospasm, mucosal edema and hypersecretion**. Eosinophilic inflammation and bronchial hyperreactivity form the basis of the disease.

Table 7.4. *Clinical forms (phenotypes) of bronchial asthma*

Shape	Features
Allergic (atopic, extrinsic)	Begins in childhood, a history of allergic diseases (eczema, rhinitis), eosinophilia in the blood and sputum, raised IgE
Nonallergic	In adults, after infections, without an allergic history
Aspirin-induced (NSAID-sensitive) asthma	The "aspirin triad": choking attacks + aspirin intolerance + polyposis of the nose and paranasal sinuses
Exercise-induced asthma	Attacks appear with physical exertion (running), especially in cold air
Late-onset asthma	In adults, more often in women

Periods of an attack of bronchial asthma

1. **Prodromal period** — sneezing, watery nasal discharge, dry cough, a feeling of chest tightness.
2. **Period of the height of the attack** — choking: inspiration is short, expiration is long and difficult, wheezes are audible at a distance. The patient is in a forced position — sitting leaning forward with the hands propped on a table or the knees; the accessory muscles take part; the face is slightly puffy, there is cyanosis, and the neck veins are distended. The chest seems fixed in the position of maximal inspiration. On percussion, a hyperresonant (box-like) note, lowered lower

borders and markedly decreased mobility; on auscultation, diminished vesicular breathing with prolonged expiration and numerous dry whistling rales. The absolute cardiac dullness is reduced; tachycardia.

3. **Period of resolution** — viscous, glassy sputum is expectorated, and the choking gradually passes.

Laboratory signs

Eosinophilia in the blood; in the sputum, **eosinophils**, **Curschmann spirals** (casts of the mucus of the small bronchi) and **Charcot–Leyden crystals** (crystals formed from the breakdown of eosinophils).

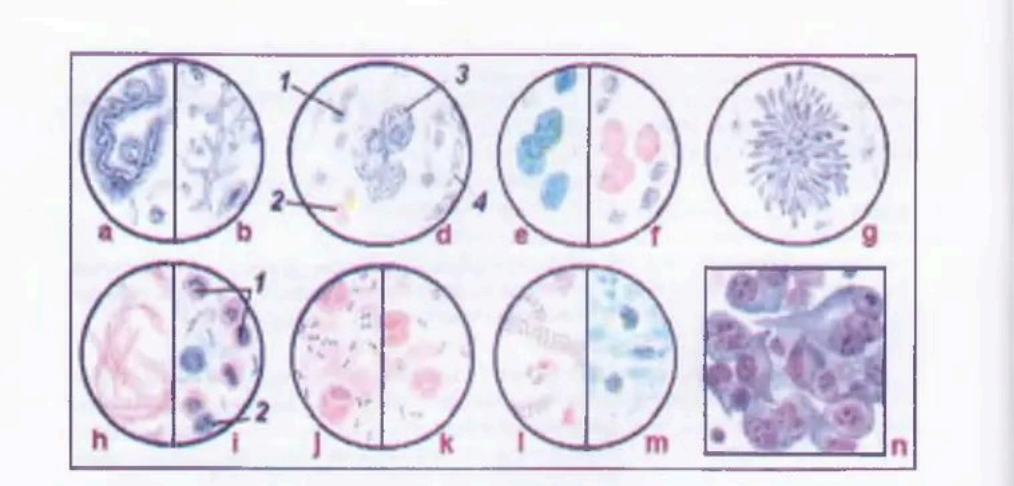


Figure 7.1. *Sputum microscopy: a — Curschmann spiral, Charcot–Leyden crystals and other elements (Gadayev A. et al., 2012, Fig. 59)*

MEMORY AID

For asthmatic sputum, **“E + C + C”**: **E**osinophils, **C**urschmann spirals, **C**harcot–Leyden crystals — “Eosinophil, Curschmann, Charcot”.

Status asthmaticus and severe exacerbation

The textbook divides status asthmaticus (prolonged severe choking that does not respond to the usual bronchodilators) into three stages: **I** — relative compensation (resistance to bronchodilators, hypocapnia); **II** — decompensation, the **“silent chest”**: the bronchi are plugged with viscous mucus, the previously heard rales disappear, hypoxemia and hypercapnia; **III** — hypercapnic coma.

CAUTION

GINA 2025 uses the concepts of a **severe and a life-threatening exacerbation** instead of the term “status asthmaticus”. Signs of a threat to life: impaired consciousness (drowsiness), a **“silent chest”**, cyanosis, bradycardia or hypotension, SpO_2 below 90 %, PEF less than 25–50 % of the predicted value. Such a patient is transferred urgently to the intensive care unit.

Functional criteria in the diagnosis of bronchial asthma (GINA 2025)

Table 7.5. *Tests confirming variable expiratory airflow*

Test	Criterion in adults
Bronchodilator test (salbutamol 200–400 µg)	An increase in FEV_1 of more than 12 % and more than 200 ml from baseline

Test	Criterion in adults
Diurnal variability of PEF (2 weeks, twice a day)	Average more than 10 %
After 4 weeks of anti-inflammatory treatment	An increase in FEV_1 of more than 12 % and 200 ml
Exercise challenge test	A fall in FEV_1 of more than 10 % and 200 ml
Bronchial provocation test (methacholine)	A fall in FEV_1 of 20 % (at a low dose)

6. The syndrome of increased air content of the lung tissue. Pulmonary emphysema

Pulmonary emphysema is a permanent enlargement of the air spaces distal to the terminal bronchioles with destruction of the alveolar walls. As a result, the lung tissue loses its elasticity, its air content increases and the gas-exchange surface decreases.

Causes and types:

- **Primary (diffuse, panacinar)** — congenital alpha-1 antitrypsin deficiency; in young people, in the lower parts of the lungs.
- **Secondary (centriacinar)** — chronic obstructive bronchitis, long-standing bronchial asthma, smoking; more often in the upper lobes.
- **Vicarious (compensatory)** — in the remaining part of the lung when part of it has been removed or is not functioning.
- **Senile emphysema** — age-related decrease in elasticity.
- **Bullous emphysema** — large air-filled blebs (bullae); if they rupture, pneumothorax.

Table 7.6. *Signs of the syndrome of increased air content of the lung tissue (emphysema)*

Method of examination	Signs
Complaints	Gradually increasing expiratory dyspnea (first on exertion, then at rest)
Inspection	Barrel chest, bulging supraclavicular fossae, horizontal ribs, obtuse epigastric angle, decreased respiratory excursion of the chest, use of the accessory muscles; exhalation through pursed lips (the “puffer”); cyanosis; clubbing may be present
Palpation	Increased resistance of the chest, vocal fremitus equally decreased on both sides
Percussion	Hyperresonant (box-like) note; increased height of the apices, widened Krönig’s isthmus, lowered lower borders, markedly decreased mobility of the lower edge; the absolute cardiac dullness reduced or undetectable, the liver dullness shifted downward
Auscultation	Vesicular breathing equally diminished on both sides; when combined with obstructive bronchitis — prolonged expiration, dry rales; muffled heart sounds, accentuation of S2 over the pulmonary artery
X-ray	Increased transparency of the lung fields, a low, flattened, poorly mobile diaphragm, horizontal ribs, a “hanging” (drop-shaped) heart, widened retrosternal space
Spirometry	FEV_1 and FEV_1/FVC decreased; total lung capacity and residual volume increased

CLINICAL SIGNIFICANCE

Two classic phenotypes of patients with COPD are distinguished: the **“pink puffer”** (emphysematous type) — thin, very breathless, exhales through pursed lips, little cyanosis; the **“blue bloater”** (bronchitic type) — heavier, coughs a lot and produces sputum, with diffuse cyanosis and early right ventricular failure (edema). This division is conventional; most patients have mixed features.

7. Examination of external respiratory function: spirometry and peak flowmetry

Spirometry is the measurement of lung volumes and airflow rates. The main parameters:

Table 7.7. *The main parameters of spirometry*

Parameter	Uzbek abbreviation	English	Meaning	Normal
Vital capacity	O‘HS	VC	Volume of air exhaled after a maximal inspiration	More than 80 % of predicted
Forced vital capacity	O‘FHS	FVC	Volume of air exhaled as fast and as completely as possible	More than 80 % of predicted
Forced expiratory volume in 1 second	NChHH1	FEV_1	Volume exhaled in the first second of forced expiration	More than 80 % of predicted
Tiffeneau index	NChHH1/O‘FHS	FEV_1/FVC	Index of obstruction	More than 0.70 (70 %)
Peak expiratory flow	NChEYuT	PEF	Maximal flow rate during forced expiration	Depends on age, sex and height

Table 7.8. *Types of ventilatory impairment*

Type	Causes	Spirometry
Obstructive	Bronchial obstruction: asthma, COPD	FEV_1/FVC less than 0.70, FEV_1 decreased, FVC normal or slightly decreased
Restrictive	Restriction of lung expansion: pneumosclerosis, fibrosis, pleural adhesions, kyphoscoliosis, a large amount of fluid, obesity	FVC (VC) decreased, FEV_1/FVC normal or increased
Mixed	Both mechanisms together	Both FVC and FEV_1/FVC decreased

Peak flowmetry is the measurement of the peak expiratory flow (PEF) with a special device (peak flow meter). Performed by the patient at home twice a day, it makes it possible to monitor the course of asthma and detect an exacerbation early.

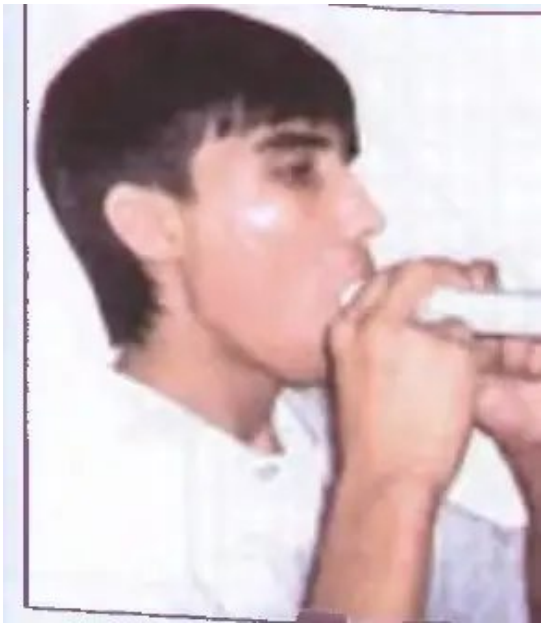


Figure 7.3. *Technique of peak flowmetry (same source, Fig. 56)*



Figure 7.2. *Peak flow meter (Gadayev A. et al., 2012, Fig. 54)*

MEMORY AID

The **"traffic light" system** of asthma control by PEF: **green zone** — PEF more than 80 % of the personal best (controlled); **yellow zone** — 50–80 % (caution, treatment needs to be stepped up); **red zone** — less than 50 % (a dangerous condition, urgent help).

8. Pathogenesis of bronchial asthma: a deeper analysis

Bronchial asthma is based on **chronic inflammation**, which persists even in the periods without symptoms. That is why the mainstay of treatment is not bronchodilators but **anti-inflammatory** agents (inhaled glucocorticosteroids).

Table 7.9. *The main participants in inflammation in asthma*

Cell or substance	Role
Mast cells	Activated via IgE, they release histamine, leukotrienes and prostaglandin D_2 — immediate bronchospasm
Eosinophils	Major basic protein and cytotoxic proteins — damage to the epithelium; eosinophilia in the blood and sputum
T helper 2 lymphocytes	IL-4 (IgE synthesis), IL-5 (eosinophils), IL-13 (mucus hypersecretion, hyperreactivity)
Leukotrienes	Powerful bronchoconstrictors (1000 times stronger than histamine), mucus secretion and edema
Bronchial smooth muscle	Hyperreactivity, hypertrophy

Airway remodeling — irreversible changes in the bronchial wall in long-untreated asthma: thickening of the basement membrane, smooth muscle hypertrophy, hyperplasia of the mucous glands, subepithelial fibrosis. As a result, the obstruction becomes partly irreversible.

Early and late asthmatic reactions: 10–20 minutes after exposure to the allergen comes the **early reaction** (bronchospasm lasting 1–2 hours), and after 4–8 hours the **late reaction** (eosinophilic inflammation, long-lasting, increasing hyperreactivity).

8.1. Assessment of asthma control (GINA 2025)

Table 7.10. *Control of asthma symptoms over the last 4 weeks*

Question	Yes / No
Daytime symptoms more than twice a week?	
Any night waking due to asthma?	
Reliever medication (short-acting bronchodilator) needed more than twice a week?	
Any activity limitation due to asthma?	

REMEMBER
Number of "yes" answers: **0 — well controlled**, **1–2 — partly controlled**, **3–4 — uncontrolled asthma**. Together with control, future risk factors (exacerbations, low FEV_1, smoking, eosinophilia) are also assessed.

8.2. Differences between bronchial asthma and cardiac asthma

Table 7.11. *Comparison of bronchial and cardiac asthma*

Feature	Bronchial asthma	Cardiac asthma
Age and history	Often young, allergy	Often elderly; IHD, hypertension, valvular defects, myocardial infarction
Type of dyspnea	Expiratory	Inspiratory or mixed
Condition	Sitting leaning forward, propped on the hands	Sitting with the legs lowered (orthopnea)
Sputum	Scanty, glassy, viscous (at the end of the attack)	Pink, frothy (in pulmonary edema)
Percussion	Hyperresonant (box-like) note	Impaired note in the lower parts
Auscultation	Dry whistling rales, prolonged expiration	Moist rales in the lower parts (sometimes also the "wheezes of cardiac asthma")
Heart	Tachycardia, accentuation of S2 over the pulmonary artery	Enlarged heart, gallop rhythm, arrhythmias
Effect of bronchodilators	Good effect	Little effect; diuretics and nitrates are effective

9. COPD: clinical forms, exacerbation and complications

An exacerbation of COPD is a worsening of dyspnea and/or cough with increased sputum within 14 days; its causes are most often viral and bacterial infections and air pollution. The **Anthonisen criteria** of exacerbation: increased dyspnea, increased sputum volume and purulent sputum; if all three signs are present, an antibiotic is indicated.

Complications of COPD:

- **Chronic respiratory failure** — hypoxemia, secondary erythrocytosis;
- **Pulmonary hypertension and chronic cor pulmonale** — alveolar hypoxia causes spasm of the pulmonary arterioles (the Euler–Liljestrand reflex); the right ventricle hypertrophies, and failure develops later;
- spontaneous pneumothorax (rupture of bullae), pneumonias, an increased risk of lung cancer.

Table 7.12. *Signs of chronic cor pulmonale*

Method of examination	Signs
Inspection	Diffuse warm cyanosis, distended neck veins, leg edema, epigastric pulsation
Palpation	Cardiac impulse, enlarged and tender liver
Auscultation	Accentuation of S2 over the pulmonary artery, the systolic murmur of tricuspid regurgitation at the base of the xiphoid process
ECG	P pulmonale (tall, peaked P in II, III and aVF, more than 2.5 mm), right ventricular hypertrophy, right axis deviation
Echocardiography	The right ventricle dilated and hypertrophied, the pulmonary artery pressure raised

10. Pathogenesis of pulmonary emphysema

The main theory of the development of emphysema is **an imbalance between proteases and antiproteases**. Under the influence of smoking and other irritants, neutrophils and macrophages accumulate in the lung and release **elastase** and other proteases, which destroy the elastic fibers in the alveolar walls. Normally this process is stopped by **alpha$_1$-antitrypsin**. In its congenital deficiency or in smoking (the smoke inactivates the antiproteases), the alveolar walls are destroyed, and the alveoli merge into large air spaces.

As a result: the elastic recoil of the lung decreases → the small bronchi collapse during expiration (**expiratory collapse**, "air trapping") → the residual volume and total lung capacity increase (**hyperinflation**) → the diaphragm flattens and works inefficiently → dyspnea increases; the gas-exchange surface and the capillaries decrease → diffusion is impaired.

> **CLINICAL SIGNIFICANCE**
> **Pursed-lip breathing** ("puffing") is a useful technique that emphysema patients discover themselves: the resistance created at the mouth raises the pressure in the airways and prevents early collapse of the small bronchi. This technique is also taught in breathing exercises.

11. Instrumental methods of examining the respiratory organs

11.1. Radiological examination



Figure 7.4. *Normal radiograph of the lungs (Gadayev A. et al., 2012, Fig. 46)*



Figure 7.5. *Pathological changes on a lung radiograph (same source, Fig. 46)*

Chest radiography is performed in two views (frontal and lateral). The following are assessed: the transparency of the lung fields, the lung pattern, the hila, the position and sharpness of the outline of the diaphragm, the costophrenic sinuses, the mediastinum and the cardiac shadow. Common syndromes: **opacity** (pneumonia, atelectasis, fluid, tumor), **lucency** (cavity, pneumothorax, emphysema, bulla), **accentuation of the lung pattern** (bronchitis, congestion, interstitial diseases), **widening of the hila** (lymphadenopathy, pulmonary hypertension).

Computed tomography (CT) shows the lung tissue, bronchi and mediastinum layer by layer with high resolution; it is the main method for detecting bronchiectasis, interstitial diseases, small tumors and pulmonary embolism (CT angiography).

11.2. Bronchoscopy



Figure 7.6. *Fiberoptic bronchoscope (Gadayev A. et al., 2012, Fig. 49)*

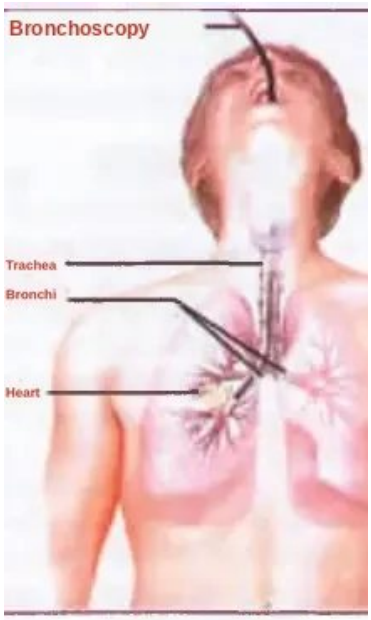


Figure 7.7. *Diagram of bronchoscopy (same source, Fig. 50)*

Bronchoscopy is the inspection of the trachea and bronchi with a flexible fiberoptic bronchoscope. Diagnostic purposes: detecting tumors and taking biopsies, finding the source of bleeding, detecting a foreign body, obtaining bronchoalveolar lavage. Therapeutic purposes: removing a foreign body and viscous sputum, sanitation of the bronchi (bronchiectasis, abscess), stopping bleeding.

11.3. Spirometry and peak flowmetry: practical aspects



Figure 7.9. *Computerized spirometer (same source, Fig. 52)*



Figure 7.8. *Spirograph (Gadayev A. et al., 2012, Fig. 51)*

Rules for performing spirometry: short-acting bronchodilators are withheld for 4–6 hours and long-acting ones for 12–24 hours before the test; no smoking for 1 hour beforehand; the patient sits, with a nose clip; after a maximal deep inspiration, he or she breathes out through the mouthpiece as forcefully as possible for at least 6 seconds; at least 3 satisfactory attempts are made. **Bronchodilator test:** the test is repeated 15–20 minutes after inhalation of salbutamol 400 µg.



Figure 7.10. *Peak flow meter scale (Gadayev A. et al., 2012, Fig. 55)*

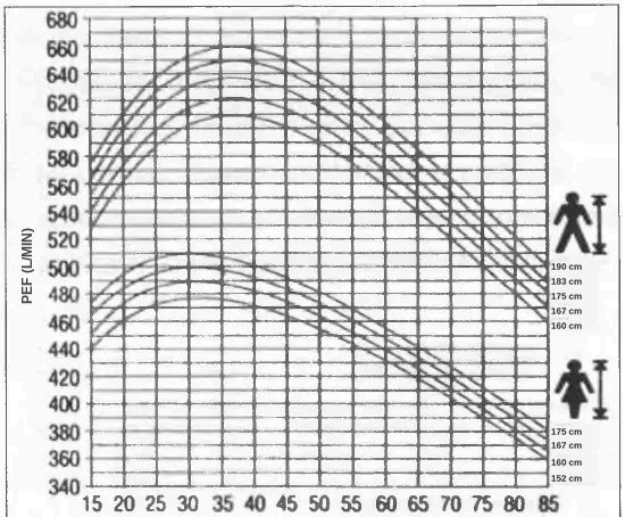


Figure 7.11. *Predicted values of PEF by age, sex and height (same source)*

REMEMBER

The **diurnal variability** of PEF = (highest PEF during the day − lowest PEF) / their mean × 100 %. If it averages **more than 10 %** over two weeks (13 % in children), it confirms the diagnosis of bronchial asthma.

11.4. Blood gases and other investigations

Arterial blood gases: PaO_2 (normal 80–100 mm Hg), $PaCO_2$ (35–45), pH (7.35–7.45). In hypoxemic (type I) respiratory failure, PaO_2 is decreased and $PaCO_2$ normal or low; in hypercapnic (type II)

respiratory failure, $PaCO_2$ is above 45 (severe COPD, a severe asthma attack, respiratory muscle fatigue).

Allergological investigations: skin tests (prick test), total and specific IgE in the blood, the blood eosinophil count; **FeNO** — fractional exhaled nitric oxide (an indicator of eosinophilic inflammation).

12. Principles of treating bronchial asthma and COPD (in brief)

Knowing the principles of treatment consolidates the student's understanding of the pathogenesis of the disease.

Table 7.13. *The main groups of drugs in asthma and COPD*

Drug group	Mechanism of action	Examples	Application
Inhaled glucocorticosteroids (ICS)	Suppress inflammation	Budesonide, fluticasone, beclomethasone	The mainstay of treatment in asthma; in COPD with eosinophilia and frequent exacerbations
Short-acting beta2-agonists	Rapidly dilate the bronchi (in 3–5 minutes)	Salbutamol (albuterol), fenoterol	Relief of symptoms
Long-acting beta2-agonists	Bronchodilator effect for 12–24 hours	Formoterol, salmeterol, indacaterol	Together with ICS (asthma); first-line in COPD
Long-acting anticholinergics	Block vagal bronchoconstriction	Tiotropium, glycopyrronium	First-line in COPD, add-on in severe asthma
Leukotriene receptor antagonists	Block the action of leukotrienes	Montelukast	Aspirin-induced and exercise-induced asthma, asthma with allergic rhinitis
Biological agents	Monoclonal antibodies against IgE, IL-5, IL-4/13	Omalizumab, mepolizumab, dupilumab	Severe eosinophilic or allergic asthma

REMEMBER

The main principle of GINA 2025: in adults and adolescents, treating asthma **with short-acting beta$_2$-agonists alone is not recommended**; every patient receives ICS-containing treatment (the preferred track is low-dose budesonide-formoterol "as needed" and as maintenance treatment). Treatment with salbutamol alone increases the risk of severe exacerbations and death.

12.1. Inhaler technique

The most common cause of treatment failure is **incorrect use** of the inhaler. For a metered-dose aerosol inhaler: shake → breathe out fully → close the lips around the mouthpiece → press the canister at the same time as starting a slow deep inspiration → hold the breath for 10 seconds → rinse the mouth after ICS (to prevent candidiasis). A **spacer** is recommended for children and the elderly.

13. Differential diagnosis of wheezing

"Not all wheezing is asthma" — this clinical rule is important.

Table 7.14. *Causes of wheezing*

Cause	Distinguishing signs
Bronchial asthma	Variability, allergy, good response to a bronchodilator
COPD	Smoking, persistent symptoms, not fully reversible obstruction
Left ventricular failure (“cardiac asthma”)	Heart disease, orthopnea, moist rales, raised NT-proBNP
Foreign body or bronchial tumor	Monophonic, in one place, unilateral
Narrowing of the larynx and trachea	Stridor (mainly during inspiration), change in voice
Paradoxical vocal fold movement	Young women, mainly during inspiration, heard over the larynx
Pulmonary embolism	Sudden dyspnea, tachycardia, leg thrombosis
Anaphylaxis	Rash, hypotension, contact with an allergen

14. When bronchiectasis, bronchial asthma and COPD coexist

In practice, diseases often coexist:

- **Asthma–COPD overlap** — in an asthma patient who smokes, or in a COPD patient showing eosinophilia and high reversibility; according to the GINA and GOLD recommendations, such patients must be given ICS.
- **COPD and bronchiectasis** — detected on CT; runs a course with frequent bacterial exacerbations and much purulent sputum.
- **COPD and heart failure** — both cause dyspnea; echocardiography, NT-proBNP and spirometry are used together for the diagnosis.

15. Complications of pulmonary emphysema and bullae

Bullae are thin-walled air spaces more than 1 cm in diameter. Large bullae (more than a third of the lung field) compress the healthy tissue (the “vanishing lung” syndrome). Complications: **spontaneous pneumothorax** (rupture of a bulla — sudden severe dyspnea in a COPD patient) and infection. On physical examination, a tympanic note and markedly diminished breath sounds over a large bulla resemble pneumothorax; they are distinguished by CT.

16. Clinical part

Comparison of bronchial asthma and COPD

Table 7.15. *Differences between bronchial asthma and COPD*

Feature	Bronchial asthma	COPD
Age at onset	Most often childhood and youth	After the age of 40
Main risk factor	Atopy, allergens, heredity	Smoking (more than 10 pack-years), occupational dusts
Symptoms	Paroxysmal, at night and early in the morning, variable	Persistent, gradually increasing
Dyspnea	Absent or slight between attacks	Persistent, worse on exertion
Concomitant allergic diseases	Many (rhinitis, eczema)	Few

Feature	Bronchial asthma	COPD
Reversibility of obstruction	Well reversible (FEV_1 increases by more than 12 % and 200 ml)	Not fully reversible
Eosinophilia in the blood and sputum	Characteristic	Uncommon

Sample case history entry

The patient is in a forced position — sitting with the hands propped on the edge of the bed. Respiratory rate 26 per minute, expiration markedly prolonged, wheezes heard at a distance, the accessory muscles take part in breathing, diffuse cyanosis. The chest is emphysematous. Vocal fremitus is symmetrically decreased. On percussion, a hyperresonant (box-like) note; the lower borders of the lungs are lowered by one rib, mobility 2 cm. On auscultation, diminished vesicular breathing with prolonged expiration and numerous dry whistling rales over the whole lung. PEF 180 l/min (40 % of the predicted value).

Conclusion: the syndromes of bronchial obstruction and increased air content of the lung tissue — characteristic of a severe attack of bronchial asthma.

17. Practical skills

Skill: performing peak flowmetry

1. Set the pointer of the peak flow meter to the beginning of the scale.
2. The patient stands or sits upright and holds the peak flow meter horizontally, with the fingers not touching the scale.
3. The patient takes a deep breath and closes the lips tightly around the mouthpiece.
4. He or she breathes out (blows) as hard and as fast as possible.
5. Record the result and return the pointer to the beginning.
6. Repeat the test 3 times and record the highest reading.
7. Compare the result with the predicted or personal best value and determine the zone.

Skill: identifying the syndrome of bronchial obstruction by physical examination

1. Establish the complaints: type of dyspnea (expiratory), attacks, character of the sputum.
2. Inspect: position, respiratory rate and ratio of the respiratory phases, accessory muscles, cyanosis, shape of the chest.
3. Palpation: resistance and vocal fremitus.
4. Percussion: comparative (hyperresonant note) and topographic (borders, mobility).
5. Auscultation: prolonged expiration, dry rales.
6. Determine the forced expiratory time.
7. Draw a syndromic conclusion and plan spirometry and peak flowmetry.

Review questions

1. List the reversible and irreversible mechanisms of bronchial obstruction.
2. What is expiratory collapse of the small bronchi, and when does it develop?
3. What are the physical signs of the syndrome of bronchial obstruction?
4. What are the causes and clinical signs of acute bronchitis?

5. How can acute bronchitis be distinguished from pneumonia?
6. What is the WHO criterion of chronic bronchitis?
7. On what criterion is the diagnosis of COPD based (GOLD 2026)? What are the grades of obstruction?
8. Give a modern definition of bronchial asthma.
9. Describe the periods of an attack of bronchial asthma and the physical signs at its height.
10. What elements are found in the sputum of an asthma patient?
11. What is the “aspirin triad”?
12. What is the “silent chest”, and what does it indicate?
13. Which functional tests are used in the diagnosis of bronchial asthma?
14. What are the types and physical signs of pulmonary emphysema?
15. How do obstructive and restrictive ventilatory impairments differ on spirometry?

Test questions

1. Which of the following belongs to the irreversible component of bronchial obstruction?
   A) Infiltration of the mucosa
   B) Increased viscous secretion
   C) Expiratory collapse of the small bronchi
   D) Mucosal edema
   E) Bronchospasm

2. For how long must cough be observed for the diagnosis of chronic bronchitis by the WHO criterion?
   A) For 1 month
   B) At least 3 months a year for at least 2 consecutive years
   C) 1 month a year for 5 years
   D) 2 weeks
   E) 6 months a year for 1 year

3. What is the spirometric criterion for the diagnosis of COPD according to GOLD 2026?
   A) FVC less than 80 %
   B) Post-bronchodilator FEV_1/FVC less than 0.70
   C) PEF less than 50 %
   D) An increase in FEV_1 of 12 %
   E) Decreased total lung capacity

4. Which elements are found in the sputum in bronchial asthma?
   A) Elastic fibers and Dittrich plugs
   B) Mycobacterium tuberculosis
   C) Atypical cells
   D) Hemosiderophages
   E) Curschmann spirals and Charcot–Leyden crystals

5. According to GINA 2025, the bronchodilator test is considered positive if FEV_1:
   A) Increases by 5 %
   B) Increases by more than 12 % and more than 200 ml
   C) Decreases by 20 %
   D) Does not change
   E) Increases by 100 ml

6. What does the “aspirin triad” include?
   A) Choking attacks, aspirin intolerance, nasal polyposis
   B) Fever, cough, dyspnea
   C) Eosinophilia, Curschmann spirals, Charcot–Leyden crystals
   D) Cyanosis, tachycardia, hypotension
   E) Eczema, rhinitis, conjunctivitis

7. What is the disappearance of the previously heard rales in a severe asthma attack called?
   A) A “silent chest”
   B) Amphoric breathing
   C) Stridor
   D) Crepitation
   E) Bronchophony

8. What note is found on percussion in pulmonary emphysema?
   A) Metallic
   B) Thigh note
   C) Dull-tympanic
   D) Dull
   E) Hyperresonant (box-like)

9. How does the absolute cardiac dullness change in pulmonary emphysema?
   A) It enlarges
   B) It shifts to the left
   C) Unchanged
   D) It decreases or cannot be determined
   E) It shifts to the right

10. What is the cause of primary (panacinar) emphysema?
    A) Tuberculosis
    B) Congenital alpha-1 antitrypsin deficiency
    C) Bronchial asthma
    D) Smoking
    E) Pneumonia

11. Which change is characteristic of restrictive ventilatory impairment?
    A) FEV_1/FVC less than 0.70
    B) FVC decreased, FEV_1/FVC normal or increased
    C) A diurnal variability of PEF of 20 %
    D) An increase in FEV_1 after a bronchodilator
    E) An increase in the residual volume of the lungs

12. What is heard on auscultation in acute bronchitis?
    A) Harsh breathing, dry and nonconsonating moist rales
    B) Bronchial breathing
    C) Pleural friction rub
    D) Crepitation
    E) Amphoric breathing

13. If PEF is 40 % of the personal best, which zone of the “traffic light” system is it?
    A) Yellow
    B) Red
    C) The zone cannot be determined
    D) Blue
    E) Green

14. What time on the forced expiratory test indicates bronchial obstruction?
    A) Less than 1 second
    B) 2 seconds
    C) Less than 4 seconds
    D) More than 6 seconds
    E) 3 seconds

15. Which sign is characteristic of the “pink puffer” phenotype?
    A) Obesity
    B) Much sputum and early right ventricular failure
    C) Diffuse cyanosis and edema
    D) Fever
    E) Thinness, severe dyspnea, exhalation through pursed lips

Clinical cases

Case 1

A 22-year-old female student complains of choking attacks at night in spring, when the apple trees are in blossom. She has had eczema since childhood, and her mother has pollinosis. During an attack the patient sits with her hands propped on the table, expiration is prolonged, and wheezes are heard at a distance. At the end of the attack a small amount of glassy, viscous sputum was expectorated. Blood eosinophils 12 %.

Questions: Which disease and which form of it can be considered? What is expected to be found in the sputum? Which functional test confirms the diagnosis?

Answer: Allergic (atopic) bronchial asthma — the seasonality, the atopic history (eczema, pollinosis in the family), the attacks of expiratory choking and the eosinophilia are characteristic of it. Eosinophils, Curschmann spirals and Charcot–Leyden crystals are expected in the sputum. The diagnosis is confirmed on spirometry by the bronchodilator test (an increase in FEV_1 of more than 12 % and 200 ml) or by a diurnal variability of PEF of more than 10 %.

Case 2

A 64-year-old man has smoked a pack of cigarettes a day for 45 years. For 20 years he has been bothered by a morning productive cough and, over the last 5 years, by increasing dyspnea. On spirometry, the post-bronchodilator FEV_1/FVC is 0.58 and FEV_1 is 42 % of the predicted value.

Questions: Calculate the smoking index. What is the diagnosis? Assess the degree of obstruction according to GOLD. What signs are expected on physical examination?

Answer: Smoking index 20 × 45 / 20 = 45 pack-years. FEV_1/FVC less than 0.70 — the diagnosis of COPD is confirmed; FEV_1 42 % — GOLD 3 (severe obstruction). On physical examination: an emphysematous chest, symmetrically decreased vocal fremitus, a hyperresonant (box-like) note, lowered lower borders and decreased mobility, diminished vesicular breathing with prolonged expiration, dry rales.

Case 3

A 35-year-old woman with bronchial asthma complains of choking that has lasted for 18 hours; salbutamol inhalations are not helping. The patient is drowsy, cannot speak, and has diffuse cyanosis. On auscultation, breath sounds are markedly diminished, and the previously heard rales are almost inaudible. SpO_2 84 %.

Questions: Assess the condition. Is the disappearance of the rales a good or a bad sign?

Answer: This is a life-threatening severe asthma exacerbation (stage II of status asthmaticus in the old terminology). The disappearance of the rales — the “silent chest” — is not an improvement but a very dangerous sign showing that the bronchi are completely plugged with viscous mucus and ventilation has fallen sharply. Together with the drowsiness, cyanosis and an SpO_2 below 90 %, it means the patient must be transferred immediately to the intensive care unit.

Quick review

Table 7.16. *Key facts on Topic 7*

Question	Answer
Mechanisms of obstruction	Spasm, edema, secretions (reversible); fibrosis, expiratory collapse (irreversible)
Signs of obstruction	Expiratory dyspnea, prolonged expiration, wheezes
Chronic bronchitis (WHO)	Productive cough for 3 months a year for 2 consecutive years
COPD (GOLD 2026)	FEV_1/FVC less than 0.70 (post-bronchodilator)
GOLD grades	1: 80 % or more; 2: 50–79 %; 3: 30–49 %; 4: less than 30 %
Asthmatic sputum	Eosinophils, Curschmann spirals, Charcot–Leyden crystals
Bronchodilator test	An increase in FEV_1 of more than 12 % and 200 ml
PEF variability	More than 10 % — asthma
A “silent chest”	Sign of a life-threatening exacerbation
Emphysema	Barrel chest, hyperresonant note, diminished breath sounds, reduced absolute cardiac dullness
Restrictive impairment	FVC decreased, FEV_1/FVC normal

TOPIC 8

The syndrome of consolidation of the lung tissue. Lobar and focal pneumonia. The syndromes of fluid and air accumulation in the pleural cavity. Hydrothorax. Pneumothorax. Dry and exudative pleurisy

Learning objectives

To teach the student the pathogenesis and physical signs of the syndromes of consolidation of the lung tissue and of fluid and air accumulation in the pleural cavity, the symptomatology of lobar and focal pneumonia, dry and exudative pleurisy, hydrothorax and pneumothorax, and how to examine and interpret pleural fluid.

By the end of the session, the student should know:

- the causes and physical signs of the syndrome of consolidation of the lung tissue;
- the stages of lobar pneumonia and the physical signs at each stage;
- the differences between lobar and focal pneumonia, the CURB-65 score;
- the syndrome of fluid accumulation in the pleural cavity, the Damoiseau line, and Garland's and Rauchfuss–Grocco triangles;
- the differences between a transudate and an exudate (Rivalta test, Light's criteria);
- the clinical signs of dry and exudative pleurisy;
- the types of pneumothorax and the syndrome of air accumulation in the pleural cavity.

The student should be able to:

- identify and distinguish the syndromes of consolidation and of fluid and air accumulation by physical examination;
- interpret the analysis of pleural fluid;
- explain the steps of pleural puncture (demonstrate them on a manikin).

1. The syndrome of consolidation of the lung tissue

Consolidation of the lung tissue is the loss of air from the lung as a result of the filling of the alveoli with exudate (inflammation), transudate, blood, tumor or connective tissue, or their collapse.

Causes: pneumonia (lobar — by lobe, focal — small foci), pulmonary infarction, tuberculous infiltrate, lung tumor, compression atelectasis, pneumosclerosis, pulmonary edema.

Table 8.1. *Signs of the syndrome of lobar consolidation of the lung tissue (bronchial patency preserved)*

Method of examination	Signs
Complaints	Cough, sputum, dyspnea; if the pleura is involved — stabbing pain
Inspection	The affected side lags behind in breathing, tachypnea
Palpation	Vocal fremitus increased over the consolidated area
Percussion	Dull or impaired note, decreased mobility of the lower edge
Auscultation	Bronchial breathing, consonating moist rales, crepitation; bronchophony increased
X-ray	Homogeneous opacity (shadow) within the borders of a segment or lobe

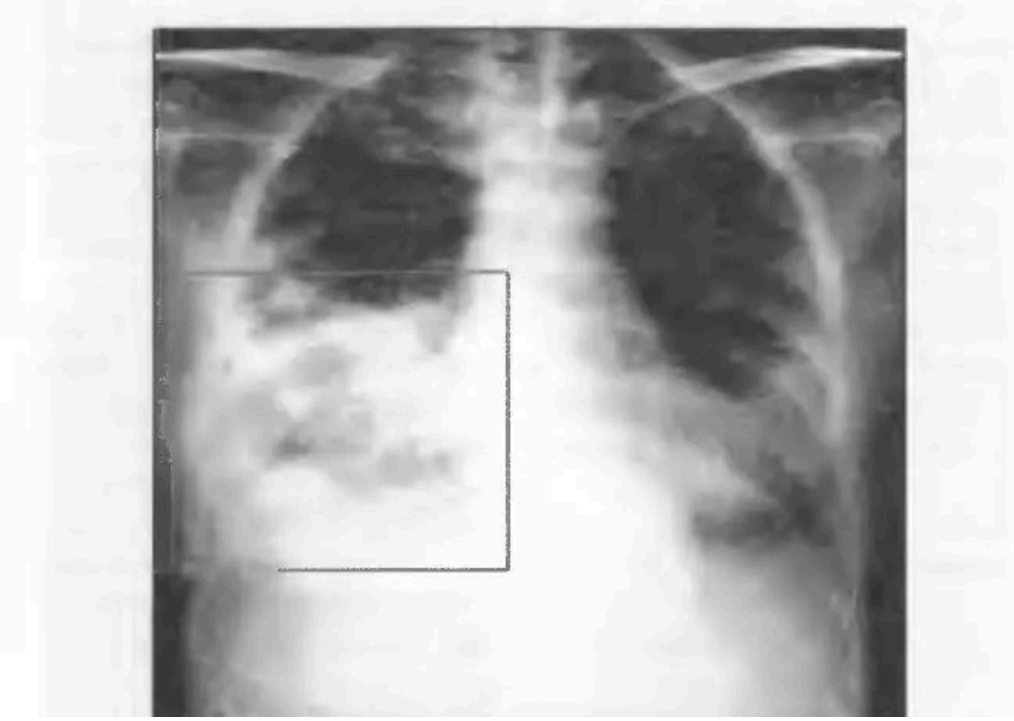


Figure 8.1. *Focal consolidation of the lung tissue: an infiltrate in the lower part of the right lung (Gadayev A. et al., 2012, Fig. 60)*

CAUTION

If the bronchus of the consolidated area is blocked by a tumor or a foreign body (**obstructive atelectasis**), then even though the percussion note is dull, vocal fremitus, bronchophony and breath sounds are **decreased or absent**, bronchial breathing is not heard, and the mediastinum shifts to the affected side. It is very important to distinguish this condition from pneumonia (especially in elderly smokers — lung cancer).

2. Pneumonias: general information

Pneumonia is an acute infectious (mainly bacterial) inflammation of the respiratory parts of the lung, accompanied by exudation into the alveoli and confirmed radiologically.

Table 8.2. *Modern classification of pneumonias*

Type	Definition	Main pathogens
Community-acquired	Developed outside hospital or within 48 hours of admission	Pneumococcus (the most common), Haemophilus influenzae, Mycoplasma, Chlamydia, Legionella, viruses
Hospital-acquired (nosocomial)	Developed 48 hours or more after admission	Pseudomonas aeruginosa, Klebsiella, Staphylococcus (MRSA), Enterobacteriaceae
Aspiration	After aspiration of oropharyngeal or gastric contents	Anaerobes, gram-negative bacteria
In patients with severely reduced immunity	HIV, chemotherapy, transplantation	Pneumocystis, fungi, cytomegalovirus

Classic propaedeutics textbooks distinguish, on morphological and clinical grounds, **lobar** and **focal (bronchopneumonia)** pneumonia.

3. Lobar pneumonia

Lobar pneumonia (pleuropneumonia) is an acute inflammation involving one or several lobes (segments) of the lung and accompanied by inflammation of the pleura. Its causative agent is most often **pneumococcus** (Streptococcus pneumoniae). Fibrinous exudate enters the alveoli.

3.1. Pathological stages and physical signs

Table 8.3. *Physical signs in the stages of lobar pneumonia*

Stage	Morphology	Vocal fremitus	Percussion	Auscultation
I. Congestion (hyperemia, “engorgement”), 1–2 days	The capillaries are engorged with blood, there is little exudate in the alveoli, air is preserved	Slightly increased	Dull-tympanic	Diminished vesicular breathing, initial crepitation (crepitatio indux)
II. Red and gray hepatization, 3–7 days	The alveoli are filled with fibrin and red cells, then leukocytes; airless	Markedly increased	Dull	Bronchial breathing, increased bronchophony, pleural friction rub
III. Resolution (recovery)	The exudate liquefies and is resorbed; air returns to the alveoli	Returns to normal	Dull-tympanic, then a clear lung note	Diminished vesicular breathing, final crepitation (crepitatio redux), consonating moist rales

MEMORY AID
The stages of lobar pneumonia: **“Crepitation — Bronchial — Crepitation”** (indux — hepatization — redux), and on percussion **“Dull-tympanic — Dull — Dull-tympanic”**.

3.2. Clinical picture

- **Sudden onset**: severe chills (shivering), the body temperature rises to 39–40 °C within a few hours.
- **Stabbing pain** on the affected side, worse on deep inspiration and coughing (inflammation of the parietal pleura). In lower lobe pneumonia the pain may radiate to the abdomen and resemble an “acute abdomen”.
- **Cough**: dry at first, with viscous **“rusty”** sputum from the 2nd–3rd day.
- **Dyspnea**, tachycardia, sometimes severe intoxication and delirium.

On inspection: a febrile face, flushing of the cheek on the affected side, **herpes** eruptions on the lips and nasal alae, movement of the nasal alae with breathing, cyanosis; the affected side lags behind in breathing.

The fever is of the continuous type (febris continua); untreated, it lasts 7–10 days and falls by **crisis** (within a few hours) or **lysis**. During a crisis there is a risk of collapse. In the present era of antibiotics, the fever usually falls by lysis within 2–3 days.

Laboratory and instrumental signs: neutrophilic leukocytosis in the blood ($15–30 \cdot 10^9$/l), a shift of the differential to the left, toxic granulation of the neutrophils, a raised ESR, raised C-reactive protein and procalcitonin; pneumococci, fibrin and red cells in the sputum; a homogeneous opacity of a lobe or segment on X-ray; febrile proteinuria in the urine.

Complications: exudative pleurisy (parapneumonic), lung abscess, infectious-toxic shock, acute respiratory failure, myocarditis, meningitis, pericarditis.

4. Focal pneumonia (bronchopneumonia)

In focal pneumonia the inflammation begins in a bronchus and involves one or several lobules or segments; the pleura is rarely involved; the exudate is serous or mucopurulent.

Table 8.4. *Differences between lobar and focal pneumonia*

Feature	Lobar pneumonia	Focal pneumonia
Onset	Sudden, with chills	Gradual, often after an ARVI or bronchitis
Fever	39–40 °C, continuous	38–39 °C, remittent or irregular
Pleuritic pain	Always present	Uncommon
Sputum	“Rusty”, viscous	Mucopurulent
Herpes	Common	Few
Extent of involvement	A lobe or segment	A lobule, small foci
Vocal fremitus	Markedly increased	Unchanged or slightly increased
Percussion	Dull (stage II)	Impaired, often unchanged
Auscultation	Bronchial breathing, crepitation	Harsh or mixed breathing, consonating fine-bubbling moist rales
Course	In stages, crisis or lysis	Without stages, lysis

Assessing the severity of pneumonia: the CURB-65 score

Table 8.5. *The CURB-65 score*

Feature	Criterion	Score
C — Confusion	Newly arisen impairment of consciousness	1
U — Urea	Blood urea above 7 mmol/l	1
R — Respiratory rate	Respiratory rate 30 per minute or more	1
B — Blood pressure	Systolic BP below 90 or diastolic BP 60 mm Hg or below	1
65	Age 65 or older	1

REMEMBER

By CURB-65, **0–1 point** — outpatient treatment is possible; **2 points** — hospital admission is recommended; **3–5 points** — severe pneumonia, intensive care is likely. Where urea cannot be measured, the **CRB-65** variant is used.

5. The syndrome of fluid accumulation in the pleural cavity

The accumulation of fluid in the pleural cavity may be **noninflammatory** (transudate — hydrothorax) or **inflammatory** (exudate — exudative pleurisy); blood (hemothorax), lymph (chylothorax) and pus (empyema, pyothorax) may also accumulate.

Table 8.6. *Signs of the syndrome of fluid accumulation in the pleural cavity*

Method of examination	Signs
Complaints	Dyspnea, a feeling of heaviness in the chest; with an exudate — fever and, at the onset, stabbing pain (the pain lessens once fluid has accumulated)
Inspection	Forced position — lying on the affected side; the affected side is enlarged, the intercostal spaces are smoothed out or bulge, and it lags behind in breathing
Palpation	Vocal fremitus markedly decreased or absent over the fluid; resistance increased
Percussion	A dull (thigh) note over the fluid; an oblique upper border — the Damoiseau line; no mobility of the lower edge
Auscultation	Breath sounds markedly diminished or absent over the fluid; soft bronchial breathing at the upper border (compressed lung); no bronchophony
Mediastinum	With a large amount of fluid (more than 1.5–2 l) it shifts to the healthy side
X-ray	A homogeneous opacity in the lower part with an oblique upper border; the costophrenic sinus is filled

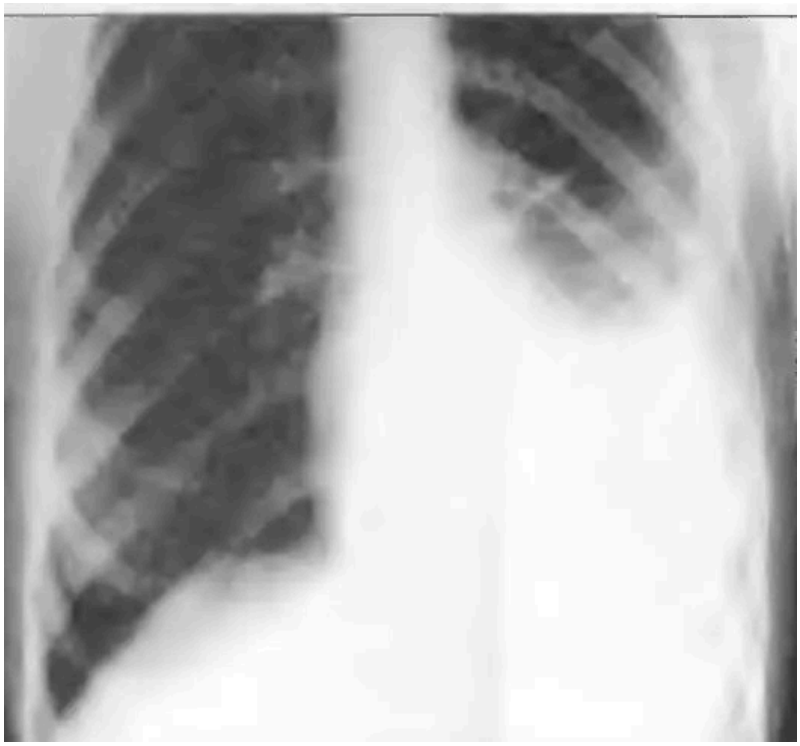


Figure 8.2. *Fluid accumulation in the pleural cavity: a homogeneous opacity on the left (Gadayev A. et al., 2012, Fig. 62)*

5.1. Percussion borders of an exudate

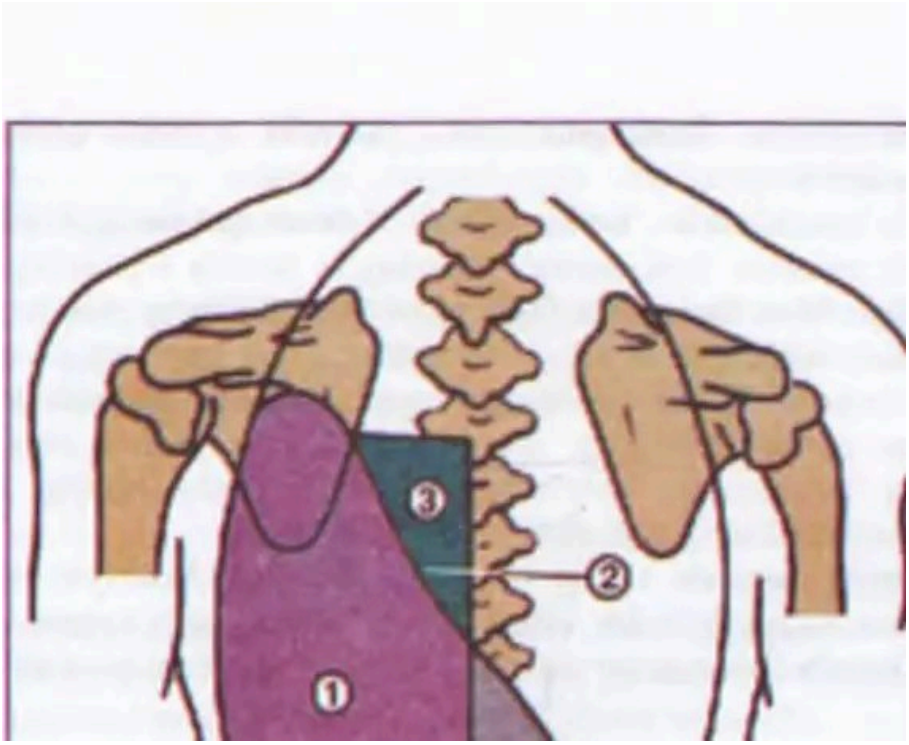


Figure 8.3. *Percussion and auscultation in left-sided exudative pleurisy: 1 — exudate; 2 — the Ellis–Damoiseau–Sokolov line; 3 — Garland's triangle; 4 — the Grocco–Rauchfuss triangle (Gadayev A. et al., 2012, Fig. 64)*

- **The Ellis–Damoiseau–Sokolov line** is the oblique upper border of an exudate: it starts at the spine, rises upward and laterally, reaches its highest point at the posterior axillary line and then descends forward. This shape arises because the fluid compresses the lung obliquely.
- **Garland's triangle** is the area on the affected side between the Damoiseau line, the spine and a straight line dropped from the highest point of the line to the spine. Because the compressed lung lies here, percussion reveals a **dull-tympanic** note and auscultation **soft bronchial breathing**.
- **The Grocco–Rauchfuss triangle** is a triangle of dullness on the **healthy** side, next to the spine. It is formed by the shift of the mediastinal organs to the healthy side; vocal fremitus is decreased here.

CAUTION

In a **transudate** (hydrothorax), the fluid moves freely and its upper border is usually **horizontal**; the border also shifts when the patient changes position. In an **exudate**, because of adhesion of the pleural layers and fibrin, the border is **oblique** (the Damoiseau line) and depends little on position. This difference is relative; they are distinguished precisely by pleural puncture and fluid analysis.

5.2. Distinguishing a transudate from an exudate

Table 8.7. *Differences between a transudate and an exudate*

Parameter	Transudate	Exudate
Mechanism of formation	Increased hydrostatic pressure or decreased oncotic pressure	Inflammation or tumor of the pleura, increased capillary permeability
Causes	Heart failure, liver cirrhosis, nephrotic syndrome, hypoalbuminemia	Pneumonia, tuberculosis, tumor, pulmonary infarction, SLE, rheumatoid arthritis

Parameter	Transudate	Exudate
Appearance	Clear, pale yellow	Turbid, serous, serofibrinous, purulent or hemorrhagic
Specific gravity	Less than 1015	More than 1015
Protein	Less than 30 g/l	More than 30 g/l
Rivalta test	Negative	Positive
Fluid protein / serum protein	0.5 or less	More than 0.5
Fluid LDH / serum LDH	0.6 or less	More than 0.6
Fluid LDH	Less than 2/3 of the upper limit of serum LDH	More than 2/3
Cells	Few, mainly mesothelial	Many: neutrophils (bacterial), lymphocytes (tuberculosis, tumor)

REMEMBER

Light's criteria (1972): the fluid is an **exudate** if **at least one** of the following is present: 1) the ratio of pleural fluid protein to serum protein is **more than 0.5**; 2) the ratio of fluid LDH to serum LDH is **more than 0.6**; 3) the fluid LDH is **more than 2/3** of the upper limit of normal for serum LDH.

CLINICAL SIGNIFICANCE

Rivalta test: a few drops of acetic acid are added to 100 ml of distilled water in a cylinder, and then one drop of pleural fluid is let fall into it. In an exudate, the drop sinks forming a white cloud like "cigarette smoke" (positive); in a transudate, it dissolves without a trace (negative). The test is based on the presence of seromucoid proteins.

6. Pleurisy

Pleurisy is inflammation of the pleural layers. **Dry (fibrinous)** and **exudative (effusive)** pleurisy are distinguished. It is most often not an independent disease but a complication or manifestation of other diseases (pneumonia, tuberculosis, tumor, rheumatism, SLE, pulmonary infarction, uremia).

6.1. Dry (fibrinous) pleurisy

Complaints: stabbing pain on the affected side of the chest, worse on deep inspiration, coughing, sneezing and bending to the healthy side; dry cough, low-grade fever. The pain of diaphragmatic pleurisy may radiate to the abdomen and shoulder.

Inspection: the patient lies on the affected side or presses the chest with the hand (to restrict movement); the affected side lags behind in breathing; breathing is shallow and rapid.

Palpation: a tender area; sometimes the pleural friction is felt with the hand.

Percussion: the note is unchanged or slightly impaired; on the affected side the mobility of the lower edge is decreased (because of pain).

Auscultation: the pleural friction rub is the main sign; breath sounds are diminished.

6.2. Exudative pleurisy

It often begins as dry pleurisy: as fluid begins to accumulate, the pain lessens and the **dyspnea** increases. Fever, weakness, sweating. The physical signs correspond to the syndrome of fluid accumulation in the pleural cavity (see section 5). During the resorption of the fluid, the pleural friction rub reappears.

> **MEMORY AID**
> The dynamics of pleurisy: **"The pain goes — the dyspnea comes"** — when dry pleurisy turns into exudative pleurisy, the fluid separates the pleural layers and the pain lessens, but the lung is compressed and the dyspnea increases.

7. The syndrome of air accumulation in the pleural cavity. Pneumothorax

Pneumothorax is the entry of air into the pleural cavity with collapse of the lung.

Table 8.8. *Types of pneumothorax*

Classification	Type	Content
By origin	Primary spontaneous	In a healthy person, most often in tall, thin young men, from rupture of subpleural bullae
	Secondary spontaneous	Against the background of lung diseases: COPD, emphysema, tuberculosis, abscess, bronchial asthma
	Traumatic and iatrogenic	Chest injury, puncture, central venous catheterization
By communication with the external environment	Closed	The opening through which air entered has closed; the amount of air does not change
	Open	The pleural cavity communicates permanently with the external environment or a bronchus
	Valvular (tension)	Air enters on inspiration but does not leave on expiration — the pressure rises and the mediastinum shifts; life-threatening

Table 8.9. *Signs of the syndrome of air accumulation in the pleural cavity*

Method of examination	Signs
Complaints	Sudden severe stabbing pain, severe dyspnea, dry cough
Inspection	The affected side is enlarged, does not take part in breathing, the intercostal spaces are smoothed out; in tension — cyanosis, distended neck veins, hypotension
Palpation	Vocal fremitus markedly decreased or absent
Percussion	Tympanic note; the lower border of the lung and its mobility cannot be determined; the mediastinum and heart are shifted to the healthy side
Auscultation	Breath sounds markedly diminished or absent; in open pneumothorax, amphoric or metallic breathing
X-ray	A lucent area without lung markings; the collapsed lung compressed toward the hilum

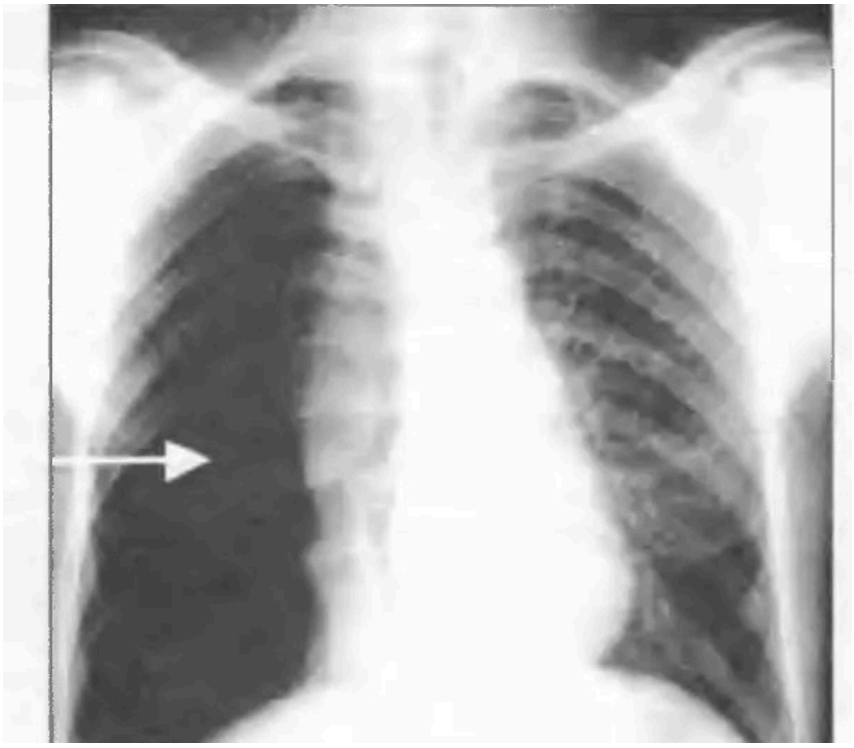


Figure 8.4. *Air accumulation in the pleural cavity: the border of the collapsed lung is shown by an arrow (Gadayev A. et al., 2012, Fig. 63)*

CLINICAL SIGNIFICANCE

Hydropneumothorax is the presence of both air and fluid in the pleural cavity. On percussion there is a tympanic note above and a dull note below; the border between them is **horizontal** and shifts when the patient changes position. When the patient is shaken (the Hippocratic test), a "succussion splash" (succussio Hippocratis) is heard.

8. Pleural puncture

Indications: diagnostic (to determine the nature of the fluid) and therapeutic (to remove fluid or air, to administer drugs).

Site: in the area where dullness on percussion is most pronounced, usually **in the VII–VIII intercostal space along the scapular or posterior axillary line**. The needle is inserted along the **upper edge** of the rib, because the intercostal vessels and nerves run along the lower edge of the rib. In pneumothorax, the puncture is performed in the II intercostal space along the midclavicular line.

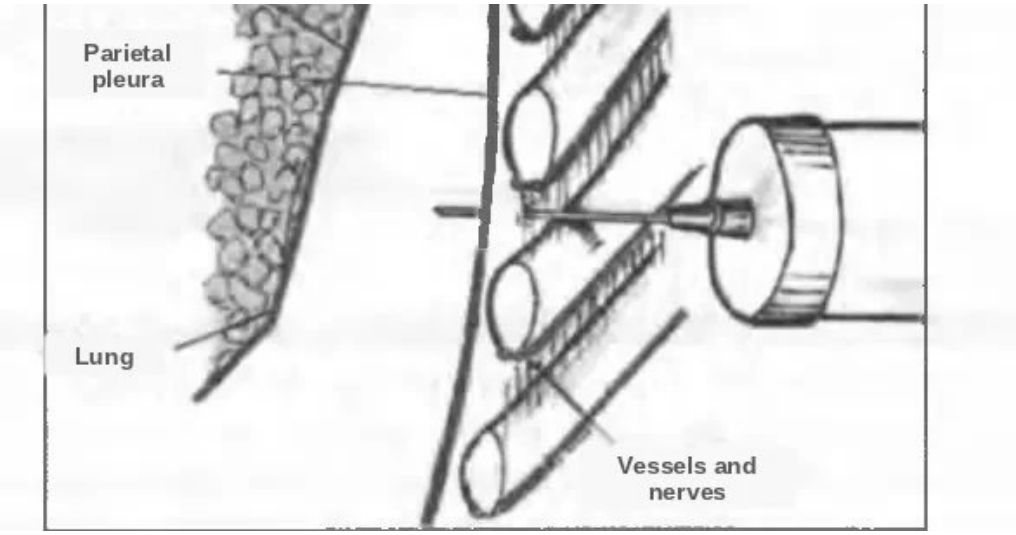


Figure 8.5. *Technique of pleural puncture: the needle is inserted along the upper edge of the rib (Gadayev A. et al., 2012, Fig. 58)*

CAUTION

Usually **no more than 1–1.5 liters** of fluid are removed in a single procedure: rapid removal of a large amount of fluid may lead to an abrupt mediastinal shift, collapse and "re-expansion" pulmonary edema. The fluid obtained is sent for physicochemical, cytological and bacteriological examination.

9. Pathogenesis and morphology of pneumonia

Normally the lower airways are almost sterile: they are protected by **mucociliary clearance** (ciliated epithelium and mucus), the **cough reflex**, alveolar macrophages, secretory IgA and surfactant. Pneumonia develops as a result of the route of entry of the microorganism and the weakening of the defense mechanisms.

Table 8.10. *Routes by which microorganisms enter the lung*

Route	Content	Examples
Microaspiration	Aspiration of oropharyngeal contents during sleep — the most common route	Pneumococcus, Haemophilus influenzae
Macroaspiration	Aspiration of a large amount of gastric or oral contents	Alcoholism, stroke, epilepsy, anesthesia — anaerobes, aspiration pneumonia and abscess
Inhalation	By the airborne route	Viruses, Mycoplasma, Legionella, Mycobacterium tuberculosis
Hematogenous	Through the blood from a distant focus	Staphylococcus (infective endocarditis, injection drug use, sepsis) — multiple foci
Direct spread	From a neighboring focus	Subhepatic abscess, mediastinitis

Table 8.11. *Factors predisposing to pneumonia*

Group	Factors
General	Old age (over 65), smoking, alcoholism, chilling, stress, malnutrition
Concomitant diseases	COPD, bronchiectasis, heart failure, diabetes mellitus, chronic kidney and liver diseases, tumors, HIV
Risk of aspiration	Impaired consciousness, impaired swallowing, stroke, epilepsy, nasogastric tube

Group	Factors
Iatrogenic	Mechanical ventilation (ventilator-associated pneumonia), immunosuppressive therapy, glucocorticoids, prolonged bed rest (hypostatic pneumonia)

9.1. “Atypical” pneumonias

Table 8.12. *Features of pneumonias caused by particular pathogens*

Pathogen	Clinical features
Pneumococcus	Classic lobar pneumonia: sudden onset, chills, pleuritic pain, rusty sputum, a lobar shadow
Mycoplasma	In young people and in groups (students, soldiers), gradual onset, prolonged dry cough, pharyngitis, few physical signs (“the X-ray is worse than the clinical picture”), myalgia
Chlamydophila	Similar to Mycoplasma, hoarseness, sinusitis
Legionella	Air conditioners and water systems, severe course, high fever and relative bradycardia, diarrhea, impaired consciousness, hyponatremia
Klebsiella	In alcoholism and diabetes mellitus, bloody viscous sputum like “red currant jelly”, in the upper lobes, rapid destruction (abscess)
Staphylococcus aureus	After influenza, severe course, multiple cavities (bullae), pyopneumothorax
Viruses (influenza, SARS-CoV-2)	Bilateral interstitial changes (“ground glass”), hypoxemia, bacterial infection is often superimposed

CAUTION

The term “atypical” does not mean a mild or unusual course of pneumonia but refers to the intracellular pathogens **insensitive to the usual antibiotics (beta-lactams)** (Mycoplasma, Chlamydophila, Legionella). Macrolides or respiratory fluoroquinolones are used for them.

9.2. Complications of pneumonia

Table 8.13. *Pulmonary and extrapulmonary complications of pneumonia*

Pulmonary complications	Extrapulmonary complications
Parapneumonic exudative pleurisy and pleural empyema	Infectious-toxic shock, sepsis
Lung abscess and gangrene	Acute respiratory distress syndrome (ARDS)
Broncho-obstructive syndrome	Myocarditis, pericarditis, arrhythmias, decompensation of heart failure
Acute respiratory failure	Meningitis, encephalopathy, delirium
Atelectasis	Acute kidney injury, DIC

CLINICAL SIGNIFICANCE

If the fever does not fall by the 3rd–5th day of treatment or rises again, the following must be excluded: **an antibiotic-resistant pathogen**, **a complication** (empyema, abscess), **a wrong diagnosis** (tuberculosis, lung cancer, pulmonary infarction, vasculitis) or **drug fever**.

10. Types of atelectasis

Atelectasis is the loss of air and collapse of lung tissue (a segment, a lobe or the whole lung).

Table 8.14. *Types of atelectasis and their physical signs*

Type	Mechanism	Causes	Physical signs
Obstructive	The bronchial lumen is blocked, and the air distal to it is absorbed	Tumor, foreign body, viscous sputum (after surgery)	The affected side is smaller and lags behind in breathing; dull note; fremitus, breath sounds and bronchophony decreased or absent; the mediastinum shifts to the affected side
Compression	The lung is compressed from outside	A large amount of fluid or air in the pleura, a large tumor	Dull-tympanic note over the compressed area, soft bronchial breathing (the bronchus is patent)
Contraction (shrinkage)	The lung shrinks as a result of fibrosis	Pneumosclerosis, tuberculosis, scars	Retracted chest, impaired note, mediastinum shifted to the affected side
Discoid (basal)	Diaphragmatic movement is restricted, and the lower parts collapse	After surgery, abdominal pain, prolonged bed rest	Often few physical signs; linear shadows on X-ray

MEMORY AID

The rule of mediastinal shift: **“Volume added — it moves away, volume lost — it is pulled toward”**. Fluid, air or a large tumor in the pleura (volume added) — the mediastinum shifts to the **healthy** side; obstructive and contraction atelectasis, pneumonectomy (volume lost) — to the **affected** side.

11. Extended interpretation of pleural fluid analysis

Table 8.15. *Additional parameters in pleural fluid analysis*

Parameter	Interpretation
Appearance	Clear, pale yellow — transudate; turbid — exudate; purulent — empyema; hemorrhagic — tumor, pulmonary infarction, injury, tuberculosis; milky — chylothorax (triglycerides more than 1.24 mmol/l)
pH	Below 7.2 — complicated parapneumonic pleurisy or empyema (an indication for drainage), as well as tumor, tuberculosis, rheumatoid pleurisy
Glucose	Below 3.3 mmol/l — empyema, rheumatoid pleurisy, tuberculosis, tumor
Cell composition	Neutrophils predominate — an acute bacterial process (parapneumonic), pulmonary infarction; lymphocytes more than 50 % — tuberculosis, tumor, lymphoma, heart failure; eosinophils more than 10 % — air or blood in the pleural space, parasitic disease, drug effect
Adenosine deaminase (ADA)	Above 40 U/l — strong suspicion of tuberculous pleurisy
Cytology	Atypical cells — metastatic pleurisy, mesothelioma
Microbiology	Gram stain, culture, PCR for Mycobacterium tuberculosis

Table 8.16. *The main causes of exudative pleurisy*

Cause	Features
Parapneumonic	With or after pneumonia, neutrophilic; the complicated type and empyema require drainage
Tuberculous pleurisy	More common in young people, low-grade fever, lymphocytic exudate, high ADA, mycobacteria rarely found

Cause	Features
Tumor (metastatic, mesothelioma)	In the elderly, often hemorrhagic, reaccumulates rapidly; lung and breast cancer, lymphomas
Pulmonary embolism	Often small in amount, sometimes hemorrhagic
Connective tissue diseases	SLE, rheumatoid arthritis — often bilateral
Abdominal diseases	Acute pancreatitis (on the left, high amylase), subhepatic abscess

REMEMBER

The most common cause of a transudate is **heart failure** (often bilateral or right-sided), and of an exudate, **pneumonia, tumor and tuberculosis**. A unilateral, large, rapidly reaccumulating hemorrhagic exudate raises suspicion of a **tumor** first of all.

12. Pathophysiology of pneumothorax and danger signs

Normally the pressure in the pleural cavity is below atmospheric (negative) and keeps the lung expanded. When air enters, the pressure equalizes and the lung collapses owing to its elastic recoil. In **valvular (tension) pneumothorax**, air enters with each inspiration but does not leave: the intrapleural pressure becomes positive, the mediastinum shifts to the healthy side, the venae cavae kink and the venous return to the heart decreases → **obstructive shock**.

CAUTION

Signs of tension pneumothorax: severe dyspnea and cyanosis, tachycardia, **hypotension**, distended neck veins, deviation of the trachea to the **healthy** side, a tympanic note and absent breath sounds on the affected side. This condition requires urgent **needle decompression** without waiting for an X-ray (the II intercostal space in the midclavicular line or the IV–V space in the anterior axillary line).

Table 8.17. *Clinical comparison of the types of pneumothorax*

Feature	Closed	Open	Valvular
Pleural pressure	Below or equal to atmospheric	Equal to atmospheric	Above atmospheric, rising
Mediastinum	Slightly shifted	“Swings” with breathing (mediastinal flutter)	Markedly shifted to the healthy side
Hemodynamics	Stable	Variable	Shock
Auscultation	Diminished breath sounds	Amphoric or metallic breathing	No breath sounds

13. Comparison of lobar pneumonia, exudative pleurisy and atelectasis

Table 8.18. *Differential diagnosis of conditions with a dull percussion note*

Feature	Lobar pneumonia	Exudative pleurisy	Obstructive atelectasis
Onset	Acute, high fever	Gradual or after pneumonia	Gradual (tumor) or sudden (foreign body)
Chest	Unchanged	The affected side enlarged	The affected side smaller
Vocal fremitus	Increased	Decreased or absent	Decreased or absent

Feature	Lobar pneumonia	Exudative pleurisy	Obstructive atelectasis
Percussion border	Within the borders of a lobe	The oblique Damoiseau line	Within the borders of a lobe
Respiration	Bronchial	Absent	Absent or markedly decreased
Mediastinum	In place	To the healthy side	To the affected side
Confirmatory investigation	X-ray, blood test	US, puncture	CT, bronchoscopy

14. Criteria for the diagnosis of pneumonia

The diagnosis of pneumonia is based on a combination of **clinical and radiological** data.

Table 8.19. *Diagnosis of community-acquired pneumonia*

Group of criteria	Signs
Clinical (at least two)	Fever of acute onset (above 38 °C); cough (productive); pleuritic chest pain; dyspnea
Physical	Signs of local consolidation: dull note, bronchial breathing, increased bronchophony and vocal fremitus, local consonating fine-bubbling rales or crepitation
Laboratory	Leukocytosis (more than $10 \cdot 10^9$/l) and/or band neutrophils more than 10 %; CRP raised
Radiological (the main confirmation)	A newly appeared focal or infiltrative change in the lung

CAUTION
If there are clinical signs but no infiltrate is found on X-ray (the initial stage, dehydration, neutropenia, a very small focus), pneumonia is considered “probable”, and the X-ray is repeated after 24–48 hours or CT is performed. Conversely, an infiltrate on X-ray is not always pneumonia: it must be distinguished from pulmonary infarction, tumor, tuberculosis and eosinophilic infiltrate.

14.1. Radiological types of pneumonia

Table 8.20. *Radiological appearances of pneumonia*

Type	Appearance	Pathogens
Lobar	Homogeneous opacity of a whole lobe, an “air bronchogram”	Pneumococcus, Klebsiella
Focal (bronchopneumonia)	Several patchy shadows	Staphylococcus, Haemophilus influenzae, mixed flora
Interstitial	Reticular changes, “ground glass”	Viruses, Mycoplasma, Pneumocystis
Destructive	Cavities within the infiltrate	Staphylococcus, Klebsiella, anaerobes

The “air bronchogram” is the visualization of air-containing bronchi as lucent bands against the background of consolidated lung tissue; it shows that the consolidation is in the alveoli (pneumonia); this sign is absent when there is fluid in the pleura.

15. Pleural empyema

Pleural empyema (pyothorax) is the accumulation of pus in the pleural cavity. It most often develops after pneumonia, rupture of a lung abscess into the pleura, chest injury and operations. Signs: hectic

fever, chills, profuse sweating, intoxication, pain and heaviness in the chest, signs of pleural fluid; sometimes swelling and redness of the chest wall. In the aspirate — thick pus, pH below 7.0, markedly low glucose, bacteria on culture. **Treatment:** drainage of the pleural cavity and antibiotics; if delayed, the pleura thickens and the lung becomes “encased in armor”.

16. Pulmonary embolism and pulmonary infarction

In the differential diagnosis of conditions with consolidation and pleuritic pain, **pulmonary embolism** (PE) must always be remembered.

Risk factors: prolonged bed rest, operations (especially orthopedic), leg injuries, tumors, pregnancy and the postpartum period, hormonal contraceptives, previous venous thrombosis, obesity, long flights.

Table 8.21. *Signs of pulmonary embolism*

Group	Signs
Complaints	Sudden dyspnea (the most common), chest pain (pleuritic or retrosternal), cough, hemoptysis, syncope
Objective	Tachypnea, tachycardia, cyanosis, distended neck veins, hypotension (massive PE); accentuation of S2 over the pulmonary artery; signs of deep vein thrombosis in the leg
In pulmonary infarction	Pleuritic pain, hemoptysis, pleural friction rub, signs of limited consolidation, a small amount of hemorrhagic exudate
Investigations	Raised D-dimer, right ventricular strain on the ECG (the SIQIIITIII pattern), right ventricular dilation on echocardiography; CT angiography is the confirmatory method

CLINICAL SIGNIFICANCE

In a patient who develops sudden dyspnea and tachycardia on the 5th–7th day after surgery, PE should be considered first of all, even if there are few physical changes in the lungs. In this situation the “mismatch between the physical signs and the degree of dyspnea” is an important diagnostic sign.

17. Hydrothorax and other noninflammatory fluids

Table 8.22. *Types of noninflammatory pleural fluid*

Type	Cause	Features
Hydrothorax (transudate)	Heart failure, liver cirrhosis (often on the right), nephrotic syndrome, hypoalbuminemia, peritoneal dialysis	Bilateral or on the right, clear, low protein
Hemothorax	Injury, rupture of an aortic aneurysm, anticoagulants, tumor	Hematocrit more than 50 % of the blood hematocrit
Chylothorax	Injury to the thoracic duct (trauma, surgery, lymphoma)	Milky, high triglycerides
Pseudochylothorax	Long-standing exudate (tuberculosis, rheumatoid)	Cholesterol crystals

MEMORY AID

The **“4C”** order of pleural fluid analysis: **C**olor (appearance), **C**hemistry (Rivalta test and protein — transudate or exudate), **C**ells (neutrophils, lymphocytes, eosinophils, atypical cells), **C**ulture (microbiology: bacteria, tuberculosis).

18. Clinical part

Distinguishing the three main syndromes

Table 8.23. *Differences between the syndromes of consolidation and of fluid and air accumulation*

Feature	Consolidation of the lung tissue	Fluid in the pleural cavity	Air in the pleural cavity
Volume of the affected side	Unchanged	Enlarged	Enlarged
Vocal fremitus	Increased	Decreased or absent	Decreased or absent
Percussion	Dull	Dull (thigh note), Damoiseau line	Tympanic
Respiration	Bronchial	Decreased or absent	Decreased or absent (amphoric)
Bronchophony	Increased	Absent	Absent
Added sounds	Crepitation, consonating moist rales	None (friction rub during resorption)	Absent
Mediastinum	In place	Shifted to the healthy side	Shifted to the healthy side

MEMORY AID

The key to distinguishing the three syndromes is **vocal fremitus**: it is **increased in consolidation and decreased in fluid and air**. Fluid and air, in turn, are distinguished by **percussion**: fluid — dull, air — tympanic.

Sample case history entry

The right side of the chest is larger than the left and lags behind in breathing; the intercostal spaces in the lower part are smoothed out. On the right below the VI rib, vocal fremitus is not detected. On percussion there is a dull note over this area with an oblique upper border rising to the V rib along the posterior axillary line (the Damoiseau line); the mobility of the lower edge of the right lung cannot be determined. On the left beside the spine, there is a triangle of dullness (Grocco–Rauchfuss). On auscultation, no breath sounds are heard on the right below the VI rib, and there is soft bronchial breathing at the upper border of the dullness.

Conclusion: the syndrome of fluid accumulation in the right pleural cavity — exudative pleurisy (or hydrothorax); pleural puncture and fluid analysis are needed.

19. Practical skills

Skill: distinguishing the syndromes of consolidation, fluid and air on physical examination

1. Inspect: the symmetry of the chest, the size of the affected side and its participation in breathing.
2. Compare vocal fremitus on the two sides: increased — consolidation; decreased — fluid, air or obstruction.
3. Perform comparative percussion: a dull (consolidation or fluid) or tympanic (air) note.
4. If dullness is found, determine the shape of its upper border (oblique — the Damoiseau line) and look for Garland’s and the Grocco–Rauchfuss triangles.
5. Auscultate: bronchial breathing (consolidation) or diminished/absent breath sounds (fluid, air).

6. Examine bronchophony.
7. Assess mediastinal shift by determining the position of the apex beat and the trachea.
8. Draw a syndromic conclusion and plan radiography, US and, if necessary, pleural puncture.

Skill: the steps of pleural puncture (on a manikin)

1. Establish the indication, obtain the patient's consent and mark the puncture site with X-ray or US.
2. Seat the patient leaning slightly forward with the arms resting on a pillow on a table.
3. Clean your hands and put on sterile gloves; treat the puncture area widely with an antiseptic and drape it with a sterile sheet.
4. Local anesthesia: inject novocaine (procaine) or lidocaine solution layer by layer.
5. Insert the needle perpendicularly along the upper edge of the rib; after the sensation of "entering a space", pull back the syringe to check for fluid.
6. Withdraw the fluid slowly (no more than 1–1.5 l at a time) and prevent air entry (a clamp on the rubber tube or a three-way tap).
7. Remove the needle and apply a sterile dressing to the puncture site.
8. Record the amount and appearance of the fluid and send samples to the laboratory; observe the patient's condition and perform a control radiograph.

Review questions

1. What are the causes and physical signs of the syndrome of consolidation of the lung tissue?
2. How do the physical signs of obstructive atelectasis differ from the consolidation syndrome?
3. State the modern classification of pneumonias.
4. What are the stages of lobar pneumonia and the percussion and auscultation findings at each stage?
5. What are the clinical signs of lobar pneumonia?
6. How do you distinguish lobar from focal pneumonia?
7. What are the components and significance of the CURB-65 score?
8. List the physical signs of the syndrome of fluid accumulation in the pleural cavity.
9. What are the Damoiseau line and Garland's and the Grocco–Rauchfuss triangles?
10. What are the differences between a transudate and an exudate? State Light's criteria.
11. How is the Rivalta test performed?
12. What are the clinical signs of dry pleurisy?
13. List the types of pneumothorax. What is the danger of valvular pneumothorax?
14. What are the physical signs of the syndrome of air accumulation in the pleural cavity?
15. Where and how is pleural puncture performed?

Test questions

1. What is the most common causative agent of lobar pneumonia?
   A) Staphylococcus
   B) Pneumococcus
   C) Klebsiella
   D) Pseudomonas aeruginosa
   E) Mycoplasma

2. What is heard on auscultation in the hepatization stage of lobar pneumonia?
   A) Dry whistling rales (wheezes)
   B) Crepitation
   C) Diminished vesicular breathing
   D) Bronchial breathing
   E) Amphoric breathing

3. At which stage is “crepitatio redux” heard?
   A) At the stage of gray hepatization
   B) At the stage of red hepatization
   C) At the stage of resolution
   D) Never
   E) At the stage of congestion

4. How does vocal fremitus change in the syndrome of consolidation of the lung tissue?
   A) It is decreased
   B) It is absent
   C) Increases
   D) Unchanged
   E) It is decreased only at the back

5. Which parameter is not included in the CURB-65 score?
   A) Impaired consciousness
   B) Urea above 7 mmol/l
   C) Respiratory rate 30 or more
   D) Body temperature above 39 °C
   E) Age 65 or older

6. What is the oblique upper border of an exudate called?
   A) The Grocco–Rauchfuss line
   B) The Krönig line
   C) The Traube line
   D) The Garland line
   E) The Ellis–Damoiseau–Sokolov line

7. Where is the Grocco–Rauchfuss triangle located?
   A) On the healthy side, beside the spine
   B) On the affected side, in the axilla
   C) On the affected side, above the Damoiseau line
   D) In the suprascapular region
   E) Over the sternum

8. What percussion note is found in Garland’s triangle?
   A) Dull
   B) Dull-tympanic
   C) Thigh note
   D) Hyperresonant (box-like)
   E) Tympanic

9. Which is a sign of an exudate according to Light’s criteria?
   A) Fluid protein / serum protein more than 0.5
   B) Specific gravity 1010
   C) Negative Rivalta test
   D) Protein 20 g/l
   E) Fluid LDH / serum LDH 0.3

10. Where does the mediastinum shift when a large amount of fluid accumulates in the pleural cavity?
    A) To the healthy side
    B) Upward
    C) It does not shift
    D) Downward
    E) To the affected side

11. What note is found on percussion in pneumothorax?
    A) Clear lung (resonant) note
    B) Tympanic
    C) Impaired
    D) Dull
    E) Hyperresonant (box-like)

12. What is the main auscultatory sign of dry pleurisy?
    A) Dry whistling rales (wheezes)
    B) Amphoric breathing
    C) Bronchial breathing
    D) Crepitation
    E) Pleural friction rub

13. How is the needle inserted in pleural puncture?
    A) Through the rib
    B) Beside the sternum
    C) Along the upper edge of the rib
    D) It does not matter
    E) Along the lower edge of the rib

14. Which type of pneumothorax is the most life-threatening?
    A) Closed
    B) Partial
    C) Limited
    D) Valvular (tension)
    E) Open

15. What is found on auscultation in obstructive atelectasis?
    A) Crepitation
    B) Amphoric breathing
    C) Bronchial breathing
    D) Breath sounds markedly diminished or inaudible
    E) Consonating moist rales

Clinical cases

Case 1

After chilling, a 38-year-old man fell ill suddenly with severe chills, a fever of 40 °C and pain on the right that increased on inspiration. On the third day he began to expectorate rust-colored sputum. On inspection: the right cheek is flushed, there is herpes on the lips. In the right infrascapular region vocal fremitus is increased, the percussion note is dull, and there is bronchial breathing and increased bronchophony. Blood leukocytes $22 \cdot 10^9$/l, band neutrophils 15 %, ESR 48 mm/h. Age 38, consciousness clear, respiratory rate 24, BP 120/80, urea 5 mmol/l.

Questions: What are the syndrome and the diagnosis? At which stage is the disease? Assess it by CURB-65.

Answer: The syndrome of consolidation of the lung tissue — right-sided lower lobe lobar pneumonia, hepatization (II) stage (dull note, bronchial breathing, increased vocal fremitus and bronchophony). Neutrophilic leukocytosis and a shift of the differential to the left confirm bacterial inflammation. CURB-65 = 0 points (clear consciousness, normal urea, respiratory rate below 30, normal BP, age below 65). However, given the high fever and extensive involvement, X-ray confirmation and the start of treatment are needed.

Case 2

2 weeks ago, a 50-year-old patient had pain on the left that increased with breathing; then the pain lessened and the dyspnea increased. On the left below the VII rib, vocal fremitus and breath sounds are not detected, and on percussion there is a dull note with an oblique border. Pleural puncture yielded 1000 ml of turbid yellowish fluid: specific gravity 1024, protein 42 g/l, Rivalta test positive, fluid/serum protein 0.68, many lymphocytes.

Questions: Is the fluid a transudate or an exudate? Explain the clinical dynamics. Which cause should be considered?

Answer: An exudate (specific gravity above 1015, protein above 30 g/l, positive Rivalta test, protein ratio above 0.5 — Light’s criterion). The replacement of pain by dyspnea shows that dry pleurisy has turned into exudative pleurisy: the fluid separates the pleural layers and lessens the pain but compresses the lung and causes dyspnea. A lymphocytic exudate raises suspicion of a tuberculous or tumor etiology — the fluid should be examined for mycobacteria and atypical cells.

Case 3

A 72-year-old patient has chronic heart failure: leg edema, an enlarged liver. There is dullness in the lower parts on both sides with an almost horizontal upper border. Puncture yielded a clear, pale yellow fluid: specific gravity 1010, protein 18 g/l, Rivalta test negative.

Questions: What is the type of fluid, and what is this condition called? By what mechanism did it develop?

Answer: A transudate; the condition is bilateral hydrothorax. In heart failure it develops as a result of a rise in venous (hydrostatic) pressure, with fluid leaving into the pleural cavity without inflammation. A transudate is most often bilateral or larger on the right, with a horizontal border that shifts with position.

Quick review

Table 8.24. *Key facts on Topic 8*

Question	Answer
Consolidation syndrome	Fremitus increased, dull note, bronchial breathing, bronchophony increased
Stages of lobar pneumonia	Congestion (crepitatio indux) → hepatization (bronchial breathing) → resolution (crepitatio redux)
Clinical picture of lobar pneumonia	Sudden chills, 39–40 °C, stabbing pain, rusty sputum, herpes
CURB-65	0–1 outpatient; 2 hospital; 3–5 severe
Fluid syndrome	No fremitus, dull note, no breath sounds, mediastinum to the healthy side
Damoiseau line	The oblique upper border of an exudate
Garland’s triangle	Affected side: dull-tympanic, soft bronchial breathing
Grocco–Rauchfuss triangle	Dullness on the healthy side
Exudate (Light)	Protein ratio more than 0.5; LDH ratio more than 0.6; LDH more than 2/3
Dry pleurisy	Stabbing pain, pleural friction rub
Pneumothorax	No fremitus, tympanic note, no breath sounds
Pleural puncture	VII–VIII intercostal space, scapular or posterior axillary line, upper edge of the rib

TOPIC 9

The syndrome of cavity formation in the lung. Lung abscess. Bronchiectasis. Patient curation. Writing case history 1

Learning objectives

To teach the student the causes and physical signs of the syndrome of cavity formation in the lung and the symptomatology of lung abscess and bronchiectasis; to develop the skills of independent curation of a patient with a respiratory disease and of writing the educational case history (case history 1).

By the end of the session, the student should know:

- the causes of cavity formation in the lung and the conditions for detecting a cavity by physical methods;
- the physical signs of the cavity syndrome;
- the stages of lung abscess and the clinical signs at each stage;
- the causes and the clinical and physical signs of bronchiectasis;
- the signs distinguishing lung abscess from bronchiectasis;
- the structure of the educational case history and the requirements for writing each section.

The student should be able to:

- take a full history from the patient and perform an objective examination;
- combine the signs found into syndromes and justify the syndromic diagnosis;
- interpret a sputum analysis and the conclusion of a radiograph;
- write the educational case history according to the outline.

1. The syndrome of cavity formation in the lung

A cavity in the lung is an air-filled space formed as a result of destruction of the lung tissue and most often communicating with a bronchus.

Causes: lung abscess (purulent destruction), tuberculous cavity, cavitating lung cancer, lung gangrene, a suppurating cyst, large bronchiectases, bullae.

REMEMBER

A cavity is detected by physical methods only under the following conditions:

- it is **more than 4–5 cm** in diameter;
- it lies **close to the chest wall (superficially)**;
- it **communicates with a bronchus** and is filled with air (the pus has drained);
- the lung tissue around it is consolidated (inflammatory infiltrate).

A small, deep-seated or pus-filled cavity gives the picture of consolidation or no change at all on physical examination.

Table 9.1. *Signs of the syndrome of cavity formation in the lung*

Method of examination	Signs
Complaints	Cough with a large amount (especially in the morning, “a mouthful”) of purulent, often foul-smelling sputum; the sputum separates into layers; hemoptysis; fever, sweating
Inspection	The affected side lags behind in breathing; in a long course, clubbing and watch-glass nails
Palpation	Vocal fremitus increased over the cavity
Percussion	Tympanic note; over a large smooth-walled cavity — metallic; when communicating with a bronchus through a narrow opening — the “cracked-pot” sound
Auscultation	Bronchial or amphoric breathing, coarse-bubbling consonating moist rales; bronchophony increased
X-ray	A rounded lucent area (cavity) with a horizontal fluid level, with infiltration around it

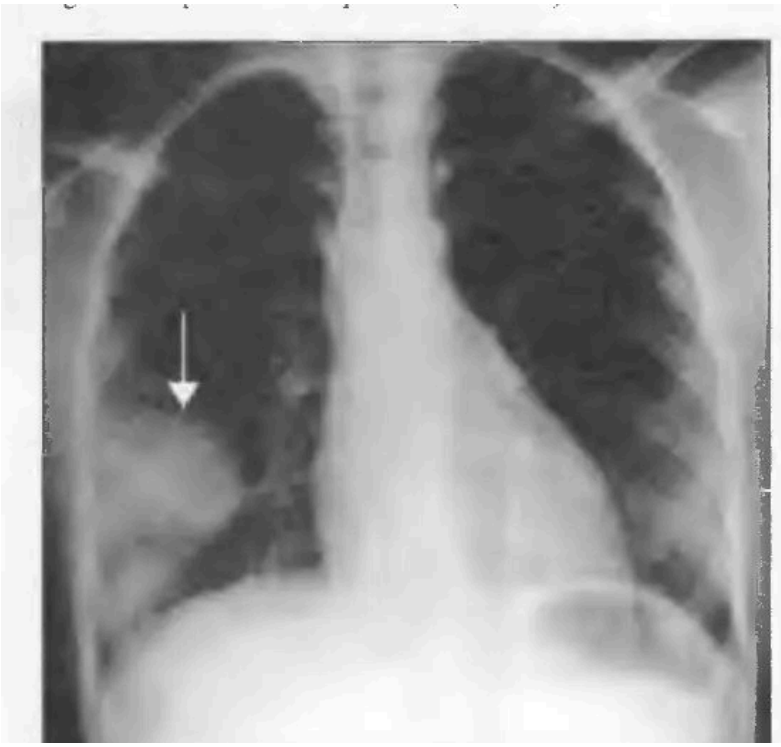


Figure 9.1. *Cavity formation in the lung: a cavity with a fluid level in the right lung (arrow) (Gadayev A. et al., 2012, Fig. 61)*

CLINICAL SIGNIFICANCE

Wintrich’s sign — the pitch of the percussion note rises when the patient opens the mouth and falls when it is closed (showing that the cavity communicates with the outside air through a bronchus). **Gerhardt’s sign** — a change in the pitch of the tympanic note when the patient changes position (with the shifting of the fluid level in the cavity).

2. Lung abscess

A lung abscess is a nonspecific purulent-necrotic destruction of the lung tissue with the formation of one or more cavities bounded by granulation tissue (a capsule). It occurs most often in men aged 30–50.

Causes and predisposing factors: aspiration (alcohol intoxication, epilepsy, anesthesia, impaired swallowing, dental caries and periodontal disease), pneumonia (Staphylococcus, Klebsiella),

bronchial obstruction (tumor, foreign body), septic embolism, diabetes mellitus, reduced immunity. Pathogens: anaerobes, Staphylococcus aureus, Klebsiella, Pseudomonas aeruginosa.

The two periods of the course of an abscess

Table 9.2. *Periods of lung abscess*

Feature	Period I — formation of the abscess (before rupture)	Period II — after rupture of the abscess into a bronchus
Duration	Usually 7–14 days	Several weeks
Fever	Hectic, chills, profuse sweating	Falls sharply
Cough and sputum	Dry cough or scanty sputum	Suddenly, “a mouthful” — a large amount (200–1000 ml a day) of purulent, foul-smelling sputum
Pain	On the affected side (pleural involvement)	Decreases
General condition	Severe, intoxication	Improves rapidly once the pus has drained
Vocal fremitus	Increased (infiltrate)	Increased over the cavity
Percussion	Dull	Tympanic (if the cavity has emptied and is superficial and large)
Auscultation	Diminished or bronchial breathing, moist rales	Amphoric breathing, coarse-bubbling consonating moist rales
X-ray	Homogeneous infiltrate	A cavity with a horizontal fluid level

Laboratory signs: marked neutrophilic leukocytosis in the blood (up to 20–30·10^9/l), a shift of the differential to the left, a raised ESR; in the sputum, many leukocytes, **elastic fibers** (a sign of destruction of the lung tissue), hematoidin and cholesterol crystals and fatty acid crystals. When left to settle, the sputum **separates into layers**.

> **CAUTION**
> The improvement of the condition and the fall in temperature after an abscess has ruptured into a bronchus may be deceptive. If the cavity drains poorly (a narrow or oblique draining bronchus), the fever and intoxication increase again. An abscess that has not healed in 2–3 months becomes a **chronic abscess**.

Complications: pyopneumothorax and pleural empyema, pulmonary hemorrhage, sepsis, brain abscess, amyloidosis, transition to a chronic abscess.

3. Bronchiectasis

Bronchiectasis is an irreversible dilation (bronchiectases) and deformity of the bronchi in which a chronic suppurative process takes place. The disease most often forms in childhood and adolescence (5–25 years) and is more common in males.

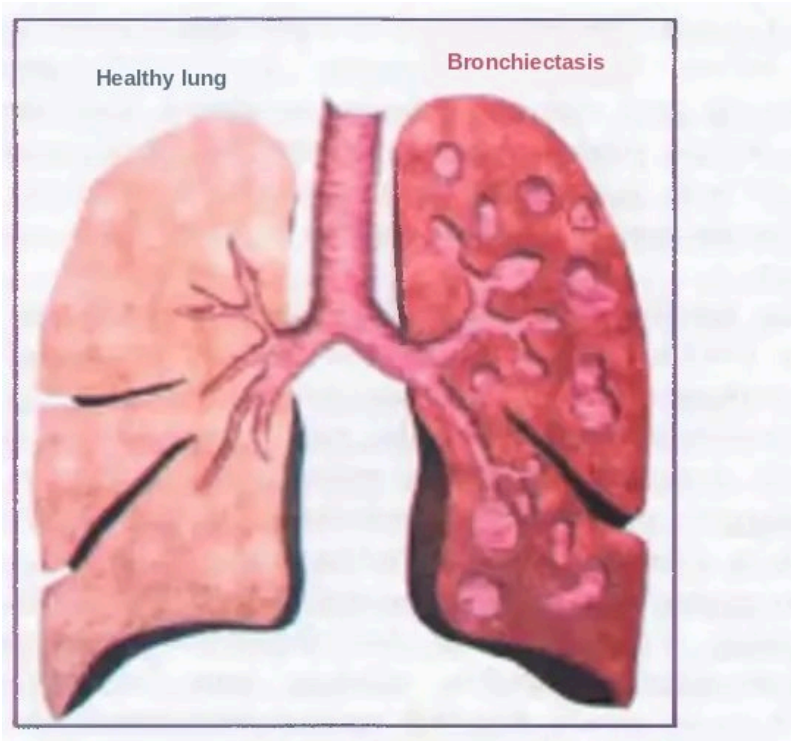


Figure 9.2. *Diagram of bronchiectases: a healthy lung (left) and bronchiectasis (right) (Gadayev A. et al., 2012, Fig. 65)*

Causes: severe pneumonias, measles and pertussis in childhood; congenital factors (developmental defects of the bronchial wall, cystic fibrosis, primary ciliary dyskinesia — Kartagener's syndrome); bronchial obstruction (foreign body, tumor); immunodeficiency.

Forms of bronchiectasis: cylindrical, saccular, fusiform, mixed. **The lower lobes of the lungs** and the lingular segments of the left lung are most often affected (their drainage is poorer).

Clinical picture:

- **Cough with a large amount of purulent sputum**, especially in the morning ("a mouthful") and in the drainage position (lying on the healthy side with the head lower); the daily amount is 50–500 ml or more; when left to settle, the sputum separates into three layers and is often foul-smelling.
- **Hemoptysis** (often recurrent), sometimes pulmonary hemorrhage.
- Fever, weakness and sweating during exacerbations; exacerbations of the disease in spring and autumn are characteristic.
- In a long course — dyspnea, weight loss, **clubbing** and watch-glass nails, cyanosis.

Physical signs: few changes during remission. During an exacerbation, over the affected area (most often the lower parts) percussion reveals an impaired note (surrounding pneumosclerosis and infiltration), and auscultation reveals harsh breathing and **medium- and coarse-bubbling moist rales**, which decrease or disappear after coughing and expectoration. When pneumosclerosis and emphysema develop, their signs appear.

Diagnosis: the "gold standard" is **high-resolution computed tomography** (a bronchial diameter larger than that of the adjacent artery, absence of tapering of the bronchus toward the periphery); bronchography was used previously. Bronchoscopy is used to find the source of pus and for sanitation.

Complications: pulmonary hemorrhage, respiratory failure, chronic cor pulmonale, amyloidosis (kidneys — nephrotic syndrome), lung abscess, brain abscess.

MEMORY AID

The main signs of bronchiectasis — **“Pus, Blood, Clubs”**: a large amount of purulent sputum (in the morning), hemoptysis, clubbing.

Differences between lung abscess and bronchiectasis

Table 9.3. *Lung abscess and bronchiectasis*

Feature	Lung abscess	Bronchiectasis
Onset	Acute, at a definite time	From childhood, gradual
Course	Acute; chronic if it does not heal in 2–3 months	Chronic, with exacerbations and remissions
Sputum	Suddenly “a mouthful” at the time of rupture	A large amount every morning, dependent on position
Location	Most often the posterior segments of the upper lobe and the superior segments of the lower lobe	Most often the lower lobes, bilateral
Elastic fibers	Present (tissue destruction)	Usually absent
X-ray and CT	A single cavity with a fluid level	Dilated, deformed bronchi, “honeycombing”

4. Patient curation and writing case history 1

Curation is the student’s independent examination and observation of a patient and keeping of the case history under the supervision of the teacher. Case history 1 is written on a patient with a respiratory disease.

4.1. Stages of curation

1. A patient is assigned by the teacher; the student gets acquainted with the patient and obtains consent.
2. History taking (complaints, anamnesis morbi, anamnesis vitae) — according to the plan of topic 1.
3. General inspection and objective examination by systems (status praesens), with the main attention on the respiratory organs (status localis).
4. Combining the signs found into syndromes.
5. Formulating and justifying the syndromic and primary diagnosis.
6. Drawing up a plan of investigation and interpreting the available test results.
7. Keeping progress notes (the daily dynamics of the patient’s condition).
8. Discharge summary (epicrisis) and recommendations.
9. Submitting the case history to the teacher and defending it.

4.2. Justifying the syndromic diagnosis

The syndromic diagnosis is written on the basis of the signs found in the following order: “On the basis of the patient’s complaints of ..., the history data ..., the signs found on inspection ..., on palpation ..., on percussion ... and on auscultation ..., the following syndromes can be identified: ...”

Table 9.4. *The main syndromes in respiratory diseases*

Syndrome	Main signs	Example diseases
Inflammation (intoxication)	Fever, weakness, leukocytosis, raised ESR	Pneumonia, abscess, pleurisy
Consolidation of the lung tissue	Increased fremitus, dull note, bronchial breathing	Lobar pneumonia, pulmonary infarction
Bronchial obstruction	Expiratory dyspnea, dry whistling rales	Bronchial asthma, COPD
Increased air content of the lungs	Barrel chest, hyperresonant note, diminished breath sounds	Emphysema
Fluid in the pleural cavity	No fremitus, dull note, no breath sounds	Exudative pleurisy, hydrothorax
Air in the pleural cavity	No fremitus, tympanic note, no breath sounds	Pneumothorax
Cavity in the lung	Increased fremitus, tympanic note, amphoric breathing, coarse-bubbling rales	Abscess, cavern
Respiratory failure	Dyspnea, cyanosis, tachypnea, decreased SpO_2	All severe lung diseases
Chronic cor pulmonale	Signs of right ventricular failure, accentuation of S2 over the pulmonary artery	COPD, bronchiectasis

CAUTION

Typical errors in the case history: listing complaints without analysis; breaking the chronology of the history; writing a general “normal” in the objective examination (in fact, each parameter is written with an exact figure: respiratory rate, pulse, BP, borders); not justifying the syndromic diagnosis with signs; “copying” the diagnosis from the medical record before talking to the patient.

4.3. Sample status localis of the respiratory system

Respiratory system. Breathing through the nose is free. The voice is unchanged. The chest is normosthenic in shape and asymmetrical: the left side lags behind in breathing. Breathing is of the mixed type, rhythmic, respiratory rate 24 per minute. The accessory muscles do not take part in inspiration.

Palpation: the chest is painless and elastic. Vocal fremitus is increased in the left infrascapular region.

Comparative percussion: a dull note in the left infrascapular region below the VII rib; a clear lung note over the other areas.

Topographic percussion: the lung apices are 3.5 cm above the clavicle in front and at the level of the VII cervical vertebra behind. Krönig’s isthmus is 5 cm on both sides. The lower borders of the lungs are normal (given in a table). The mobility of the lower edge of the lung along the mid-axillary line is 7 cm on the right and 3 cm on the left.

Auscultation: in the left infrascapular region, bronchial breathing, crepitation at the end of inspiration and consonating fine-bubbling moist rales; bronchophony increased. Vesicular breathing over the other areas.

Syndromic conclusion: the syndromes of inflammation and of consolidation of the lung tissue in the lower lobe of the left lung. **Primary diagnosis:** community-acquired left-sided lower lobe pneumonia.

5. Differential diagnosis of diseases forming cavities in the lung

Table 9.5. *Differences between cavities in the lung*

Feature	Lung abscess	Tuberculous cavity	Cavitating lung cancer	Suppurating cyst
Onset	Acute, high fever	Gradual, low-grade fever	Slow, often asymptomatic	Often found incidentally
Sputum	Abundant, purulent, foul-smelling	Scanty or moderate, blood-stained	Scanty, blood-stained, “raspberry jelly”	Abundant on rupture
Location	The posterior segments of the upper lobe and the superior segments of the lower lobe (aspiration)	The apicoposterior segments of the upper lobes	Any	Any
X-ray	Thick-walled, with a fluid level inside and infiltration around it	Thin-walled, with foci around it (“seeding foci”)	Thick, irregular, with a rough inner wall (“eroded”)	Thin, smooth-walled
Sputum analysis	Elastic fibers, anaerobes	Mycobacterium tuberculosis	Atypical cells	Nonspecific
Age and history	Alcoholism, aspiration	Contact with tuberculosis, social factors	Elderly smokers	—

CAUTION
In a smoker over 40, any cavity in the lung, especially one with a thick, irregular wall, is considered a **cavitating lung cancer** until proven otherwise: bronchoscopy and biopsy are performed. “Treating an abscess” with antibiotics may lead to a loss of time.

5.1. Lung gangrene

Lung gangrene is a poorly demarcated, rapidly spreading putrefactive necrosis of the lung tissue. It differs from an abscess in that the patient’s condition is very severe (sepsis, shock), the sputum is very abundant (up to 1 liter a day), **very foul-smelling** (noticed even by those nearby), brownish-gray and multilayered; there is unbounded destruction in the lung with multiple cavities. Mortality is high, and surgical treatment is often needed.

6. Pathogenesis and diagnosis of bronchiectasis

Bronchiectases develop according to a **“vicious circle”** (P. Cole’s hypothesis): impaired mucociliary clearance → accumulation of mucus → bacterial colonization → chronic neutrophilic inflammation → neutrophil elastase and other enzymes destroy the elastic and muscle fibers of the bronchial wall → the bronchus dilates and becomes deformed → clearance is further impaired.

Table 9.6. *Forms and causes of bronchiectases*

Classification	Types
By shape	Cylindrical (uniformly dilated), varicose (irregular, beaded), saccular (cystic — the most severe)
By origin	Congenital (developmental defect of the bronchial wall, cystic fibrosis, Kartagener’s syndrome) and acquired (childhood infections, obstruction, immunodeficiency, allergic bronchopulmonary aspergillosis)
By extent	Localized (one lobe) and diffuse (bilateral)

Classification	Types
By course	Periods of exacerbation and remission

CLINICAL SIGNIFICANCE

Kartagener's syndrome is the triad of **bronchiectasis + chronic sinusitis + reversed position of the internal organs** (situs inversus — the heart on the right). It is based on immobility of the ciliated epithelium (primary ciliary dyskinesia); infertility is also observed in men. If the heart sounds of a patient are heard on the right, this syndrome should be remembered.



Figure 9.3. *Bronchography: the normal structure of the bronchi (Gadayev A. et al., 2012, Fig. 48)*



Figure 9.4. *Bronchography: dilated bronchi (same source, Fig. 48)*

Formerly, **bronchography** (radiography after a contrast agent has been introduced into the bronchi) was used to confirm bronchiectasis; it has now been replaced by **high-resolution CT**. CT signs: an internal bronchial diameter larger than the diameter of the adjacent artery (the "signet ring" sign), absence of tapering of the bronchus toward the periphery, visibility of bronchi within 1 cm of the pleura.

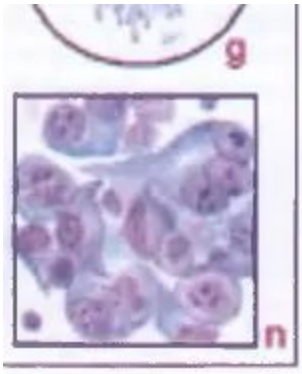


Figure 9.5. *Atypical (tumor) cells in the sputum (Gadayev A. et al., 2012, Fig. 59, n)*

7. Writing the main sections of the case history: a detailed sample

7.1. Status praesens — general inspection

The status praesens (present status) is written systematically, with each parameter stated precisely:

1. **General condition** (satisfactory, moderately severe, severe), **consciousness** (clear, GCS score), **position** (active, passive, forced — which), **body build** (constitutional type, epigastric angle), **height, weight, BMI**, **body temperature**.
2. **Skin:** color, moisture, turgor, rashes, scars, vascular changes; **visible mucous membranes**; **hair and nails**.
3. **Subcutaneous fat:** development, skinfold thickness, edema (location, character).
4. **Lymph nodes:** by groups (size, consistency, tenderness, mobility).
5. **Muscles:** development, tone, strength, tenderness.
6. **Bones:** deformity, pain.
7. **Joints:** shape, range of movement, pain.
8. **Thyroid gland.**

7.2. The respiratory system — in detail

1. **Inspection:** breathing through the nose, the voice; the shape and symmetry of the chest; the type, rate, depth and rhythm of breathing; use of the accessory muscles; participation of both sides in breathing.
2. **Palpation:** tender areas, resistance, vocal fremitus.
3. **Comparative percussion:** the character of the note by areas.
4. **Topographic percussion:** height of the apices (in front and behind), Krönig's isthmus, the lower borders (table), the mobility of the lower edge (along three lines).
5. **Auscultation:** the main breath sound, added sounds, bronchophony.

Table 9.7. *Form for recording the results of topographic percussion*

Topographic line	Right lung	Left lung
Parasternal	...	—
Midclavicular	...	—
Anterior axillary	...	...
Mid-axillary	...	...
Posterior axillary	...	...
Scapular	...	...
Paravertebral	...	...

7.3. The other systems (in brief)

The cardiovascular system (inspection, apex beat, borders, heart sounds, murmurs, pulse, BP), the digestive system (tongue, inspection of the abdomen, superficial and deep palpation, the liver by Kurlov, the spleen), the urinary system (lumbar region, Pasternatsky's sign, urination), the nervous system (reflexes, sleep, mental state).

7.4. Identifying the syndromes and justifying the primary diagnosis

The justification formula: **"On the basis of the complaints (...) + the history (...) + the objective findings (...), the following syndromes are identified: 1) ...; 2) ...; 3) .... The combination of these syndromes and the acute onset (or course) of the disease is characteristic of ... disease. Primary diagnosis: ..."**

7.5. Sample progress note

Progress note (date, time): The patient's condition is moderately severe, with positive dynamics. Complaints: the cough has decreased, the sputum is mucopurulent, up to 20 ml a day, and dyspnea occurs only on climbing stairs. Body temperature 36.9 °C in the morning and 37.3 °C in the evening. Objectively: the skin is pink and moist. Respiratory rate 20 per minute. In the right infrascapular region the impaired percussion note persists; on auscultation breath sounds are diminished and crepitation (crepitatio redux) is heard. Heart sounds are clear and rhythmic, pulse 84/min, BP 120/80 mm Hg. The abdomen is soft and painless. Bowel movements and urination are normal. Treatment is continued.

7.6. Structure of the discharge summary

Briefly, the discharge summary gives: the patient's full name and age, the date and reason for admission, the main complaints, the main objective and laboratory-instrumental findings that confirmed the diagnosis, the clinical diagnosis, the treatment given, the result of treatment and the patient's condition at discharge, capacity for work, and further recommendations (regimen, diet, medicines, follow-up).

MEMORY AID

It is convenient to write progress notes and conclusions according to the **SOAP** system of the English-language literature: **S**ubjective — the patient's complaints; **O**bjective — objective findings; **A**ssessment — assessment (dynamics, diagnosis); **P**lan — plan (investigations and treatment).

8. Sputum examination: extended information

Rules for collecting sputum: in the morning, before eating, after rinsing the mouth with boiled water, by coughing deeply, into a sterile container (it must be sputum, not saliva). For bacteriological examination it is taken before antibiotics are started; for Mycobacterium tuberculosis it is collected on 3 consecutive days.

Table 9.8. *Types of sputum examination*

Examination	Content
Macroscopic	Amount, color, odor, consistency, separation into layers, admixtures (blood, Dittrich plugs, Curschmann spirals)
Microscopic (native)	Leukocytes, red cells, epithelium, elastic fibers, Curschmann spirals, Charcot–Leyden crystals, cholesterol crystals
Stained smear	Gram stain (bacteria), Ziehl–Neelsen stain (Mycobacterium tuberculosis), Romanowsky stain (eosinophils), cytology (atypical cells)
Bacteriological	Culture and antibiotic sensitivity
Molecular	PCR (Mycobacterium tuberculosis and rifampicin resistance — GeneXpert, viruses)

9. Postural drainage

Postural drainage is placing the patient in a special position so that sputum drains out of the affected bronchi. It is used in bronchiectasis, lung abscess and chronic purulent bronchitis. The procedure is performed 2–3 times a day for 15–30 minutes, before meals or 2 hours after them; it is more effective combined with vibratory massage (percussion) of the chest.

Table 9.9. *Drainage positions according to the affected lobe*

Affected area	Drainage position
The apical segments of the upper lobes	Sitting, leaning slightly backward or forward
The anterior segments of the upper lobe	Lying on the back
The right middle lobe	On the left side, turned slightly onto the back, with the foot of the bed raised by 15–20 cm
The left lingular segments	On the right side, turned slightly onto the back, with the foot of the bed raised
The basal segments of the lower lobes	Lying on the healthy side or on the abdomen, with the foot of the bed raised by 30–40 cm (“head down”)

CAUTION
Postural drainage is **contraindicated** in pulmonary hemorrhage, severe heart failure, high arterial hypertension, acute cerebrovascular events and in the elderly in a severe condition.

10. Principles of treating lung abscess and bronchiectasis

- **Lung abscess:** prolonged (4–6 weeks or more) antibiotic therapy (drugs active against anaerobes); postural drainage; bronchoscopic sanitation; if ineffective — percutaneous drainage or surgery (removal of a lobe of the lung).
- **Bronchiectasis:** ensuring airway clearance (physiotherapy, drainage, mucolytics), antibiotics during exacerbations, long-term macrolides for frequent exacerbations; surgery for localized bronchiectasis; vaccination (influenza, pneumococcus).

11. Lung cancer: an important cause of the cavity and consolidation syndromes

Lung cancer is one of the most important diseases in modern pulmonology, and it may present as various syndromes (consolidation, atelectasis, pleural fluid, cavity).

Table 9.10. *Clinical manifestations of lung cancer*

Group	Signs
Local (endobronchial growth)	Prolonged cough, a change in the character of the cough (in a smoker), hemoptysis (“raspberry jelly”), recurrent pneumonias in the same place, a monophonic wheeze, obstructive atelectasis
Local spread	Chest pain (pleura, ribs), hoarseness (recurrent laryngeal nerve), superior vena cava syndrome, diaphragmatic paralysis, hemorrhagic pleural fluid, difficulty swallowing
Pancoast tumor (lung apex)	Severe pain in the shoulder and arm, atrophy of the arm muscles, Horner’s triad
Metastases	Brain (headache, seizures), bones (pain), liver, adrenal glands
Paraneoplastic syndromes	Clubbing and hypertrophic osteoarthropathy, hypercalcemia, Cushing’s syndrome, hyponatremia, myasthenic syndrome, thrombophlebitis
General	Weight loss, weakness, loss of appetite, anemia, low-grade fever

REMEMBER

The most important risk factor for lung cancer is **smoking** (80–90 % of cases). In a smoker over 40 with a prolonged (more than 3 weeks) cough, hemoptysis or pneumonia recurring in the same place, chest CT and bronchoscopy are mandatory.

12. Parameters for following the patient's condition during curation

Table 9.11. *Indicators of dynamics in respiratory diseases*

Parameter	Positive dynamics	Negative dynamics
Body temperature	Falls and normalizes	Persists or becomes hectic
Respiratory rate and SpO_2	The respiratory rate decreases and SpO_2 rises	Tachypnea increases and SpO_2 falls
Sputum	Decreases and becomes mucoid	Increases, becomes purulent, foul-smelling, blood-stained
Physical signs	Crepitation in place of bronchial breathing, then vesicular breathing	The area of consolidation enlarges, pleural fluid appears
Blood test	Leukocytes and CRP decrease	Leukocytosis increases, the left shift becomes more pronounced
X-ray	The infiltrate resolves (2–6 weeks)	Appearance of a cavity or fluid

MEMORY AID

The **"5 parameters"** followed daily in the progress notes: **temperature, respiratory rate, SpO_2, sputum, auscultation**. Their improvement shows that the treatment is effective.

13. Clinical part

Interpretation of sputum analysis

Table 9.12. *Elements detected in sputum analysis*

Element	Diagnostic significance
Elastic fibers	Destruction of the lung tissue: abscess, tuberculous cavity, cavitating tumor
Dittrich plugs	Yellowish-gray foul-smelling granules: bronchiectasis, lung gangrene, abscess
Curschmann spirals, Charcot–Leyden crystals, eosinophils	Bronchial asthma
Hemosiderin-laden macrophages ("heart failure cells")	Congestion in the pulmonary circulation (mitral stenosis)
Atypical cells	Lung cancer
Mycobacterium tuberculosis (Ziehl–Neelsen stain)	Pulmonary tuberculosis
Numerous leukocytes and bacteria	A suppurative process (bronchitis, abscess, bronchiectasis)

14. Practical skills

Skill: identifying the cavity syndrome in the lung

1. Establish the complaints: the amount (daily), character and odor of the sputum, its separation into layers, its increase in the morning and dependence on position.
2. Inspect the fingers: clubbing, watch-glass nails.
3. Compare vocal fremitus over symmetrical areas.
4. Perform comparative percussion: tympanic note, metallic note, “cracked-pot” sound; perform Wintrich’s test.
5. Auscultate: amphoric breathing, coarse-bubbling consonating moist rales; have the patient cough and observe changes in the rales.
6. Interpret the results of sputum analysis (elastic fibers) and the radiograph (a cavity with a fluid level).

Skill: writing case history 1

1. Fill in the passport section.
2. Divide the complaints into main and additional groups and describe each in detail.
3. Write the history of present illness in chronological order.
4. Write the life history according to the plan (with the smoking index).
5. Status praesens: general inspection, all systems briefly, the respiratory system in detail (with numerical parameters and the topographic percussion table).
6. Combine the signs into syndromes and justify the syndromic and primary diagnosis.
7. Draw up a plan of investigation and interpret the results.
8. Write the progress notes and discharge summary and list the literature used.

Review questions

1. List the main causes of cavity formation in the lung.
2. Under what conditions is a cavity detected by physical methods?
3. What are the palpation, percussion and auscultation signs of the cavity syndrome?
4. How are amphoric breathing and the “cracked-pot” sound produced?
5. What are Wintrich’s and Gerhardt’s signs?
6. What are the causes and predisposing factors of lung abscess?
7. How do the clinical signs differ in periods I and II of an abscess?
8. What elements are found in the sputum in an abscess?
9. What are the causes of bronchiectasis?
10. What are the clinical signs and physical findings of bronchiectasis?
11. Which investigation is considered the “gold standard” for confirming the diagnosis of bronchiectasis?
12. How can lung abscess and bronchiectasis be distinguished?
13. What stages does curation consist of?
14. What main syndromes are identified in respiratory diseases?
15. What errors are common in writing the case history?

Test questions

1. What must the minimum diameter of a cavity be for it to be detected by physical methods?
   A) 0.5 cm
   B) 1 cm
   C) 2 cm
   D) 4–5 cm
   E) 10 cm

2. What breathing is heard over a large cavity communicating with a bronchus?
   A) Vesicular
   B) Diminished vesicular
   C) Amphoric
   D) Saccadic
   E) Harsh

3. What note is found on percussion over a cavity?
   A) Dull
   B) Clear lung (resonant) note
   C) Thigh note
   D) Tympanic
   E) Hyperresonant (box-like)

4. Which fever is characteristic of period I of a lung abscess?
   A) Inverse
   B) No fever
   C) Low-grade (subfebrile)
   D) Undulant
   E) Hectic

5. Which sign is observed when an abscess ruptures into a bronchus?
   A) Vocal fremitus disappears
   B) Dyspnea increases
   C) A large amount of purulent, foul-smelling sputum is suddenly expectorated
   D) The fever rises
   E) The dry cough increases

6. What does the finding of elastic fibers in the sputum indicate?
   A) Destruction of the lung tissue
   B) Heart failure
   C) Bronchitis
   D) Bronchial asthma
   E) Pulmonary edema

7. Which parts of the lungs are most often affected in bronchiectasis?
   A) Only the right lung
   B) The middle lobe
   C) The apices
   D) The lower lobes
   E) The upper lobes

8. When is much sputum expectorated in bronchiectasis?
   A) In the morning and in the drainage position
   B) After meals
   C) In the evening
   D) On physical exertion
   E) Only with fever

9. What is the “gold standard” for confirming the diagnosis of bronchiectasis?
   A) Bronchoscopy
   B) High-resolution CT
   C) Sputum analysis
   D) Plain radiography
   E) Spirometry

10. What is the rise in the pitch of the percussion note when the patient opens the mouth called?
    A) Krönig’s sign
    B) Traube’s sign
    C) Damoiseau’s sign
    D) Wintrich’s sign
    E) Gerhardt’s sign

11. In which disease is clubbing common?
    A) Bronchial asthma
    B) Acute pneumonia
    C) Acute bronchitis
    D) Dry pleurisy
    E) Bronchiectasis

12. What is an abscess that has not healed within 2–3 months called?
    A) A cavern
    B) Gangrene
    C) A chronic abscess
    D) An acute abscess
    E) Empyema

13. Kartagener’s syndrome may be the cause of which disease?
    A) Emphysema
    B) Bronchiectasis
    C) Lobar pneumonia
    D) Bronchial asthma
    E) Pneumothorax

14. What develops in the kidneys as a severe complication of bronchiectasis?
    A) Kidney stones
    B) Pyelonephritis
    C) Polycystic disease
    D) Glomerulonephritis
    E) Amyloidosis

15. Which added sound is heard on auscultation over a cavity?
    A) Pleural friction rub
    B) Coarse-bubbling consonating moist rales
    C) Whistling dry rales
    D) Crepitation
    E) Nonconsonating fine-bubbling rales

Clinical cases

Case 1

A 42-year-old man abuses alcohol. 2 weeks ago he fell asleep in the street while drunk. Afterward, a hectic fever, chills, profuse sweating and pain on the right appeared. Today, after a bout of severe coughing, he expectorated “a mouthful” — 400 ml of purulent, foul-smelling sputum, and his temperature fell to 37.2 °C. In the right infrascapular region vocal fremitus is increased, there is a tympanic note on percussion, and amphoric breathing and coarse-bubbling moist rales are heard.

Questions: What are the syndrome and the diagnosis? In which period is the disease? What is found in the sputum? What is expected on X-ray?

Answer: The syndromes of cavity formation in the lung and of inflammation — an abscess of the right lung, period II (after rupture into a bronchus). Aspiration (intoxication) is the predisposing factor. When left to settle, the sputum separates into layers, and microscopy reveals elastic fibers and many leukocytes. A cavity with a horizontal fluid level is expected on X-ray.

Case 2

A 28-year-old woman has complained since childhood of producing a large amount (up to 200 ml a day) of purulent sputum in the morning and of occasional hemoptysis. At the age of 5 she had severe measles and pneumonia. The sputum increases when she lies on her right side. There is clubbing of the fingers. Over the lower part of the left lung there are medium- and coarse-bubbling moist rales, which decrease after coughing.

Questions: Which disease can be considered? Explain the dependence of the sputum on position. Which investigation confirms the diagnosis?

Answer: Bronchiectasis (measles and pneumonia in childhood, abundant purulent sputum in the morning, hemoptysis, clubbing, moist rales over the lower lobe). Pus from the bronchiectases in the left lower lobe drains out through the draining bronchi when she lies on her right side (the healthy side) — that is why the sputum increases. The diagnosis is confirmed by high-resolution CT.

Case 3

A student wrote the objective examination in the case history as follows: “Lungs normal on percussion and auscultation. Diagnosis: pneumonia.” The patient has fever and cough, and when the teacher examined him, consonating fine-bubbling moist rales were found in the right infrascapular region.

Questions: What errors did the student make? What should the correct entry be?

Answer: 1) the results of the objective examination were written as a general “normal” — precise data must be written for each method (comparative and topographic percussion, the main and added breath sounds, the areas); 2) the pathological signs (consonating moist rales) were not detected — the examination was not performed completely; 3) the diagnosis was not based on syndromes and

contradicts the objective data. The correct entry: describe the changes in the right infrascapular region precisely, identify the syndromes of inflammation and focal consolidation, and then justify the primary diagnosis.

Quick review

Table 9.13. *Key facts on Topic 9*

Question	Answer
Conditions for detecting a cavity	Larger than 4–5 cm, superficial, communicating with a bronchus, air-filled
Cavity syndrome	Increased fremitus, tympanic note, amphoric breathing, coarse-bubbling consonating rales
Abscess, period I	Hectic fever, intoxication, signs of an infiltrate
Abscess, period II	Purulent sputum “a mouthful”, improvement of the condition, signs of a cavity
Sputum of an abscess	Separates into layers, elastic fibers
Bronchiectasis	From childhood, abundant purulent sputum in the morning, hemoptysis, clubbing, lower lobes
Diagnosis of bronchiectasis	High-resolution CT
Wintrich’s sign	The pitch of the percussion note rises when the mouth is opened
Curation	History → examination → syndromes → diagnosis → plan → progress notes → discharge summary

TOPIC 10

Methods of examining patients with cardiovascular diseases. History taking, inspection. Inspection of the heart region and peripheral vessels. Palpation of the heart region

Learning objectives

To teach the student to identify and analyze the main complaints in cardiovascular diseases, the signs characteristic of heart disease on general inspection, the inspection of the heart region and peripheral vessels, and the detection by palpation of the apex beat, the cardiac impulse and the “cat’s purr” (thrill).

By the end of the session, the student should know:

- the projection of the heart and great vessels onto the chest and the main phases of the cardiac cycle;
- the main complaints in cardiovascular diseases (pain, dyspnea, choking attacks, palpitations, edema) and their analysis;
- the differences between chest pain in angina pectoris, myocardial infarction, pericarditis and other causes;
- the NYHA functional classes of chronic heart failure;
- the signs detected on inspection: forced position, cyanosis, edema, neck veins, pulsations;
- the normal features of the apex beat and its changes;
- the types of “cat’s purr” (thrill) and their diagnostic significance.

The student should be able to:

- take the history of a patient with a cardiovascular disease according to a plan;
- inspect the heart region and the peripheral vessels;
- palpate the apex beat and assess its location, area, height, force and resistance;
- detect the cardiac impulse, epigastric pulsation and chest thrills.

1. Brief anatomy and physiology of the heart

The heart is a four-chambered muscular organ located in the anterior inferior part of the mediastinum, two-thirds of it to the left of the midline. The base of the heart faces upward, to the right and backward, and the apex downward, to the left and forward. The anterior surface of the chest is touched mainly by the **right ventricle**; the left ventricle forms the left contour only as a narrow strip.

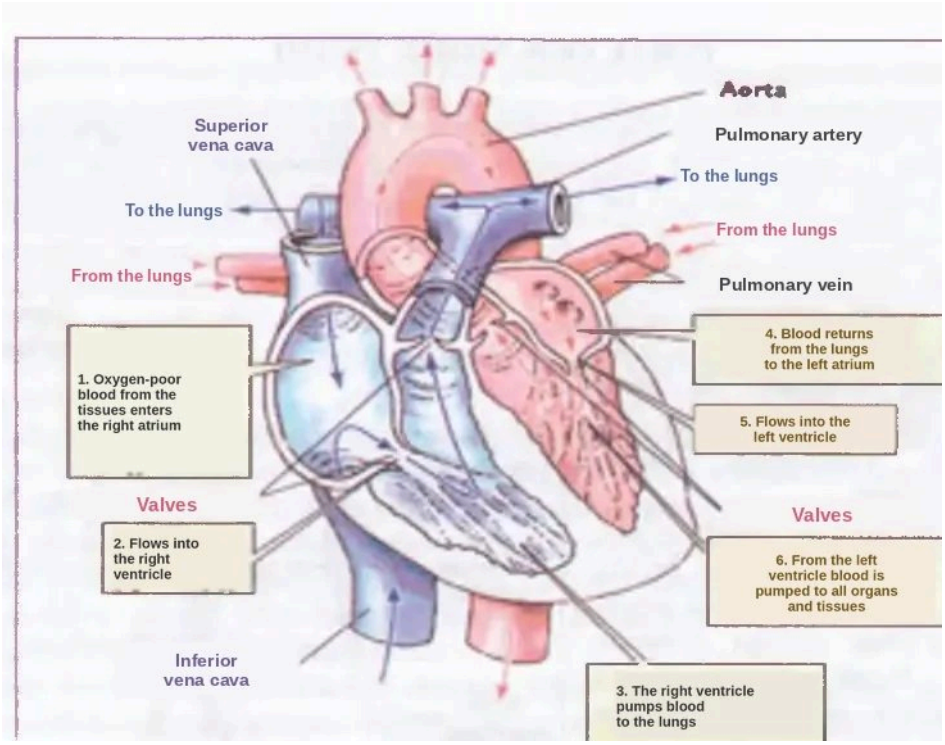


Figure 10.1. *General structure of the heart and the direction of blood flow (Gadayev A. et al., 2012, Fig. 2)*

Table 10.1. *The heart valves*

Valve	Location	Function
Mitral (bicuspid)	Between the left atrium and the left ventricle	Prevents the return of blood to the left atrium during systole
Tricuspid	Between the right atrium and the right ventricle	Prevents the return of blood to the right atrium during systole
Aortic (semilunar)	Between the left ventricle and the aorta	Prevents the return of blood to the left ventricle during diastole
Pulmonary valve (semilunar)	Between the right ventricle and the pulmonary artery	Prevents the return of blood to the right ventricle during diastole

The cardiac cycle consists of systole and diastole. Ventricular systole: the periods of isovolumetric contraction (all valves closed) and ejection; diastole: the periods of isovolumetric relaxation, rapid and slow filling, and atrial systole. At a heart rate of 72 per minute, one cycle lasts about 0.8 seconds (systole 0.3, diastole 0.5 seconds).

> **REMEMBER**
> The atrioventricular (mitral and tricuspid) valves close **at the beginning of systole** (the first heart sound, S1), and the semilunar (aortic and pulmonary) valves close **at the beginning of diastole** (the second heart sound, S2). This rule is the key to understanding heart sounds and murmurs.

2. History taking: the main complaints

2.1. Pain in the heart region and in the chest

Pain is the most important sign of heart disease. It is analyzed in detail according to a plan: site, character, radiation, circumstances of onset, duration, what relieves it, associated symptoms.

Table 10.2. *Differences in chest pain of various causes*

Feature	Angina pectoris	Myocardial infarction	Pericarditis	Aortic dissection	Cardialgia (functional)
Site	Behind the sternum	Retrosternal, more widespread	Retrosternal, in the heart region	Retrosternal, interscapular	In the region of the cardiac apex, at a small point
Character	Squeezing, pressing, burning	Very severe, squeezing, “like a dagger stab”	Sharp, stabbing	Very severe, tearing	Stabbing, smarting, aching
Spread	Left arm, left shoulder blade, neck, lower jaw	The same, more widespread	Neck, left shoulder	Along the back, lower back, abdomen	Radiates little
Onset	On physical or emotional exertion	Even at rest	Worse with breathing, coughing and lying down; lessens when leaning forward	Sudden, against a background of arterial hypertension	After excitement, fatigue
Duration	1–15 minutes (most often 2–5)	More than 20–30 minutes, for hours	For hours, for days	For hours	For hours, days or seconds
Nitroglycerin	Relieves it within 1–3 minutes	No effect	No effect	No effect	No effect
Associated symptoms	Fear, dyspnea	Cold sweat, fear, dyspnea, arrhythmia, hypotension	Fever, pericardial friction rub	Asymmetry of the pulses in the limbs	Nervousness, weakness

MEMORY AID
The three features of anginal pain — **“Squeezes, on Exertion, eased by Nitro”**. If all three are present, it is typical angina; if two, atypical; if one or none, cardialgia (non-anginal pain).

2.2. Dyspnea and choking attacks (cardiac asthma)

In heart diseases, dyspnea most often develops as a result of **left ventricular failure** and congestion in the pulmonary circulation. It appears first on physical exertion, then on little exertion and at rest; it worsens on lying down and lessens on sitting up (**orthopnea**). A sudden choking attack at night is **cardiac asthma** (paroxysmal nocturnal dyspnea); in severe cases, **pulmonary edema** with pink frothy sputum develops.

Table 10.3. *Functional classes of chronic heart failure (NYHA)*

Class	Definition
I	No limitation of physical activity: ordinary exertion does not cause dyspnea, weakness or palpitations
II	Slight limitation of physical activity: no symptoms at rest; dyspnea and weakness on ordinary exertion
III	Marked limitation of physical activity: no symptoms at rest; symptoms appear on less than ordinary exertion
IV	No physical activity can be carried out without symptoms; symptoms are present even at rest

2.3. Palpitations and skipped beats

Palpitations are the sensation of the heartbeat. They are observed in tachycardia, extrasystole, atrial fibrillation, fever, anemia, thyrotoxicosis and neuroses. **Skipped beats** (the feeling that "the heart has stopped") occur in extrasystole and AV blocks. Attacks of palpitations that begin and end suddenly are characteristic of paroxysmal tachycardia.

2.4. Edema

Cardiac edema is a sign of **right ventricular failure** (congestion in the systemic circulation): it first appears in the evening on the feet and ankles and lessens by the morning, then becomes permanent and spreads upward (thighs, lower back, abdominal wall); fluid accumulates in the body cavities (ascites, hydrothorax) — **anasarca**. In bedridden patients it is seen over the sacrum and lower back.



Figure 10.2. *Leg edema (Gadayev A. et al., 2012, Fig. 10)*



Figure 10.3. *Anasarca (same source, Fig. 11)*

2.5. Other complaints

- **Heaviness and pain in the right hypochondrium** — liver enlargement and stretching of its capsule (right ventricular failure).
- **Cough** (dry, often on lying down) and **hemoptysis** — congestion in the pulmonary circulation (mitral stenosis).
- **Headache, dizziness, tinnitus, "floaters" before the eyes** — arterial hypertension; **syncope** — aortic stenosis, severe bradyarrhythmias (Morgagni–Adams–Stokes attack).
- **Leg pain on walking, limping** ("intermittent claudication") — atherosclerosis of the leg arteries.
- **Weakness, easy fatigability, fever** — rheumatism, infective endocarditis, myocarditis.

2.6. Features of the history

In the history of present illness, the following are established: when the pain began, how it has changed (increasing frequency of angina attacks — unstable angina), hospital admissions, ECG and echocardiography results, the medicines being taken. In the life history: **tonsillitis and joint pain in childhood** (rheumatism), risk factors — smoking, arterial hypertension, diabetes mellitus, dyslipidemia, obesity, physical inactivity, psychological stress; **heredity** — premature ischemic heart disease in close relatives (men under 55, women under 65), sudden death.

3. General inspection

Table 10.4. *Signs found on general inspection in cardiovascular diseases*

Feature	Content	Diagnostic significance
Orthopnea position	Sitting with the legs lowered	Left ventricular failure
Sitting leaning forward	The “prayer position”	Exudative pericarditis
Acrocyanosis (peripheral cyanosis)	Lips, tip of the nose, earlobes and fingertips blue, skin cold	Heart failure (slowed blood flow)
Central (diffuse) cyanosis	The whole skin and mucous membranes blue, warm	Congenital “blue” defects, cor pulmonale
Facies mitralis	Bluish-red patches on the cheeks	Mitral stenosis
Pallor	Pale skin	Aortic valve defects; “café au lait” color — infective endocarditis
Edema	In the legs, anasarca	Right ventricular failure
Clubbing	Thickened fingertips	Infective endocarditis, “blue” congenital defects
Xanthomas, xanthelasmas	Yellow spots on the skin and eyelids	Hypercholesterolemia, atherosclerosis
Nutmeg-colored yellowish skin	Yellowish-cyanotic	Long-standing heart failure, “cardiac” liver cirrhosis

4. Inspection of the heart region and the peripheral vessels

4.1. Inspection of the heart region

The patient is inspected lying down or standing, with the light falling from the side. The following are assessed:

1. **The cardiac hump** (gibbus cardiacus) — a bulging of the chest over the heart region; it develops as a result of cardiac enlargement from childhood (congenital and early acquired defects), while the chest is not yet ossified.
2. **The apex beat** — normally visible in more than half of healthy people, on the left in the V intercostal space.
3. **The cardiac impulse** — a diffuse pulsation at the left sternal border in the III–IV intercostal spaces and below the xiphoid process — right ventricular hypertrophy and dilation.
4. **A negative (retracting) apex beat** — indrawing of the area of the apex beat during systole — **adhesive pericarditis** (the heart is adherent to the pericardium and the chest wall).
5. **Pulsations in the region of the base of the heart:** in the II intercostal space to the right of the sternum — dilation or aneurysm of the aorta, aortic regurgitation; in the II intercostal space on the left — dilation of the pulmonary artery (pulmonary hypertension, mitral stenosis); behind the manubrium (in the jugular notch) — dilation of the aortic arch.
6. **Epigastric pulsation** — see section 6.

4.2. Inspection of the peripheral vessels

Arteries:

- **The "carotid dance"** — strong pulsation of the carotid arteries; **de Musset's sign** — nodding of the head in time with the pulse; **Quincke's capillary pulse** — rhythmic alternation of redness and pallor under the nail; **pulsation of the pupils** (Landolfi's sign). These are characteristic of **aortic regurgitation**.
- Tortuous, thickened temporal arteries — atherosclerosis, giant cell arteritis.

Veins:

- **Distension of the neck veins** — right ventricular failure, exudative and constrictive pericarditis, tricuspid valve defects, compression of the superior vena cava.
- **Pulsation of the neck veins:** normally a visible negative venous pulse (corresponding to atrial systole); a **positive venous pulse** (the vein fills at the same time as ventricular systole) — **tricuspid regurgitation**.
- **Kussmaul's sign** — increased distension of the neck veins on inspiration (normally it decreases) — constrictive pericarditis, severe right ventricular failure.
- **Hepatojugular reflux** — increased distension of the neck veins on pressing the right hypochondrium for 10–15 seconds — right ventricular failure.
- Varicose veins of the legs; an infiltrate along a vessel with redness and pain — thrombophlebitis.

> **CAUTION**
> When pulsation is seen in the neck, arterial and venous pulsation must be distinguished. **Venous pulsation** is soft, undulating, biphasic and visible but not palpable; it disappears when the vein is pressed above the clavicle and changes with the patient's position and breathing. **Arterial pulsation** is abrupt, monophasic, easily palpable and does not disappear with pressure.

5. Palpation of the heart region

Palpation detects the apex beat, the cardiac impulse, chest thrills (the "cat's purr"), other pulsations and the pericardial friction rub.

5.1. The apex beat

The apex beat is the striking of the apex of the left ventricle against the chest wall during systole (the contracting heart makes a rotary movement, and its apex rises forward).

Method of palpation: the patient lies on the back (with the head slightly raised) or stands. The physician places the palm of the right hand between the III and VI ribs, with the base of the palm toward the sternum and the fingers toward the axilla (orienting palpation). Once the impulse has been located, its exact site and features are assessed with the fingertips (2–3 fingers, bent). In women, the breast is moved upward and to the right.



Figure 10.4. *Orienting location of the apex beat with the palm (Gadayev A. et al., 2012, Fig. 13)*



Figure 10.5. *Precise assessment with the fingertips (same source, Fig. 14)*

Table 10.5. *Features of the apex beat*

Property	Normal	Changes and causes
Site	On the left in the V intercostal space, 1–1.5 cm medial to the left midclavicular line	To the left — left ventricular dilation (mitral regurgitation, aortic valve defects, hypertension); to the left and downward (to the VI–VII space) — aortic regurgitation; with marked enlargement of the right ventricle — to the left (the left ventricle is pushed aside); fluid or air — to the healthy side; atelectasis, pleural adhesions — to the affected side; with a high diaphragm (ascites, pregnancy, obesity) — upward and to the left; with a low diaphragm (emphysema, asthenic build) — downward and medially
Area (width)	1–2 cm^2 (diameter 1.5–2 cm)	Broad (more than 2 cm, diffuse) — left ventricular dilation; limited — emphysema, obesity, pleural fluid
Pitch	Average	High — strong physical exertion, thyrotoxicosis, fever, left ventricular dilation; low — obesity, emphysema, pericardial fluid
Severity	Average	Increased — left ventricular hypertrophy; decreased — myocardial damage, pericarditis
Resistance	Elastic, moderately firm	Resistant (“dome-shaped”, heaving) — concentric left ventricular hypertrophy (aortic stenosis, arterial hypertension)

REMEMBER

Displacement of the apex beat to the left and its widening are signs of **left ventricular dilation**, and its being resistant is a sign of **hypertrophy**. In aortic regurgitation the left ventricle is both dilated and hypertrophied — the impulse is **displaced to the left and downward, broad, high, forceful and resistant** (“dome-shaped”).

MEMORY AID

The 5 features of the apex beat — **“Location–Area–Height–Force–Resistance”**: “**L**azy **A**nts **H**ate **F**ast **R**unning”.

5.2. The cardiac impulse

The cardiac impulse is a diffuse pulsation felt with the palm at the left sternal border in the III–IV intercostal spaces and in the epigastric region in hypertrophy and dilation of the right ventricle. It is

not detected in a healthy person (except in children and thin people). It is observed in mitral stenosis, pulmonary hypertension, chronic cor pulmonale and congenital defects.



Figure 10.6. *Detecting the cardiac impulse (Gadayev A. et al., 2012, Fig. 15)*

5.3. Chest thrill (the "cat's purr")

The "cat's purr" (frémissement cataire) is a low-frequency vibration felt over the chest wall as a result of turbulence of the blood flow passing through a narrow opening; it resembles the purring of a stroked cat. It is detected by placing the palm over the points of cardiac auscultation.



Figure 10.7. *Detecting a chest thrill (Gadayev A. et al., 2012, Fig. 17)*

Table 10.6. *Types of the "cat's purr"*

Site	Phase	Disease
At the cardiac apex	Diastolic (presystolic)	Mitral stenosis
In the II intercostal space to the right of the sternum (over the aorta), radiating to the neck vessels	Systolic	Aortic stenosis
In the II intercostal space to the left of the sternum	Systolic	Pulmonary artery stenosis
At the left sternal border in the III–IV spaces	Systolic	Ventricular septal defect

Site	Phase	Disease
In the II space to the left of the sternum	Systolic-diastolic (“machinery”)	Patent ductus arteriosus (ductus Botalli)

CLINICAL SIGNIFICANCE

The “cat’s purr” is always a sign of an **organic** defect (it does not occur with functional murmurs). To determine which phase of the cardiac cycle it corresponds to, the pulse in the carotid artery is palpated at the same time: a thrill coinciding with the pulse is systolic, and one coinciding with the pause after the pulse is diastolic.

6. Other pulsations in and around the heart region

Epigastric pulsation is pulsation below the xiphoid process. It arises from three causes:

Table 10.7. *Differentiation of epigastric pulsation*

Cause	Method of determination
Right ventricular hypertrophy	When a finger is inserted under the xiphoid process directed upward (toward the heart), the pulsation strikes the fingertip and increases on inspiration
Pulsation of the abdominal aorta (in thin people, aneurysm)	The pulsation strikes the palmar surface of the finger (from above downward) and decreases on inspiration
Pulsation of the liver	When the liver is held with both hands, the whole liver expands and contracts (true pulsation — tricuspid regurgitation) or only moves forward (transmitted pulsation)

The pericardial friction rub is sometimes detected on palpation at the left sternal border as a rough vibration felt by the hand (dry pericarditis).

7. A detailed analysis of the circulations and the cardiac cycle

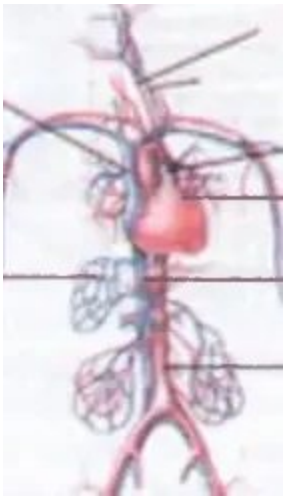


Figure 10.8. *General diagram of the cardiovascular system (Gadayev A. et al., 2012, Fig. 1)*



Figure 10.9. *The systemic and pulmonary circulations (same source, Fig. 3)*

The systemic circulation begins from the left ventricle with the aorta, delivers arterial blood to all the organs and ends via the venae cavae in the right atrium. **The pulmonary circulation** begins from

the right ventricle with the pulmonary artery; in the pulmonary capillaries the blood is saturated with oxygen and drains through the four pulmonary veins into the left atrium. The pulmonary circulation is a low-pressure system: the systolic pressure in the pulmonary artery is normally **15–30 mm Hg** (100–140 in the aorta).

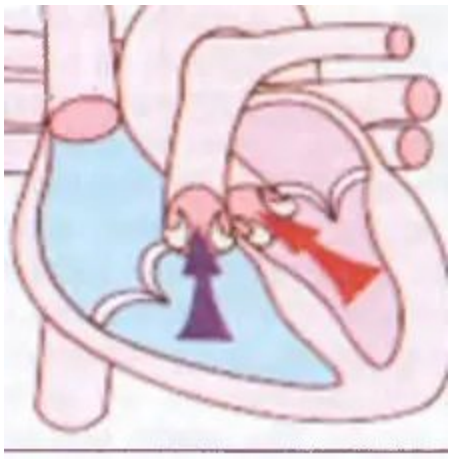


Figure 10.10. *Ventricular diastole (Gadayev A. et al., 2012, Fig. 5)*

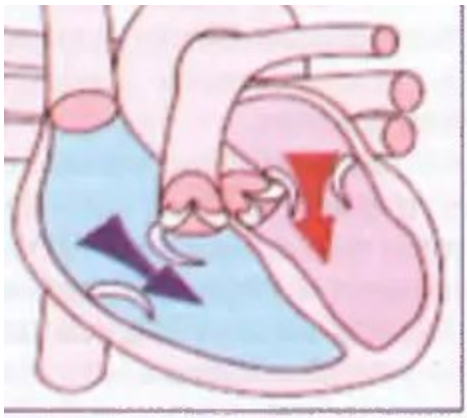


Figure 10.11. *Ventricular systole (same source, Fig. 5)*

Table 10.8. *Phases of the cardiac cycle and their auscultatory expression*

Period	State of the valves	Event	On auscultation
Isovolumetric contraction	All valves closed	The pressure in the ventricle rises rapidly	S1 (closure of the AV valves)
Ejection (rapid and slow)	Semilunar valves open	Blood is ejected into the aorta and pulmonary artery	Systole — the interval between S1 and S2 (systolic murmurs occur here)
Protodiastole and isovolumetric relaxation	All valves closed	The pressure in the ventricle falls	S2 (closure of the semilunar valves)
Rapid filling	AV valves open	Blood flows from the atria into the ventricles	S3, mitral opening snap, protodiastolic murmurs
Slow filling (diastasis)	AV valves open	Slow filling	Mid-diastolic murmurs
Atrial systole	AV valves open	An additional 20–25 % of blood into the ventricles	S4, presystolic murmur

7.1. Projection of the heart and great vessels onto the chest

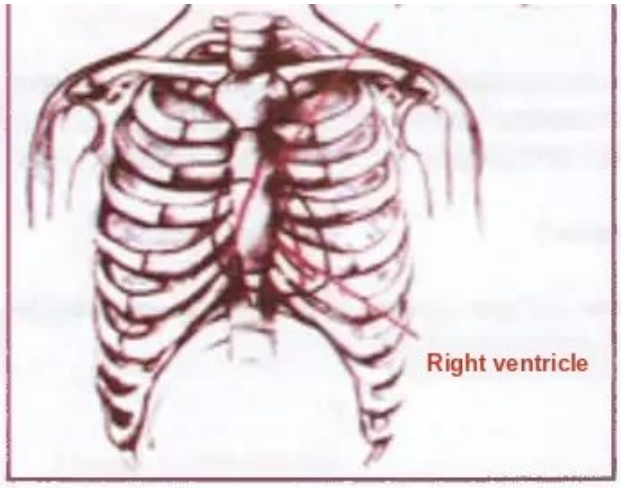


Figure 10.12. *Projection of the ventricles onto the anterior surface of the chest (Gadayev A. et al., 2012, Fig. 6)*

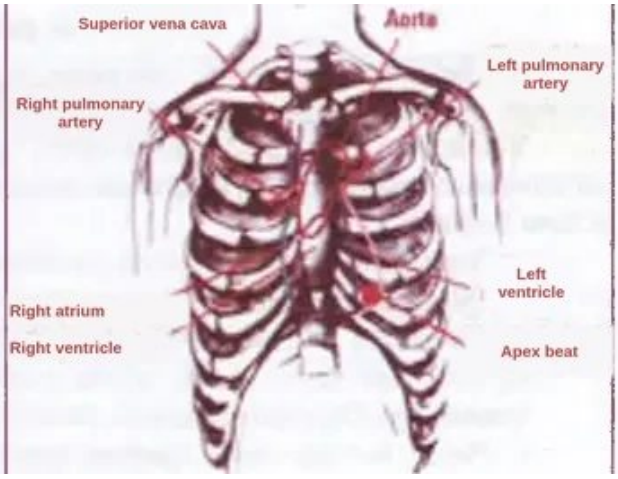


Figure 10.13. *Projection of the atria and great vessels (same source, Fig. 7)*

The right border of the heart (1–1.5 cm lateral to the right sternal edge) is formed by the **right atrium**, the left border by the **left ventricle**, and the upper border by the **left atrial appendage and the conus of the pulmonary artery**. Most of the anterior surface is occupied by the **right ventricle**. The cardiac apex corresponds to the V intercostal space and the base to the level of the III rib.

8. Heart failure: pathogenesis and signs

Heart failure is a clinical syndrome in which, as a result of a disturbance of the structure or function of the heart, the heart cannot meet the oxygen demands of the tissues or does so only at the cost of raised filling pressures (ESC, 2021).

Compensatory mechanisms: the Frank–Starling mechanism (increased filling strengthens contraction), activation of the sympathetic nervous system (tachycardia, vasoconstriction), the renin–angiotensin–aldosterone system (retention of sodium and water, edema), myocardial hypertrophy and remodeling. These mechanisms help at first but later aggravate the disease — modern treatment is aimed precisely at blocking them.

Table 10.9. *Signs of left and right ventricular failure*

Feature	Left ventricular failure (congestion in the pulmonary circulation)	Right ventricular failure (congestion in the systemic circulation)
Main complaints	Dyspnea, orthopnea, nocturnal choking attacks (cardiac asthma), cough, hemoptysis	Leg edema, heaviness in the right hypochondrium, abdominal enlargement, loss of appetite
Inspection	Orthopnea position, acrocyanosis	Distended neck veins, edema, ascites, acrocyanosis
Palpation	Apex beat displaced to the left	Enlarged and tender liver, hepatojugular reflux, cardiac impulse
Auscultation	Moist rales in the lower parts of the lungs, gallop rhythm, accentuation of S2 over the pulmonary artery	The murmur of tricuspid regurgitation at the base of the xiphoid process
Main causes	IHD, myocardial infarction, arterial hypertension, aortic and mitral valve defects, cardiomyopathy	Left ventricular failure (the most common), mitral stenosis, cor pulmonale, tricuspid valve defects, pulmonary embolism



Figure 10.14. *A patient with severe heart failure: the orthopnea position (Gadayev A. et al., 2012, Fig. 122)*

REMEMBER

The stages of chronic heart failure (the classification of N.D. Strazhesko and V.Kh. Vasilenko): **I** — latent, signs only on physical exertion; **IIA** — congestion in one circulation, signs even at rest; **IIB** — congestion in both circulations; **III** — the final, dystrophic stage: irreversible changes in the organs, cachexia. This classification is used together with the NYHA functional classes.

9. Angina pectoris: classification and risk factors

Table 10.10. *Functional classes of angina pectoris (Canadian Cardiovascular Society)*

Class	Definition
I	Ordinary physical activity (walking, climbing stairs) does not cause an attack; attacks occur only with very strenuous, rapid or prolonged exertion
II	Slight limitation of ordinary activity: an attack on walking fast or climbing stairs rapidly, after meals, in the cold, with excitement; on walking more than 200 m or climbing more than one flight of stairs
III	Marked limitation of ordinary activity: an attack on walking 100–200 m or climbing one flight of stairs
IV	An attack with any physical activity or at rest

Table 10.11. *Risk factors for cardiovascular diseases*

Nonmodifiable	Modifiable
Age (men over 45, women over 55)	Smoking
Male sex	Arterial hypertension
Heredity (premature IHD in close relatives)	Dyslipidemia (raised LDL cholesterol)
—	Diabetes mellitus
—	Abdominal obesity
—	Physical inactivity, unhealthy diet
—	Psychoemotional stress, alcohol abuse

CLINICAL SIGNIFICANCE

Signs of **unstable angina**: angina appearing for the first time (within the last 2 months), attacks becoming more frequent, more severe and longer, appearing with less exertion or at rest, and an increasing need for nitroglycerin. This condition belongs to the group of **acute coronary syndromes** and requires urgent hospital admission.

10. Differential diagnosis of syncope and palpitations

Table 10.12. *The main causes of syncope*

Group	Causes	Characteristic signs
Reflex (vasovagal)	Excitement, pain, prolonged standing, a hot room	Preceded by nausea, sweating, dimming of vision; rapid recovery; common in young people
Orthostatic	Drugs, dehydration, blood loss, autonomic neuropathy	On rising quickly from a lying position
Cardiogenic — arrhythmic	Complete AV block (Morgagni–Adams–Stokes attack), ventricular tachycardia, sick sinus syndrome	Sudden, without warning, even when lying down; injuries; a history of heart disease
Cardiogenic — structural	Aortic stenosis, hypertrophic cardiomyopathy, pulmonary embolism, myxoma	During physical exertion

CAUTION

Syncope **during physical exertion**, palpitations preceding it, syncope in the lying position and sudden death of young people in the family are signs of cardiogenic syncope and are life-threatening; such a patient urgently undergoes ECG, echocardiography and Holter monitoring.

11. Heart-related types of cyanosis

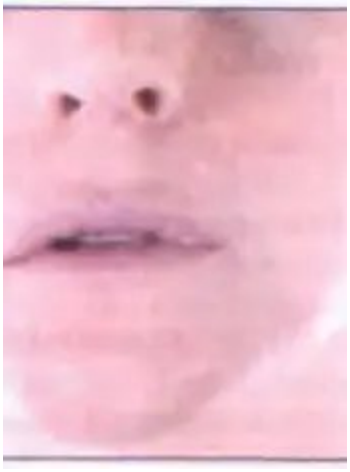


Figure 10.15. *Cyanosis of the lips (Gadayev A. et al., 2012, Fig. 9)*



Figure 10.16. *Cyanosis of the neck and face (same source, Fig. 123)*

Table 10.13. *Differences between central and peripheral cyanosis in heart diseases*

Feature	Central cyanosis	Peripheral cyanosis (acrocyanosis)
Mechanism	The arterial blood is not fully saturated with oxygen (right-to-left shunt, impaired gas exchange in the lungs)	The blood flow slows, and the tissues extract more oxygen
Location	The whole skin, tongue, lips, mucous membranes	Lips, tip of the nose, earlobes, fingers
Skin temperature	Warm	Cold
On massage or warming	Unchanged	Decreases
When oxygen is given	Decreases if the cause is pulmonary; changes little with a shunt	Changes little
Examples	“Blue” congenital defects, cor pulmonale	Heart failure, mitral stenosis, cardiogenic shock

12. Pulsations in the heart region and additional methods of palpating the apex beat



Figure 10.17. *Diagram of the features of the apex beat in left ventricular hypertrophy (Gadayev A. et al., 2012)*



Figure 10.18. *Detecting pulsation in the region of the base of the heart (Gadayev A. et al., 2012, Fig. 17, d)*



Figure 10.19. *Detecting epigastric pulsation (same source, Fig. 17, e)*

Causes of a nonpalpable apex beat: the apex lying behind a rib, obesity, a thick chest wall, the breasts in women, pulmonary emphysema, left-sided exudative pleurisy, exudative pericarditis. In such cases the patient is laid on the **left side** (the impulse shifts 2–3 cm to the left and is felt better) or is examined **leaning forward**.

MEMORY AID

Extracardiac causes of displacement of the apex beat — **"fluid and air push, adhesions and atelectasis pull"**: fluid or air in the pleura — the impulse shifts to the healthy side; pleural adhesions and atelectasis — to the affected side; a high diaphragm (ascites, pregnancy, meteorism) — upward and to the left; a low diaphragm (emphysema, ptosis) — downward and medially.

13. Examination of the peripheral arteries and veins

Signs of atherosclerosis of the leg arteries: intermittent claudication (calf pain after walking a certain distance that passes on stopping), pale, cold, dry skin of the legs with hair loss, thickened nails, trophic ulcers; weakened or absent pulses in the dorsalis pedis and posterior tibial arteries. **The ankle-brachial index** (the ratio of the systolic pressure at the ankle to the systolic pressure in the arm) is normally **1.0–1.4**; **below 0.9** indicates peripheral arterial disease.

Veins: varicose veins of the legs, signs of chronic venous insufficiency (edema, hyperpigmentation, trophic ulcers — most often in the region of the medial malleolus); signs of deep vein thrombosis — swelling and pain of the calf, a calf circumference more than 3 cm larger than that of the healthy leg.

14. "Red flags" and emergencies in chest pain

In a patient presenting with chest pain, **life-threatening** causes must be excluded first.

Table 10.14. *Life-threatening causes of chest pain*

Disease	Main signs	Urgent investigation
Acute coronary syndrome (myocardial infarction)	Squeezing pain lasting more than 20 minutes, sweating, fear, arrhythmia	ECG (within 10 minutes), troponins
Aortic dissection	Sudden "tearing" pain radiating to the back, asymmetry of BP and pulses in the arms	CT angiography
Pulmonary embolism	Sudden dyspnea, tachycardia, pleuritic pain, leg thrombosis	D-dimer, CT angiography
Tension pneumothorax	Sudden pain and dyspnea, absent breath sounds on one side, hypotension	Clinical diagnosis, needle decompression
Cardiac tamponade	Dyspnea, hypotension, distended neck veins, muffled heart sounds (Beck's triad), paradoxical pulse	Echocardiography
Esophageal rupture	Severe pain after vomiting, subcutaneous emphysema	CT, contrast X-ray

REMEMBER

Beck's triad (cardiac tamponade): **arterial hypotension + distended neck veins + muffled heart sounds**. Suspicion of **aortic dissection** — a difference in systolic BP between the two arms of **more than 20 mm Hg** and sudden severe back pain.

15. Investigations confirming heart failure

Table 10.15. *Additional investigations in the diagnosis of heart failure*

Examination	Significance
Natriuretic peptides (BNP, NT-proBNP)	To exclude chronic HF: if NT-proBNP is below 125 pg/ml and BNP below 35 pg/ml, HF is very unlikely

Examination	Significance
Echocardiography	Ejection fraction, chamber size, valves, diastolic function, pulmonary artery pressure — the main confirmation
ECG	With a normal ECG, HF is unlikely; signs of previous infarction, hypertrophy, arrhythmias, blocks
Chest X-ray	Cardiomegaly, pulmonary venous congestion, Kerley lines, pleural fluid
Blood tests	Anemia, kidney function, electrolytes, thyroid hormones

16. Assessment of palpitations

Table 10.16. *Types of palpitations*

The patient's description	Probable cause
"My heart stopped, then gave a strong beat"	Extrasystole (a premature beat and a compensatory pause)
Fast, regular beating that starts and stops suddenly; stops with vagal maneuvers	Paroxysmal supraventricular tachycardia
Fast, irregular beating	Atrial fibrillation
Regular beating that speeds up gradually and slows down gradually	Sinus tachycardia (excitement, fever, anemia, thyrotoxicosis)
"Pulsation" in the neck, regular fast beating	AV nodal reentrant tachycardia

CAUTION

If palpitations are accompanied by **syncope, chest pain, dyspnea** or structural heart disease, this may be a sign of a dangerous arrhythmia (ventricular tachycardia) and requires urgent investigation.

17. Differential diagnosis of the edematous syndrome

Table 10.17. *Distinguishing the cardiac, renal, hepatic and other causes of edema*

Feature	Heart	Renal (nephrotic)	Liver cirrhosis	Hypothyroidism
Location	Legs, from below upward	Face, eyelids, then the whole body	Ascites predominates, less in the legs	Face, limbs, firm
Time	In the evening	Morning	Continuous	Continuous
Skin	Cyanotic, cold	Pale, warm	Yellowish, "spider angiomas"	Dry, yellowish, cold
Pitting on pressure	Present	Present (soft)	Present	Absent
Associated symptoms	Dyspnea, distended neck veins, hepatomegaly	Proteinuria (more than 3.5 g/day), hypoalbuminemia	Jaundice, splenomegaly, dilated abdominal veins	Bradycardia, hoarseness, raised TSH

18. Clinical part

Sample case history entry

Cardiovascular system. The patient is in the orthopnea position. Acrocyanosis. Firm, cyanotic, cold edema of the lower legs and ankles. The neck veins are distended and visible even in the upright position; the hepatojugular reflux is positive. On inspection, the heart region is unchanged, and the apex beat is not visible. On palpation, the apex beat is in the VI intercostal space 2 cm lateral to the left midclavicular line, with an area of 3 cm^2, high, forceful and resistant. The cardiac impulse and epigastric pulsation are not detected. There is no “cat’s purr”.

Conclusion: signs of left ventricular hypertrophy and dilation; heart failure with congestion in the systemic circulation.

19. Practical skills

Skill: palpation of the apex beat

1. Undress the patient to the waist, lying on the back (with the head raised by 30°) or standing.
2. Place the palm of the right hand over the region of the III–VI ribs, with the base of the palm toward the sternum and the fingers toward the axillary line.
3. Place the tips of 2–3 fingers (bent) perpendicularly over the place where the impulse is felt and find its exact site — the most lateral and lowest point.
4. Count the intercostal space (starting from the II rib — next to the sternal angle) and measure the distance from the midclavicular line.
5. Assess its area, height, force and resistance.
6. If the impulse is not detected, turn the patient onto the left side (the impulse shifts 2–3 cm to the left) and examine again; ask the patient to breathe out deeply and hold the breath.
7. Record the result.

Skill: detecting the “cat’s purr”

1. Place the palm in turn over the cardiac apex, the II intercostal space to the right and left of the sternum, and the left sternal border (III–IV space).
2. At the same time, palpate the pulse in the carotid artery with the other hand.
3. Determine the presence, site and phase (systolic or diastolic) of the thrill.
4. To feel the thrill better in mitral stenosis, lay the patient on the left side.

Review questions

1. Which parts of the heart are in contact with the anterior surface of the chest?
2. What phases does the cardiac cycle consist of? With the closure of which valves are S1 and S2 associated?
3. How do you distinguish the pain of angina pectoris from that of myocardial infarction?
4. What are the features of the pain of pericarditis?
5. How does cardiac asthma develop, and how does it manifest itself?
6. State the NYHA functional classes of heart failure.
7. What are the features of cardiac edema?
8. What signs are found on general inspection in heart diseases?

9. What is the cardiac hump, and when does it form?
10. In which disease is a negative apex beat observed?
11. What signs are observed in the peripheral vessels in aortic regurgitation?
12. What do a positive venous pulse and Kussmaul’s sign indicate?
13. What are the normal features of the apex beat, and in which conditions do they change?
14. What is the “cat’s purr”, and what types are there?
15. How do you distinguish the causes of epigastric pulsation?

Test questions

1. Which part of the heart is most in contact with the anterior surface of the chest?
   A) The aorta
   B) The right atrium
   C) The left atrium
   D) The left ventricle
   E) The right ventricle

2. How soon after taking nitroglycerin does the pain of angina usually stop?
   A) Within 1–3 minutes
   B) Within 15–20 minutes
   C) Within 1 hour
   D) It does not stop
   E) Within 30 minutes

3. Which condition is characterized by severe retrosternal pain lasting more than 30 minutes, not relieved by nitroglycerin, with cold sweat and fear?
   A) Myocardial infarction
   B) Dry pleurisy
   C) Intercostal neuralgia
   D) Cardialgia
   E) Angina pectoris

4. Which disease is characterized by retrosternal pain that lessens on leaning forward and increases with breathing and coughing?
   A) Pericarditis
   B) Mitral stenosis
   C) Aortic dissection
   D) Myocardial infarction
   E) Angina pectoris

5. According to NYHA, which functional class is it if symptoms appear on less than ordinary exertion?
   A) I
   B) II
   C) III
   D) IV
   E) The class cannot be determined

6. Where is the apex beat normally found?
   A) In the IV intercostal space on the midclavicular line
   B) In the V intercostal space 1–1.5 cm medial to the left midclavicular line
   C) In the VI intercostal space on the anterior axillary line
   D) In the V intercostal space at the sternal border
   E) In the III intercostal space on the left

7. What is the normal area of the apex beat?
   A) 0.5 cm^2
   B) 1–2 cm^2
   C) 4–5 cm^2
   D) 6 cm^2
   E) 10 cm^2

8. Which valvular defect is characterized by displacement of the apex beat to the left and downward (to the VI–VII space), with a broad, high and resistant impulse?
   A) Tricuspid regurgitation
   B) Pulmonary artery stenosis
   C) Aortic regurgitation
   D) Mitral stenosis
   E) Exudative pericarditis

9. In which condition is a negative (retracting) apex beat observed?
   A) Emphysema
   B) Mitral stenosis
   C) Adhesive pericarditis
   D) Aortic stenosis
   E) Exudative pericarditis

10. Which defect is a diastolic “cat’s purr” at the cardiac apex characteristic of?
    A) Mitral regurgitation
    B) Mitral stenosis
    C) Tricuspid regurgitation
    D) Aortic regurgitation
    E) Aortic stenosis

11. Which defect is a systolic “cat’s purr” in the II intercostal space to the right of the sternum characteristic of?
    A) Pulmonary artery stenosis
    B) Mitral stenosis
    C) Mitral regurgitation
    D) Aortic stenosis
    E) Patent ductus arteriosus

12. Which defect are the “carotid dance”, de Musset’s sign and Quincke’s capillary pulse characteristic of?
    A) Tricuspid stenosis
    B) Mitral regurgitation
    C) Aortic stenosis
    D) Mitral stenosis
    E) Aortic regurgitation

13. In which defect is a positive venous pulse observed?
    A) Aortic stenosis
    B) Mitral stenosis
    C) Mitral regurgitation
    D) Tricuspid regurgitation
    E) Aortic regurgitation

14. Which condition is the cardiac impulse a sign of?
    A) Left ventricular hypertrophy
    B) Right ventricular hypertrophy and dilation
    C) Left atrial enlargement
    D) Aortic aneurysm
    E) Pericarditis

15. If epigastric pulsation strikes the fingertip (from above) and increases on inspiration, what is its cause?
    A) Stomach
    B) Pulsation of the liver
    C) Normal state
    D) Right ventricular hypertrophy
    E) Abdominal aorta

Clinical cases

Case 1

A 56-year-old man has complained for 2 years of squeezing retrosternal pain appearing on walking fast and climbing stairs. The pain radiates to the left arm, lasts 3–5 minutes and passes 1–2 minutes after he stops to rest or takes nitroglycerin. This morning severe pain appeared at rest; it has lasted for 40 minutes, nitroglycerin has not helped, the patient is drenched in cold sweat and has a feeling of fear.

Questions: Interpret the pain over the 2 years and today's pain. What urgent investigation is needed?

Answer: The pain over the 2 years is characteristic of typical exertional angina (squeezing, on exertion, relieved by nitroglycerin). Today's pain began at rest, has lasted more than 20–30 minutes, does not respond to nitroglycerin and is accompanied by cold sweat and fear — acute coronary syndrome (myocardial infarction) is suspected. An ECG (within 10 minutes) and cardiac troponins are performed immediately, and the patient is admitted to the cardiac intensive care unit.

Case 2

In a 48-year-old woman, the neck veins are distended even in the upright position, and they distend further on inspiration. There is leg edema and an enlarged liver. On inspection of the heart region, indrawing of the area of the apex beat is observed during systole.

Questions: What are the signs found called, and which disease are they characteristic of?

Answer: Increased distension of the neck veins on inspiration is Kussmaul's sign; indrawing of the area of the apex beat during systole is a negative apex beat. Together with the signs of congestion in the systemic circulation, these are characteristic of constrictive (adhesive) pericarditis: the pericardium thickens and restricts the filling of the heart in diastole.

Case 3

A 25-year-old woman has dyspnea and hemoptysis. She had rheumatism in childhood. There are bluish-red patches on her cheeks. On palpation with the palm at the cardiac apex, a thrill is felt in diastole, and there is a diffuse pulsation at the left sternal border in the III–IV spaces.

Questions: What signs have been found? Which defect are they characteristic of? Why has the cardiac impulse appeared?

Answer: Facies mitralis, a diastolic "cat's purr" at the apex and a cardiac impulse have been found. These signs are characteristic of mitral stenosis. When the mitral orifice narrows, the pressure in the left atrium rises, and congestion in the pulmonary circulation and pulmonary hypertension develop; the right ventricle hypertrophies and dilates to overcome the resistance — that is why the cardiac impulse appears.

Quick review

Table 10.18. *Key facts on Topic 10*

Question	Answer
Angina pectoris	Squeezing, on exertion, 1–15 minutes, relieved by nitroglycerin
Myocardial infarction	More than 20–30 minutes, at rest, nitroglycerin has no effect
Cardiac asthma	Choking at night, orthopnea, moist rales in the lungs
NYHA	I — no limitation; II — slight; III — marked; IV — symptoms at rest
Cardiac edema	In the evening, in the legs, cyanotic, from below upward
Normal apex beat	V intercostal space, 1–1.5 cm medial to the midclavicular line, 1–2 cm^2
Resistant impulse	Left ventricular hypertrophy (aortic stenosis, hypertension)
Negative impulse	Adhesive pericarditis
Cardiac impulse	Right ventricular hypertrophy
Diastolic "cat's purr" at the apex	Mitral stenosis
Systolic "cat's purr" over the aorta	Aortic stenosis
Positive venous pulse	Tricuspid regurgitation
"Carotid dance", de Musset	Aortic regurgitation

TOPIC 11

Percussion of the heart: the borders of relative and absolute dullness. Configuration of the heart. The pulse and its features. Analysis of radiographs

Learning objectives

To teach the student to determine the borders of relative and absolute cardiac dullness and the width of the vascular bundle in a healthy person and in cardiovascular diseases, to assess the configuration of the heart, to examine the arterial pulse and interpret its features, and to analyze the cardiac shadow on a chest radiograph.

By the end of the session, the student should know:

- the rules and sequence of cardiac percussion;
- the normal borders of relative and absolute cardiac dullness and the width of the vascular bundle;
- the causes of displacement of the cardiac borders (cardiac and noncardiac);
- the normal, mitral, aortic and trapezoid configurations of the heart;
- the features of the arterial pulse (symmetry, rhythm, rate, filling, tension, amplitude, shape) and their changes;
- the structures forming the cardiac contours on a frontal radiograph and the cardiothoracic ratio.

The student should be able to:

- determine the borders of relative and absolute cardiac dullness and draw the configuration of the heart;
- examine the pulse in the radial artery and assess all its features;
- detect the pulse in other arteries (carotid, femoral, popliteal, dorsalis pedis);
- assess the shape of the cardiac shadow on a radiograph and calculate the cardiothoracic ratio.

1. Percussion of the heart

Part of the heart is covered by the lungs, and part is in direct contact with the chest wall. That is why two kinds of dullness are distinguished:

- **Relative cardiac dullness** is the **true projection** of the heart onto the anterior surface of the chest (including the parts covered by the lungs). It is determined by percussion of moderate strength, by the change from the clear lung note to an impaired note.
- **Absolute cardiac dullness** is the part of the heart not covered by the lungs and in direct contact with the chest wall (mainly the right ventricle). It is determined by **the quietest** percussion, by the change from an impaired note to a completely dull note.

1.1. Rules of percussion

1. The patient stands or lies on the back (always in the same position, so that the results can be compared).
2. The pleximeter finger is placed **parallel** to the border being determined.
3. Percussion proceeds from the resonant note toward the dull note.
4. The border is marked along the edge of the pleximeter facing the resonant note.

5. Sequence: the right border is determined first, then the upper border, then the left border.

1.2. Determining the borders of relative cardiac dullness

The right border. First the lower border of the right lung is determined along the midclavicular line (normally the VI rib). Then the plexímeter is moved one intercostal space higher — to the **IV intercostal space** — and percussion proceeds medially with the plexímeter parallel to the sternum.



Figure 11.1. *Determining the right border of relative cardiac dullness (Gadayev A. et al., 2012, Fig. 18)*

The left border. The apex beat is found first. Percussion is performed **in the V intercostal space**, starting from the anterior axillary line, with the plexímeter placed parallel to the border (vertically), and proceeds medially. If the apex beat cannot be found, it starts in the V space from the mid-axillary line.

The upper border. The plexímeter is placed 1 cm lateral to the left sternal edge (along the left parasternal line), parallel to the ribs, and moved downward from the I intercostal space.

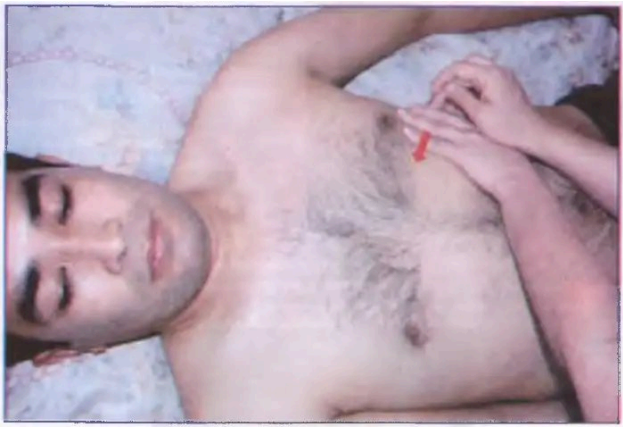


Figure 11.2. *Determining the left border (Gadayev A. et al., 2012, Fig. 21)*



Figure 11.3. *The left border in a normal person (same source, Fig. 22)*

Table 11.1. *Borders of relative and absolute cardiac dullness in a healthy person*

Border	Relative dullness	The part forming it	Absolute dullness
Right	In the IV intercostal space, 1–1.5 cm lateral to the right sternal edge	The right atrium	Along the left sternal edge
High	The III rib along the left parasternal line	The left atrial appendage and the conus of the pulmonary artery	The IV rib
Left	In the V intercostal space, 1–1.5 cm medial to the left midclavicular line (coincides with the apex beat)	The left ventricle	1–2 cm medial to the left border of relative dullness

REMEMBER

Normal borders of relative cardiac dullness: **right — IV intercostal space, 1–1.5 cm lateral to the right sternal edge; upper — III rib; left — V intercostal space, 1–1.5 cm medial to the midclavicular line**. The transverse diameter of the heart is **11–13 cm** (the right part 3–4 cm, the left part 8–9 cm).

MEMORY AID

Relative dullness: **“4-3-5”** — the right border in the IV space, the upper border at the III rib, the left border in the V space. Absolute dullness: **“Sternum-4”** — the right border is the left sternal edge, the upper border the IV rib.

1.3. Determining the width of the vascular bundle

Percussion is performed **in the II intercostal space** from the right and left midclavicular lines toward the sternum. Normally the borders of the vascular bundle coincide with the sternal edges, and its width is **5–6 cm**. It is formed on the right by the aorta and the superior vena cava and on the left by the pulmonary artery. It widens in atherosclerosis and aneurysm of the aorta, aortic regurgitation and arterial hypertension.

1.4. Changes in the cardiac borders

Table 11.2. *Causes of displacement of the borders of relative cardiac dullness*

Displacement	Related to enlargement of the cardiac chambers	Extracardiac causes
Right border to the right	Dilation of the right atrium and right ventricle (tricuspid regurgitation, mitral stenosis, cor pulmonale)	Fluid or air in the pleura on the left, adhesions or atelectasis on the right
Upper border upward	Enlargement of the left atrium (mitral stenosis, mitral regurgitation)	—
Left border to the left	Left ventricular dilation (aortic valve defects, mitral regurgitation, arterial hypertension, cardiomyopathies); also with marked enlargement of the right ventricle	Fluid or air in the pleura on the right; a high diaphragm (the heart lies horizontally)
In all directions	Diffuse enlargement (“ox heart” — cor bovinum), exudative pericarditis	—

Enlargement of the absolute cardiac dullness — right ventricular dilation, pericardial fluid, shrinkage of the lung edges; **its reduction or disappearance** — pulmonary emphysema, left-sided pneumothorax, an attack of bronchial asthma.

2. Configuration of the heart

Once the cardiac borders have been determined, the contour of the heart is drawn along them. **The cardiac waist** is the concavity on the left contour between the vascular bundle and the left ventricle (at the level of the III rib); it is formed by the pulmonary artery and the left atrial appendage.

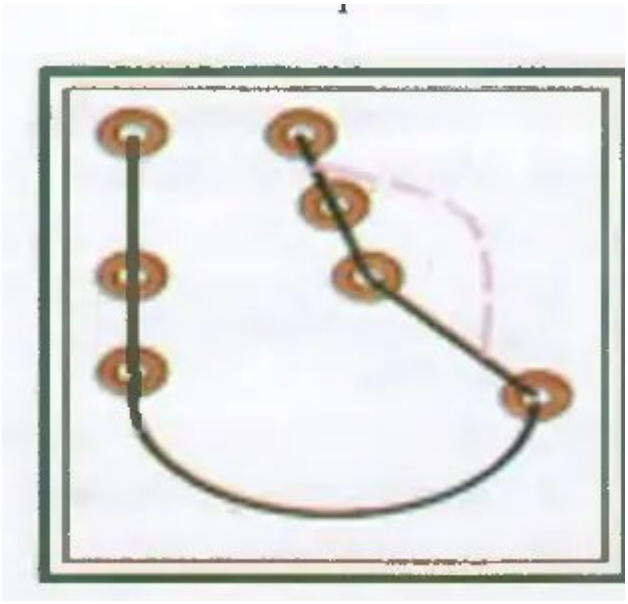


Figure 11.4. *Normal configuration: the cardiac waist forms an obtuse angle (Gadayev A. et al., 2012, Fig. 28)*

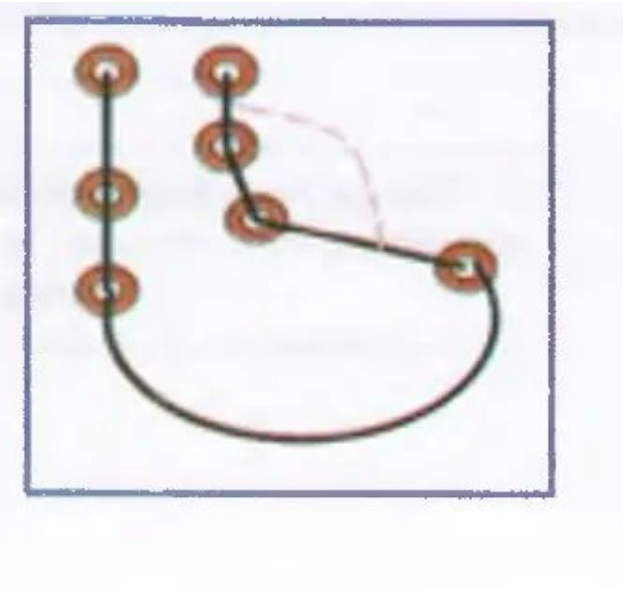


Figure 11.6. *Aortic configuration: the cardiac waist is deepened (same source)*

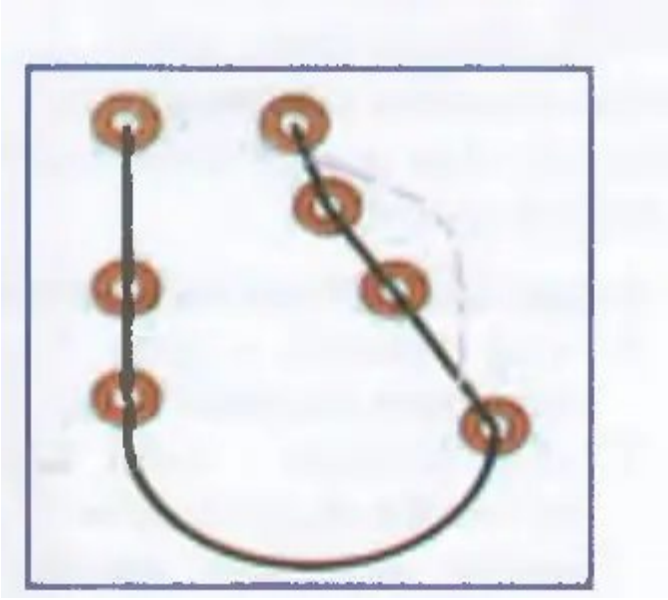


Figure 11.5. *Mitral configuration: the cardiac waist is smoothed out (same source)*

Table 11.3. *Configurations of the heart*

Configuration	Signs	Cause	Diseases
Normal	The cardiac waist is moderately expressed, an obtuse angle	—	Healthy person
Mitral	The cardiac waist is smoothed out or bulges, the upper border is displaced upward and the right border to the right	Enlargement of the left atrium (and pulmonary artery), right ventricular dilation	Mitral stenosis, mitral regurgitation

Configuration	Signs	Cause	Diseases
Aortic (“sitting duck”, “boot” shape)	The cardiac waist is deepened (close to a right angle), the left border is displaced to the left and downward	Left ventricular enlargement	Aortic regurgitation, aortic stenosis, arterial hypertension
Trapezoid (triangular)	The borders are widened to both sides, the base is broad below	Pericardial fluid	Exudative pericarditis
“Ox heart”	All the borders markedly widened	Dilation of all the chambers	Combined valvular defects, dilated cardiomyopathy

CAUTION

Percussion assesses the size of the heart only approximately: the error is large in obesity, emphysema and chest deformities. **Echocardiography** and radiography are used to assess the size of the cardiac chambers precisely.

3. The arterial pulse

The arterial pulse is the rhythmic oscillation of the arterial walls resulting from the ejection of blood into the aorta during cardiac systole.

3.1. Examining the pulse in the radial artery

The pulse is palpated between the styloid process of the radius and the tendon of the inner muscle of the forearm, on both arms at the same time. The physician holds the patient’s hand from the palmar side, places the index, middle and ring fingers over the artery, with the thumb on the back of the forearm.



Figure 11.7. *Feeling the pulse on both arms at the same time (Gadayev A. et al., 2012, Fig. 48)*



Figure 11.8. *Assessing the pulse in the radial artery (same source, Fig. 49)*

Table 11.4. *Features of the pulse*

Property	Normal	Changes	Causes
Symmetry (equality)	The same on both arms (pulsus aequalis)	Pulsus differens — smaller and delayed on one side	Compression of the subclavian artery by the enlarged left atrium in mitral stenosis (Popov–Savelyev sign), aneurysm of the aortic arch, arteritis, tumors
Rhythm	Regular (regularis)	Irregular (irregularis)	Extrasystole, atrial fibrillation, blocks; respiratory arrhythmia (in young people, a variant of normal)

Property	Normal	Changes	Causes
Rate	60–80 (60–90) per minute	Tachycardia (more than 90), bradycardia (less than 60)	Fever, anemia, thyrotoxicosis, heart failure / athletes, hypothyroidism, AV block, jaundice
Pulse deficit	Absent (pulse = heart rate)	The heart rate is higher than the pulse rate	Atrial fibrillation, frequent extrasystole (some contractions do not eject enough blood into the aorta)
Filling	Well filled (plenus)	Poorly filled (vacuus)	Blood loss, shock, collapse, aortic stenosis
Tension	Average	Tense, hard (durus) / soft (mollis)	Arterial hypertension / hypotension
Size	Average	Large (magnus) / small (parvus), thready (filiformis)	Aortic regurgitation / aortic stenosis, mitral stenosis; shock
Shape (speed)	Average	Rapid and high (celer et altus) / slow and small (tardus et parvus)	Aortic regurgitation / aortic stenosis

REMEMBER

In aortic regurgitation the pulse is rapid, high and large (pulsus celer, altus, magnus); **in aortic stenosis** it is slow and small (pulsus tardus, parvus). **Pulse deficit** is a typical sign of atrial fibrillation; to detect it, one person counts the heartbeats on auscultation and another counts the pulse at the same time for one minute.

Table 11.5. *Special types of pulse*

Type	Definition	Cause
Paradoxical pulse (pulsus paradoxus)	On inspiration the pulse waves become markedly smaller or disappear (the systolic pressure falls by more than 10 mm Hg)	Cardiac tamponade, exudative and constrictive pericarditis, a severe attack of asthma
Alternating pulse (pulsus alternans)	Large and small pulse waves alternate rhythmically	Severe damage to the left ventricular myocardium
Dicrotic pulse	A small additional wave after each pulse wave	Reduced tone of the peripheral vessels (severe infections, fever)

CAUTION

The normal pulse rate differs between sources: in Russian and Uzbek textbooks it is **60–80** (or 60–90), and in international cardiology guidelines (ESC) **60–100** beats. More than 100 per minute is tachycardia. The manual gives 60–80 as normal, but the range 80–100 should not be regarded as pathological.

3.2. Detecting the pulse in other arteries

Table 11.6. *Sites for detecting the pulse*

Artery	Site of palpation
Temporal	In front of the ear, over the temporal bone
Carotid	At the level of the thyroid cartilage, at the inner edge of the sternocleidomastoid muscle (one side at a time, never press both at once!)
Subclavian	In the supraclavicular fossa

Artery	Site of palpation
Axillary	In the axillary fossa, over the head of the humerus
Brachial	On the inner surface of the arm, at the inner edge of the biceps muscle
Femoral	At the middle of the inguinal ligament, below the inguinal fold
Popliteal	In the popliteal fossa, with the leg slightly bent at the knee
Dorsalis pedis artery	On the dorsum of the foot, between the I and II metatarsal bones
Posterior tibial	Behind the medial malleolus

CLINICAL SIGNIFICANCE

A weakened or absent pulse in the leg arteries (femoral, popliteal, dorsalis pedis) is a sign of **atherosclerosis obliterans of the leg arteries**; a pulse and BP markedly lower in the legs than in the arms are characteristic of **coarctation of the aorta**.

4. Analysis of the cardiac shadow on a chest radiograph

The heart and great vessels are examined in the frontal (posteroanterior), left lateral and oblique projections.



Figure 11.9. *The right and left contours of the heart and vessels on a frontal radiograph (Gadayev A. et al., 2012, Fig. 90)*

Table 11.7. *Contours (arcs) of the cardiac shadow in the frontal projection*

Contour	Arcs (from top to bottom)
Right contour (2 arcs)	1 — superior vena cava and ascending aorta; 2 — right atrium
Left contour (4 arcs)	1 — aortic arch; 2 — pulmonary trunk; 3 — left atrial appendage; 4 — left ventricle

REMEMBER

The cardiothoracic ratio (CTR) is the ratio of the greatest transverse diameter of the cardiac shadow to the greatest internal transverse diameter of the chest. Normally it **does not exceed 0.5** (50 %); more than 0.5 is cardiomegaly.

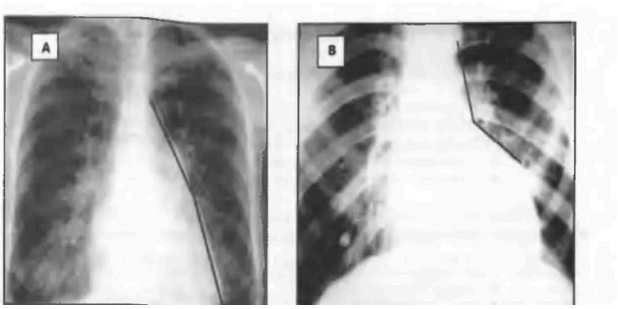


Figure 11.10. *Configurations of the heart on a radiograph: A — mitral; B — aortic (Gadayev A. et al., 2012, Fig. 93)*

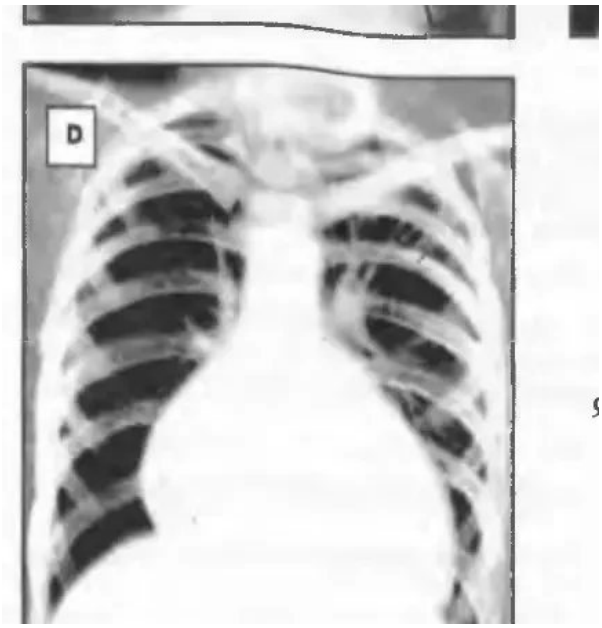


Figure 11.11. *Spherical (trapezoid) cardiac shadow (Gadayev A. et al., 2012, Fig. 93, D)*

Table 11.8. *Radiographic signs of enlargement of the cardiac chambers*

Enlarged chamber	Radiographic sign
The left ventricle	The 4th arc of the left contour is lengthened and extended to the left and downward, the apex is rounded, the cardiac waist is deepened (“aortic shape”)
The left atrium	The 3rd arc of the left contour bulges, the cardiac waist is smoothed out (“mitral shape”); in the lateral projection the contrast-filled esophagus is displaced backward along an arc of small radius
The right ventricle	The heart is widened to the right and to the left; in the lateral projection the retrosternal space narrows
The right atrium	The lower arc of the right contour bulges to the right
The aorta	The 1st arc of the right contour and the aortic arch (1st arc of the left contour) are widened and lengthened
Pulmonary artery	The 2nd arc of the left contour bulges (pulmonary hypertension)

5. Technique of cardiac percussion: a step-by-step illustration

5.1. The right and upper borders of relative dullness



Figure 11.13. *The normal right border (same source, Fig. 20)*



Figure 11.12. *Moving the pleximeter parallel to the sternum when determining the right border (Gadayev A. et al., 2012, Fig. 19)*



Figure 11.15. *The normal upper border — the III rib (same source)*



Figure 11.14. *Determining the upper border (Gadayev A. et al., 2012, Fig. 23)*

5.2. The transverse diameter of the heart



Figure 11.16. *Measuring the transverse diameter of the heart (Gadayev A. et al., 2012, Fig. 24)*

The transverse diameter of the heart is the sum of the distances from the right and left borders of relative dullness to the anterior midline. Normally the right part is **3–4 cm**, the left part **8–9 cm**, and the total transverse diameter **11–13 cm**. The longitudinal diameter (from the point of the right border at the level of the III–IV rib to the apex) is **13–14 cm**.

5.3. The vascular bundle



Figure 11.18. *The normal width of the vascular bundle is 5–6 cm (same source, Fig. 27)*



Figure 11.17. *Determining the borders of the vascular bundle (Gadayev A. et al., 2012, Fig. 26)*

5.4. Absolute cardiac dullness



Figure 11.20. *Determining the upper border of absolute dullness (same source, Fig. 32)*



Figure 11.19. *Determining the right border of absolute dullness (Gadayev A. et al., 2012, Fig. 29)*



Figure 11.21. *The normal upper border of absolute dullness — the IV rib (Gadayev A. et al., 2012, Fig. 34)*



Figure 11.22. *The normal left border of absolute dullness (same source, Fig. 35)*

Absolute dullness is determined by **the quietest** (threshold) percussion, percussing from the border of relative dullness toward the center of the heart; the point where the impaired note changes to a completely dull (flat) note is marked. Normally the area of absolute dullness corresponds to the right ventricle of the heart.

> **CLINICAL SIGNIFICANCE**
> **Equalization of the absolute cardiac dullness with the relative dullness** (that is, the two borders almost coincide) is seen when **fluid accumulates in the pericardium**: the fluid pushes the lung edges aside, and the whole heart region comes into direct contact with the chest wall. This sign helps to distinguish exudative pericarditis from myocardial dilation.

6. Arterial pressure and its measurement

Arterial pressure (BP) is the pressure exerted by the blood on the walls of the arteries. **Systolic** (upper) pressure is the highest pressure during ventricular systole; **diastolic** (lower) pressure is the lowest pressure in diastole; **pulse pressure** is the difference between them (normally 40–50 mm Hg); **mean arterial pressure** ≈ diastolic + 1/3 of the pulse pressure.



Figure 11.25. *Technique of measuring arterial blood pressure (same source, Fig. 64)*



Figure 11.24. *Correct placement of the cuff (same source, Fig. 61)*

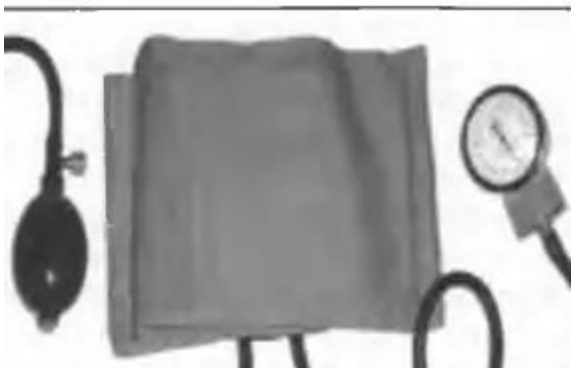


Figure 11.23. *Sphygmomanometer (Gadayev A. et al., 2012, Fig. 59)*

6.1. Rules for measuring BP by the Korotkoff method

1. The patient sits quietly for 5 minutes; for 30 minutes beforehand he does not smoke, drink coffee or exert himself physically; the bladder is empty.
2. The patient sits leaning back, legs uncrossed, arm resting on a table, the middle of the upper arm **at the level of the heart**.
3. The cuff is wrapped around the upper arm, its lower edge **2–3 cm above** the elbow crease, so that one finger fits between the cuff and the skin. The cuff width should cover **40 %** of the arm circumference, and the bladder length **80 %** (a large cuff for obese patients).
4. First, by the palpatory method, air is pumped in until the pulse in the radial artery disappears; then the pressure is raised a further **20–30 mm Hg** above this level.
5. The membrane of the stethoscope is placed over the brachial artery in the cubital fossa.
6. The air is released at a rate of **2–3 mm Hg per second**.
7. The reading at which **the first Korotkoff sound** appears is the systolic pressure; the reading at which the sounds **disappear completely** (phase V) is the diastolic pressure.
8. The measurement is repeated 2–3 times at intervals of 1–2 minutes, and the average is taken; at the first visit BP is measured on both arms (the difference should not exceed 10–15 mm Hg).

Table 11.9. *Phases of the Korotkoff sounds*

Phase	Description	Significance
I	Appearance of clear, repetitive sounds	Systolic pressure
II	The sounds take on a murmur-like character	—
III	The sounds become clear and loud again	—
IV	The sounds abruptly become softer and muffled	Diastolic pressure in children, pregnant women and in states with an “infinite sound”
V	Complete disappearance of the sounds	Diastolic pressure in adults

CAUTION

The “auscultatory gap” is a temporary disappearance of the sounds between phases I and II (most often in hypertension and atherosclerosis). If the cuff is not first checked by the palpatory method, the lower limit of the gap is taken as the systolic pressure by mistake, and the systolic pressure is underestimated by 20–40 mm Hg. **The “infinite sound”** — the sounds are heard down to zero (aortic regurgitation, thyrotoxicosis, pregnancy, children).

6.2. Classification of arterial pressure

Table 11.10. *Classification of BP in adults (ESH, 2023)*

Category	Systolic, mm Hg		Diastolic, mm Hg
Optimal	Below 120	and	Below 80
Normal	120–129	and/or	80–84
High normal	130–139	and/or	85–89
Grade 1 hypertension	140–159	and/or	90–99
Grade 2 hypertension	160–179	and/or	100–109
Grade 3 hypertension	180 or higher	and/or	110 or higher
Isolated systolic hypertension	140 or higher	and	Below 90

REMEMBER

The ESC 2024 guideline proposed a simplified classification: **non-elevated BP** — below 120/70; **elevated BP** — 120–139/70–89; **hypertension** — 140/90 or higher. In both classifications the threshold of hypertension is **140/90 mm Hg** when measured in the clinic and **135/85 mm Hg** when measured at home.

Arterial hypotension — BP below 100/60 (in women 95/60) mm Hg: physiological (athletes, young asthenic people) or pathological (shock, blood loss, adrenal insufficiency, hypothyroidism, drug effects). **Orthostatic hypotension** — a fall in systolic BP of more than 20 mm Hg or in diastolic BP of more than 10 mm Hg within 3 minutes of standing up.

7. Detecting the pulse in the peripheral arteries



Figure 11.26. *Pulse in the carotid artery (Gadayev A. et al., 2012, Fig. 51)*



Figure 11.27. *Carotid pulse in a lying patient (same source)*



Figure 11.28. *Pulse in the brachial artery (Gadayev A. et al., 2012, Fig. 52)*



Figure 11.29. *Pulse in the femoral artery (same source, Fig. 53)*



Figure 11.30. *Pulse in the popliteal artery (Gadayev A. et al., 2012, Fig. 54)*



Figure 11.32. *Pulse in the posterior tibial artery (same source, Fig. 56)*



Figure 11.31. *Pulse in the dorsalis pedis artery (same source, Fig. 55)*

CAUTION

The pulses in the carotid arteries must not be palpated **on both sides at the same time** or with strong pressure: this may reduce cerebral blood flow and, through the carotid sinus reflex, lead to bradycardia and syncope (especially in the elderly). When the signs of clinical death are being determined, the pulse is checked **in the carotid artery** (within 10 seconds).

State of the arterial wall: after the pulse has been stopped by pressure, the artery is palpated distally; normally it is not felt or is felt as a soft elastic tube. In atherosclerosis the artery is hard, tortuous and “beaded” (calcified).

8. The venous pulse

The venous pulse visible in the neck veins reflects pressure changes in the right atrium. Normally there is a **negative venous pulse**: the vein empties during ventricular systole (the blood flows into the right atrium). Waves on the phlebogram: **a** — atrial systole; **c** — bulging of the tricuspid valve toward the atrium; **x** descent — atrial relaxation; **v** — filling of the atrium; **y** descent — opening of the tricuspid valve.

Table 11.11. *Changes in the venous pulse*

Change	Cause
Large “a” waves	Tricuspid stenosis, pulmonary hypertension, pulmonary valve stenosis
Cannon a waves	Complete AV block (the atrium contracts against a closed tricuspid valve)
Absence of the “a” wave	Atrial fibrillation
Large “v” waves — positive venous pulse	Tricuspid regurgitation
Steep “y” descent	Constrictive pericarditis

9. Additional projections in the radiographic examination of the heart

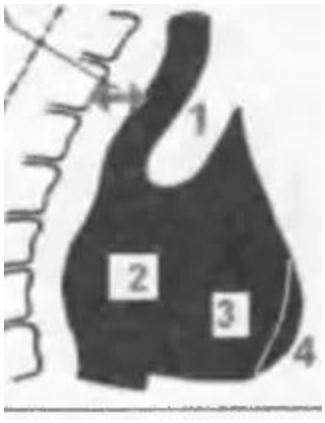


Figure 11.33. *Parts of the heart in the first (right anterior) oblique projection (Gadayev A. et al., 2012, Fig. 91)*



Figure 11.34. *Parts of the heart in the second (left anterior) oblique projection (same source, Fig. 92)*

Table 11.12. *Diagnostic possibilities of the radiographic projections*

Projection	What is seen well
Frontal (posteroanterior)	General shape and size, CTR, lung pattern, configuration
Left lateral	Left atrium (with a contrast-filled esophagus), retrosternal space (right ventricle), aorta
First oblique (right anterior)	Left atrium (backward displacement of the contrast-filled esophagus), conus of the pulmonary artery
Second oblique (left anterior)	Left ventricle, aortic arch, right atrium

MEMORY AID

By the displacement of the contrast-filled esophagus: **“small radius — Mitral stenosis, large radius — Mitral regurgitation”**. In mitral stenosis the left atrium is moderately enlarged (the esophagus is displaced along an arc of small radius), whereas in mitral regurgitation it may be very large (an arc of large radius).

10. Echocardiography — the main method of assessing the size of the heart

In modern practice, the size of the cardiac chambers, the wall thickness and contractility are determined by **echocardiography** (echo).

Table 11.13. *Normal values of the main echocardiographic parameters (adults, approximate)*

Parameter	Normal
Left ventricular end-diastolic diameter	Men 42–58 mm, women 38–52 mm
Thickness of the interventricular septum and posterior wall	6–11 mm (12 mm or more — hypertrophy)
Left atrium (anteroposterior diameter)	Up to 40 mm
Aortic root	Up to 37 mm
Left ventricular ejection fraction	52–55 % or more (below 50 % — systolic dysfunction)
Pulmonary artery systolic pressure	Up to 35 mm Hg

REMEMBER

The ejection fraction (EF) is the ratio of the blood ejected from the left ventricle with each systole to the diastolic volume. By EF, heart failure is divided into three types (ESC, 2021): **reduced EF** (40 % or less), **mildly reduced EF** (41–49 %) and **preserved EF** (50 % or more).

11. Arterial hypertension: basics of semiotics

Arterial hypertension is a BP of 140/90 mm Hg or higher on repeated measurements in the clinic. In 90–95 % of cases it is **primary (essential) hypertension**; in 5–10 % it is **secondary (symptomatic)** hypertension.

Table 11.14. *Main causes of symptomatic arterial hypertension*

Group	Diseases	Signs that raise suspicion
Renal parenchymal	Glomerulonephritis, pyelonephritis, polycystic kidney disease, diabetic nephropathy	Urinary changes, edema, raised creatinine
Renovascular	Atherosclerosis or fibromuscular dysplasia of the renal artery	A systolic bruit around the umbilicus, treatment-resistant BP, a rise in creatinine after an ACE inhibitor
Endocrine	Primary hyperaldosteronism, pheochromocytoma, Cushing's syndrome, thyrotoxicosis	Hypokalemia; BP crises with headache, sweating and palpitations; Cushingoid features
Hemodynamic	Coarctation of the aorta	A young patient, BP high in the arms and low in the legs, a delayed femoral pulse, an interscapular murmur
Others	Sleep apnea syndrome, drugs (NSAIDs, glucocorticoids, contraceptives), alcohol	Snoring, daytime sleepiness

Target-organ damage in essential hypertension: left ventricular hypertrophy (a resistant apex beat, an accentuated S2 over the aorta, ECG and echo signs), retinopathy on the fundus, the kidneys (microalbuminuria, raised creatinine), the brain (stroke, encephalopathy), the peripheral arteries.

REMEMBER

Physical signs characteristic of essential hypertension: a **resistant** apex beat slightly displaced to the left, a widened left cardiac border, an **aortic configuration**, an **accentuated S2** over the aorta (sometimes with a metallic tinge), a **tense (hard)** pulse.

12. Special changes of the pulse and their clinical significance

Table 11.15. *Rare but important changes of the pulse*

Change	Content	Cause
Bisferiens pulse (pulsus bisferiens)	Two peaks in each pulse wave	Severe aortic regurgitation, aortic stenosis with regurgitation, hypertrophic cardiomyopathy
Relative bradycardia	A pulse slow relative to the degree of fever	Typhoid fever, legionellosis, brucellosis, drug fever
Radio-femoral delay	When the radial and femoral pulses are palpated at the same time, the femoral pulse is delayed and weak	Coarctation of the aorta

Change	Content	Cause
Absent or asymmetric pulse in the arms	No pulse in one or both arms	Takayasu arteritis (“pulseless disease”), aortic dissection, atherosclerosis of the subclavian artery
Pulse deficit	The heart rate exceeds the pulse rate	Atrial fibrillation, frequent extrasystole

13. Changes in the cardiac borders and configuration in valvular defects: a summary table

Table 11.16. *Percussion findings in valvular heart defects*

Defect	Right border	Upper border	Left border	Configuration	Vascular bundle
Mitral stenosis	To the right	Upward (II rib)	Unchanged	Mitral	Unchanged
Mitral regurgitation	To the right (late)	Upward	To the left	Mitral	Unchanged
Aortic stenosis	Unchanged	Unchanged	To the left (moderately)	Aortic	Slightly widened
Aortic regurgitation	Unchanged	Unchanged	To the left and downward (markedly)	Aortic (“sitting duck”)	Widened
Tricuspid regurgitation	To the right (markedly)	Unchanged	To the left (pushed by the right ventricle)	Trapezoid-like	Unchanged
Exudative pericarditis	To the right	Upward	To the left	Trapezoid	Widened

MEMORY AID

The cardiac borders and the chambers: **“Right — Right atrium, Upper — Left atrium, Left — Left ventricle”**. Whichever border is displaced, that chamber is enlarged.

14. Typical errors in measuring BP and special situations

Table 11.17. *Errors in measuring BP*

Error	Result
Cuff too small (in an obese patient)	BP falsely high
Cuff too large	BP falsely low
Arm below the level of the heart	BP falsely high
The patient talks, crosses his legs, the arm is unsupported	BP higher by 5–15 mm Hg
Releasing the air too fast	The systolic pressure is underestimated and the diastolic pressure overestimated
Ignoring the auscultatory gap	The systolic pressure is markedly underestimated
Measuring over clothing	Inaccurate result

“White-coat hypertension” — BP is high in the clinic but normal at home and on ambulatory monitoring. **Masked hypertension** is the opposite — normal in the clinic, high at home. To detect

them, **24-hour ambulatory BP monitoring** (ABPM; normal — below 135/85 during the day, below 120/70 at night, and a 24-hour average below 130/80) or home measurement is used.

In atrial fibrillation BP should be measured several times and the average taken; automatic devices may be inaccurate.

15. Clinical part

Percussion, pulse and radiographic findings in the main valvular defects

Table 11.18. *Changes in percussion and pulse in valvular heart defects*

Defect	Percussion	Configuration	Pulse
Mitral stenosis	Upper border upward, right border to the right	Mitral	Small, often irregular (atrial fibrillation, pulse deficit), pulsus differens
Mitral regurgitation	Upper border upward, left border to the left	Mitral	Unchanged or irregular
Aortic stenosis	Left border to the left (hypertrophy — slight displacement)	Aortic	Slow, small (tardus, parvus), tendency to bradycardia
Aortic regurgitation	Left border markedly displaced to the left and downward, vascular bundle widened	Aortic	Rapid, high, large (celer, altus, magnus)
Exudative pericarditis	The borders widened in all directions, absolute dullness enlarged	Trapezoid	Small, paradoxical

Sample case history entry

Percussion. Borders of relative cardiac dullness: right — in the IV intercostal space, 1 cm lateral to the right sternal edge; upper — III rib; left — in the V intercostal space, 2 cm lateral to the left midclavicular line. The cardiac waist is deepened, aortic configuration. The width of the vascular bundle is 7 cm. **The pulse** is the same on both arms, regular, 84 per minute, well filled, tense, large, rapid and high. There is no pulse deficit.

Conclusion: left ventricular enlargement, an aortic configuration and a rapid, high pulse — characteristic of aortic valve regurgitation.

16. Practical skills

Skill: determining the borders of relative cardiac dullness

1. Have the patient undress to the waist, standing or lying down.
2. **Right border:** find the lower border of the right lung along the midclavicular line, move one intercostal space higher, place the plexímeter in the IV space parallel to the sternum and percuss medially with moderate force; mark the point where the dullness begins (along the outer edge of the plexímeter).
3. **Left border:** locate the apex beat; percuss in the V space from the anterior axillary line medially, with the plexímeter held vertically; mark the border and measure the distance to the midclavicular line.
4. **Upper border:** percuss downward from the I space along the left parasternal line, with the plexímeter parallel to the ribs.

5. **Vascular bundle:** percuss in the II space from the right and from the left toward the sternum and measure the width.
6. Draw the cardiac contour and determine the configuration.

Skill: examining the pulse in the radial artery

1. The patient holds the arms relaxed, at the level of the heart.
2. Palpate both radial arteries at the same time and compare their symmetry.
3. Assess the rhythm on one arm; if the pulse is regular, count for 30 seconds and multiply by 2; if it is irregular, count for 1 full minute.
4. If the pulse is irregular, count the heart rate on auscultation and determine the pulse deficit.
5. Assess the filling (compress the artery completely, then release it and feel it fill), the tension (the force needed to stop the pulse), the amplitude and the shape.
6. Record the result: “The pulse is the same on both arms, regular, 72 per minute, full, not tense, of moderate amplitude and shape.”

Review questions

1. What are relative and absolute cardiac dullness? With what force are they determined?
2. State the rules and sequence of cardiac percussion.
3. Where are the borders of relative cardiac dullness in a healthy person?
4. Which parts of the heart form the borders of relative cardiac dullness?
5. What are the borders of absolute cardiac dullness and the causes of their changes?
6. How is the width of the vascular bundle determined, and what is its normal value?
7. Name the causes (cardiac and noncardiac) of displacement of the cardiac borders.
8. What is the cardiac waist? Describe the mitral and aortic configurations.
9. Which features of the pulse are assessed?
10. What is pulse deficit, and how is it determined?
11. In which conditions is pulsus differens seen?
12. What is the pulse like in aortic stenosis and in aortic regurgitation?
13. What do the paradoxical and alternating pulses indicate?
14. Which structures form the right and left contours of the cardiac shadow in the frontal projection?
15. How is the cardiothoracic ratio calculated, and what is its normal value?

Test questions

1. Where is the right border of relative cardiac dullness normally located?
   A) Along the left sternal edge
   B) In the IV intercostal space, 1–1.5 cm lateral to the right sternal edge
   C) In the V intercostal space on the right midclavicular line
   D) At the level of the III rib on the right parasternal line
   E) In the II intercostal space at the right sternal edge

2. Where is the upper border of relative cardiac dullness normally located?
   A) I rib
   B) II rib
   C) III rib
   D) The IV rib
   E) V rib

3. Which part of the heart forms the left border of relative cardiac dullness?
   A) The left ventricle
   B) The left atrium
   C) The right atrium
   D) The right ventricle
   E) The aorta

4. Where is the right border of absolute cardiac dullness normally located?
   A) Right parasternal line
   B) Right sternal edge
   C) Left sternal edge
   D) Midline
   E) Left midclavicular line

5. How does absolute cardiac dullness change in pulmonary emphysema?
   A) It enlarges
   B) Unchanged
   C) It becomes smaller or disappears
   D) It shifts to the right
   E) It moves upward

6. What is the normal width of the vascular bundle?
   A) 2–3 cm
   B) 5–6 cm
   C) 8–9 cm
   D) 10–12 cm
   E) 1–2 cm

7. What is the cardiac waist like in the mitral configuration?
   A) Deepened
   B) Smoothed out or bulging
   C) Cannot be determined
   D) It forms a right angle
   E) Unchanged

8. Enlargement of which part of the heart is associated with the aortic configuration?
   A) The right atrium
   B) The left atrium
   C) Pulmonary artery
   D) The left ventricle
   E) The right ventricle

9. Which condition is the trapezoid configuration of the heart characteristic of?
    A) Exudative pericarditis
    B) Arterial hypertension
    C) Mitral stenosis
    D) Emphysema
    E) Aortic stenosis

10. By what mechanism does pulsus differens arise in mitral stenosis?
    A) Compression of the left subclavian artery by the enlarged left atrium
    B) Narrowing of the aortic valve
    C) Atrial fibrillation
    D) Myocardial weakness
    E) Pericardial adhesions

11. In which condition is pulse deficit most often seen?
    A) Respiratory arrhythmia
    B) Sinus tachycardia
    C) Sinus bradycardia
    D) Arterial hypertension
    E) Atrial fibrillation

12. What is the pulse like in aortic regurgitation?
    A) Alternating
    B) Thready
    C) Paradoxical
    D) Slow, small
    E) Rapid, high, large

13. Which condition is a marked decrease of the pulse waves on inspiration (paradoxical pulse) characteristic of?
    A) Thyrotoxicosis
    B) Arterial hypertension
    C) Mitral regurgitation
    D) Aortic regurgitation
    E) Cardiac tamponade

14. What forms the 3rd arc of the left contour of the cardiac shadow in the frontal projection?
    A) The right atrium
    B) Aortic arch
    C) The left ventricle
    D) Left atrial appendage
    E) Pulmonary trunk

15. What is the normal value of the cardiothoracic ratio?
    A) Less than 0.3
    B) Does not exceed 0.5
    C) 0,6–0,7
    D) 0,8
    E) 1,0

Clinical cases

Case 1

In a 32-year-old woman, on percussion the upper border of relative cardiac dullness is at the II rib, the right border 3 cm lateral to the right sternal edge, and the left border normal. The pulse is irregular, 96 per minute, and the heart rate on auscultation is 118. The pulse on the left arm is smaller than on the right.

Questions: Interpret the percussion findings and determine the configuration. What do the features of the pulse indicate?

Answer: Upward displacement of the upper border shows enlargement of the left atrium, and displacement of the right border to the right shows dilation of the right ventricle and right atrium; the cardiac waist is smoothed out — a mitral configuration (mitral stenosis). The pulse is irregular with a pulse deficit of 22 (118 − 96) — atrial fibrillation. The smaller pulse on the left arm is pulsus differens (the enlarged left atrium compresses the left subclavian artery).

Case 2

In a 40-year-old man, the apex beat is in the VI intercostal space, 2 cm lateral to the midclavicular line. The left cardiac border is at this point, the cardiac waist is deepened, the vascular bundle is 7 cm. The pulse is rapid and high, BP 160/40 mm Hg.

Questions: Determine the configuration and the character of the pulse. Which defect may be suspected?

Answer: Aortic configuration (the left ventricle is markedly enlarged, the cardiac waist is deepened), the vascular bundle is widened; the pulse is rapid, high and large (pulsus celer et altus); the pulse pressure is very large (120 mm Hg), and the diastolic pressure is low. These signs are characteristic of aortic valve regurgitation.

Case 3

In a 55-year-old patient, on a frontal radiograph the transverse diameter of the cardiac shadow is 17 cm and the internal transverse diameter of the chest is 30 cm. The cardiac shadow is trapezoid, and pulsation of its contours is reduced. Clinically there is a paradoxical pulse and distension of the neck veins.

Questions: Calculate the cardiothoracic ratio. Which condition should be considered? How will you confirm the diagnosis?

Answer: CTR = 17 / 30 = 0.57 — cardiomegaly. A trapezoid shadow, a paradoxical pulse and distended neck veins are characteristic of exudative pericarditis with a risk of cardiac tamponade. The diagnosis is confirmed by echocardiography (fluid in the pericardium).

Quick review

Table 11.19. *Key facts on Topic 11*

Question	Answer
Relative dullness: right	IV space, 1–1.5 cm lateral to the right sternal edge (right atrium)
Relative dullness: upper	III rib (left atrial appendage, pulmonary artery)
Relative dullness: left	V space, 1–1.5 cm medial to the midclavicular line (left ventricle)

Question	Answer
Absolute dullness	Right — left sternal edge; upper — IV rib; left — 1–2 cm medial to the relative border
Vascular bundle	II space, 5–6 cm
Mitral configuration	Cardiac waist smoothed out — mitral defects
Aortic configuration	Cardiac waist deepened — aortic defects, hypertension
Normal pulse	Symmetric, regular, 60–80 (ESC: 60–100), full, of moderate tension
Pulse deficit	Atrial fibrillation
Pulsus celer et altus	Aortic regurgitation
Pulsus tardus et parvus	Aortic stenosis
Pulsus paradoxus	Tamponade, pericarditis
CTR	Does not exceed 0.5

TOPIC 12

Auscultation of the heart. Heart sounds in health and disease. Splitting and reduplication of the sounds. Extra sounds. Heart murmurs

Learning objectives

To teach the student the rules and points of cardiac auscultation, the mechanism of formation and normal features of the heart sounds, the accentuation, weakening, splitting and reduplication of the sounds in disease, the extra sounds, and also the mechanism of formation, classification and features of heart murmurs.

By the end of the session, the student should know:

- the projection of the heart valves onto the chest and the points and order of auscultation;
- the mechanism of formation of S1 and S2 and their distinguishing features;
- the causes of accentuation and weakening of the sounds, accentuation of S2;
- physiological and pathological splitting of S2;
- S3 and S4, the “quail rhythm”, the gallop rhythm, the pericardial knock;
- the classification of murmurs, the differences between organic and functional murmurs;
- the plan for describing a murmur (phase, epicenter, radiation, intensity, shape, timbre).

The student should be able to:

- perform cardiac auscultation in the correct order and in different positions;
- distinguish S1 and S2 and assess their features;
- detect extra sounds and murmurs and describe them according to the plan.

1. General rules of cardiac auscultation

1. The room should be quiet and warm.
2. The patient is auscultated standing, sitting and lying down, and if necessary on the left side (mitral defects) and leaning forward (aortic regurgitation).
3. Auscultation is performed first during quiet breathing, then while the breath is held after inspiration and after expiration (so that lung sounds do not interfere).
4. Sometimes auscultation is performed after physical exertion (sitting down and standing up 10–15 times) — the murmur of mitral stenosis becomes louder.
5. **The diaphragm** is used for high-frequency sounds (S1 and S2, the murmur of aortic regurgitation), and **the bell** for low-frequency sounds (S3 and S4, the diastolic murmur of mitral stenosis).
6. To distinguish the sounds, the carotid pulse or the apex beat is palpated at the same time: they coincide with **S1**.

2. Projection of the heart valves and auscultation points

Because the anatomical projections of the valves onto the chest lie close to each other, the valves are auscultated at the sites where sound is best conducted along the blood flow.

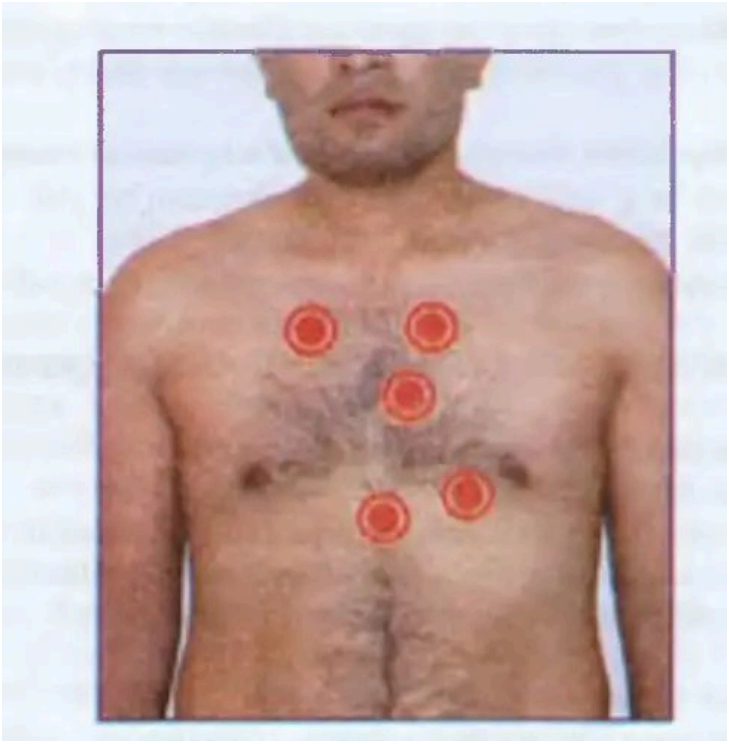


Figure 12.1. *The main points of cardiac auscultation (Gadayev A. et al., 2012, Fig. 45)*

Table 12.1. *Projection of the heart valves and auscultation points*

Order	Valve	Anatomical projection	Auscultation point
1	Mitral	On the left at the junction of the III rib with the sternum	Cardiac apex (V intercostal space, 1–1.5 cm medial to the midclavicular line)
2	Aortic	Behind the sternum at the level of the III ribs	In the II intercostal space to the right of the sternum
3	Pulmonary artery	On the left in the II intercostal space	In the II intercostal space to the left of the sternum
4	Tricuspid	In the middle of the sternum, on the line joining the left III and the right V costal cartilages	At the base of the xiphoid process, at the lower end of the sternum (slightly to the right)
5	Aortic (additional)	—	Botkin–Erb point: to the left of the sternum, where the III–IV ribs join the sternum

MEMORY AID

The order of listening follows the frequency of valve involvement: **“Mitral — Aortic — Pulmonary — Tricuspid — Botkin”** — “**M**y **A**unt **P**olly **T**akes **B**uses”. By site: apex → right II → left II → xiphoid → Botkin–Erb.

3. Heart sounds

3.1. Mechanism of formation of the sounds

S1 (systolic) arises at the beginning of ventricular systole and consists of four components:

1. **The valvular component** (the main one) — closure and vibration of the atrioventricular (mitral and tricuspid) valves;

2. **The muscular component** — vibration of the ventricular myocardium during isovolumetric contraction;
3. **The vascular component** — vibration of the initial segments of the aorta and pulmonary artery as they fill with blood;
4. **The atrial component** — atrial contraction (at the onset of S1).

S2 (diastolic) arises at the beginning of ventricular diastole from **closure of the semilunar (aortic and pulmonary) valves** and vibration of the walls of the aorta and pulmonary artery. It consists of two components: **aortic (A_2)** and **pulmonary (P_2)**.

S3 — vibration of the ventricular walls during rapid ventricular filling in early diastole (0.12–0.18 seconds after S2); it may be heard in healthy children and adolescents and in thin young people.

S4 — vibration of the ventricular walls in presystole (before S1), when the atria contract and drive blood into the ventricles; it is usually not heard in healthy adults.

3.2. Distinguishing S1 and S2

Table 12.2. *Differences between S1 and S2*

Feature	S1	S2
When it is heard	After the long pause (after the long diastole)	After the short pause (after the short systole)
Where it is heard best	Cardiac apex and base of the xiphoid process	Base of the heart (II space on the right and left)
Duration	Longer (0.09–0.12 s)	Shorter (0.05–0.07 s)
Pitch (timbre)	Lower	Higher
With the apex beat and the carotid pulse	Coincides	Does not coincide

REMEMBER
In a healthy person **S1 is louder than S2 at the apex** (S1 is about 1.5–2 times stronger), and **S2 is louder than S1 at the base of the heart**. In young people S2 over the pulmonary artery is slightly louder than over the aorta; after the age of 40 it is the opposite — louder over the aorta.

4. Changes in the heart sounds

4.1. Simultaneous change of both sounds

- **Accentuation of both sounds**: extracardiac causes — thinness, a thin chest wall, shrinkage of the lung edges; cardiac causes — thyrotoxicosis, physical exertion, excitement, fever, anemia, tachycardia.
- **Weakening (muffling) of both sounds**: extracardiac causes — obesity, a thick chest wall, pulmonary emphysema, left-sided exudative pleurisy, pericardial fluid; cardiac causes — myocardial damage (reduced contractility in myocarditis, myocardial infarction, cardiosclerosis), collapse, shock.

4.2. Changes in S1

Table 12.3. *Changes in S1*

Change	Mechanism	Diseases
Accentuation (a "snapping" S1 at the apex)	Incomplete filling of the ventricle and rapid closure of the valve leaflets from a widely open position; thickening of the leaflets	Mitral stenosis, tachycardia, extrasystoles, the "cannon" sound in complete AV block (Strazhesko)
Diminished	Incomplete closure of the valve leaflets (no period of closed valves)	Mitral regurgitation, tricuspid regurgitation, aortic regurgitation (overfilling of the ventricle)
	Reduced myocardial contractility	Myocarditis, cardiosclerosis, myocardial infarction
	Slowed contraction with ventricular hypertrophy	Aortic stenosis

CLINICAL SIGNIFICANCE

Strazhesko's "cannon sound" is a very loud S1 heard in complete AV block when atrial and ventricular systole happen to coincide. The pulse is then slow (30–40 per minute), and the intensity of the sounds varies.

4.3. Changes in S2. Accentuation of S2

Accentuation of S2 is an increase in the loudness of S2 over the aorta or the pulmonary artery compared with the symmetric point.

Table 12.4. *Changes in S2*

Change	Mechanism	Diseases
Accentuation over the aorta	Raised pressure in the aorta, thickening of the aortic wall	Arterial hypertension, atherosclerosis of the aorta (an S2 with a metallic tinge)
Accentuation over the pulmonary artery	Raised pressure in the pulmonary circulation (pulmonary hypertension)	Mitral stenosis, mitral regurgitation, chronic cor pulmonale, left ventricular failure
Weakening or disappearance over the aorta	Destruction of the valve leaflets or loss of their mobility	Aortic regurgitation, aortic stenosis
Weakening over the pulmonary artery	—	Pulmonary valve regurgitation and stenosis

4.4. Splitting and reduplication of the sounds

When the components of a sound do not arise at the same time but are heard at a certain interval, this is called **splitting** (an interval of 0.02–0.04 s — the sound is "cleft") or **reduplication** (an interval of more than 0.05 s — two separate sounds).

Reduplication of S1 — in bundle branch block (the ventricles do not contract at the same time) and in ventricular extrasystole.

Physiological reduplication of S2 — on inspiration the negative pressure in the chest increases, the blood flow to the right ventricle rises, right ventricular systole lengthens and the pulmonary valve closes later. This reduplication is heard **only on inspiration** and disappears on expiration; it is found over the pulmonary artery in young people.

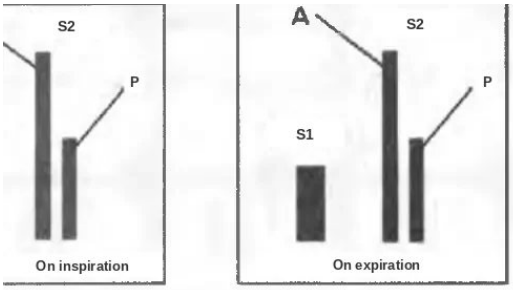


Figure 12.3. *Pathological reduplication of S2 (same source, Fig. 40)*

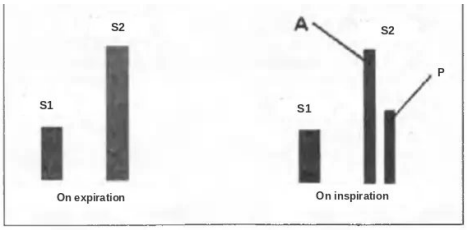


Figure 12.2. *Physiological reduplication of S2: A — aortic, P — pulmonary component (Gadayev A. et al., 2012, Fig. 39)*

Pathological reduplication of S2 is a constant reduplication that does not depend on the phase of respiration: pulmonary hypertension (mitral stenosis), increased right ventricular load, atrial septal defect (a strict, “fixed” reduplication), right bundle branch block.

CAUTION

Physiological splitting of S2 appears on inspiration and disappears on expiration. **Pathological** splitting is constant and does not depend on the phases of respiration. To distinguish them, the patient is auscultated while holding the breath after inspiration and after expiration.

5. Extra sounds and three-part rhythms

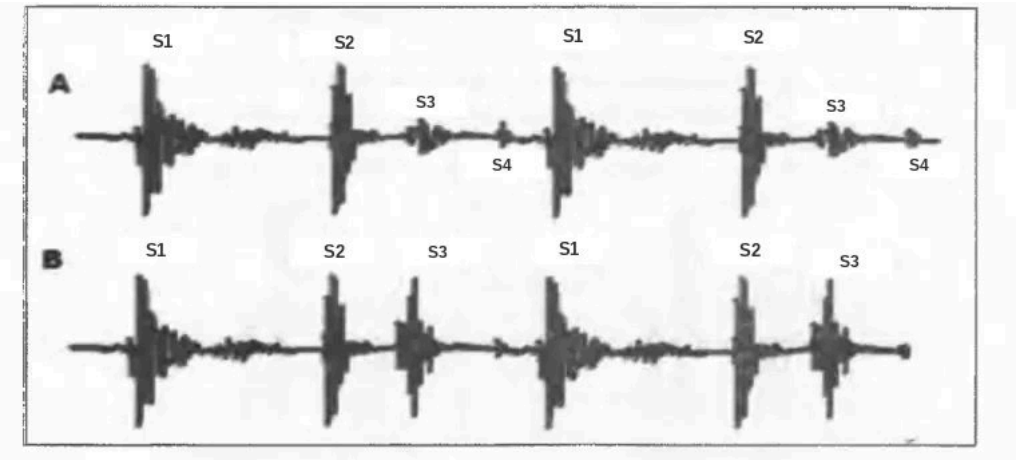


Figure 12.4. *Heart sounds on the phonocardiogram: A — the normal four sounds; B — accentuation of S3 (gallop rhythm) (Gadayev A. et al., 2012, Fig. 41)*

Table 12.5. *Extra sounds and three-part rhythms*

Rhythm or sound	Mechanism and description	Diseases
Gallop rhythm — protodiastolic	A pathological S3: vibration of a ventricle with reduced myocardial tone during rapid filling; with tachycardia, a three-part rhythm resembling a “galloping horse”	Severe myocardial damage: myocarditis, myocardial infarction, dilated cardiomyopathy, heart failure — a poor prognostic sign
Presystolic gallop	A pathological S4: filling of a hypertrophied, poorly compliant ventricle during atrial systole	Arterial hypertension, aortic stenosis, hypertrophic cardiomyopathy, myocardial ischemia
Mitral valve opening snap	Sudden tensing of the sclerosed mitral valve leaflets in diastole (0.07–0.13 s after S2), heard at the apex and at the Botkin–Erb point	Mitral stenosis

Rhythm or sound	Mechanism and description	Diseases
“Quail rhythm”	Snapping S1 + S2 + mitral opening snap (“SNAP — dub — click”)	Mitral stenosis
Pericardial knock	The thickened pericardium abruptly stops the expansion of the ventricle in diastole	Constrictive pericarditis
Systolic “click”	Bulging of a mitral valve leaflet into the atrium during systole	Mitral valve prolapse
Early systolic (ejection) sound	Vibration of a dilated aorta or pulmonary artery during systole	Aortic stenosis, aortic dilation

MEMORY AID

“Quail rhythm” = “Snapping S1 + S2 + opening snap” — only in mitral stenosis. Remember it by the rhythm “**SNAP — dub — click**”: “snap” is the snapping S1, “dub” S2 and “click” the opening snap.

6. Heart murmurs

6.1. Mechanism of formation of murmurs

Normally the blood flow in the heart and vessels is **laminar** (in layers) and silent. A murmur appears when the blood flow becomes **turbulent** (eddying).

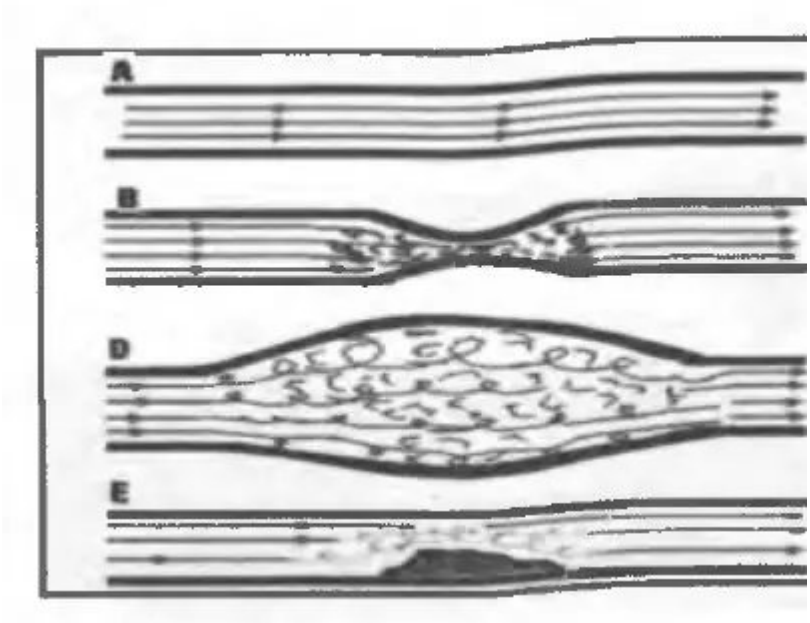


Figure 12.5. *Mechanisms of murmur formation: A — normal laminar flow; B — turbulent flow through a narrowing; D — turbulent flow in a dilated vessel; E — irregularity of the wall (Gadayev A. et al., 2012, Fig. 43)*

Causes of turbulent flow:

1. **Narrowing of an orifice** (stenosis) — the blood passes through the narrow site at high speed.
2. **Valvular regurgitation** — the blood flows back in the opposite direction through a narrow slit.
3. **Dilation of a vessel or cavity** — eddies form as blood passes from a narrow site into a wider one.

4. **Reduced blood viscosity** and **increased blood flow velocity** (anemia, fever, thyrotoxicosis) — functional murmurs.
5. Pathological shunts (ventricular septal defect, patent ductus arteriosus).

6.2. Classification of murmurs

Table 12.6. *Classification of heart murmurs*

Group	Type	Examples
Intracardiac	Organic	Acquired defects (stenosis, regurgitation), congenital defects
	Functional	Anemic murmurs, murmurs in fever and thyrotoxicosis, “innocent” murmurs in young people; murmurs of relative regurgitation (dilation of the valve ring — mitral, tricuspid); the Austin Flint murmur, the Graham Steell murmur
Extracardiac	Pericardial friction rub	Dry pericarditis, uremia
	Pleuropericardial rub	Inflammation of the pleura adjacent to the heart
	Cardiopulmonary murmur	Compression and expansion of the adjacent parts of the lung with the cardiac contractions

By phase, **systolic** (between S1 and S2) and **diastolic** (between S2 and the next S1) murmurs are distinguished. Diastolic murmurs in turn are **protodiastolic** (at the beginning of diastole), **mesodiastolic** (in the middle) and **presystolic** (at the end).

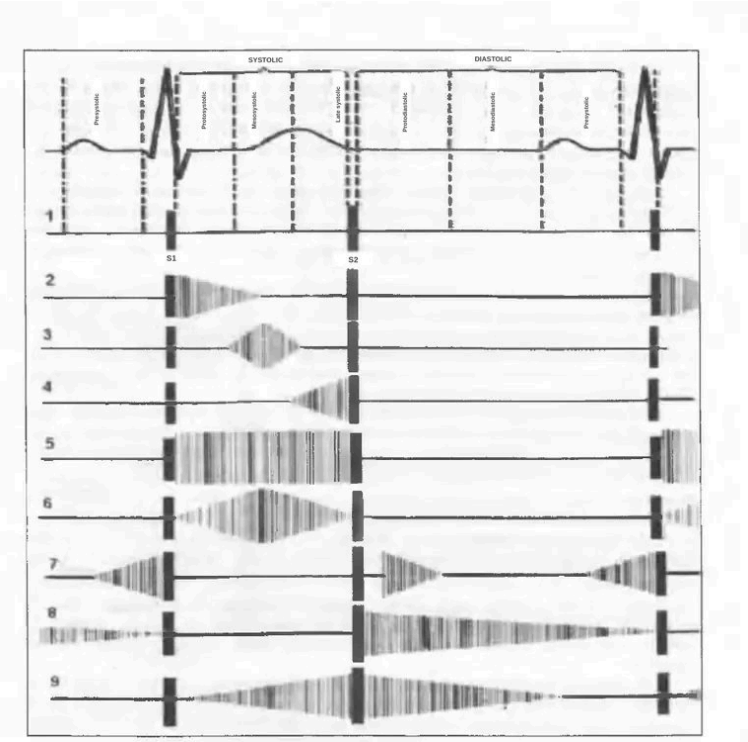


Figure 12.6. *Diagram of some intracardiac murmurs (Gadayev A. et al., 2012, Fig. 47)*

6.3. Plan for describing a murmur

1. **Phase** — systolic or diastolic (determined by palpating the apex beat or the carotid pulse at the same time).
2. **Relation to the sounds** — whether it begins after a sound or “swallows” the sound.

3. **Epicenter** — the point where it is heard best (shows which valve it belongs to).
4. **Radiation** — in the direction of the blood flow.
5. **Intensity (loudness)** — grades 1–6 on the Levine scale.
6. **Shape** — increasing (crescendo), decreasing (decrescendo), diamond-shaped, band-shaped.
7. **Timbre** — harsh, saw-like, blowing, soft, musical.
8. **Duration** — how much of the phase it occupies.
9. **Dependence on position, respiration and exertion**.

Table 12.7. *Assessing the intensity of a murmur (Levine scale)*

Grade	Definition
1	Very faint, heard only with careful attention
2	Faint, but heard immediately
3	Moderately loud, without a thrill
4	Loud, with a thrill
5	Very loud, heard even when only the edge of the stethoscope touches the skin
6	Heard even with the stethoscope off the skin

Table 12.8. *Murmurs in the main acquired defects*

Defect	Phase	Epicenter	Spread	Character
Mitral regurgitation	Systolic	Cardiac apex	To the left axillary region	Decreasing or band-shaped, merges with S1, louder in the left lateral position
Mitral stenosis	Diastolic (mesodiastolic and presystolic)	Cardiac apex	Hardly radiates	Low-pitched, rumbling, louder in presystole; heard well in the left lateral position and after exertion
Aortic stenosis	Systolic	In the II space to the right of the sternum	To the carotid arteries, the clavicle, the interscapular region	Harsh, saw-like, diamond-shaped, loud
Aortic regurgitation	Diastolic (protodiastolic)	Botkin–Erb point, II space on the right	To the cardiac apex	Soft, blowing, decreasing; heard well when leaning forward and holding the breath after expiration
Tricuspid regurgitation	Systolic	Base of the xiphoid process	To the right	Louder on inspiration (Rivero-Carvallo sign)

MEMORY AID
To hear a diastolic murmur well: **mitral stenosis — “lie on the left side”** (bell, at the apex); **aortic regurgitation — “lean forward and breathe out”** (diaphragm, at the Botkin–Erb point).

6.4. Differences between organic and functional murmurs

Table 12.9. *Organic and functional murmurs*

Feature	Organic murmur	Functional murmur
Phase	Systolic and diastolic	Mostly systolic

Feature	Organic murmur	Functional murmur
Timbre	Harsh, saw-like	Soft, blowing
Duration	Long, occupies most of systole	Short
Severity	Often loud	Faint (grade 1–2)
Spread	Radiates widely along the blood flow	Radiates little, localized
Relation to the sounds	The sounds are changed (weakening of S1, etc.)	The sounds are unchanged
Dependence on position and exertion	Changes little	Variable with position, respiration and exertion, sometimes disappears
Thrill (“cat’s purr”)	May be present	Never present
Cardiac borders	Often widened	Unchanged

CAUTION

Any **diastolic murmur** is almost always regarded as **organic** (the Austin Flint and Graham Steell murmurs are exceptions — but they too arise against the background of an organic defect). If a diastolic murmur is found, echocardiography is always performed.

6.5. Extracardiac murmurs

The pericardial friction rub is heard in systole and diastole, best over the area of absolute cardiac dullness (at the left sternal edge); it does not radiate anywhere, becomes louder when pressed with the stethoscope and when leaning forward, and persists when the breath is held. **The pleuropericardial murmur**, by contrast, becomes louder on inspiration and decreases or disappears when the breath is held.

7. Anatomical basis of the valve projections and auscultation points

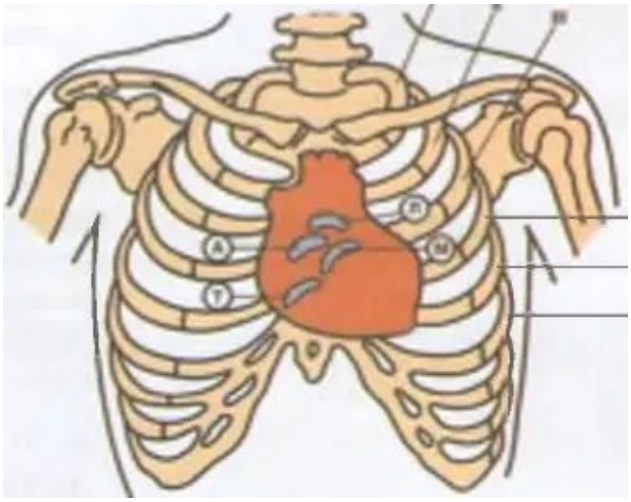


Figure 12.7. *Projection of the heart valves onto the chest (Gadayev A. et al., 2012, Fig. 36)*

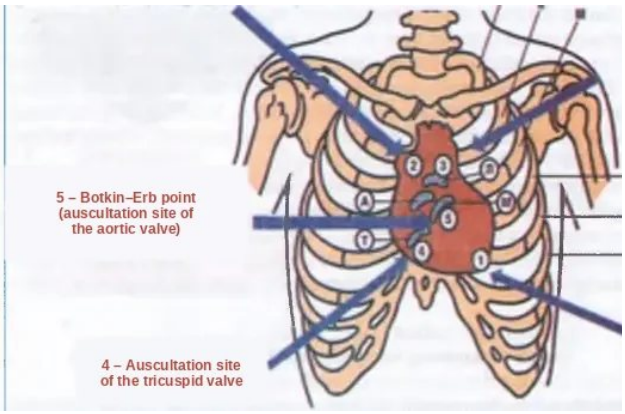


Figure 12.8. *Points of cardiac auscultation (same source, Fig. 37)*

The valves lie very close to each other on the chest (within an area of about 3–4 cm). That is why the sounds are heard not at the valve projections but where the sound is best conducted **in the direction of the blood flow**: the sounds of the mitral valve pass through the left ventricle to the apex, those of the aortic valve through the ascending aorta to the right II intercostal space, those of the pulmonary

artery to the left II space, and those of the tricuspid valve through the right ventricle to the base of the xiphoid process.

REMEMBER

In modern practice, in addition to the points, it is recommended to listen over the whole heart region by **"inching" auscultation** (moving the stethoscope from the apex upward along the left sternal edge, then to the right); otherwise murmurs between the points are missed.

8. Phonocardiography and the physical characteristics of the sounds

Phonocardiography (PCG) is the graphic recording of heart sounds with a microphone. It shows precisely the timing, duration and shape of the sounds and murmurs, but has now largely been replaced by echocardiography.

Table 12.10. *Physical parameters of the heart sounds*

Sound	Duration	Frequency	Timing relative to the ECG
S1	0.09–0.12 s	30–150 Hz	0.02–0.04 s after the onset of the QRS complex
S2	0.05–0.07 s	Up to 150 Hz and higher	Coincides with the end of the T wave
S3	0.03–0.05 s	Low (20–50 Hz)	0.12–0.18 s after S2
S4	0.03–0.05 s	Low	After the P wave, before S1

9. Dynamic auscultation of murmurs

Some murmurs change with position, respiration and special maneuvers; this helps to determine their origin.

Table 12.11. *Effect of various maneuvers on murmurs*

Maneuver	Physiological effect	Effect on murmurs
Inspiration	Venous return to the right heart increases	Right-heart murmurs (tricuspid, pulmonary) become louder — Rivero-Carvallo sign
Expiration	Blood flow to the left heart increases relatively	Left-heart murmurs (mitral, aortic) become louder
Lying on the left side	The apex comes closer to the chest wall	The murmurs of mitral stenosis and regurgitation and S3 and S4 are heard well
Sitting leaning forward	The base of the heart comes closer to the chest wall	The murmur of aortic regurgitation and the pericardial rub become louder
Valsalva maneuver (straining)	Venous return and left ventricular filling decrease	Most murmurs become softer; the murmurs of hypertrophic obstructive cardiomyopathy and mitral prolapse become louder
Standing up from squatting	Venous return decreases	The murmurs of hypertrophic cardiomyopathy and mitral prolapse become louder
Squatting from standing	Venous return and peripheral resistance increase	Most murmurs become louder; the murmur of hypertrophic cardiomyopathy becomes softer
Clenching the fist tightly (isometric exercise)	Peripheral resistance increases	The murmurs of mitral and aortic regurgitation become louder, the murmur of aortic stenosis becomes softer

10. Differential diagnosis of systolic and diastolic murmurs

Table 12.12. *Main causes of systolic murmurs*

Cause	Epicenter	Radiation	Character
Aortic stenosis	II space on the right	To the carotid arteries	Diamond-shaped, harsh, S2 weakened
Mitral regurgitation	Apex	Left axilla	Band-shaped (holosystolic), blowing, S1 weakened
Tricuspid regurgitation	Base of the xiphoid process	To the right of the sternum	Louder on inspiration
Ventricular septal defect	III–IV space at the left sternal edge	Along the sternum, transversely across the chest	Very harsh, with a thrill
Hypertrophic obstructive cardiomyopathy	III–IV space at the left sternal edge and the apex	Little to the carotid arteries	Louder with Valsalva
Mitral valve prolapse	Apex	—	Midsystolic click + late systolic murmur
Pulmonary artery stenosis	II space on the left	To the left clavicle	Diamond-shaped, S2 weakened
Functional (“innocent”)	II–III space on the left, apex	Does not radiate	Short, soft, grade 1–2, changes with position

Table 12.13. *Main causes of diastolic murmurs*

Cause	Phase	Epicenter	Character
Aortic regurgitation	Protodiastolic	Botkin–Erb point, II space on the right	Soft, blowing, decreasing
Pulmonary valve regurgitation (Graham Steell)	Protodiastolic	II space on the left	In pulmonary hypertension
Mitral stenosis	Mesodiastolic and presystolic	Apex	Low-pitched, rumbling, louder in presystole
Tricuspid stenosis	Mesodiastolic and presystolic	Base of the xiphoid process	Louder on inspiration
Austin Flint murmur	Presystolic	Apex	In aortic regurgitation, the regurgitant flow pushes the mitral leaflet
Carey Coombs murmur	Mesodiastolic	Apex	Mitral valvulitis in ARF

10.1. Types of functional murmurs

1. **The “innocent” (physiological) murmur of young people** — in children and adolescents and in thin young people; Still’s murmur (“musical”, midway between the apex and the left sternal edge).
2. **Flow murmurs** — anemia (blood viscosity is reduced), fever, thyrotoxicosis, pregnancy, physical exertion.

3. **Murmurs of relative valvular regurgitation** — in left ventricular dilation the valve ring widens and the leaflets do not close completely (relative mitral and tricuspid regurgitation); they become softer when the dilation decreases with treatment.
4. **Venous hum** (a continuous murmur in the jugular veins) — in children and in anemia; it disappears when the head is turned or the vein is compressed.

11. Auscultatory signs of rhythm disturbances

Table 12.14. *Auscultatory and pulse findings in arrhythmias*

Arrhythmia	Auscultation	Pulse
Sinus tachycardia	Rapid, regular sounds; diastole shortens	Rapid, regular
Extrasystole	A premature contraction (an accentuated S1) followed by a long compensatory pause	Occasional “dropped” beats
Atrial fibrillation	Completely irregular sounds, the intensity of S1 varies (“delirium cordis”)	Irregular, of varying filling, pulse deficit
Complete AV block	A slow rhythm (30–40/min), with an occasional Strazhesko “cannon sound”	Slow, regular
Embryocardia	At a rapid rate the intensity of S1 and S2 becomes equal and the pauses become equal (“pendulum rhythm”)	Rapid

CLINICAL SIGNIFICANCE

Embryocardia (pendulum rhythm) — the intensity of the sounds and the pauses become equal, resembling the fetal heart rhythm; it may be a sign of severe myocardial damage (myocardial infarction, myocarditis, shock).

12. Echocardiography — confirming the auscultatory findings

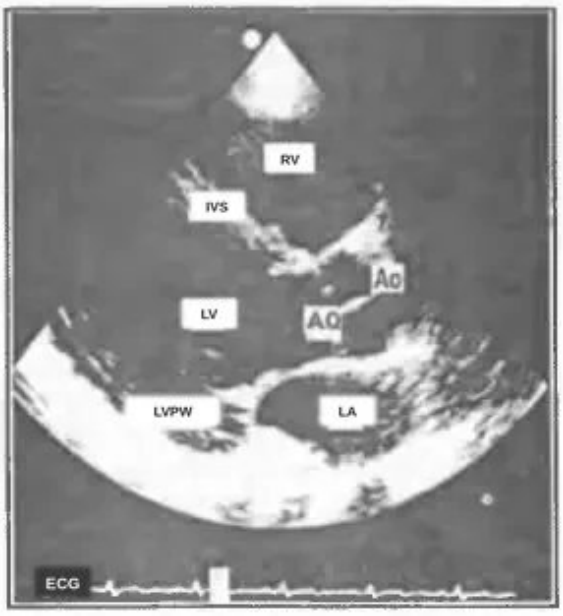


Figure 12.9. *Two-dimensional echocardiogram: RV — right ventricle, IVS — interventricular septum, LV — left ventricle, Ao — aorta (Gadayev A. et al., 2012, Fig. 82)*

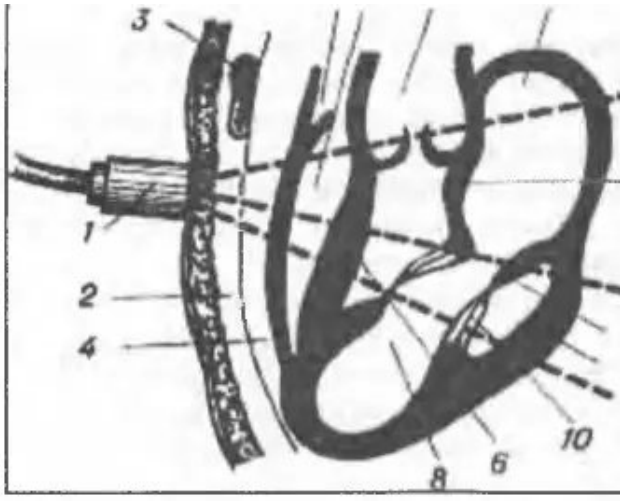


Figure 12.10. *Diagram of the propagation of ultrasound pulses (same source, Fig. 83)*

Echocardiography shows the structure of the valves, the movement of the leaflets, the area of the orifices, the regurgitant jet (color Doppler) and the pressure gradients. **Any diastolic murmur, a systolic murmur louder than grade 3, a murmur accompanied by symptoms, or a murmur combined with ECG and radiographic changes** is an indication for echocardiography.

> **CAUTION**
> The ability to auscultate requires regular practice. According to studies, students and young physicians correctly identify only about half of heart murmurs. It is therefore recommended to listen to every patient, compare the result with the echocardiogram and practice with recorded heart sounds.

13. Auscultation after valve replacement and heart surgery

- **Mechanical prosthetic valves** — distinct, loud, metallic closing (and opening) “clicks” (with a mitral prosthesis in place of S1, with an aortic prosthesis in place of S2); their disappearance or weakening may be a sign of thrombosis.
- **Biological valves** — the sounds are close to normal.
- With an aortic prosthesis a mild systolic murmur is normal; any new **diastolic** murmur raises the suspicion of prosthetic dysfunction or infective endocarditis.

14. Auscultatory signs in congenital heart defects

Although the propaedeutics course mainly studies acquired defects, it is necessary to know the auscultatory signs of the congenital defects seen in adults.

Table 12.15. *Common congenital defects*

Defect	Auscultation	Other signs
Ventricular septal defect	A very harsh pansystolic murmur at the left sternal edge in the III–IV space, radiating across the chest; a thrill	With a small defect the murmur is loud, but the hemodynamics are good (“much ado about nothing”)
Atrial septal defect	Fixed reduplication of S2 — independent of the phases of respiration; a soft systolic murmur in the II space on the left	Often detected in adults; right ventricular enlargement
Patent ductus arteriosus (ductus Botalli)	A continuous systolic-diastolic “machinery” murmur in the II space on the left and below the clavicle	Wide pulse pressure
Coarctation of the aorta	A systolic murmur in the interscapular region (on the left), accentuation of S2 over the aorta	BP high in the arms and low in the legs, delayed femoral pulse; on the radiograph notching of the lower edges of the ribs (usuration)
Pulmonary artery stenosis	A harsh diamond-shaped systolic murmur in the II space on the left, S2 weakened	Right ventricular hypertrophy
Tetralogy of Fallot	Systolic murmur, S2 over the pulmonary artery weakened	Central cyanosis, “drumstick” fingers, squatting position

15. Distinguishing the pericardial friction rub from other murmurs

Table 12.16. *Distinguishing features of the pericardial friction rub*

Feature	Pericardial rub	Intracardiac murmur	Pleuropericardial rub
Relation to the cardiac cycle	In systole and diastole, often three-component	In a particular phase	With the cardiac contractions
Relation to respiration	Persists when the breath is held	Persists	Becomes softer or disappears when the breath is held
Location	In the area of absolute dullness, does not radiate	Radiates from the epicenter along the blood flow	Next to the left cardiac border
Pressure with the stethoscope and leaning forward	Increases	Unchanged	Becomes slightly louder
“Superficial” quality	Right under the ear, very close	From deeper	Superficial

CLINICAL SIGNIFICANCE

To diagnose **acute pericarditis** (ESC), at least two of the following four signs are required: **pericardial chest pain** (reduced on leaning forward), **a pericardial friction rub**, **widespread ST elevation on the ECG** (or PR depression), **new or increasing pericardial fluid**.

16. Useful rules for recording and teaching murmurs and sounds

1. Listen **to the sounds first, then to the murmurs**: the phase of a murmur cannot be determined without placing it relative to S1 and S2.
2. **Attend to one sound at a time**: listen “selectively” first only to S1, then only to S2, then to systole, then to diastole.
3. **In tachycardia** the duration of systole and diastole becomes equal — determine the phases by the carotid pulse.
4. **In atrial fibrillation** the loudness of a murmur varies with the cycle: in the cycle after a long pause the murmur of aortic stenosis becomes louder, while the murmur of mitral regurgitation does not change.
5. Always record the result in the same form: **phase — epicenter — radiation — intensity (Levine) — shape — timbre — relation to the sounds — dynamic maneuvers**.

MEMORY AID

A short formula for remembering a murmur — **“WHERE, WHEN, WHERE TO, HOW LOUD”**: where it is heard best (epicenter), when (systole or diastole), where it radiates, and how loud it is (Levine grade).

17. Auscultatory signs of tricuspid valve defects

Tricuspid regurgitation is most often **relative** (right ventricular dilation due to mitral stenosis, pulmonary hypertension or cor pulmonale), less often organic (infective endocarditis — in injecting drug users, rheumatism, Ebstein anomaly). Signs: a **pansystolic murmur** at the base of the xiphoid process and the lower part of the left sternal edge that **becomes louder on inspiration (Rivero-Carvallo sign)**; a **positive venous pulse**, systolic pulsation of the liver, hepatomegaly, ascites, edema.

Tricuspid stenosis is rare and almost always rheumatic and combined with mitral stenosis; a diastolic murmur at the base of the xiphoid process that becomes louder on inspiration, large “a” waves in the neck veins, marked peripheral edema and ascites (even though pulmonary congestion is slight).

18. Clinical part

Sample case history entry

Auscultation. The heart sounds are regular, 78 per minute. At the apex S1 is weakened, and a systolic murmur is heard at the apex: it merges with S1, occupies the whole of systole, has a blowing timbre, is grade 3 on the Levine scale, radiates to the left axillary region and becomes louder in the left lateral position. Accentuation of S2 over the pulmonary artery. S2 over the aorta is preserved. No extra sounds.

Conclusion: weakening of S1 at the apex, a systolic murmur radiating to the left axilla and accentuation of S2 over the pulmonary artery — characteristic of mitral valve regurgitation.

19. Practical skills

Skill: auscultation of the heart

1. Have the patient lie down or sit, undressed to the waist; ensure quiet in the room.
2. Locate the apex beat by palpation (the first point).
3. Listen to the points in order: apex → II space to the right of the sternum → II space to the left of the sternum → base of the xiphoid process → Botkin–Erb point.
4. At each point: distinguish S1 and S2 (using the carotid pulse), compare the rhythm and rate and the loudness of the sounds (S1 should be louder at the apex and S2 louder at the base).
5. Compare S2 over the aorta and over the pulmonary artery and determine any accentuation.
6. Look for splitting, reduplication and extra sounds (on inspiration and on expiration).
7. If there is a murmur, describe it according to the plan; to check its radiation, move the stethoscope from the epicenter in different directions.
8. Listen again at the apex with the patient lying on the left side and at the Botkin–Erb point with the patient leaning forward.
9. Record the result in the case history.

Review questions

1. State the general rules of cardiac auscultation.
2. Where are the anatomical projections of the heart valves and their auscultation points?
3. In what order is cardiac auscultation performed, and why?
4. What are the components of S1?
5. How is S2 formed?
6. By which signs do you distinguish S1 and S2?
7. What causes weakening of both sounds?
8. Why is S1 accentuated in mitral stenosis?
9. What is accentuation of S2? When is accentuation seen over the aorta and over the pulmonary artery?
10. How do you distinguish physiological and pathological reduplication of S2?

11. What are the types and significance of the gallop rhythm?
12. What does the “quail rhythm” consist of, and which defect is it characteristic of?
13. List the mechanisms of murmur formation.
14. By what plan is a murmur described? State the Levine scale.
15. What are the differences between organic and functional murmurs?

Test questions

1. Where is the mitral valve auscultated?
   A) At the base of the xiphoid process
   B) In the II intercostal space to the left of the sternum
   C) At the cardiac apex
   D) In the II intercostal space to the right of the sternum
   E) At the Botkin–Erb point

2. Where is the Botkin–Erb point located?
   A) In the II intercostal space to the right of the sternum
   B) To the left of the sternum, where the III–IV ribs join the sternum
   C) At the cardiac apex
   D) At the base of the xiphoid process
   E) In the II intercostal space to the left of the sternum

3. What is the main (valvular) component of S1 associated with?
   A) Closure of the atrioventricular valves
   B) Atrial contraction
   C) Rapid ventricular filling
   D) Closure of the pulmonary valve
   E) Closure of the aortic valves

4. Which sign is most reliable for distinguishing S1 from S2?
   A) Timbre
   B) Dependence on respiration
   C) Duration
   D) Coincidence with the apex beat and the carotid pulse
   E) Pitch

5. Which defect is a “snapping” S1 at the apex characteristic of?
   A) Aortic regurgitation
   B) Aortic stenosis
   C) Tricuspid regurgitation
   D) Mitral stenosis
   E) Mitral regurgitation

6. In which condition is accentuation of S2 over the aorta seen?
   A) Mitral stenosis
   B) Cor pulmonale
   C) Aortic regurgitation
   D) Mitral regurgitation
   E) Arterial hypertension

7. What does accentuation of S2 over the pulmonary artery indicate?
   A) Anemia
   B) Raised pressure in the pulmonary circulation
   C) Aortic stenosis
   D) Thyrotoxicosis
   E) Arterial hypertension

8. When is physiological reduplication of S2 heard?
   A) Only on expiration
   B) Continuous
   C) Never
   D) During physical exertion
   E) Only on inspiration

9. What does the “quail rhythm” consist of?
   A) S1 + S2 + S3
   B) Snapping S1 + S2 + mitral opening snap
   C) S1 + S4 + S2
   D) S1 + systolic click + S2
   E) S1 + S2 + pericardial knock

10. The appearance of which sound underlies the protodiastolic gallop rhythm?
    A) A pathological S3
    B) S1
    C) S2
    D) S4
    E) Opening snap

11. Which murmur is almost always regarded as organic?
    A) A faint systolic murmur
    B) A soft, short systolic murmur
    C) The murmur in anemia
    D) A murmur that changes with position
    E) A diastolic murmur

12. Where does the systolic murmur of aortic stenosis radiate?
    A) To the cardiac apex
    B) Does not radiate
    C) To the xiphoid process
    D) To the carotid arteries
    E) To the left axillary region

13. Where does the systolic murmur of mitral regurgitation radiate?
    A) To the neck
    B) To the interscapular region
    C) To the left axillary region
    D) To the carotid arteries
    E) To the right sternal edge

14. Which Levine grade corresponds to a loud murmur accompanied by a thrill?
    A) 1
    B) 2
    C) 3
    D) 4
    E) 6

15. Which sign is characteristic of the pericardial friction rub?
    A) It is heard only in the left lateral position
    B) It disappears when the breath is held
    C) It does not radiate anywhere and becomes louder when pressed with the stethoscope
    D) It is heard only in diastole
    E) It radiates to the carotid arteries

Clinical cases

Case 1

In a 20-year-old student at a medical examination, reduplication of S2 was heard over the pulmonary artery on inspiration; the reduplication disappears on expiration. At the apex a short, soft, grade 1 systolic murmur is heard at the beginning of systole; it disappears on standing and does not radiate anywhere. The cardiac borders are normal, and the sounds are unchanged.

Questions: Interpret the changes found. Are they pathological?

Answer: Reduplication of S2 that appears on inspiration and disappears on expiration is physiological reduplication (on inspiration right ventricular filling increases and the pulmonary valve closes later). A short, soft, faint murmur that does not radiate and varies with position, with unchanged sounds and cardiac borders, is characteristic of a functional (“innocent”) murmur. Both are variants of normal in young people; if in doubt, echocardiography is performed.

Case 2

In a 30-year-old woman, a snapping S1 at the apex, an extra sound after S2 and a low-pitched diastolic murmur increasing in presystole are heard; the murmur is heard well in the left lateral position. Accentuation and reduplication of S2 over the pulmonary artery.

Questions: Interpret the auscultatory findings. What is the rhythm called? Which defect is involved?

Answer: Snapping S1 + S2 + mitral opening snap — the “quail rhythm”; a diastolic murmur at the apex increasing in presystole; accentuation and reduplication of S2 over the pulmonary artery (pulmonary hypertension). All of this is characteristic of mitral stenosis.

Case 3

A 62-year-old man complains of fainting and retrosternal pain on exertion. A harsh, saw-like, diamond-shaped systolic murmur is heard in the II intercostal space to the right of the sternum; it radiates to the carotid arteries, and a systolic thrill is felt in this area. S2 over the aorta is weakened.

Questions: Grade the murmur on the Levine scale. Which defect may be suspected?

Answer: A loud murmur accompanied by a thrill is at least grade 4. A harsh diamond-shaped systolic murmur over the aorta radiating to the carotid arteries, weakening of S2, syncope and angina are characteristic of aortic stenosis; the diagnosis is confirmed by echocardiography.

Quick review

Table 12.17. *Key facts on Topic 12*

Question	Answer
Order of listening	Apex (mitral) → right II (aortic) → left II (pulmonary) → xiphoid (tricuspid) → Botkin–Erb
S1	Closure of the AV valves; loud at the apex; coincides with the pulse
S2	Closure of the semilunar valves; loud at the base
Snapping S1	Mitral stenosis
Weakened S1	Mitral regurgitation, myocardial damage
Accentuated S2 over the aorta	Arterial hypertension, atherosclerosis
Accentuated S2 over the pulmonary artery	Pulmonary hypertension (mitral defects)
Physiological reduplication	Only on inspiration
“Quail rhythm”	Mitral stenosis
Gallop rhythm	Severe myocardial damage (S3)
Levine scale	1–6; grade 4 and above — with a thrill
A diastolic murmur	Almost always organic

TOPIC 13

Symptoms and initial diagnosis of rheumatism (acute rheumatic fever) and primary rheumatic carditis. Mitral defects: mitral valve regurgitation and mitral stenosis

Learning objectives

To teach the student the etiology, the major and minor clinical features of acute rheumatic fever, the symptoms of primary rheumatic carditis and the modern diagnostic criteria (the Jones criteria revised by the AHA in 2015), and also the hemodynamics and the clinical and physical signs of mitral valve regurgitation and mitral stenosis.

By the end of the session, the student should know:

- the definition, etiology and pathogenesis of acute rheumatic fever;
- the major and minor manifestations of rheumatism, the Jones criteria (2015);
- the signs of primary rheumatic carditis;
- the laboratory parameters that determine the activity of the rheumatic process;
- the hemodynamic changes in mitral regurgitation and mitral stenosis;
- the findings of inspection, palpation, percussion, auscultation, radiography, ECG and echocardiography in each defect;
- the grades of severity of mitral stenosis (ESC/EACTS 2025).

The student should be able to:

- take the history from and examine a patient with suspected rheumatism;
- justify the diagnosis by applying the Jones criteria;
- detect mitral defects by physical examination and distinguish them from each other.

1. Acute rheumatic fever (rheumatism)

Acute rheumatic fever (ARF) is a systemic inflammatory disease that develops **2–3 weeks** after tonsillitis or pharyngitis caused by group A beta-hemolytic streptococcus in genetically predisposed individuals and mainly involves the cardiovascular system, the joints, the central nervous system and the skin. It most often occurs in children and adolescents aged **7–15 (20) years**, more often in girls.

Rheumatic heart disease is a chronic disease formed after ARF and characterized by permanent damage to the heart valves (a defect).

> **CAUTION**
> The older literature used the term “rheumatism”. International and CIS classifications now use the terms **“acute rheumatic fever”** and **“chronic rheumatic heart disease”**. The term “rheumatism” in the course syllabus is understood in this sense.

1.1. Etiology and pathogenesis

The main cause is **group A beta-hemolytic streptococcus** (GABHS). **Molecular mimicry** plays the main role in the pathogenesis: the streptococcal M protein and other antigens resemble antigens of the heart tissues (myosin, valves), the synovium of the joints and the basal ganglia of the brain.

The antibodies and T lymphocytes produced against the streptococcus damage the body's own tissues (autoimmune inflammation). The morphological basis is disorganization of the connective tissue (mucoid swelling → fibrinoid changes → granulomatosis — **Aschoff–Talalayev granulomas** → sclerosis).

1.2. Clinical picture

The disease usually begins 2–3 weeks after tonsillitis or pharyngitis: fever (38–39 °C), weakness, sweating, joint pain.

Table 13.1. *Main manifestations of acute rheumatic fever*

Manifestation	Frequency	Features
Carditis	50–70 %	The most important sign, determines the outcome of the disease; endocarditis (valvulitis) predominates, most often of the mitral and aortic valves
Polyarthritis	60–75 %	Large joints (knee, ankle, elbow, shoulder, wrist), migratory, symmetric, severe pain, swelling, redness; resolves completely without deformity; responds rapidly to NSAIDs
Minor chorea (Sydenham chorea)	6–30 %	More often in girls; involuntary, irregular movements of the facial and limb muscles, disturbed handwriting, emotional lability; disappears during sleep; may appear late (after 1–6 months)
Erythema marginatum	4–17 %	Pale pink annular rashes with pale centers on the trunk and proximal parts of the limbs; not itchy, change quickly
Rheumatic nodules	1–7 %	Painless, firm nodules of 0.5–2 cm over the joints, bony prominences and the occipital region; occur together with severe carditis

CLINICAL SIGNIFICANCE

Rheumatic arthritis in adults often takes the form of **arthralgia** and is mild. A classic rule: "**Rheumatism licks the joints but bites the heart**" (Lasègue–Bouillaud) — the joint changes pass without a trace, while the heart is damaged with a permanent defect.

1.3. Primary rheumatic carditis

Rheumatic carditis is rheumatic inflammation of all layers of the heart (endocardium, myocardium, sometimes pericardium); in primary rheumatic carditis **endomyocarditis** is most often seen.

Table 13.2. *Signs of primary rheumatic carditis*

Method of examination	Signs
Complaints	Discomfort or pain in the heart region, palpitations, skipped beats, dyspnea on exertion, weakness
Pulse	Tachycardia out of proportion to the fever (sometimes bradycardia), arrhythmia, a soft and small pulse
Percussion	Widening of the cardiac borders to the left (myocarditis, dilation)
Auscultation	Weakening of S1 at the apex; an S3 or a protodiastolic gallop; a soft blowing systolic murmur at the apex (valvulitis or relative mitral regurgitation); sometimes a mesodiastolic murmur at the apex (Carey Coombs murmur — swelling of the mitral valve leaflets); a protodiastolic murmur over the aorta (aortic valvulitis); in pericarditis — a pericardial friction rub

Method of examination	Signs
ECG	Prolongation of the PQ interval (first-degree AV block), extrasystole, changes in the ST segment and T wave, reduced voltage
Echocardiography	Swelling and thickening of the valves, pathological regurgitation (mitral, aortic), restricted leaflet mobility, pericardial fluid — subclinical carditis is also detected

1.4. Laboratory findings

- In the blood, neutrophilic leukocytosis, **raised ESR** (in severe cases 50–70 mm/h), appearance and rise of **C-reactive protein**;
- dysproteinemia: decreased albumin, increased alpha2- and gamma-globulins, raised fibrinogen, raised seromucoid and sialic acids;
- a rise in the titers of antistreptococcal antibodies: **antistreptolysin O (ASO)**, antistreptokinase, antihyaluronidase;
- a throat swab culture for group A streptococcus or a rapid antigen test.

1.5. Diagnostic criteria: the Jones criteria (AHA, 2015)

Table 13.3. *The revised Jones criteria (2015)*

Criterion	Low-risk populations	Moderate- and high-risk populations
Major criteria	Carditis (clinical and/or subclinical); polyarthritis; chorea; erythema marginatum; subcutaneous nodules	Carditis (clinical and/or subclinical); monoarthritis, polyarthritis or polyarthralgia; chorea; erythema marginatum; subcutaneous nodules
Minor criteria	Polyarthralgia; fever of 38.5 °C or higher; ESR of 60 mm/h and/or CRP of 3.0 mg/dl or higher; prolonged PR interval (unless carditis is counted as a major criterion)	Monoarthralgia; fever of 38.0 °C or higher; ESR of 30 mm/h and/or CRP of 3.0 mg/dl or higher; prolonged PR interval (unless carditis is counted as a major criterion)

REMEMBER

Diagnosis of **initial ARF**: **2 major** or **1 major + 2 minor** criteria together with evidence of a preceding group A streptococcal infection. **Recurrent ARF** (with a history of ARF or rheumatic heart disease): 2 major, 1 major + 2 minor, or **3 minor** criteria. Evidence of streptococcal infection — a raised or rising ASO titer, a positive throat culture for streptococcus or a positive rapid antigen test. Chorea alone is also sufficient for the diagnosis.

MEMORY AID

The major Jones criteria — **“JONES”**: **J** — Joints (polyarthritis); **O** — the heart-shaped “O” (carditis); **N** — Nodules (subcutaneous nodules); **E** — Erythema marginatum; **S** — Sydenham chorea.

CAUTION

The textbook gives three grades of activity of the rheumatic process (maximal — III, moderate — II, minimal — I) (A.I. Nesterov classification). This classification is partly retained in CIS practice, but in international practice the diagnosis is made by the Jones criteria, and carditis is assessed by echocardiography (including subclinical carditis).

2. General information on acquired heart defects

A heart defect is a permanent anatomical change of the valves or orifices that disturbs the blood flow. If a valve does not close completely, it is **regurgitation** (insufficiency); if an orifice is narrowed, it is **stenosis**. If both are present on one valve, it is a **combined (complex)** defect; if several valves are involved, it is a **multivalvular** defect.

Causes: rheumatic heart disease (the main cause in developing countries), infective endocarditis, atherosclerosis and degenerative calcification of the valves (in the elderly), connective tissue diseases, syphilis, mitral valve prolapse, cardiac injuries.

Stages of the course of defects: compensation (the circulation is maintained through hypertrophy and dilation of the cardiac chambers, no complaints) → subcompensation → decompensation (heart failure).

3. Mitral valve regurgitation

Mitral regurgitation is incomplete closure of the mitral valve leaflets during systole with backflow of blood from the left ventricle into the left atrium (regurgitation).

Types: organic (wrinkling and shortening of the leaflets and chordae: rheumatism, endocarditis, prolapse, chordal rupture) and **functional (relative)** — dilation of the valve ring and the left ventricle (dilated cardiomyopathy, myocarditis, aortic defects), papillary muscle dysfunction (myocardial infarction).

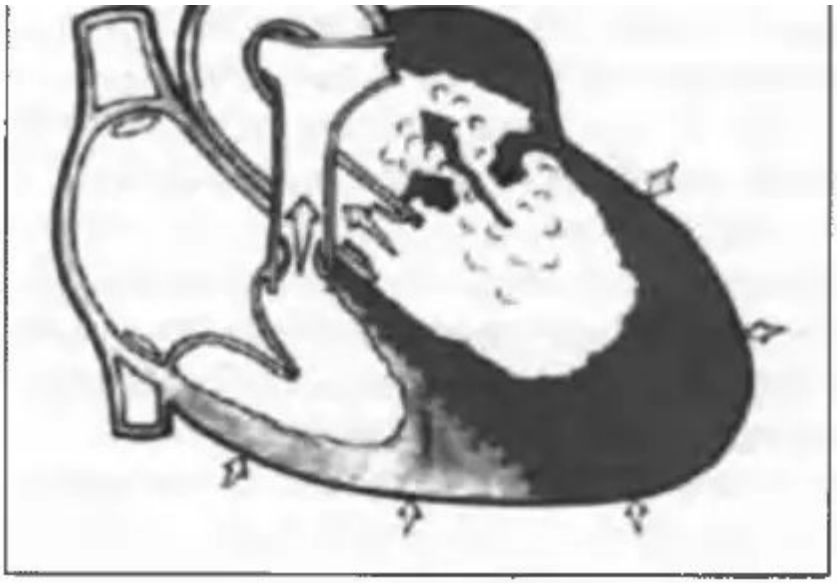


Figure 13.1. *Hemodynamics in mitral valve regurgitation (Gadayev A. et al., 2012, Fig. 120)*

Hemodynamics: in systole part of the blood returns to the left atrium → the left atrium is overfilled with blood and dilates → in diastole excess blood also enters the left ventricle → **dilation and hypertrophy of the left ventricle** (volume overload). Compensation is maintained for a long time; in decompensation the pressure in the left atrium rises, and congestion in the pulmonary circulation, pulmonary hypertension and later right ventricular failure develop.

Table 13.4. *Signs of mitral valve regurgitation*

Method of examination	Signs
Complaints	None in compensation; later dyspnea on exertion, palpitations, weakness; in decompensation — orthopnea, cardiac asthma, edema
Inspection	The apex beat is visible and increased; in decompensation, acrocyanosis
Palpation	The apex beat is displaced to the left (sometimes downward), diffuse, increased, resistant
Percussion	The cardiac borders are widened upward (left atrium) and to the left (left ventricle); mitral configuration (cardiac waist smoothed out)
Auscultation	At the apex S1 is weakened or inaudible; a systolic murmur at the apex — merging with S1, decreasing or band-shaped (holosystolic), blowing, radiating to the left axillary region, louder in the left lateral position and on expiration; accentuation of S2 over the pulmonary artery; often an S3
Pulse and BP	Unchanged; in decompensation, atrial fibrillation
X-ray	The left atrium and left ventricle are enlarged; the contrast-filled esophagus is displaced backward along an arc of large radius; the “rocker” (seesaw) sign of the left atrium in systole
ECG	Signs of left atrial enlargement (P mitrale — a bifid, widened P wave), left ventricular hypertrophy, atrial fibrillation
Echocardiography	The regurgitant jet (color Doppler), its severity, the leaflet changes and the size of the left atrium and ventricle — confirm the diagnosis

REMEMBER

The auscultatory triad of mitral regurgitation: **weakening of S1 at the apex + a systolic murmur merging with S1 and radiating to the left axilla + accentuation of S2 over the pulmonary artery**.

4. Mitral stenosis (narrowing of the left atrioventricular orifice)

Mitral stenosis is narrowing of the left atrioventricular orifice resulting from fusion of the mitral valve leaflets along the commissures, their thickening and calcification, and shortening of the chordae. It is almost always of **rheumatic** origin and is more common in women.

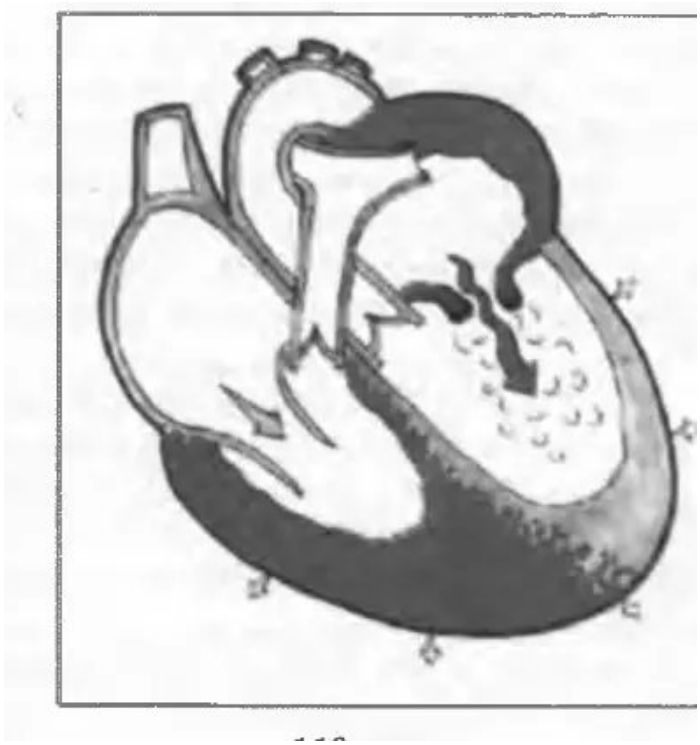


Figure 13.2. *Hemodynamics in mitral stenosis (Gadayev A. et al., 2012, Fig. 119)*

Hemodynamics: the normal mitral orifice area is **4–6 cm^2**. When the orifice narrows, blood cannot pass completely from the left atrium into the left ventricle in diastole → **the pressure in the left atrium rises**, and the atrium hypertrophies and dilates → the pressure is transmitted to the pulmonary veins and capillaries (**"passive" or venous pulmonary hypertension**) → reflex spasm of the pulmonary arterioles (**the Kitaev reflex** — "active", arterial pulmonary hypertension, the "second barrier") → **hypertrophy and dilation of the right ventricle** → right ventricular failure. The left ventricle, being underfilled, remains unchanged or becomes smaller.

Table 13.5. *Grades of severity of mitral stenosis (echocardiography, ESC/EACTS 2025)*

Grade	Mitral orifice area
Normal	4–6 cm^2
Clinically significant (moderate and severe)	1.5 cm^2 or less
Extremely severe	Less than 1.0 cm^2

Table 13.6. *Signs of mitral stenosis*

Method of examination	Signs
Complaints	Dyspnea on exertion (appears early), cough, hemoptysis, palpitations, attacks of cardiac asthma, fatigue; hoarseness (compression of the recurrent nerve by the enlarged left atrium — Ortner syndrome)
Inspection	Facies mitralis (bluish-red patches on the cheeks, cyanosis of the lips); if present from childhood, a "cardiac hump"; cardiac impulse and epigastric pulsation (right ventricle); lag in physical development ("mitral nanism")
Palpation	The apex beat is not displaced and is weak (small left ventricle); a diastolic (presystolic) thrill at the apex; a cardiac impulse at the left sternal edge
Percussion	The cardiac borders are widened upward (left atrium) and to the right (right ventricle, right atrium); mitral configuration

Method of examination	Signs
Auscultation	A “snapping” S1 at the apex; a mitral opening snap (after S2) — the “quail rhythm”; a diastolic murmur at the apex — mesodiastolic, low-pitched, rumbling, increasing in presystole, not radiating, heard well in the left lateral position and after exertion; accentuation and reduplication of S2 over the pulmonary artery; in pulmonary hypertension, the Graham Steell diastolic murmur over the pulmonary artery
Pulse	Small; pulsus differens (smaller on the left arm); often atrial fibrillation and pulse deficit (in which case the presystolic accentuation disappears)
X-ray	The left atrium is enlarged (the contrast-filled esophagus is displaced backward along an arc of small radius), the cardiac waist is smoothed out, the pulmonary artery bulges, the right ventricle is enlarged; venous congestion in the lung roots
ECG	P mitrale, right ventricular hypertrophy, right axis deviation, atrial fibrillation
Echocardiography	The leaflets are thickened, move in one direction (in parallel), and their diastolic opening is restricted; the orifice area and pressure gradient; the left atrium is enlarged, sometimes with a thrombus

REMEMBER

The auscultatory signs of mitral stenosis: **snapping S1 + mitral opening snap (“quail rhythm”) + a diastolic murmur at the apex increasing in presystole + accentuation of S2 over the pulmonary artery**.

MEMORY AID

Mitral stenosis — **“Face, Blood, Quail”**: facies mitralis, hemoptysis, the “quail rhythm”. Mitral regurgitation — **“Soft S1, murmur to the Axilla”**.

Complications: pulmonary edema, pulmonary hemorrhage, atrial fibrillation, thrombus formation in the left atrium and arterial thromboembolism (brain, kidneys, spleen, leg vessels), right ventricular failure, infective endocarditis.

5. Acute rheumatic fever: extended information

5.1. Epidemiology and risk factors

ARF and rheumatic heart disease remain the main cause of acquired heart defects among young people worldwide, especially in developing countries. According to the WHO, **more than 40 million** people in the world live with rheumatic heart disease. Risk factors: crowded and poor living conditions, delayed medical care, inadequate treatment of streptococcal pharyngitis, hereditary predisposition (familial cases, certain HLA antigens, the D8/17 marker of B lymphocytes).

CAUTION

Streptococcal **pharyngitis and tonsillitis** lead to ARF, whereas streptococcal **skin infections** (impetigo) usually do not (they are associated more with glomerulonephritis). That is why, in a child with a sore throat and fever, detecting the streptococcal infection and giving a full 10-day course of antibiotics — **primary prevention** — is extremely important.

5.2. Pathomorphology

Rheumatic inflammation of the connective tissue proceeds in four stages:

1. **Mucoid swelling** (1–2 weeks) — reversible changes;

2. **Fibrinoid changes** (2–4 weeks) — irreversible;
3. **Granulomatosis** — **Aschoff–Talalayev granulomas** (clusters of large basophilic cells in the heart tissue) — 1–3 months;
4. **Sclerosis** — scar formation; in the valves — thickening, fusion and wrinkling of the leaflets, shortening of the chordae → a defect.

Rheumatic valvulitis most often affects the mitral valve (70–75 %), then the aortic valve; the tricuspid and pulmonary valves are rarely involved (and usually together with others).

5.3. Differences between rheumatic and other arthritides

Table 13.7. *Differential diagnosis of rheumatic arthritis*

Feature	Acute rheumatic fever	Rheumatoid arthritis	Reactive arthritis
Age	7–15 (20) years	30–50 years, women	Young men
Preceding infection	Tonsillitis, pharyngitis (2–3 weeks before)	Absent	Intestinal or urogenital infection
Joints involved	Large, migratory	Small (metacarpophalangeal, proximal interphalangeal), symmetric	Large joints of the legs, asymmetric
Course	Resolves without a trace in 1–4 weeks	Chronic, deformity	Weeks to months, sometimes chronic
Heart	Carditis	Few	Few
Laboratory	ASO raised	Rheumatoid factor, anti-CCP antibodies	HLA-B27, chlamydia, etc.
Effect of NSAIDs	Very rapid (within 1–2 days)	Partial	Partial

5.4. Rheumatic chorea

Minor chorea (Sydenham chorea) is autoimmune damage to the subcortical nuclei of the brain. It most often occurs in girls aged 5–15 years, **1–6 months** after the streptococcal infection (so other signs and laboratory activity may be absent). Signs: involuntary, irregular, purposeless movements (facial expressions, limbs), muscle hypotonia, impaired coordination, emotional lability. The movements become stronger with excitement and disappear during sleep. Clinical tests: **the “choreic hand”** (when the arms are stretched forward, the wrist flexes and the fingers extend), **the “milkmaid’s grip”** (rhythmic relaxation when squeezing the physician’s fingers), inability to keep the tongue protruded.

5.5. Prevention of ARF

Table 13.8. *Prevention of ARF*

Type	Purpose	Measures
Primary	Prevention of the first attack	Timely detection of streptococcal pharyngitis and tonsillitis and treatment with penicillin (or amoxicillin) for 10 days
Secondary	Prevention of recurrences	Long-acting benzathine benzylpenicillin intramuscularly every 3–4 weeks

REMEMBER

Duration of secondary prevention (AHA): without carditis — **5 years or until the age of 21** (whichever is longer); carditis without residual valve disease — **10 years or until the age of 21**; carditis with residual valve disease — **10 years or until the age of 40**, sometimes lifelong.

6. Mitral defects: extended information

6.1. Stages of the course of mitral stenosis

The textbook and the Russian literature (the classification of A.N. Bakulev and Ye.A. Damir) distinguish clinical stages of mitral stenosis:

Table 13.9. *Stages of mitral stenosis (after Bakulev–Damir)*

Stage	Content
I — full compensation	No complaints; the defect is detected only on objective examination
II — relative insufficiency	Dyspnea on physical exertion, signs of hypertension in the pulmonary circulation
III — predominance of congestion in the pulmonary circulation	Dyspnea at rest, hemoptysis, attacks of cardiac asthma, slight enlargement of the liver
IV — dystrophic	Congestion in both circulations, atrial fibrillation, enlarged liver, edema
V — terminal	Irreversible changes, no response to treatment

CLINICAL SIGNIFICANCE

In mitral stenosis, patients often deteriorate sharply for the first time against the background of **pregnancy**, **the onset of atrial fibrillation** or **fever**: in these situations the heart rate rises and diastole shortens — there is not enough time for the left ventricle to fill through the narrowed orifice, the pressure in the left atrium rises sharply, and pulmonary edema may develop.

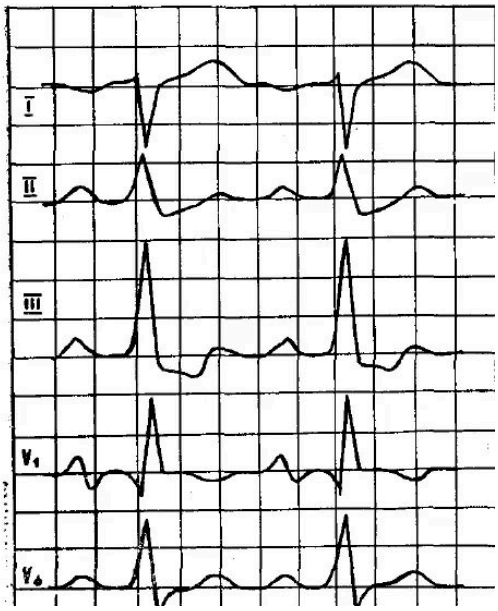


Figure 13.3. *ECG in mitral stenosis: P mitrale, right ventricular hypertrophy (Kamolov N.M., Qo‘yliyev M.U., 1991, Fig. 2)*

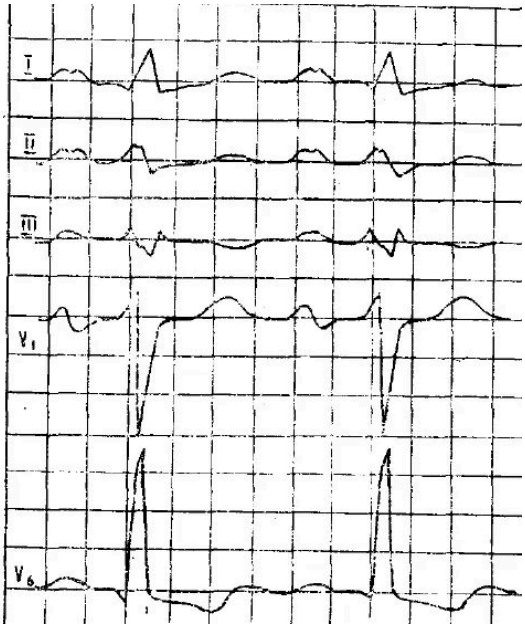


Figure 13.4. *ECG in mitral valve regurgitation: signs of left atrial and left ventricular hypertrophy (Kamolov N.M., Qo‘yliyev M.U., 1991, Fig. 1)*

6.2. Etiological types of mitral regurgitation

Table 13.10. *Types of mitral regurgitation (modern classification)*

Type	Mechanism	Causes
Primary (organic, degenerative)	The valve apparatus itself is damaged	Rheumatic disease, mitral valve prolapse (myxomatous degeneration), chordal rupture, infective endocarditis, congenital defects
Secondary (functional)	The valve structure is unchanged; the left ventricle and the ring are dilated or the papillary muscles are displaced	Dilated cardiomyopathy, ischemic heart disease (after myocardial infarction), dilation of the left atrium in atrial fibrillation
Acute	Develops suddenly	Papillary muscle rupture (myocardial infarction), chordal rupture, infective endocarditis — acute pulmonary edema

6.3. Mitral valve prolapse

Mitral valve prolapse is bulging of the mitral valve leaflets (most often the posterior leaflet) into the cavity of the left atrium by more than 2 mm during systole. It occurs in 2–3 % of the population, more often in asthenic women and in connective tissue dysplasia (Marfan syndrome). In most cases it is asymptomatic; sometimes there is stabbing pain in the heart region, palpitations, weakness. **Auscultation:** a **midsystolic "click"** at the apex followed by a **late systolic murmur**; on standing and during the Valsalva maneuver the click moves closer to S1 and the murmur lengthens. Complications: significant mitral regurgitation, arrhythmias, infective endocarditis (if there is regurgitation).

6.4. Combined mitral defect

In rheumatic disease, mitral stenosis and regurgitation often occur together. It is important to determine which one predominates:

Table 13.11. *Determining the predominant lesion in a combined mitral defect*

Feature	Stenosis predominates	Regurgitation predominates
Apex beat	Weak, not displaced	Increased, displaced to the left
Left border	Unchanged	Widened to the left
S1 at the apex	Snapping	Decreased
Opening snap	Distinct	Poorly expressed or absent
Predominant murmur	Diastolic	Systolic
ECG	Right ventricular hypertrophy	Left ventricular hypertrophy

7. Principles of treatment of mitral defects (in brief)

Knowing the principles of treatment helps the student understand the essence of the disease:

- **Mitral stenosis:** in severe symptomatic stenosis — **percutaneous balloon mitral commissurotomy** (if the leaflets are suitable) or surgical valve replacement; slowing the heart rate with drugs (beta-blockers), diuretics; in atrial fibrillation — **anticoagulants** (warfarin) are mandatory.
- **Mitral regurgitation:** in severe primary regurgitation — valve **repair** (plasty) or replacement; in the secondary type — full treatment of heart failure, sometimes transcatheter edge-to-edge

repair of the leaflets.

- In all patients — secondary prevention of rheumatism, sanitation of foci of infection, dental hygiene.

MEMORY AID
To remember the changes in the cardiac chambers in mitral defects: **"In stenosis — the left atrium and the right heart; in regurgitation — the left atrium and the left ventricle"**. The left atrium enlarges in both defects; the difference lies in the ventricles.

8. Chronic rheumatic heart disease and pregnancy

Rheumatic defects often occur in young women, so pregnancy has a great effect on their course. During pregnancy the circulating blood volume increases by **30–50 %**, and the heart rate and cardiac output rise.

- **In mitral stenosis** pregnancy is most dangerous: the heart rate increases and diastole shortens, the pressure in the left atrium rises — **pulmonary edema** (most often in the II–III trimester and during delivery), atrial fibrillation. In severe mitral stenosis, balloon commissurotomy may be performed before or during pregnancy.
- **In mitral and aortic regurgitation** pregnancy usually goes relatively well (the fall in peripheral resistance reduces the regurgitation).
- **In severe aortic stenosis** the risk is high.

CAUTION
Every pregnant woman with a heart defect should be followed jointly by **a cardiologist and an obstetrician** and monitored by echocardiography. In pregnant women with a mechanical prosthetic valve, anticoagulant therapy is managed according to a special protocol.

9. ECG and echocardiographic signs of rheumatic defects: a summary table

Table 13.12. *Instrumental signs of mitral defects*

Examination	Mitral stenosis	Mitral regurgitation
ECG	P mitrale (a wide, bifid P wave in I, II and aVL, longer than 0.12 s), a biphasic P wave in V_1, right ventricular hypertrophy, right axis deviation; atrial fibrillation	P mitrale, left ventricular hypertrophy, atrial fibrillation
X-ray	Mitral configuration, enlarged left atrium (the esophagus is displaced along an arc of small radius), bulging pulmonary artery, venous congestion in the lungs, Kerley lines	The left atrium and left ventricle are enlarged, the esophagus is displaced along an arc of large radius
Echocardiography	The leaflets are thickened, the commissures fused, the anterior and posterior leaflets move in one direction, the anterior leaflet has a "hockey stick" shape in diastole; orifice area 1.5 cm^2 or less; the left atrium is enlarged	The regurgitant jet (color Doppler), its width and volume; the left atrium and ventricle are enlarged; the leaflet structure
PCG	A loud S1, an opening snap 0.07–0.12 s after S2, a mesodiastolic and presystolic murmur	Reduced amplitude of S1, a systolic murmur merging with S1

REMEMBER

In mitral stenosis, the shorter **“S2 — opening snap” interval**, the more severe the stenosis (the pressure in the left atrium is high, and the valve opens earlier). If this interval on the PCG is less than 0.07 s, the stenosis is severe.

10. Assessing the severity of carditis

Table 13.13. *Severity of rheumatic carditis*

Grade	Clinical and instrumental signs
Subclinical	No clinical signs; only pathological regurgitation (mitral or aortic) on echocardiography
Mild	Tachycardia, weakening of S1, a soft systolic murmur, prolonged PQ; the heart is not enlarged
Average	The above + widening of the cardiac borders, definite valvulitis, mild heart failure
Severe	Cardiomegaly, heart failure, often pancarditis (pericardial rub or fluid), severe regurgitation

11. Differential diagnosis of acute rheumatic fever

Table 13.14. *Distinguishing ARF from other diseases*

Disease	Features distinguishing it from ARF
Reactive arthritis	After an intestinal or urogenital infection, asymmetric arthritis of the legs, conjunctivitis, urethritis, HLA-B27; carditis is rare
Rheumatoid arthritis (juvenile)	Small joints, prolonged morning stiffness, deformity, slow response to NSAIDs
Systemic lupus erythematosus	“Butterfly” rash, nephritis, cytopenias, antinuclear antibodies
Infective endocarditis	Positive blood cultures, vegetations, emboli, splenomegaly
Viral myocarditis	A preceding viral infection, ASO not raised, no arthritis
Poststreptococcal reactive arthritis	Within 10 days of the streptococcal infection, small joints also involved, prolonged course, poor response to NSAIDs, carditis is rare
Lyme disease	Tick bite, erythema migrans, AV block

12. Natural course of mitral defects

Mitral stenosis: it usually takes **10–20 years** after ARF for the defect to form; symptoms often appear at the age of 30–40. Once symptoms have appeared, the 10-year survival of untreated patients is about 50–60 %. Causes of death: progressive heart failure, thromboembolism (stroke), pulmonary edema, infective endocarditis.

Chronic mitral regurgitation remains compensated for many years; left ventricular dilation and a fall in the ejection fraction (below 60 %) are indications for surgery, because the results of the operation are good if it is performed before irreversible myocardial dysfunction develops.

MEMORY AID

Complications of mitral stenosis — **“RASH”**: **R**ight ventricular failure, **A**trial fibrillation, **S**troke and other thromboembolism, **H**emoptysis and pulmonary edema.

13. Clinical part

Comparison of the mitral defects

Table 13.15. *Differences between mitral regurgitation and mitral stenosis*

Feature	Mitral regurgitation	Mitral stenosis
Face	Unchanged	Facies mitralis
Apex beat	Displaced to the left, increased, diffuse	Not displaced, weak
Thrill (“cat’s purr”)	Few	Diastolic, at the apex
Cardiac impulse	Appears late	Early and pronounced
Cardiac borders	Upward and to the left	Upward and to the right
S1 at the apex	Decreased	Snapping
Extra sounds	S3	Mitral opening snap (“quail rhythm”)
Sound	Systolic, radiates to the left axilla	Diastolic (increasing in presystole), does not radiate
S2 over the pulmonary artery	Accentuated (late)	Accentuated and reduplicated (early)
Enlarged chambers	Left atrium, left ventricle	Left atrium, right ventricle

Sample case-history entry (mitral stenosis)

Bluish-red patches on the cheeks, the lips cyanotic. A cardiac impulse at the left sternal edge in the III–IV intercostal spaces, epigastric pulsation. The apex beat is in the V space, 1 cm medial to the midclavicular line, weak, with an area of 1 cm^2. A diastolic thrill at the apex. Borders of relative cardiac dullness: right — 2.5 cm lateral to the sternum, upper — II rib, left — 1 cm medial to the midclavicular line; mitral configuration. At the apex a snapping S1, a mitral opening snap (“quail rhythm”) and a diastolic murmur increasing in presystole; accentuation and reduplication of S2 over the pulmonary artery. The pulse is irregular, 104/min, pulse deficit 18, the pulse on the left arm smaller.

Conclusion: chronic rheumatic heart disease: mitral stenosis; atrial fibrillation; pulmonary hypertension.

14. Practical skills

Skill: justifying the diagnosis of ARF by the Jones criteria

1. Establish whether there was tonsillitis or pharyngitis 2–3 weeks earlier.
2. Look for the major criteria: carditis (complaints, auscultation, ECG, echocardiography), arthritis (location, migratory character), chorea, erythema marginatum, subcutaneous nodules.
3. Determine the minor criteria: fever, arthralgia, ESR and CRP, the PR interval.
4. Check for evidence of streptococcal infection: ASO titer, throat swab.
5. Count the criteria taking into account the risk group of the population and draw a conclusion.

Skill: hearing the diastolic murmur in mitral stenosis

1. Have the patient lie on the left side.

2. Find the apex beat by palpation.
3. Place the bell of the stethoscope lightly over the apex (do not press).
4. Ask the patient to breathe out and hold the breath.
5. Listen for the opening snap after S2, the diastolic murmur and the presystolic accentuation before S1.
6. If the murmur is faint, have the patient sit down and stand up 10–15 times and listen again.

Review questions

1. Define acute rheumatic fever. Which pathogen causes it?
2. What is the role of molecular mimicry in the pathogenesis of ARF?
3. List and describe the main clinical manifestations of ARF.
4. What are the features of rheumatic polyarthritis?
5. What are the auscultatory and ECG signs of primary rheumatic carditis?
6. What are the laboratory indicators of ARF activity?
7. State the Jones criteria (2015). How are initial and recurrent ARF diagnosed?
8. What are the main causes of acquired heart defects?
9. How do the hemodynamics change in mitral regurgitation?
10. List the physical signs of mitral regurgitation.
11. How do the hemodynamics change in mitral stenosis? What is the “second barrier”?
12. Which signs are found on inspection and palpation in mitral stenosis?
13. What are the auscultatory signs of mitral stenosis?
14. What are the grades of severity of mitral stenosis by echocardiography?
15. What are the radiographic and ECG changes in mitral defects?

Test questions

1. Which is the causative agent of acute rheumatic fever?
   A) Group A beta-hemolytic streptococcus
   B) Viridans streptococcus
   C) Pneumococcus
   D) Staphylococcus aureus
   E) Enterococcus

2. How long after the streptococcal infection does ARF develop?
   A) 1–2 days
   B) 2–3 weeks
   C) 3–6 months
   D) 1 year
   E) At the same time

3. Which is the characteristic feature of rheumatic polyarthritis?
   A) Symmetric involvement of the small joints and deformity
   B) Chronic involvement of a single joint
   C) Migratory involvement of the large joints that resolves without deformity
   D) Formation of tophi
   E) Involvement of the spine

4. Which of these is a major Jones criterion?
   A) Prolonged PR interval
   B) Raised ESR
   C) Fever
   D) Chorea
   E) Arthralgia (in a low-risk population)

5. What combination of criteria is needed to diagnose initial ARF?
   A) 1 major criterion
   B) 2 major or 1 major + 2 minor criteria (with evidence of streptococcal infection)
   C) 3 minor criteria
   D) A raised ASO alone
   E) 1 minor criterion

6. Which is the most characteristic ECG change in primary rheumatic carditis?
   A) A pathological Q wave
   B) Delta wave
   C) Shortened QT
   D) ST-segment elevation
   E) Prolongation of the PQ (PR) interval

7. What is the Carey Coombs murmur?
   A) A functional systolic murmur
   B) Pericardial friction rub
   C) The murmur of tricuspid regurgitation
   D) The systolic murmur of aortic stenosis
   E) A mesodiastolic murmur at the apex associated with swelling of the mitral leaflets in ARF

8. How does S1 at the apex change in mitral regurgitation?
   A) Snapping
   B) Reduplicated
   C) Unchanged
   D) Decreased or absent
   E) Metallic

9. Where does the systolic murmur of mitral regurgitation radiate?
   A) To the right sternal edge
   B) To the left axillary region
   C) Does not radiate
   D) To the xiphoid process
   E) To the carotid arteries

10. What is the normal mitral orifice area?
    A) 1–2 cm^2
    B) 4–6 cm^2
    C) 8–10 cm^2
    D) 0.5 cm^2
    E) 15 cm^2

11. According to ESC/EACTS 2025, what is the orifice area of clinically significant mitral stenosis?
    A) Less than 3 cm^2
    B) 1.5 cm^2 or less
    C) 2.5 cm^2
    D) 4 cm^2
    E) 5 cm^2

12. In which directions do the cardiac borders widen in mitral stenosis?
    A) Upward and to the right
    B) Unchanged
    C) Only to the left
    D) To the left and downward
    E) In all directions

13. Which murmur is heard at the apex in mitral stenosis?
    A) Systolic-diastolic
    B) Systolic, radiating to the axilla
    C) Diastolic, increasing in presystole
    D) Pericardial friction rub
    E) Protodiastolic, blowing

14. In mitral stenosis, which component of the diastolic murmur disappears when atrial fibrillation develops?
    A) None
    B) Protodiastolic
    C) Systolic
    D) The presystolic accentuation
    E) Mesodiastolic

15. What is Ortner syndrome (hoarseness) in mitral stenosis due to?
    A) Compression of the left recurrent nerve by the enlarged left atrium
    B) Pulmonary edema
    C) Enlargement of the thyroid gland
    D) Inflammation of the larynx
    E) Bronchial obstruction

Clinical cases

Case 1

A 14-year-old girl had tonsillitis 3 weeks ago. For 5 days she has had a fever of 38.6 °C and swelling and pain in the knee joint; two days later the pain moved to the ankle joint and then to the elbow. There are pink annular rashes on the trunk. Pulse 118/min, S1 at the apex is weakened, a soft systolic murmur. ESR 56 mm/h, CRP markedly raised, ASO 800 IU/ml. ECG: PQ 0.24 s.

Questions: Analyze according to the Jones criteria. What is the diagnosis?

Answer: Major criteria: carditis (tachycardia, weakening of S1, a systolic murmur at the apex), migratory polyarthritis, erythema marginatum — 3. Minor criteria: fever, raised ESR and CRP (the prolonged PR is not counted because carditis is a major criterion). Evidence of a preceding streptococcal infection: tonsillitis and a raised ASO. Diagnosis: acute rheumatic fever, primary

rheumatic carditis (mitral valvulitis), polyarthritis, erythema marginatum. Echocardiography for confirmation.

Case 2

In a 45-year-old man, the apex beat is in the VI space, 2 cm lateral to the midclavicular line, diffuse and strong. The cardiac borders are widened upward and to the left. At the apex S1 is inaudible; a blowing murmur merging with S1 and occupying the whole of systole radiates to the left axillary region. Accentuation of S2 over the pulmonary artery.

Questions: Which defect is this? Which cardiac chambers are enlarged, and why?

Answer: Mitral valve regurgitation. Because blood returns to the left atrium in systole, the left atrium dilates (displacement of the upper border); because excess blood enters the left ventricle in diastole, left ventricular dilation and hypertrophy develop (displacement of the apex beat and the left border to the left). Accentuation of S2 shows raised pressure in the pulmonary circulation.

Case 3

A 32-year-old woman complains of dyspnea on exertion, hemoptysis and palpitations. Echocardiography: mitral orifice area 1.1 cm^2, left atrium 52 mm, a thrombus in the left atrial appendage. ECG: atrial fibrillation.

Questions: Assess the severity of the stenosis. What is expected on auscultation? What complication is she at risk of?

Answer: An orifice area of less than 1.5 cm^2 — clinically significant (severe) mitral stenosis. On auscultation: a snapping S1, an opening snap, a diastolic murmur at the apex (without presystolic accentuation because of the fibrillation), accentuation of S2 over the pulmonary artery; irregular sounds and a pulse deficit. The thrombus in the left atrium and the fibrillation sharply increase the risk of arterial thromboembolism (especially stroke); anticoagulant therapy and surgery (commissurotomy or valve replacement) are considered.

Quick review

Table 13.16. *Key facts on Topic 13*

Question	Answer
Causative agent of ARF	Group A beta-hemolytic streptococcus; 2–3 weeks after tonsillitis
Major Jones criteria	Carditis, polyarthritis, chorea, erythema marginatum, subcutaneous nodules
Minor Jones criteria	Arthralgia, fever, raised ESR/CRP, prolonged PR
Diagnosis	2 major or 1 major + 2 minor (+ evidence of streptococcal infection)
Rheumatic carditis	Tachycardia, weakened S1, a soft systolic murmur at the apex, prolonged PQ
Mitral regurgitation	S1 weakened, a systolic murmur radiating to the axilla; the left atrium and left ventricle enlarged
Mitral stenosis	Facies mitralis, snapping S1, “quail rhythm”, presystolic murmur; the left atrium and right ventricle enlarged
Mitral orifice	Normal 4–6 cm^2; significant stenosis 1.5 cm^2 or less
ECG	P mitrale; atrial fibrillation

TOPIC 14

Symptoms of septic (infective) endocarditis. Aortic defects: aortic valve regurgitation and narrowing of the aortic orifice

Learning objectives

To teach the student the etiology, predisposing factors, clinical signs and modern diagnostic criteria of infective (septic) endocarditis (Duke-ISCVID 2023, ESC 2023), and also the hemodynamics and the clinical and physical signs of aortic valve regurgitation and aortic stenosis.

By the end of the session, the student should know:

- the definition, causative agents and risk factors of infective endocarditis;
- the general, cardiac, vascular (embolic) and immune signs of infective endocarditis;
- the major and minor Duke-ISCVID (2023) criteria;
- the hemodynamic changes in aortic valve regurgitation and aortic stenosis;
- the peripheral signs of aortic regurgitation;
- the findings of inspection, palpation, percussion, auscultation, radiography, ECG and echocardiography in aortic defects;
- the severity criteria of aortic stenosis (ESC/EACTS 2025).

The student should be able to:

- examine a patient with suspected infective endocarditis and identify the main signs;
- explain the rules for taking blood cultures;
- detect and distinguish aortic defects by physical examination.

1. Infective (septic) endocarditis

Infective endocarditis (IE) is microbial inflammation of the heart valves or the parietal endocardium (and also of prosthetic valves, intracardiac devices and the endothelium of large vessels), accompanied by the formation of **vegetations**, destruction of the valves, emboli and immune complications.

> **CAUTION**
> The term **“septic endocarditis”** in the course syllabus is outdated; the term **“infective endocarditis”** is now used (in the textbook, “infectious endocarditis”). The older literature distinguished, by course, **acute** (develops over a few days to weeks, most often staphylococcal) and **subacute** (endocarditis lenta, most often viridans streptococci) types.

1.1. Etiology and risk factors

Table 14.1. *Main causative agents of infective endocarditis*

Pathogen	Features
Staphylococcus aureus	The most common pathogen today; acute, severe course; injecting drug users, vascular catheters, prosthetic valves
Viridans (oral) streptococci	After dental procedures; subacute course (classic endocarditis lenta)
Enterococci	After urogenital and intestinal procedures, in the elderly

Pathogen	Features
Streptococcus gallolyticus (bovis)	Associated with tumors of the colon — colonoscopy is needed
Coagulase-negative staphylococci	Prosthetic valves and devices
HACEK group, fungi, Coxiella, Bartonella	Rare; “culture-negative” endocarditis

Risk factors (cardiac): previous IE, prosthetic valves, rheumatic and degenerative valve defects, congenital heart defects, hypertrophic cardiomyopathy, mitral valve prolapse (with regurgitation), intracardiac devices (pacemaker). **Extracardiac factors:** injecting drug use (the tricuspid valve is most often involved), hemodialysis, long-term venous catheters, dental and other invasive procedures, immunodeficiency, diabetes mellitus.

1.2. Pathogenesis

Bacteremia (tooth extraction, catheter, injection) → microbes settle on the fibrin-platelet layer of the damaged endocardium → **vegetation** (microbes, fibrin, platelets) → destruction and perforation of the valve leaflets (**valvular regurgitation** — most often aortic and mitral), abscesses → detachment of fragments of the vegetation (**emboli**: brain, kidneys, spleen, coronary arteries, lungs — in tricuspid IE) → formation of immune complexes (**vasculitis, glomerulonephritis**, Osler nodes, Roth spots).

1.3. Clinical picture

Table 14.2. *Clinical signs of infective endocarditis*

Group	Signs
General (infectious-toxic)	Fever (the most constant sign, up to 90 %) — often with chills and profuse sweating, hectic or irregular; weakness, loss of appetite, weight loss, joint and muscle pain
Skin	Pallor, a pale yellowish “café-au-lait” skin color (anemia and hemolysis); petechiae (skin, conjunctiva — Lukin–Libman sign, oral mucosa); splinter hemorrhages under the nails
Fingers and palms	Osler nodes — painful, red, pea-sized nodules on the fingertips (immune vasculitis); Janeway lesions — painless hemorrhagic spots on the palms and soles (septic microembolism); in a prolonged course, “drumstick” fingers and “watch-glass” nails
Eyes	Roth spots — hemorrhages with pale centers on the fundus
Heart	Signs of valve damage: a newly appeared regurgitant murmur (most often the diastolic murmur of aortic regurgitation or the systolic murmur of mitral regurgitation) or a change in the character of an existing murmur; heart failure; conduction disturbances (perivalvular abscess)
Vascular (emboli)	Stroke, renal infarction (low back pain, hematuria), splenic infarction (pain under the left costal margin), embolism of the leg arteries, mycotic aneurysms; in tricuspid IE — septic infarcts and foci in the lungs
Internal organs	Splenomegaly, hepatomegaly, glomerulonephritis (proteinuria, hematuria)



Figure 14.1. *Fingers: the "watch-glass" sign (A) — one of the signs of long-standing infective endocarditis (Gadayev A. et al., 2012, Fig. 11)*

CLINICAL SIGNIFICANCE

Fever + a newly appeared heart murmur = infective endocarditis must be ruled out. In any patient with a valve defect, a prolonged fever (more than 1 week) of unknown cause also raises the suspicion of IE. In such a patient, blood cultures are taken **before** antibiotics are started.

MEMORY AID

The peripheral signs of IE — **"Osler — Ouch, Janeway — painless"**: Osler nodes are painful (immune vasculitis), Janeway lesions are painless (microembolism); Roth spots — on the fundus, Lukin spots — on the conjunctiva.

1.4. Laboratory and instrumental investigations

- **Blood cultures** are the main investigation: they are taken before antibiotics are started, **at least 3 times**, from different veins, at 30-minute intervals (each time into aerobic and anaerobic bottles); there is no need to wait for a temperature spike.
- Complete blood count: normochromic **anemia**, leukocytosis (or leukopenia), **markedly raised ESR** (50–70 mm/h); raised CRP and procalcitonin; rheumatoid factor may be positive.
- Urine: proteinuria, microhematuria (glomerulonephritis, renal infarction).
- **Echocardiography** — transthoracic and, if necessary, **transesophageal** (higher sensitivity): vegetations, valve perforation, abscess, new regurgitation.
- Additional: cardiac CT, ^{18}F-FDG PET/CT (especially with prosthetic valves), brain MRI (to detect emboli).

1.5. Diagnostic criteria: Duke-ISCVID (2023)

Table 14.3. *Major Duke-ISCVID (2023) criteria*

Group	Criterion
Microbiological	Isolation of microorganisms typical of IE (Staphylococcus aureus, viridans streptococci, S. gallolyticus, HACEK, enterococci) from 2 or more separate blood cultures; isolation of nontypical microbes from 3 or more cultures; PCR or serological confirmation in the blood for Coxiella burnetii, Bartonella, Tropheryma whipplei
Imaging	On echocardiography and/or cardiac CT: a vegetation, valve perforation or aneurysm, an abscess, a pseudoaneurysm, an intracardiac fistula; new significant valvular regurgitation; new partial dehiscence of a prosthetic valve; metabolic activity on ^{18}F-FDG PET/CT in the region of prosthetic valves and devices
Surgical	Direct visualization of IE during surgery

Table 14.4. *Minor Duke-ISCVID (2023) criteria*

Criterion	Content
Predisposition	Previous IE, a prosthetic valve, valve repair, a congenital heart defect, more than moderate regurgitation or stenosis, intracardiac devices, hypertrophic obstructive cardiomyopathy, injecting drug use
Fever	Above 38.0 °C
Vascular phenomena	Arterial emboli, septic infarcts in the lungs, brain or splenic abscess, mycotic aneurysm, intracranial hemorrhage, conjunctival hemorrhages, Janeway lesions, purulent purpura
Immunological phenomena	Positive rheumatoid factor, Osler nodes, Roth spots, immune-complex glomerulonephritis
Microbiological	A positive blood culture that does not meet the major criterion, or a microbe consistent with IE from another sterile site
Imaging	Activity on ^{18}F-FDG PET/CT within 3 months of valve or device implantation
Physical examination	When echocardiography is unavailable, a new regurgitant murmur on auscultation (an increase in an existing murmur does not count)

REMEMBER

Definite IE: 2 major criteria, or 1 major + 3 minor criteria, or 5 minor criteria (or pathological confirmation). **Possible IE**: 1 major + 1 minor or 3 minor criteria. If the criteria are insufficient or another diagnosis is clear, **IE is rejected**.

2. Aortic valve regurgitation

Aortic regurgitation is incomplete closure of the aortic valve leaflets during diastole with backflow of blood from the aorta into the left ventricle.

Causes: rheumatic heart disease, **infective endocarditis**, atherosclerosis and degenerative changes, a bicuspid aortic valve, dilation of the aortic root (Marfan syndrome, arterial hypertension, aortic dissection), syphilitic aortitis, ankylosing spondylitis, rheumatoid arthritis, SLE.

Hemodynamics: in diastole part of the blood returns from the aorta to the left ventricle → the left ventricle receives blood both from the atrium and from the aorta → **volume overload of the left ventricle, eccentric hypertrophy and marked dilation** (the largest heart — “ox heart”) → in systole a large stroke volume is ejected into the aorta (**the systolic pressure rises**), while in diastole the pressure in the aorta falls sharply because of the backflow (**the diastolic pressure falls**) → **wide pulse pressure**. After a long period of compensation, left ventricular failure develops.

Table 14.5. *Signs of aortic valve regurgitation*

Method of examination	Signs
Complaints	None for a long time; then palpitations, a sensation of pulsation in the neck vessels, dizziness, a tendency to faint (poor blood supply to the brain), angina (reduced coronary perfusion), dyspnea
Inspection	The skin is pale; pulsation of the vessels: “carotid dance”, de Musset sign (nodding of the head), Quincke capillary pulse, pupillary pulsation (Landolfi), pulsation of the uvula (Müller); the apex beat is visible
Palpation	The apex beat is displaced to the left and downward (VI–VII space), diffuse, high, strong, resistant (“dome-shaped”)

Method of examination	Signs
Percussion	The left cardiac border is markedly widened to the left and downward; aortic configuration (cardiac waist deepened); the vascular bundle is widened
Auscultation	S2 over the aorta is weakened or absent; a diastolic (protodiastolic) murmur — begins immediately after S2, soft, blowing, decreasing, heard best at the Botkin–Erb point and in the II space on the right, radiates to the apex; heard well when the patient leans forward and holds the breath after expiration; S1 at the apex is weakened; the Austin Flint murmur at the apex (functional narrowing of the mitral orifice — presystolic); a functional systolic murmur of relative stenosis over the aorta
Auscultation of the vessels	Traube's double sound over the femoral artery; Duroziez's double murmur when the femoral artery is compressed with the stethoscope
Pulse	Rapid, high and large (pulsus celer, altus, magnus) — the "water-hammer" (Corrigan) pulse
Arterial pressure	The systolic pressure is raised, the diastolic pressure markedly lowered (sometimes to 0 — the "infinite sound"), wide pulse pressure (for example, 160/40 mm Hg); the systolic pressure in the legs exceeds that in the arms by more than 20 mm Hg (Hill sign)
X-ray	The left ventricle is markedly enlarged, aortic configuration ("boot"), the ascending aorta is dilated, the amplitude of pulsation is increased
ECG	Left ventricular hypertrophy, left axis deviation
Echocardiography	The diastolic regurgitant jet (color Doppler) and its severity, the size of the left ventricle, the state of the valve leaflets and the aortic root

REMEMBER

The main signs of aortic regurgitation: **aortic configuration + weakening of S2 over the aorta + a protodiastolic murmur at the Botkin–Erb point + a rapid and high pulse + a wide pulse pressure (low diastolic pressure) + peripheral vascular signs**.

MEMORY AID

The peripheral signs of aortic regurgitation — **"Everything pulsates"**: the head (de Musset), the carotid arteries ("carotid dance"), the nail-bed capillaries (Quincke), the pupils (Landolfi), the uvula (Müller); and the femoral artery "speaks" (Traube's double sound, Duroziez's double murmur).

3. Narrowing of the aortic orifice (aortic stenosis)

Aortic stenosis is narrowing of the aortic orifice resulting from fusion, thickening and calcification of the aortic valve leaflets, which impedes the outflow of blood from the left ventricle in systole.

Causes: in the elderly, **degenerative (calcific) stenosis** — now the most common cause; a **congenital bicuspid aortic valve** (at the age of 40–60); rheumatic heart disease (often together with a mitral defect).

Hemodynamics: the normal aortic orifice area is **3–4 cm^2**. When it narrows, a systolic **pressure gradient** appears between the left ventricle and the aorta → pressure overload of the left ventricle → **concentric hypertrophy** (the wall thickens without dilation of the cavity) → long compensation; then increased myocardial oxygen demand and reduced coronary perfusion (angina), limited stroke volume (syncope), and finally left ventricular dilation and failure.

REMEMBER

The classic **symptom triad** of aortic stenosis: **angina, fainting (syncope) and dyspnea (heart failure)**. Once symptoms appear, the average survival without surgery shortens sharply (about 5 years with angina, 3 years with syncope and 2 years with heart failure) — that is why symptomatic severe aortic stenosis is an indication for valve replacement.

Table 14.6. *Signs of aortic stenosis*

Method of examination	Signs
Complaints	None for a long time; then angina on exertion, dizziness and fainting (on exertion), dyspnea, fatigue
Inspection	The skin is pale; the apex beat is barely visible
Palpation	The apex beat is slightly displaced to the left, strong, resistant (concentric hypertrophy); a systolic thrill in the II space to the right of the sternum, transmitted to the carotid arteries
Percussion	The left border is displaced to the left (more marked in decompensation), aortic configuration
Auscultation	A harsh, saw-like, loud, diamond-shaped (crescendo-decrescendo) systolic murmur in the II space on the right, radiating to the carotid arteries, the clavicles and the interscapular region; S2 over the aorta is weakened (it may disappear in severe stenosis); S1 at the apex is weakened; an S4; with a bicuspid valve — an early systolic ejection sound
Pulse	Slow and small (pulsus tardus et parvus), often with a tendency to bradycardia
Arterial pressure	Systolic pressure low or normal, narrow pulse pressure
X-ray	Rounding of the left ventricular contour, aortic configuration, poststenotic dilation of the ascending aorta, valve calcification
ECG	Signs of left ventricular hypertrophy and systolic overload (ST depression, negative T in V_5–V_6), left bundle branch block
Echocardiography	The leaflets are thickened and calcified, their opening restricted; the orifice area, peak velocity and mean gradient — determine the severity

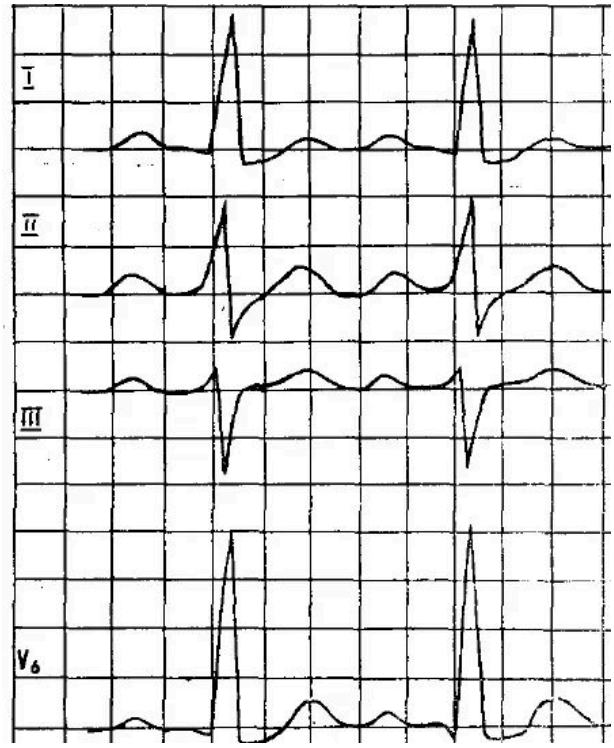


Figure 14.2. *ECG in aortic stenosis: signs of left ventricular hypertrophy and systolic overload (Kamolov N.M., Qo‘yliyev M.U., 1991, Fig. 4)*

Table 14.7. *Criteria of severe aortic stenosis (echocardiography, ESC/EACTS 2025)*

Parameter	Severe stenosis
Aortic orifice area	Less than 1.0 cm^2 (or less than 0.6 cm^2/m^2 of body surface area)
Peak transvalvular flow velocity	4.0 m/s or more
Mean pressure gradient	40 mm Hg or more

MEMORY AID

The numbers of severe aortic stenosis — “1–4–40”: orifice smaller than 1 cm^2, velocity greater than 4 m/s, gradient greater than 40 mm Hg.

4. Infective endocarditis: extended information

4.1. Classification

Table 14.8. *Modern classification of infective endocarditis (ESC, 2023)*

Basis of classification	Types
Location and prosthetic material	Native valve endocarditis; prosthetic valve endocarditis (early — within 1 year, late — after 1 year); endocarditis of intracardiac devices (pacemaker, defibrillator)
Side of the heart	Left-sided (mitral, aortic — more common) and right-sided (tricuspid — injecting drug users, catheters)
Mode of acquisition	Community-acquired; health care–associated (inside and outside hospital); associated with injecting drug use
Activity	Active, healed, relapse (within 6 months with the same pathogen) and reinfection (a different pathogen)

4.2. Differences between the acute and subacute course

Table 14.9. *Types of course of IE*

Feature	Acute IE	Subacute (protracted) IE
Pathogen	Staphylococcus aureus, pneumococcus, gonococcus	Viridans streptococci, enterococci, HACEK
State of the valve	Often a previously healthy valve	Often a previously damaged valve
Onset	Sudden, high fever, chills, sepsis	Gradual, low-grade fever, weakness, weight loss
Duration	Days and weeks	Weeks and months
Valve destruction	Rapid, severe regurgitation	Slow
Immune signs	Few	Many (Osler nodes, glomerulonephritis, splenomegaly, “drumstick” fingers)
Metastatic abscesses	Many (brain, spleen, kidneys)	Few

4.3. Complications of IE

1. **Cardiac complications:** acute valvular regurgitation and **heart failure** (the most common complication and the main cause of death), perivalvular abscess and fistulas, AV block (when the abscess extends to the conduction system), myocarditis, pericarditis, coronary embolism — myocardial infarction.
2. **Emboli:** to the central nervous system (stroke — 20–40 %), spleen, kidneys, mesentery, leg arteries; in right-sided IE — multiple septic infarcts in the lungs.
3. **Mycotic aneurysms** — of the brain and the abdominal aorta.
4. **Renal complications:** immune-complex glomerulonephritis, renal infarction, renal failure.
5. **Sepsis and metastatic abscesses.**

> **CAUTION**
> If a vegetation is **larger than 10 mm**, especially on the mitral valve and mobile, the risk of embolism increases sharply — this is one of the indications for early surgery. Other indications: severe heart failure, uncontrolled infection (abscess, persistent bacteremia), a fungal or multidrug-resistant pathogen.

4.4. Prevention of IE

Antibiotic prophylaxis is recommended only for **high-risk** patients before **high-risk dental procedures** (manipulations of the gingival tissue or the periapical region of the teeth, or perforation of the oral mucosa) (ESC, 2023).

Table 14.10. *High-risk group for IE*

Group	Examples
Prosthetic valves and prosthetic material used for valve repair	Mechanical and biological prostheses, transcatheter valves, annuloplasty rings
Previous IE	—
Congenital heart defects	Untreated cyanotic defects; defects repaired with prosthetic material (within 6 months or if there is a residual shunt)

Group	Examples
Ventricular assist devices	—

REMEMBER
Standard prophylaxis: **amoxicillin 2 g orally 30–60 minutes before** the procedure (children 50 mg/kg); in penicillin allergy — antibiotics of another group. But the most important prevention is **oral hygiene** and dental check-ups 1–2 times a year, as well as proper care of intravenous catheters.

5. Aortic valve regurgitation: extended information

5.1. Acute and chronic aortic regurgitation

Table 14.11. *Differences between acute and chronic aortic regurgitation*

Feature	Chronic	Acute
Causes	Rheumatism, bicuspid valve, aortic root dilation, syphilis	Infective endocarditis, aortic dissection, trauma
The left ventricle	Gradually dilates and hypertrophies (adapts)	Has no time to dilate; the filling pressure rises sharply
Clinical picture	Asymptomatic for many years	Acute pulmonary edema, cardiogenic shock
Peripheral signs	Pronounced	Poorly expressed
Pulse pressure	Wide	Not very wide
A diastolic murmur	Long	Short (the pressures equalize quickly)

5.2. Peripheral signs of aortic regurgitation (full list)

Table 14.12. *Eponymous signs of aortic regurgitation*

Feature	Content
Corrigan pulse (“water hammer”)	A pulse in the radial artery that rises and falls rapidly; felt best when the arm is raised
“Carotid dance”	Visible strong pulsation of the carotid arteries
De Musset’s sign	Nodding of the head with each heartbeat
Quincke capillary pulse	Rhythmic alternation of reddening and blanching under the nail or when the lip mucosa is pressed with a glass slide
Landolfi sign	Rhythmic dilation and constriction of the pupils
Müller sign	Pulsation of the uvula
Becker sign	Pulsation of the retinal arteries on the fundus
Traube’s double sound	Two sounds over the femoral artery resembling a “pistol shot”
Duroziez’s double murmur	Systolic and diastolic murmurs when the femoral artery is compressed with the stethoscope
Hill sign	The systolic pressure in the legs exceeds that in the arms by more than 20 mm Hg (by more than 60 mm in severe regurgitation)
Austin Flint murmur	A presystolic murmur at the apex (the regurgitant jet from the aorta pushes the anterior mitral leaflet and creates a functional mitral stenosis)

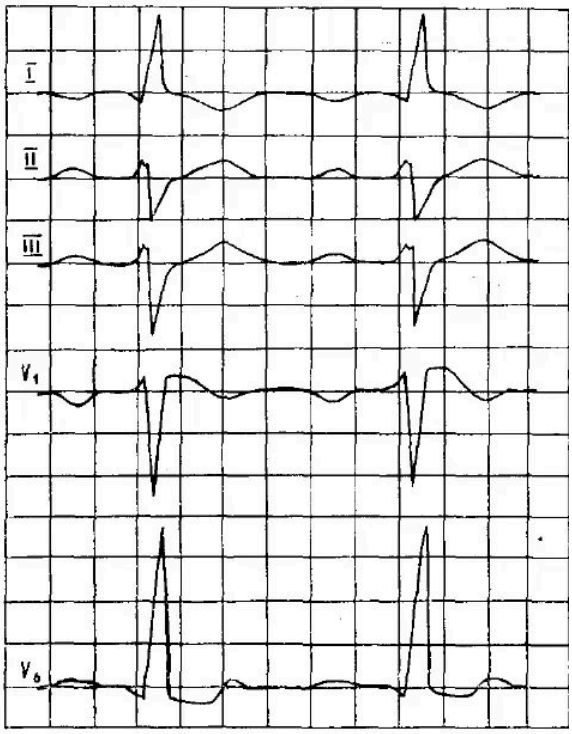


Figure 14.3. *ECG in aortic valve regurgitation: signs of left ventricular hypertrophy (Kamolov N.M., Qo'yliyev M.U., 1991, Fig. 3)*

6. Aortic stenosis: extended information

6.1. Etiology and age-related features

Table 14.13. *Causes of aortic stenosis and age*

Cause	Usual age	Features
Congenital (unicuspid)	Childhood	Severe, manifests early
Bicuspid aortic valve	40–60 years	In 1–2 % of the population; calcifies rapidly; may be accompanied by aortic dilation; early systolic ejection sound
Rheumatic	30–50 years	Usually together with a mitral defect
Degenerative (calcific)	Over 65 years	The most common form; the risk factors resemble those of atherosclerosis (age, hypertension, cholesterol, smoking, diabetes mellitus)

6.2. Distinguishing aortic stenosis from other systolic murmurs

Table 14.14. *Aortic stenosis, hypertrophic cardiomyopathy and mitral regurgitation*

Feature	Aortic stenosis	Hypertrophic obstructive cardiomyopathy	Mitral regurgitation
Epicenter	II space on the right	III–IV space at the left sternal edge	Apex
Radiation	To the carotid arteries	Little to the carotid arteries	To the left axilla
Valsalva maneuver	It is decreased	Increases	It is decreased
Squatting	Increases	It is decreased	Increases
S2 over the aorta	Decreased	Preserved	Preserved
Pulse	Slow, small	Rapidly rising, sometimes bisferiens	Normal

Feature	Aortic stenosis	Hypertrophic obstructive cardiomyopathy	Mitral regurgitation
Echocardiography	Valve calcified, opening restricted	Interventricular septum markedly thickened, valve normal	Regurgitant jet

6.3. Natural course of aortic stenosis and principles of treatment

Aortic stenosis gradually progresses over the years: the orifice area decreases on average by **0.1 cm^2** per year, and the gradient increases by **7 mm Hg** per year. The asymptomatic stage lasts a long time, but the prognosis worsens sharply once symptoms appear.

- **Treatment:** there is no effective drug therapy; in severe symptomatic stenosis — **valve replacement**: surgical (open-heart) or **transcatheter (TAVI)** — in elderly patients and those at high surgical risk; according to the ESC/EACTS 2025 guideline, the choice is based on age and risk.
- **Precautions:** in severe aortic stenosis, potent vasodilators (high-dose nitrates) and strenuous physical exertion are dangerous — they may cause syncope and sudden death.

CLINICAL SIGNIFICANCE

In severe aortic stenosis there is a risk of **sudden cardiac death** (mainly in symptomatic patients). That is why, if an elderly patient has syncope or angina on exertion and a harsh systolic murmur is heard over the aorta, echocardiography should be performed immediately.

7. Mixed and multivalvular aortic defects

A mixed aortic defect (stenosis + regurgitation) occurs in rheumatic and degenerative diseases. On auscultation, **systolic and diastolic murmurs** are heard together over the aorta (a “sawing” quality). Determining the predominant lesion: if stenosis predominates — the pulse is slow and small, the pulse pressure narrow, the thrill distinct; if regurgitation predominates — the pulse is rapid and high, the diastolic pressure low, the peripheral signs pronounced.

Multivalvular mitral-aortic defects (often rheumatic): for example, when mitral stenosis and aortic regurgitation coexist, the peripheral signs of aortic regurgitation are poorly expressed because the mitral stenosis limits the blood flow into the left ventricle — each defect “masks” the signs of the other. In such cases the diagnosis is established by echocardiography.

MEMORY AID

Changes in the left ventricle in aortic defects: **“Stenosis — the wall thickens (pressure overload), regurgitation — the cavity dilates (volume overload)”**. That is why the heart is largest in aortic regurgitation (“ox heart”).

8. The relationship of aortic diseases to aortic valve defects

Aortic valve regurgitation often develops not because of the valve itself but as a result of **dilation of the aortic root and ascending aorta**.

Table 14.15. *Diseases with dilation of the aortic root*

Disease	Features
Marfan syndrome	Tall stature, arachnodactyly, dislocation of the lens; aneurysm and dissection of the aortic root

Disease	Features
Aortopathy associated with a bicuspid aortic valve	Dilation of the ascending aorta, stenosis or regurgitation
Arterial hypertension and atherosclerosis	In the elderly, an S2 with a metallic tinge over the aorta
Syphilitic aortitis (mesaortitis)	Late syphilis, aneurysm of the ascending aorta, narrowing of the coronary ostia
Ankylosing spondylitis (Bechterew disease)	Involvement of the aortic root and valve, AV block
Aortic dissection (acute)	Sudden back pain, acute aortic regurgitation, a BP difference between the arms

CLINICAL SIGNIFICANCE

Signs of **an aneurysm of the ascending aorta**: retrosternal pain, pulsation in the II intercostal space to the right of the sternum, widening of the vascular bundle, hoarseness (an aneurysm of the aortic arch compresses the recurrent nerve), downward tugging of the trachea (Oliver–Cardarelli sign), difficulty swallowing. Confirmation — echocardiography and CT angiography.

9. Instrumental signs of aortic defects: a summary table

Table 14.16. *Instrumental investigations in aortic defects*

Examination	Aortic stenosis	Aortic regurgitation
ECG	Left ventricular hypertrophy and systolic overload (ST depression and negative T in V_5–V_6), left bundle branch block, AV block (if the calcification extends to the conduction system)	Left ventricular hypertrophy and diastolic overload (deep Q, tall R and tall positive T in V_5–V_6)
X-ray	Heart size normal for a long time, rounded apex, poststenotic dilation of the aorta, valve calcification (in the lateral projection)	The left ventricle markedly enlarged (“boot” shape), the aorta dilated, the amplitude of pulsation large
Echocardiography	The leaflets thickened and calcified, the opening amplitude small; the orifice area, gradient and velocity; the left ventricular wall thickened	The diastolic regurgitant jet and its width; the left ventricle dilated; diastolic “fluttering” of the anterior mitral leaflet
PCG	A diamond-shaped systolic murmur; S2 of small amplitude and delayed	A decreasing diastolic murmur beginning after S2

10. Infective endocarditis and aortic valve defects: the relationship

Infective endocarditis often involves **the aortic valve** and causes acute or rapidly progressive **aortic regurgitation**. In a patient with IE the following are dangerous signs:

- a **new** or rapidly increasing aortic diastolic murmur;
- a sharp fall in the diastolic BP;
- acute left ventricular failure (pulmonary edema);
- prolongation of the PR interval and AV block — a sign of **an aortic root abscess** (the abscess lies close to the conduction system).

REMEMBER

In IE, involvement of the aortic valve and heart failure are an indication for **urgent surgery**. A patient with endocarditis should have daily cardiac auscultation and ECG monitoring: the appearance of a new murmur or block indicates a complication.

11. Differential diagnosis of endocarditis, rheumatism and other febrile conditions

Table 14.17. *Conditions presenting with fever and a heart murmur*

Feature	Infective endocarditis	Acute rheumatic fever	Functional murmur in fever
Age	Anyone, most often those with a valve defect or a risk factor	Children and adolescents	Any
Preceding event	Dental procedure, injection, catheter	Tonsillitis (2–3 weeks before)	Any infection
Sound	Regurgitant murmur, new or changing	Soft systolic at the apex, Carey Coombs	Short, soft systolic, disappears when the fever falls
Emboli	Frequent	Absent	Absent
Splenomegaly	Often	Few	Absent
Blood culture	Positive	Negative	Depends on the underlying infection
ASO	Normal or slightly raised	High	Normal
Echocardiography	Vegetations, valve destruction	Valvulitis, regurgitation, valve swelling	Normal

12. Principles of treatment of aortic defects (in brief)

- **Aortic regurgitation:** in severe regurgitation, if symptoms appear or, even without symptoms, if there is left ventricular dilation (end-systolic diameter more than 50 mm) and a fall in the ejection fraction (50 % or less) — **valve replacement**; if the aortic root is dilated — together with the aorta. Drugs (vasodilators) reduce symptoms in inoperable patients.
- **Aortic stenosis:** in severe symptomatic stenosis — **surgical or transcatheter (TAVI)** valve replacement; in asymptomatic severe stenosis — regular follow-up and exercise testing, and in some cases early surgery.
- **Infective endocarditis:** intravenous bactericidal antibiotics for **4–6 weeks** according to the blood culture result; if indicated — early surgery. Treatment is managed by the “endocarditis team” (cardiologist, cardiac surgeon, infectious disease specialist, microbiologist).

MEMORY AID

To remember the surgical signal in aortic defects — **“Symptoms or Damage”**: the appearance of **S**ymptoms (angina, syncope, dyspnea) or **d**amage to the left ventricle (dilation, a fall in the ejection fraction) means that it is time to operate.

13. Clinical part

Comparison of the aortic defects

Table 14.18. *Differences between aortic regurgitation and aortic stenosis*

Feature	Aortic regurgitation	Aortic stenosis
Skin	Pale	Pale
Peripheral pulsations	Pronounced (“carotid dance”, de Musset, Quincke)	Absent
Apex beat	Markedly displaced to the left and downward, diffuse, high, resistant	Slightly to the left, strong, resistant, localized
Thrill (“cat’s purr”)	Absent	Systolic, over the aorta
S2 over the aorta	Decreased or absent	Decreased
Sound	Protodiastolic, at the Botkin–Erb point, radiates to the apex, soft	Systolic, in the II space on the right, radiates to the carotid arteries, harsh
Pulse	Rapid, high, large	Slow, small
Arterial pressure	Systolic raised, diastolic markedly lowered, wide pulse pressure	Systolic low, narrow pulse pressure
The left ventricle	Eccentric hypertrophy and marked dilation	Concentric hypertrophy

Sample case-history entry (aortic regurgitation)

The skin is pale. Strong pulsation of the carotid arteries (“carotid dance”), nodding of the head with the pulse (de Musset sign), Quincke capillary pulse positive. The apex beat is in the VI intercostal space, 2 cm lateral to the left midclavicular line, with an area of 3 cm^2, high, strong, resistant. The left cardiac border is in the VI space, 2 cm lateral to the midclavicular line, the cardiac waist is deepened, the vascular bundle is 7 cm. S2 over the aorta is weakened; at the Botkin–Erb point there is a soft protodiastolic murmur beginning after S2 and radiating to the apex, which becomes louder when the patient leans forward. Traube’s double sound over the femoral artery. Pulse 88/min, regular, rapid, high, large. BP 165/40 mm Hg.

Conclusion: aortic valve regurgitation; left ventricular dilation and hypertrophy.

14. Practical skills

Skill: detecting the peripheral signs of aortic regurgitation

1. Observe the patient sitting: nodding of the head (de Musset), pulsation of the carotid arteries in the neck (“carotid dance”).
2. Press lightly on the tip of a nail and observe the rhythmic alternation of reddening and blanching under the nail (Quincke capillary pulse).
3. Palpate the radial artery; raise the patient’s arm and feel the pulse “knocking” (Corrigan pulse).
4. Measure the arterial pressure and calculate the pulse pressure.
5. Auscultate the femoral artery (Traube’s double sound); press slightly with the stethoscope and look for Duroziez’s double murmur.

6. With the patient leaning forward and holding the breath after expiration, listen for the diastolic murmur at the Botkin–Erb point.

Skill: rules for taking blood cultures (when IE is suspected)

1. They are taken before antibiotics are started (or after a 2–3-day break).
2. At least 3 samples are taken from different peripheral veins at 30-minute intervals.
3. Each sample of 10 ml is inoculated into aerobic and anaerobic bottles.
4. The skin is treated in strict accordance with antiseptic rules; samples are not taken from a catheter.
5. There is no need to wait for a temperature spike (the bacteremia is continuous).

Review questions

1. Define infective endocarditis. Why is the term “septic endocarditis” outdated?
2. What are the main causative agents and risk factors of IE?
3. What is the role of vegetations in the pathogenesis of IE?
4. List the general, skin and finger signs of IE.
5. What are Osler nodes, Janeway lesions, Roth spots and the Lukin–Libman sign?
6. What changes in the heart are found in IE?
7. By what rules are blood cultures taken?
8. State the major and minor Duke-ISCVID (2023) criteria. How is definite IE diagnosed?
9. What are the causes and hemodynamics of aortic regurgitation?
10. List the peripheral signs of aortic regurgitation.
11. How do auscultation, the pulse and the arterial pressure change in aortic regurgitation?
12. What are the causes and hemodynamics of aortic stenosis?
13. What is the classic symptom triad of aortic stenosis?
14. What are the auscultatory and palpatory findings in aortic stenosis?
15. State the echocardiographic criteria of severe aortic stenosis.

Test questions

1. Which is the most common causative agent of infective endocarditis today?
   A) Viridans streptococcus
   B) Staphylococcus aureus
   C) Pneumococcus
   D) Fungi
   E) Enterococcus

2. Which valve does IE most often involve in injecting drug users?
   A) Mitral
   B) Aortic
   C) Tricuspid
   D) Pulmonary artery
   E) All equally

3. What are the painful red nodules on the fingertips called?
   A) Janeway lesions
   B) Roth spots
   C) Osler nodes
   D) Lukin spots
   E) Heberden nodes

4. What are hemorrhages with pale centers on the fundus called?
   A) Quincke sign
   B) Lukin–Libman sign
   C) Osler nodes
   D) Janeway lesions
   E) Roth spots

5. How are blood cultures taken in IE?
   A) Before antibiotics, at least 3 times, from different veins
   B) Only at the temperature peak, once
   C) From a catheter, 1 time
   D) Only from an artery
   E) After antibiotics have been started, once

6. By Duke-ISCVID, which combination of criteria is sufficient for a diagnosis of definite IE?
   A) 1 major criterion
   B) 2 major criteria
   C) 2 minor criteria
   D) 1 major + 1 minor criterion
   E) 3 minor criteria

7. Which pulse is seen in aortic regurgitation?
   A) Slow, small
   B) Alternating
   C) Thready
   D) Rapid, high, large
   E) Paradoxical

8. Where is the diastolic murmur of aortic regurgitation heard best?
   A) At the Botkin–Erb point
   B) In the axillary region
   C) At the cardiac apex
   D) At the base of the xiphoid process
   E) In the II space to the left of the sternum

9. How does the arterial pressure change in aortic regurgitation?
   A) Narrow pulse pressure
   B) Diastolic raised
   C) Systolic raised, diastolic markedly lowered
   D) Systolic and diastolic lowered
   E) Unchanged

10. What is the double sound heard over the femoral artery called?
    A) Austin Flint murmur
    B) Duroziez murmur
    C) Strazhesko sound
    D) Traube’s double sound
    E) Graham Steell murmur

11. What is the functional presystolic murmur heard at the apex in aortic regurgitation called?
    A) Carey Coombs
    B) Traube
    C) Graham Steell
    D) Duroziez
    E) Austin Flint

12. What does the classic symptom triad of aortic stenosis include?
    A) Hemoptysis, dyspnea, edema
    B) Angina, syncope, dyspnea
    C) Cough, fever, weight loss
    D) Headache, vomiting, convulsions
    E) Fever, murmur, anemia

13. Where does the systolic murmur of aortic stenosis radiate?
    A) To the left axillary region
    B) To the carotid arteries
    C) To the cardiac apex
    D) To the epigastric region
    E) Does not radiate

14. According to ESC/EACTS 2025, what is the mean pressure gradient in severe aortic stenosis?
    A) 10 mm Hg
    B) 20 mm Hg
    C) 40 mm Hg or more
    D) 5 mm Hg
    E) 15 mm Hg

15. How does the left ventricle change in aortic stenosis?
    A) Eccentric hypertrophy and marked dilation
    B) Dilation only
    C) Atrophy
    D) Unchanged
    E) Concentric hypertrophy

Clinical cases

Case 1

A 38-year-old man who had rheumatism in childhood is being followed up for mitral regurgitation. A tooth was extracted a month ago. For two weeks his temperature has risen to 38.5–39 °C in the evenings, with chills, profuse sweating and weakness; he has lost 4 kg. On examination the skin is pale yellowish, there are petechiae on the left conjunctiva and painful red nodules on the fingertips. A

diastolic murmur that was not present before has appeared at the Botkin–Erb point. The spleen is palpable. Hb 98 g/l, ESR 62 mm/h.

Questions: Which disease is suspected? Which signs have been found? What should be done to confirm the diagnosis?

Answer: Infective endocarditis (probably viridans streptococcal, after a dental procedure, on the background of a rheumatic defect). Signs: fever, weight loss, “café-au-lait” skin and anemia, the Lukin–Libman sign (conjunctival petechiae), Osler nodes, a new murmur of aortic regurgitation (destruction of the aortic valve), splenomegaly, raised ESR. To confirm: at least 3 blood cultures before antibiotics and echocardiography (transthoracic, then transesophageal) to detect vegetations; assessment according to the Duke-ISCVID criteria.

Case 2

A 70-year-old man reports squeezing retrosternal pain on climbing stairs over the past year and two episodes of fainting on exertion. A harsh systolic murmur is heard in the II space to the right of the sternum, radiating to the carotid arteries, with a systolic thrill at the same site. S2 over the aorta is markedly weakened. Pulse 58/min, slow and small. BP 105/80 mm Hg.

Questions: Which defect is this? Calculate and interpret the pulse pressure. Which echocardiographic parameters indicate severe stenosis?

Answer: Aortic stenosis (in old age most often degenerative calcification) — symptomatic (angina, syncope). Pulse pressure 105 − 80 = 25 mm Hg — narrow (limited stroke volume); the pulse is slow and small (tardus et parvus). Severe stenosis: orifice area less than 1.0 cm^2, peak velocity 4 m/s or more, mean gradient 40 mm Hg or more. Symptomatic severe aortic stenosis is an indication for valve replacement (surgical or transcatheter TAVI).

Case 3

A 28-year-old man (tall, with long limbs and long thin fingers) was referred because of a BP of 150/45 mm Hg found incidentally at a medical examination. Strong pulsation of the carotid arteries, Quincke pulse positive. A diastolic murmur over the aorta.

Questions: Which defect is this, and what is its probable cause? Why is the diastolic pressure low?

Answer: Aortic valve regurgitation; the body build (arachnodactyly, tall stature) suggests Marfan syndrome — valve regurgitation resulting from dilation of the aortic root. The diastolic pressure is low because in diastole blood returns from the aorta to the left ventricle and the pressure in the aorta falls sharply; the large stroke volume raises the systolic pressure — the pulse pressure is wide. On echocardiography the size of the aortic root and the severity of the regurgitation are assessed.

Quick review

Table 14.19. *Key facts on Topic 14*

Question	Answer
Main causative agent of IE	Staphylococcus aureus (today), viridans streptococcus (dental procedures)
Main signs of IE	Fever + a new regurgitant murmur; emboli, splenomegaly, anemia
Peripheral signs	Osler (painful), Janeway (painless), Roth (fundus), Lukin–Libman (conjunctiva)
Blood culture	Before antibiotics, 3 times, from different veins

Question	Answer
Definite IE (Duke-ISCVID)	2 major; 1 major + 3 minor; 5 minor
Aortic regurgitation	Protodiastolic murmur at the Botkin–Erb point; rapid, high pulse; BP 160/40; de Musset, Quincke, Traube
Aortic stenosis	Harsh systolic murmur in the II space on the right, radiating to the carotid arteries; slow, small pulse; triad: angina, syncope, dyspnea
Severe aortic stenosis	Orifice less than 1 cm^2; velocity 4 m/s or more; gradient 40 mm Hg or more
The left ventricle	In regurgitation — eccentric hypertrophy and dilation; in stenosis — concentric hypertrophy

Answers to the test questions

Topic 1: 1-C; 2-E; 3-B; 4-C; 5-C; 6-C; 7-A; 8-E; 9-D; 10-D; 11-B; 12-E; 13-D; 14-B; 15-A.

Topic 2: 1-C; 2-B; 3-D; 4-C; 5-A; 6-B; 7-C; 8-C; 9-E; 10-A; 11-D; 12-A; 13-D; 14-E; 15-E.

Topic 3: 1-D; 2-E; 3-A; 4-C; 5-B; 6-A; 7-D; 8-B; 9-B; 10-B; 11-E; 12-C; 13-B; 14-C; 15-B.

Topic 4: 1-B; 2-D; 3-D; 4-B; 5-C; 6-B; 7-E; 8-C; 9-C; 10-D; 11-E; 12-C; 13-E; 14-A; 15-C.

Topic 5: 1-D; 2-B; 3-B; 4-C; 5-A; 6-B; 7-E; 8-E; 9-B; 10-B; 11-C; 12-C; 13-C; 14-C; 15-D.

Topic 6: 1-E; 2-C; 3-B; 4-E; 5-A; 6-A; 7-D; 8-D; 9-C; 10-A; 11-B; 12-C; 13-B; 14-E; 15-C.

Topic 7: 1-C; 2-B; 3-B; 4-E; 5-B; 6-A; 7-A; 8-E; 9-D; 10-B; 11-B; 12-A; 13-B; 14-D; 15-E.

Topic 8: 1-B; 2-D; 3-C; 4-C; 5-D; 6-E; 7-A; 8-B; 9-A; 10-A; 11-B; 12-E; 13-C; 14-D; 15-D.

Topic 9: 1-D; 2-C; 3-D; 4-E; 5-C; 6-A; 7-D; 8-A; 9-B; 10-D; 11-E; 12-C; 13-B; 14-E; 15-B.

Topic 10: 1-E; 2-A; 3-A; 4-A; 5-C; 6-B; 7-B; 8-C; 9-C; 10-B; 11-D; 12-E; 13-D; 14-B; 15-D.

Topic 11: 1-B; 2-C; 3-A; 4-C; 5-C; 6-B; 7-B; 8-D; 9-A; 10-A; 11-E; 12-E; 13-E; 14-D; 15-B.

Topic 12: 1-C; 2-B; 3-A; 4-D; 5-D; 6-E; 7-B; 8-E; 9-B; 10-A; 11-E; 12-D; 13-C; 14-D; 15-C.

Topic 13: 1-A; 2-B; 3-C; 4-D; 5-B; 6-E; 7-E; 8-D; 9-B; 10-B; 11-B; 12-A; 13-C; 14-D; 15-A.

Topic 14: 1-B; 2-C; 3-C; 4-E; 5-A; 6-B; 7-D; 8-A; 9-C; 10-D; 11-E; 12-B; 13-B; 14-C; 15-E.

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